

問題# 1

A large mobile network operating company is building a machine learning model to predict customers who are likely to unsubscribe from the service. The company plans to offer an incentive for these customers as the cost of churn is far greater than the cost of the incentive. The model produces the following confusion matrix after evaluating on a test dataset of 100 customers: Based on the model evaluation results, why is this a viable model for production?

一家大型行動網路營運公司正在建立一個機器學習模型，以預測哪些客戶可能會取消訂閱服務。該公司計劃為這些客戶提供獎勵，因為客戶流失的成本遠高於獎勵成本。該模型在對100 名客戶的測試資料集進行評估後，產生了以下混淆矩陣：根據模型評估結果，為什麼這是一個可行的生產模型？

n= 100	PREDICTED CHURN	
	Yes	No
ACTUAL Churn Yes	10	4
Actual No	10	76

A. The model is 86% accurate and the cost incurred by the company as a result of false negatives is less than the false positives.

A. 此模型的準確率為86%，且公司因假陰性而產生的成本低於假陽性。

B. The precision of the model is 86%, which is less than the accuracy of the model.

B. 此模型的精確率為86%，低於模型的準確率。

C. The model is 86% accurate and the cost incurred by the company as a result of false positives is less than the false negatives.

C. 此模型的準確率為86%，且公司因假陰性而產生的成本低於假陰性。

D. The precision of the model is 86%, which is greater than the accuracy of the model.

D. 此模型的精確率為86%，高於模型的準確率。

問題# 2

A Machine Learning Specialist is designing a system for improving sales for a company. The objective is to use the large amount of information the company has on users' behavior and product preferences to predict which products users would like based on the users' similarity to other users. What should the Specialist do to meet this objective?

一位機器學習專家正在設計一個系統來提高一家公司的銷售額。其目標是利用公司掌握的大量用戶

行為和產品偏好訊息，根據用戶與其他用戶的相似性來預測用戶會喜歡哪些產品。專家該怎麼做才能實現這個目標？

A. Build a content-based filtering recommendation engine with Apache Spark ML on Amazon EMR.

A. 使用Amazon EMR 上的Apache Spark ML 建立基於內容的過濾推薦引擎。

B. Build a collaborative filtering recommendation engine with Apache Spark ML on Amazon EMR.

B. 使用Amazon EMR 上的Apache Spark ML 建立協同過濾推薦引擎。

C. Build a model-based filtering recommendation engine with Apache Spark ML on Amazon EMR.

C. 使用Amazon EMR 上的Apache Spark ML 建立基於模型的過濾推薦引擎。

D. Build a combinative filtering recommendation engine with Apache Spark ML on Amazon EMR.

D. 使用Amazon EMR 上的Apache Spark ML 建置組合過濾推薦引擎。

問題# 3

A Mobile Network Operator is building an analytics platform to analyze and optimize a company's operations using Amazon Athena and Amazon S3. The source systems send data in CSV format in real time. The Data Engineering team wants to transform the data to the Apache Parquet format before storing it on Amazon S3. Which solution takes the LEAST effort to implement?

一家行動網路營運商正在建立一個分析平台，使用Amazon Athena 和Amazon S3 分析和優化公司的營運。來源系統即時發送.CSV 格式的資料。資料工程團隊希望將資料儲存在Amazon S3 上之前，將其轉換為Apache Parquet 格式。哪一種解決方案實施起來最省力？

A. Ingest CSV data using Apache Kafka Streams on Amazon EC2 instances and use Kafka Connect S3 to serialize data as Parquet.

A. 使用Amazon EC2 執行個體上的Apache Kafka Streams 擷取.CSV 數據，並使用Kafka Connect S3 將資料序列化為Parquet。

B. Ingest CSV data from Amazon Kinesis Data Streams and use Amazon Glue to convert data into Parquet.

B. 從Amazon Kinesis Data Streams 提取.CSV 數據，並使用Amazon Glue 將數據轉換為Parquet。

C. Ingest CSV data using Apache Spark Structured Streaming in an Amazon EMR cluster and use Apache Spark to convert data into Parquet.

C. 使用Amazon EMR 叢集中的Apache Spark Structured Streaming 提取.CSV 數據，並使用Apache Spark 將資料轉換為Parquet。

D. Ingest CSV data from Amazon Kinesis Data Streams and use Amazon Kinesis Data Firehose to convert data into Parquet.

D. 從Amazon Kinesis Data Streams 提取.CSV 數據，並使用Amazon Kinesis Data Firehose 將數據轉換為Parquet。

問題# 4

A city wants to monitor its air quality to address the consequences of air pollution. A Machine Learning Specialist needs to forecast the air quality in parts per million of contaminants for the next 2 days in the city. As this is a prototype, only daily data from the last year is available. Which model is MOST likely to provide the best results in Amazon SageMaker?

某城市希望監測其空氣質量，以應對空氣污染造成的後果。機器學習專家需要預測該城市未來2天的空氣品質（以百萬分率計算）。由於這是一個原型，因此只有去年的每日數據可用。哪種模型最有可能在Amazon SageMaker 中提供最佳結果？

A. Use the Amazon SageMaker k-Nearest-Neighbors (kNN) algorithm on the single time series consisting of the full year of data with a predictor_type of regressor.

A. 在由全年資料組成的單一時間序列上使用Amazon SageMaker k-Nearest-Neighbors (kNN) 演算法，其predictor_type 為regressor。

B. Use Amazon SageMaker Random Cut Forest (RCF) on the single time series consisting of the full year of data.

B. 在由全年資料組成的單一時間序列上使用Amazon SageMaker Random Cut Forest (RCF)。

C. Use the Amazon SageMaker Linear Learner algorithm on the single time series consisting of the full year of data with a predictor_type of regressor.

C. 在全年資料組成的單一時間序列上使用Amazon SageMaker Linear Learner 演算法，其predictor_type 為regressor。

D. Use the Amazon SageMaker Linear Learner algorithm on the single time series consisting of the full year of data with a predictor_type of classifier.

D. 在全年資料組成的單一時間序列上使用Amazon SageMaker Linear Learner 演算法，其predictor_type 為classifier。

問題# 5

A Data Engineer needs to build a model using a dataset containing customer credit card information. How can the Data Engineer ensure the data remains encrypted and the credit card information is secure?

資料工程師需要使用包含客戶信用卡資訊的資料集來建立模型。資料工程師如何確保資料保持加密並且信用卡資訊安全？

A. Use a custom encryption algorithm to encrypt the data and store the data on an Amazon SageMaker instance in a VPC. Use the SageMaker DeepAR algorithm to randomize the credit card numbers.

A. 使用自訂加密演算法加密數據，並將資料儲存在VPC 中的Amazon SageMaker 實例上。使用SageMaker DeepAR 演算法隨機化信用卡號碼。

eMaker DeepAR 演算法隨機化信用卡號。

B. Use an IAM policy to encrypt the data on the Amazon S3 bucket and Amazon Kinesis to automatically discard credit card numbers and insert fake credit card numbers.

B. 使用IAM 策略加密Amazon S3 儲存桶上的數據，並使用Amazon Kinesis 自動丟棄信用卡號並插入假信用卡號。

C. Use an Amazon SageMaker launch configuration to encrypt the data once it is copied to the SageMaker instance in a VPC. Use the SageMaker principal component analysis (PCA) algorithm to reduce the length of the credit card numbers.

C. 使用Amazon SageMaker 啟動配置在將資料複製到VPC 中的SageMaker 實例後對其進行加密。使用SageMaker 主成分分析(PCA) 演算法縮短信用卡號的長度。

D. Use AWS KMS to encrypt the data on Amazon S3 and Amazon SageMaker, and redact the credit card numbers from the customer data with AWS Glue.

D. 使用AWS KMS 加密Amazon S3 和Amazon SageMaker 上的數據，並使用AWS Glue 從客戶資料中刪除信用卡號。

問題# 6

A Machine Learning Specialist is using an Amazon SageMaker notebook instance in a private subnet of a corporate VPC. The ML Specialist has important data stored on the Amazon SageMaker notebook instance's Amazon EBS volume, and needs to take a snapshot of that EBS volume. However, the ML Specialist cannot find the Amazon SageMaker notebook instance's EBS volume or Amazon EC2 instance within the VPC. Why is the ML Specialist not seeing the instance visible in the VPC?

機器學習專家正在企業VPC 的私有子網路中使用Amazon SageMaker 筆記本實例。機器學習專家在Amazon SageMaker 筆記本實例的Amazon EBS 磁碟區上儲存了重要數據，需要對該EBS 磁碟區進行快照。但是，機器學習專家無法在VPC 中找到Amazon SageMaker 筆記本實例的EBS 磁碟區或Amazon EC2 執行個體。為什麼機器學習專家在VPC 看不到該實例？

A. Amazon SageMaker notebook instances are based on the EC2 instances within the customer account, but they run outside of VPCs.

A. Amazon SageMaker 筆記型電腦執行個體是基於客戶帳戶內的EC2 執行個體，但它們在VPC 之外運作。

B. Amazon SageMaker notebook instances are based on the Amazon ECS service within customer accounts.

B. Amazon SageMaker 筆記本執行個體是基於客戶帳戶內的Amazon ECS 服務。

C. Amazon SageMaker notebook instances are based on EC2 instances running within AWS service accounts.

C. Amazon SageMaker 筆記型電腦執行個體是基於在AWS 服務帳戶內執行的EC2 執行個體。

D. Amazon SageMaker notebook instances are based on AWS ECS instances running within AWS service accounts.

D. Amazon SageMaker 筆記本執行個體是基於在AWS 服務帳戶內執行的AWS ECS 執行個體。

問題# 7

A Machine Learning Specialist is building a model that will perform time series forecasting using Amazon SageMaker. The Specialist has finished training the model and is now planning to perform load testing on the endpoint so they can configure Auto Scaling for the model variant. Which approach will allow the Specialist to review the latency, memory utilization, and CPU utilization during the load test?

機器學習專家正在建立一個模型，該模型將使用Amazon SageMaker 執行時間序列預測。專家已完成模型訓練，現在正計劃在終端節點上執行負載測試，以便他們可以為模型變體配置Auto Scaling。哪種方法可以讓專家在負載測試期間查看延遲、記憶體利用率和CPU 使用率？

A. Review SageMaker logs that have been written to Amazon S3 by leveraging Amazon Athena and Amazon QuickSight to visualize logs as they are being produced.

A. 利用Amazon Athena 和Amazon QuickSight 查看已寫入Amazon S3 的SageMaker 日誌，以便在產生日誌時進行視覺化。

B. Generate an Amazon CloudWatch dashboard to create a single view for the latency, memory utilization, and CPU utilization metrics that are outputted by Amazon SageMaker.

B. 產生Amazon CloudWatch 儀表板，以建立Amazon SageMaker 輸出的延遲、記憶體利用率和CPU 使用率指標的單一視圖。

C. Build custom Amazon CloudWatch Logs and then leverage Amazon ES and Kibana to query and visualize the log data as it is generated by Amazon SageMaker.

C. 建立自訂Amazon CloudWatch Logs，然後利用Amazon ES 和Kibana 查詢和視覺化Amazon SageMaker 產生的日誌資料。

D. Send Amazon CloudWatch Logs that were generated by Amazon SageMaker to Amazon ES and use Kibana to query and visualize the log data.

D. 將Amazon SageMaker 產生的Amazon CloudWatch Logs 傳送至Amazon ES，並使用Kibana 查詢和視覺化日誌資料。

問題# 8

A manufacturing company has structured and unstructured data stored in an Amazon S3 bucket. A Machine Learning Specialist wants to use SQL to run queries on this data. Which solution requires the LEAST effort to be able to query this data?

一家製造公司在Amazon S3 儲存桶中儲存了結構化和非結構化資料。機器學習專家希望使用SQL 對這些資料執行查詢。哪種解決方案需要最少的努力才能查詢這些資料？

A. Use AWS Data Pipeline to transform the data and Amazon RDS to run queries.

A. 使用AWS Data Pipeline 轉換資料並使用Amazon RDS 執行查詢。

B. Use AWS Glue to catalogue the data and Amazon Athena to run queries.

B. 使用AWS Glue 對資料進行分類並使用Amazon Athena 執行查詢。

C. Use AWS Batch to run ETL on the data and Amazon Aurora to run the queries.

C. 使用AWS Batch 對資料執行ETL 並使用Amazon Aurora 執行查詢。

D. Use AWS Lambda to transform the data and Amazon Kinesis Data Analytics to run queries.

D. 使用AWS Lambda 轉換資料並使用Amazon Kinesis Data Analytics 執行查詢。

問題# 9

A Machine Learning Specialist is developing a custom video recommendation model for an application. The dataset used to train this model is very large with millions of data points and is hosted in an Amazon S3 bucket. The Specialist wants to avoid loading all of this data onto an Amazon SageMaker notebook instance because it would take hours to move and will exceed the attached 5 GB Amazon EBS volume on the notebook instance. Which approach allows the Specialist to use all the data to train the model?

機器學習專家正在為應用程式開發自訂影片推薦模型。用於訓練此模型的資料集非常龐大，包含數百萬個資料點，並託管在Amazon S3 儲存桶中。專家希望避免將所有這些資料載入到Amazon SageMaker 筆記本實例上，因為移動這些資料需要數小時，並且會超出筆記本執行個體上附加的5 GB Amazon EBS 磁碟區。哪種方法允許專家使用所有資料來訓練模型？

A. Load a smaller subset of the data into the SageMaker notebook and train locally. Confirm that the training code is executing and the model parameters seem reasonable. Initiate a SageMaker training job using the full dataset from the S3 bucket using Pipe input mode.

A. 將一小部分資料載入到SageMaker 筆記本中並在本地進行訓練。確認訓練程式碼正在執行，且模型參數似乎合理。使用管道輸入模式，使用S3 儲存桶中的完整資料集啟動SageMaker 訓練作業。

B. Launch an Amazon EC2 instance with an AWS Deep Learning AMI and attach the S3 bucket to the instance. Train on a small amount of the data to verify the training code and hyperparameters. Go back to Amazon SageMaker and train using the full dataset.

B. 使用AWS Deep Learning AMI 啟動Amazon EC2 實例，並將S3 儲存桶附加到該執行個體。使用少量資料進行訓練以驗證訓練代碼和超參數。返回Amazon SageMaker 並使用完整資料集進行訓練。

C. Use AWS Glue to train a model using a small subset of the data to confirm that the data will be compatible with Amazon SageMaker. Initiate a SageMaker training job using the full dataset from the S3 bucket using Pipe input mode.

C. 使用AWS Glue 使用一小部分資料訓練模型，以確認資料與Amazon SageMaker 相容。使用管道輸入模式，使用S3 儲存桶中的完整資料集啟動SageMaker 訓練作業。

D. Load a smaller subset of the data into the SageMaker notebook and train locally. Confirm that the training code is executing and the model parameters seem reasonable. Launch an Amazon EC2 instance with an AWS Deep Learning AMI and attach the S3 bucket to train the full dataset.

D. 將一小部分資料載入到SageMaker 筆記本中並在本地進行訓練。確認訓練程式碼正在執行，且模型參數似乎合理。使用AWS Deep Learning AMI 啟動Amazon EC2 實例，並附加S3 儲存桶以訓練完整資料集。

問題# 10

A Machine Learning Specialist has completed a proof of concept for a company using a small data sample, and now the Specialist is ready to implement an end- to-end solution in AWS using Amazon SageMaker. The historical training data is stored in Amazon RDS. Which approach should the Specialist use for training a model using that data?

一位機器學習專家使用小型資料樣本為一家公司完成了概念驗證，現在該專家已準備好使用Amazon SageMaker 在AWS 中實施端到端解決方案。歷史訓練資料儲存在Amazon RDS 中。專家應使用哪種方法使用該資料訓練模型？

A. Write a direct connection to the SQL database within the notebook and pull data in.

A. 在筆記本中寫入與SQL 資料庫的直接連線並擷取資料。

B. Push the data from Microsoft SQL Server to Amazon S3 using an AWS Data Pipeline and provide the S3 location within the notebook.

B. 使用AWS Data Pipeline 將資料從Microsoft SQL Server 推送到Amazon S3，並在筆記本中提供S3 位置。

C. Move the data to Amazon DynamoDB and set up a connection to DynamoDB within the notebook to pull data in.

C. 將資料移至Amazon DynamoDB，並在筆記本中設定與DynamoDB 的連接以提取資料。

D. Move the data to Amazon ElastiCache using AWS DMS and set up a connection within the notebook to pull data in for fast access.

D. 使用AWS DMS 將數據移至Amazon ElastiCache，並在筆記本中設定連接以提取數據，以實現快速存取。

問題# 11

A Machine Learning Specialist receives customer data for an online shopping website. The data includes demographics, past visits, and locality information. The Specialist must develop a machine learning approach to identify the customer shopping patterns, preferences, and trends to enhance the website for better service and smart recommendations. Which solution should the Specialist recommend?

機器學習專家收到線上購物網站的客戶資料。數據包括人口統計、過去訪問和位置資訊。專家必須開發一種機器學習方法來識別客戶的購物模式、偏好和趨勢，以增強網站，提供更好的服務和智慧

推薦。專家應該推薦哪一種解決方案？

A. Latent Dirichlet Allocation (LDA) for the given collection of discrete data to identify patterns in the customer database.

A. 對給定的離散資料集合使用潛在狄利克雷分配(LDA) 來識別客戶資料庫中的模式。

B. A neural network with a minimum of three layers and random initial weights to identify patterns in the customer database.

B. 具有至少三層和隨機初始權重的神經網路來識別客戶資料庫中的模式。

C. Collaborative filtering based on user interactions and correlations to identify patterns in the customer database.

C. 基於使用者互動和相關性的協同過濾來識別客戶資料庫中的模式。

D. Random Cut Forest (RCF) over random subsamples to identify patterns in the customer database.

D. 對隨機子樣本使用隨機砍伐森林(RCF) 來識別客戶資料庫中的模式。

問題# 12

A Machine Learning Specialist is working with a large company to leverage machine learning within its products. The company wants to group its customers into categories based on which customers will and will not churn within the next 6 months. The company has labeled the data available to the Specialist. Which machine learning model type should the Specialist use to accomplish this task?

一位機器學習專家正在與一家大型公司合作，在其產品中利用機器學習。該公司希望根據未來6個月內哪些客戶會流失，哪些不會流失，將客戶分為幾類。該公司已標記了可供專家使用的數據。專家應該使用哪種機器學習模型來完成此任務？

A. Linear regression.

A. 線性迴歸。

B. Classification.

B. 分類。

C. Clustering.

C. 聚類。

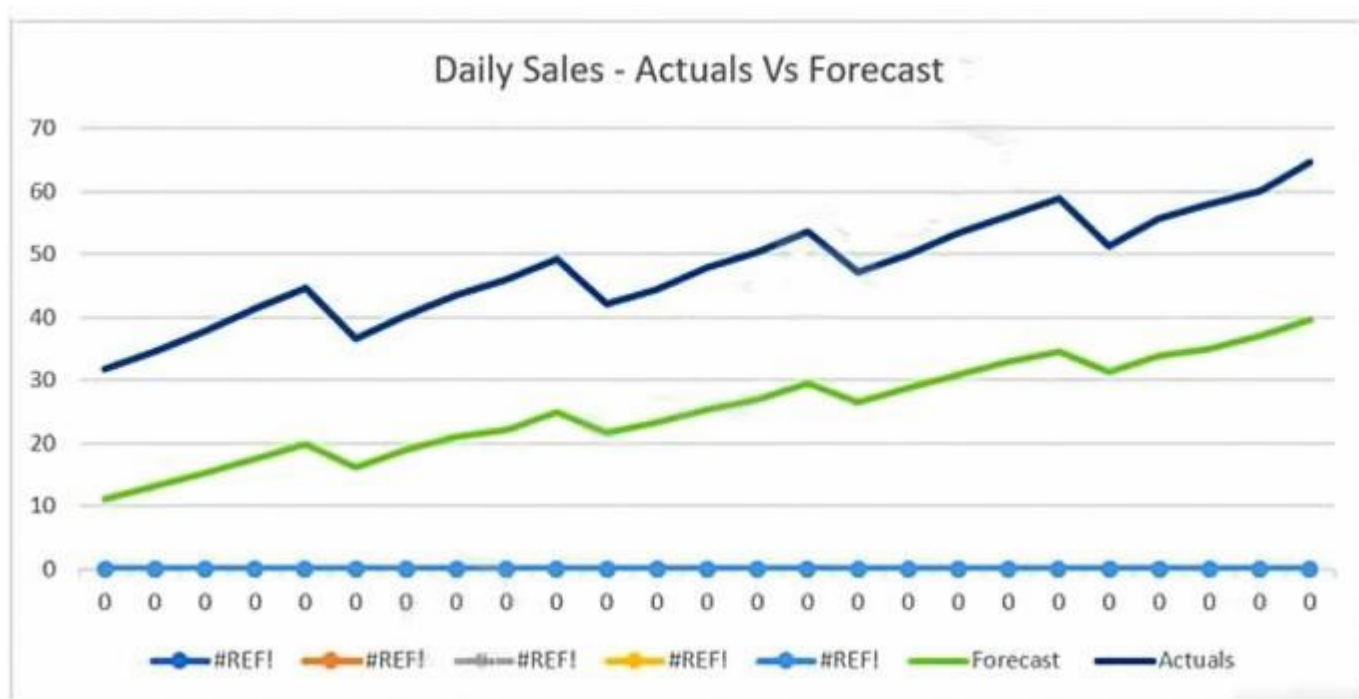
D. Reinforcement learning.

D. 強化學習。

問題# 13

The displayed graph is from a forecasting model for testing a time series. Considering the graph only, which conclusion should a Machine Learning Specialist make about the behavior of the model?

顯示的圖表來自用於測試時間序列的預測模型。只考慮圖表，機器學習專家應該對模型的行為得出什麼結論？



A. The model predicts both the trend and the seasonality well.

A. 此模型可以很好地預測趨勢和季節性。

B. The model predicts the trend well, but not the seasonality.

B. 此模型可以很好地預測趨勢，但不能預測季節性。

C. The model predicts the seasonality well, but not the trend.

C. 此模型可以很好地預測季節性，但不能預測趨勢。

D. The model does not predict the trend or the seasonality well.

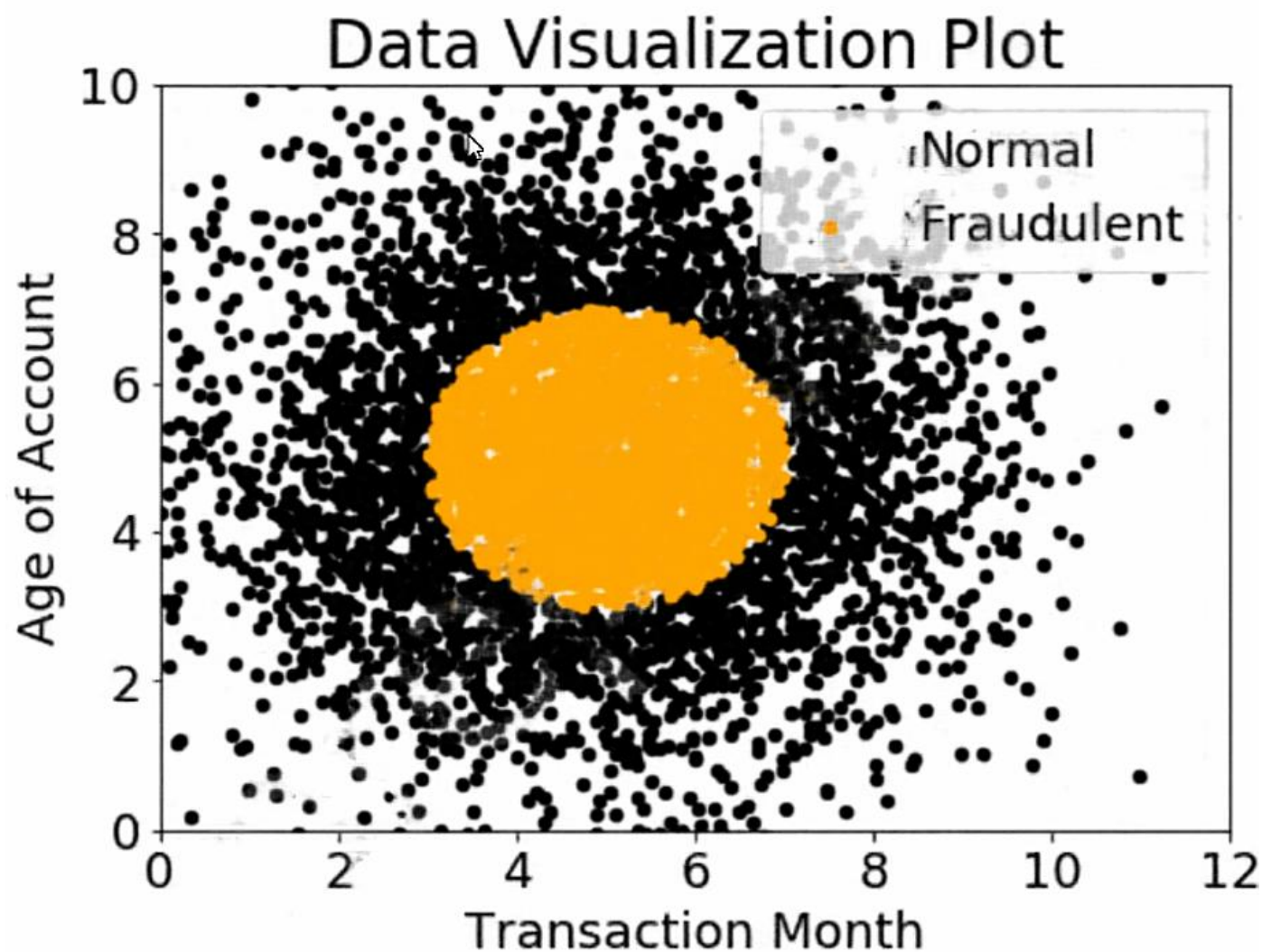
D. 此模型不能很好地預測趨勢或季節性。

問題# 14

A company wants to classify user behavior as either fraudulent or normal. Based on internal research, a Machine Learning Specialist would like to build a binary classifier based on two features: age of account and transaction month. The class distribution for these features is illustrated in the figure provided. Based on this information, which model would have the HIGHEST accuracy?

一家公司希望將用戶行為歸類為詐欺行為或正常行為。根據內部研究，機器學習專家希望基於兩個特徵來建立二元分類器：帳戶年齡和交易月份。提供的圖表顯示了這些特徵的類別分佈。根據此信

息，哪個模型的準確率最高？



A. Long short-term memory (LSTM) model with scaled exponential linear unit (SELU).

A. 具有縮放指數線性單元(SELU) 的長短期記憶(LSTM) 模型。

B. Logistic regression.

B. 邏輯迴歸。

C. Support vector machine (SVM) with non-linear kernel.

C. 具有非線性核的支援向量機(SVM)。

D. Single perceptron with tanh activation function.

D. 具有tanh 激活函數的單感知器。

問題# 15

A Machine Learning Specialist at a company sensitive to security is preparing a dataset for model training. The dataset is stored in Amazon S3 and contains Personally Identifiable Information (PII). The dataset: Must be accessible from a VPC only. Must not traverse the public internet. How can these requirements be satisfied?

一家對安全敏感的公司的機器學習專家正在準備一個資料集用於模型訓練。此資料集儲存在Amazon S3 中，包含個人識別資訊(PII)。此資料集：只能從VPC 存取。不得遍歷公共互聯網。如何滿足這些要求？

A. Create a VPC endpoint and apply a bucket access policy that restricts access to the given VPC endpoint and the VPC.

A. 建立一個VPC 終端節點並應用儲存桶存取策略，以限制對給定VPC 終端節點和VPC 的存取。

B. Create a VPC endpoint and apply a bucket access policy that allows access from the given VPC endpoint and an Amazon EC2 instance.

B. 建立一個VPC 終端節點並應用儲存桶存取策略，以允許從給定VPC 終端節點和Amazon EC2 執行個體進行存取。

C. Create a VPC endpoint and use Network Access Control Lists (NACLs) to allow traffic between only the given VPC endpoint and an Amazon EC2 instance.

C. 建立一個VPC 終端節點並使用網路存取控制清單(NACL) 以僅允許給定VPC 終端節點和Amazon EC2 執行個體之間的流量。

D. Create a VPC endpoint and use security groups to restrict access to the given VPC endpoint and an Amazon EC2 instance.

D. 建立一個VPC 終端節點並使用安全群組來限制對給定VPC 終端節點和Amazon EC2 執行個體的存取。

問題# 16

During mini-batch training of a neural network for a classification problem, a Data Scientist notices that training accuracy oscillates. What is the MOST likely cause of this issue?

在對神經網路進行分類問題的小批量訓練期間，資料科學家注意到訓練準確度會出現波動。導致此問題的最可能原因是什麼？

A. The class distribution in the dataset is imbalanced.

A. 資料集中的類別分佈不平衡。

B. Dataset shuffling is disabled.

B. 資料集改組已停用。

C. The batch size is too big.

C. 批量大小太大。

D. The learning rate is very high.

D. 學習率非常高。

問題# 17

An employee found a video clip with audio on a company's social media feed. The language used in the video is Spanish. English is the employee's first language, and they do not understand Spanish. The employee wants to do a sentiment analysis. What combination of services is the MOST efficient to accomplish the task?

一名員工在公司的社群媒體上發現了一段帶有音訊的影片片段。影片中使用的語言是西班牙語。英語是該員工的母語，他們不懂西班牙語。該員工想要進行情緒分析。哪種服務組合最能有效地完成任務？

A. Amazon Transcribe, Amazon Translate, and Amazon Comprehend.

A. Amazon Transcribe、Amazon Translate 和 Amazon Comprehend。

B. Amazon Transcribe, Amazon Comprehend, and Amazon SageMaker seq2seq.

B. Amazon Transcribe、Amazon Comprehend 和 Amazon SageMaker seq2seq。

C. Amazon Transcribe, Amazon Translate, and Amazon SageMaker Neural Topic Model (NTM).

C. Amazon Transcribe、Amazon Translate 和 Amazon SageMaker 神經主題模型(NTM)。

D. Amazon Transcribe, Amazon Translate and Amazon SageMaker BlazingText.

D. Amazon Transcribe、Amazon Translate 和 Amazon SageMaker BlazingText。

問題# 18

A Machine Learning Specialist is packaging a custom ResNet model into a Docker container so the company can leverage Amazon SageMaker for training. The Specialist is using Amazon EC2 P3 instances to train the model and needs to properly configure the Docker container to leverage the NVIDIA GPUs. What does the Specialist need to do?

機器學習專家正在將自訂 ResNet 模型打包到 Docker 容器中，以便公司可以利用 Amazon SageMaker 進行訓練。專家正在使用 Amazon EC2 P3 執行個體來訓練模型，並且需要正確配置 Docker 容器以利用 NVIDIA GPU。專家需要做什麼？

A. Bundle the NVIDIA drivers with the Docker image.

A. 將 NVIDIA 驅動程式與 Docker 映像捆綁在一起。

B. Build the Docker container to be NVIDIA-Docker compatible.

B. 建置Docker 容器以與NVIDIA-Docker 相容。

C. Organize the Docker container's file structure to execute on GPU instances.

C. 組織Docker 容器的檔案結構以在GPU 實例上執行。

D. Set the GPU flag in the Amazon SageMaker CreateTrainingJob request body.

D. 在Amazon SageMaker CreateTrainingJob 請求正文中設定GPU 標誌。

問題# 19

A Machine Learning Specialist is building a logistic regression model that will predict whether or not a person will order a pizza. The Specialist is trying to build the optimal model with an ideal classification threshold. What model evaluation technique should the Specialist use to understand how different classification thresholds will impact the model's performance?

一位機器學習專家正在建立一個邏輯回歸模型，該模型將預測一個人是否會訂購披薩。該專家正在嘗試建立具有理想分類閾值的最佳模型。該專家應該使用哪種模型評估技術來了解不同的分類閾值將如何影響模型的表現？

A. Receiver operating characteristic (ROC) curve.

A. 接收者操作特性(ROC) 曲線

B. Misclassification rate.

B. 錯誤分類率

C. Root Mean Square Error (RMSE).

C. 均方根誤差(RMSE)

D. L1 norm.

D. L1 範數。

問題# 20

An interactive online dictionary wants to add a widget that displays words used in similar contexts. A Machine Learning Specialist is asked to provide word features for the downstream nearest neighbor model powering the widget. What should the Specialist do to meet these requirements?

一個互動式線上字典想要添加一個小部件，用於顯示類似上下文中使用的單字。機器學習專家被要求為支援該小部件的下游最近鄰模型提供單字特徵。專家該怎麼做才能滿足這些要求？

A. Create one-hot word encoding vectors.

A. 建立獨熱詞編碼向量。

B. Produce a set of synonyms for every word using Amazon Mechanical Turk.

B. 使用Amazon Mechanical Turk 為每個單字產生一組同義詞。

C. Create word embedding vectors that store edit distance with every other word.

C. 建立儲存與其他每個單字的編輯距離的單字嵌入向量。

D. Download word embeddings pre-trained on a large corpus.

D. 下載在大型語料庫上預先訓練的單字嵌入。

問題# 21

A Machine Learning Specialist is configuring Amazon SageMaker so multiple Data Scientists can access notebooks, train models, and deploy endpoints. To ensure the best operational performance, the Specialist needs to be able to track how often the Scientists are deploying models, GPU and CPU utilization on the deployed SageMaker endpoints, and all errors that are generated when an endpoint is invoked. Which services are integrated with Amazon SageMaker to track this information? (Choose two.)

機器學習專家正在配置Amazon SageMaker，以便多位資料科學家可以存取筆記本、訓練模型和部署終端節點。為了確保最佳營運績效，專家需要能夠追蹤科學家部署模型的頻率、已部署SageMaker 終端節點上的GPU 和CPU 使用率以及呼叫終端節點時產生的所有錯誤。哪些服務與Amazon SageMaker 整合以追蹤此資訊？（選兩個。）

A. AWS CloudTrail.

A.AWS CloudTrail。

B. AWS Health.

B.AWS 健康。

C. AWS Trusted Advisor.

C.AWS Trusted Advisor。

D. Amazon CloudWatch.

D.Amazon CloudWatch。

E. AWS Config.

E.AWS Config。

問題# 22

A retail chain has been ingesting purchasing records from its network of 20,000 stores to Amazon S3 using Amazon Kinesis Data Firehose. To support training an improved machine learning model, training records will require new but simple transformations, and some attributes will be combined. The model needs to be retrained daily. Given the large number of stores and the legacy data ingestion, which change will require the LEAST amount of development effort?

一家零售連鎖店一直使用Amazon Kinesis Data Firehose 將來自其20,000 家門市網路的購買記錄導入到Amazon S3。為了支援訓練改進的機器學習模型，訓練記錄將需要新的但簡單的轉換，並且一些屬性將被組合。此模型需要每天重新訓練。鑑於門市數量眾多且資料導入遺留，哪種變更需要最少的開發工作量？

A. Require that the stores to switch to capturing their data locally on AWS Storage Gateway for loading into Amazon S3, then use AWS Glue to do the transformation.

A. 要求商店切換到在AWS Storage Gateway 上本地捕獲其資料以載入到Amazon S3，然後使用AWS Glue 進行轉換。

B. Deploy an Amazon EMR cluster running Apache Spark with the transformation logic, and have the cluster run each day on the accumulating records in Amazon S3, outputting new/transformed records to Amazon S3.

B. 部署一個運行Apache Spark 的Amazon EMR 叢集並採用轉換邏輯，讓叢集每天在Amazon S3 中累積的記錄上運行，將新的/轉換後的記錄輸出到Amazon S3。

C. Spin up a fleet of Amazon EC2 instances with the transformation logic, have them transform the data records accumulating on Amazon S3, and output the transformed records to Amazon S3.

C. 啟動一組採用轉換邏輯的Amazon EC2 實例，讓它們轉換在Amazon S3 上累積的資料記錄，並將轉換後的記錄輸出到Amazon S3。

D. Insert an Amazon Kinesis Data Analytics stream downstream of the Kinesis Data Firehose stream that transforms raw record attributes into simple transformed values using SQL.

D. 在Kinesis Data Firehose 流的下游插入Amazon Kinesis Data Analytics 流，使用SQL 將原始記錄屬性轉換為簡單的轉換值。

問題# 23

A Machine Learning Specialist is building a convolutional neural network (CNN) that will classify 10 types of animals. The Specialist has built a series of layers in a neural network that will take an input image of an animal, pass it through a series of convolutional and pooling layers, and then finally pass it through a dense and fully connected layer with 10 nodes. The Specialist would like to get an output from the neural network that is a probability distribution of how likely it is that the input image belongs to each of the 10 classes. Which function will produce the desired output?

一位機器學習專家正在建構一個卷積神經網路(CNN)，用於對10 種動物進行分類。該專家在神經網路中建構了一系列層，這些層將獲取動物的輸入影像，將其通過一系列卷積層和池化層，最後將其通過具有10 個節點的密集全連接層。該專家希望從神經網路中獲得一個輸出，該輸出是輸入影像屬於這10 個類別中的每一個的可能性的機率分佈。哪個函數會產生所需的輸出？

A. Dropout.

A. Dropout。

B. Smooth L1 loss.

B. 平滑L1 損失。

C. Softmax.

C. Softmax。

D. Rectified linear units (ReLU).

D. 整流線性單元(ReLU)。

問題# 24

A Machine Learning Specialist trained a regression model, but the first iteration needs optimizing. The Specialist needs to understand whether the model is more frequently overestimating or underestimating the target. What option can the Specialist use to determine whether it is overestimating or underestimating the target value?

機器學習專家訓練了一個迴歸模型，但第一次迭代需要最佳化。專家需要了解模型是否更頻繁地高估或低估目標。專家可以使用什麼選項來確定它是高估還是低估目標值？

A. Root Mean Square Error (RMSE).

A. 均方根誤差(RMSE)。

B. Residual plots.

B. 殘差圖。

C. Area under the curve.

C. 曲線下面積。

D. Confusion matrix.

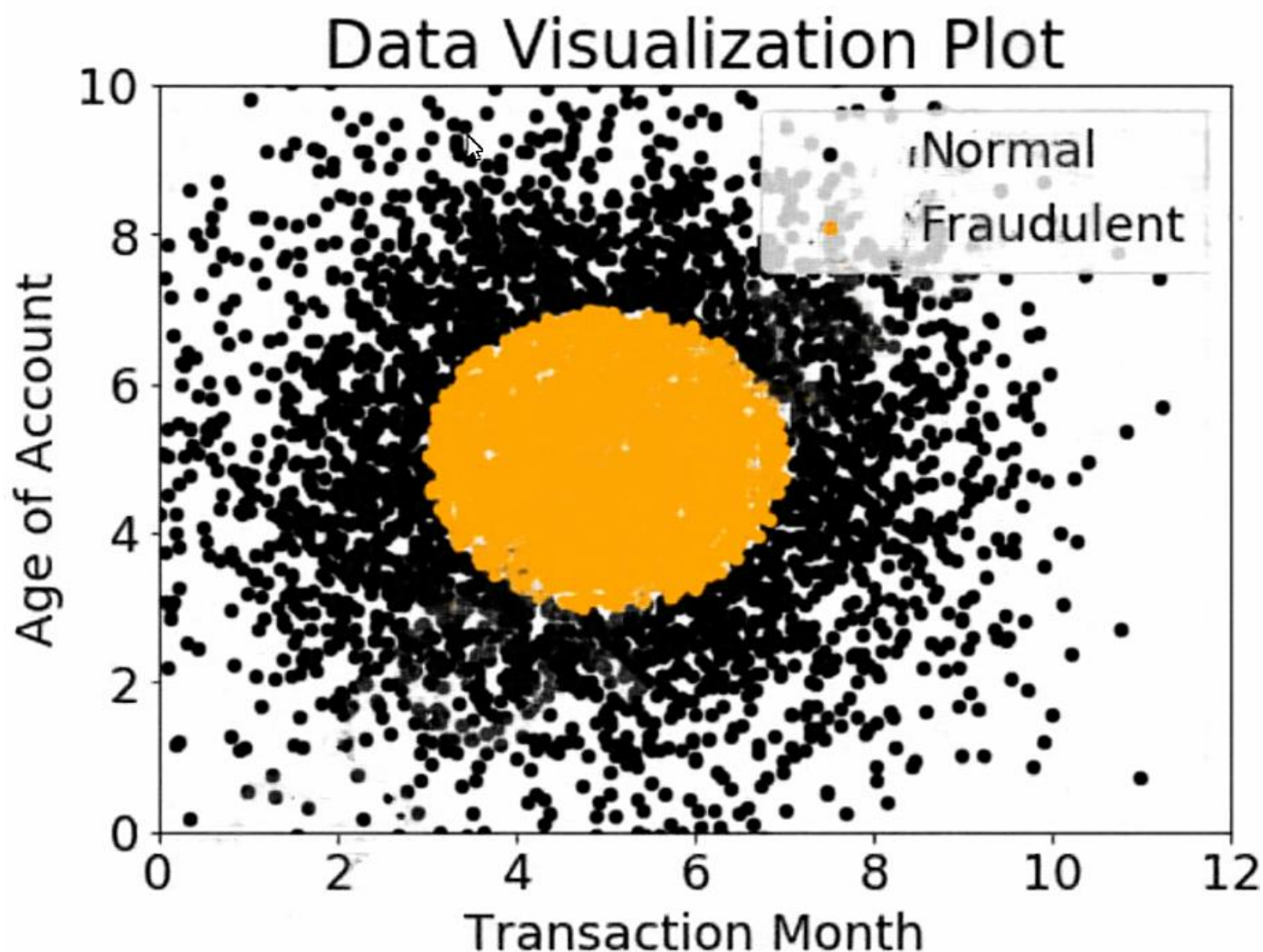
D. 混淆矩陣。

問題# 25

A company wants to classify user behavior as either fraudulent or normal. Based on internal research, a Machine Learning Specialist would like to build a binary classifier based on two features: age of account and transaction month. The class distribution for these features is illustrated in the figure provided. Based on this information, which model would have the HIGHEST recall with respect to the fraudulent class?

一家公司希望將用戶行為歸類為詐欺行為或正常行為。根據內部研究，機器學習專家希望基於兩個

特徵來建立二元分類器：帳戶年齡和交易月份。提供的圖表顯示了這些特徵的類別分佈。根據此信息，哪個模型在欺詐類別方面具有最高的召回率？



A. Decision tree.

A. 決策樹。

B. Linear support vector machine (SVM).

B. 線性支援向量機(SVM)。

C. Naive Bayesian classifier.

C. 樸素貝葉斯分類器。

D. Single Perceptron with sigmoidal activation function.

D. 具有S形激活函數的單一感知器。

問題# 26

A Machine Learning Specialist kicks off a hyperparameter tuning job for a tree-based ensemble model using Amazon SageMaker with Area Under the ROC Curve(AUC) as the objective metric. This workflow will eventually be deployed in a pipeline that retrains and tunes hyperparameters each night to model click-through on data that goes stale every 24 hours. With the goal of decreasing the amount of time it takes to train these models, and ultimately to decrease costs, the Specialist wants to reconfigure the input hyperparameter range(s). Which visualization will accomplish this?

機器學習專家使用Amazon SageMaker 啟動基於樹狀圖的整合模型的超參數調整作業，以ROC 曲線下面積(AUC) 作為客觀指標。此工作流程最終將部署在一個管道中，該管道每晚重新訓練和調整超參數，以對每24 小時過期一次的資料進行點擊建模。為了減少訓練這些模型所需的時間，並最終降低成本，專家希望重新配置輸入超參數範圍。哪種視覺化可以實現此目標？

A. A histogram showing whether the most important input feature is Gaussian.

A. 直方圖顯示最重要的輸入特徵是否為高斯分佈。

B. A scatter plot with points colored by target variable that uses t-Distributed Stochastic Neighbor Embedding (t-SNE) to visualize the large number of input variables in an easier-to-read dimension.

B. 散點圖，其中的點按目標變數著色，使用t 分佈隨機鄰域嵌入(t-SNE) 在更易於閱讀的維度中可視化大量輸入變數。

C. A scatter plot showing the performance of the objective metric over each training iteration.

C. 散佈圖顯示每次訓練迭代中目標指標的表現。

D. A scatter plot showing the correlation between maximum tree depth and the objective metric.

D. 散佈圖顯示最大樹深度與目標指標之間的相關性。

問題# 27

A Machine Learning Specialist is creating a new natural language processing application that processes a dataset comprised of 1 million sentences. The aim is to then run Word2Vec to generate embeddings of the sentences and enable different types of predictions. Here is an example from the dataset: "The quck BROWN FOX jumps over the lazy dog." Which of the following are the operations the Specialist needs to perform to correctly sanitize and prepare the data in a repeatable manner? (Choose three.)

機器學習專家正在創建一個新的自然語言處理應用程序，該應用程式可處理由100 萬個句子組成的資料集。目標是運行Word2Vec 來產生句子的嵌入並實現不同類型的預測。以下是資料集中的一個範例：「敏捷的棕狐跳過了懶狗。」以下哪些是專家需要執行的操作，以便以可重複的方式正確清理和準備資料？（選三個。）

A. Perform part-of-speech tagging and keep the action verb and the nouns only.

A. 執行詞性標註並只保留動作動詞和名詞。

B. Normalize all words by making the sentence lowercase.

B. 透過將句子變成小寫來規範化所有單字。

C. Remove stop words using an English stopword dictionary.

C. 使用英文停用詞字典刪除停用詞。

D. Correct the typography on "quck" to "quick.€.

D. 將「quck」的排版更正為「quick.€

E. One-hot encode all words in the sentence.

E. 對句子中的所有單字進行獨熱編碼。

F. Tokenize the sentence into words.

F. 將句子標記為單字。

問題# 28

A company is using Amazon Polly to translate plaintext documents to speech for automated company announcements. However, company acronyms are being mispronounced in the current documents. How should a Machine Learning Specialist address this issue for future documents?

一家公司正在使用Amazon Polly 將純文字文件翻譯成語音，以便自動發佈公司公告。但是，當前文檔中的公司縮寫發音錯誤。機器學習專家應如何解決未來文件中的此問題？

A. Convert current documents to SSML with pronunciation tags.

A. 將目前文件轉換為帶有發音標籤的SSML。

B. Create an appropriate pronunciation lexicon.

B. 創建適當的發音字典。

C. Output speech marks to guide in pronunciation.

C. 輸出引號來指導發音。

D. Use Amazon Lex to preprocess the text files for pronunciation.

D. 使用Amazon Lex 對文字檔案進行發音預處理。

問題# 29

An insurance company is developing a new device for vehicles that uses a camera to observe drivers' behavior and alert them when they appear distracted. The company created approximately 10,000 training images in a controlled environment that a Machine Learning Specialist will use to train and evaluate machine learning models. During the model evaluation, the Specialist notices that the training error rate diminishes faster as the number of epochs increases and the model is not accurately inferring on the unseen test images. Which of the following should be used to resolve this issue? (Choose two.)

一家保險公司正在開發一種新型車輛設備，該設備使用攝影機觀察駕駛員的行為，並在駕駛員分心時向他們發出警報。該公司在受控環境中創建了大約10,000 張訓練影像，而機器學習專家將使用這些影像來訓練和評估機器學習模型。在模型評估過程中，專家注意到，隨著週期數的增加，訓練錯誤率會下降得更快，並且模型無法準確地推斷未見過的測試影像。應使用下列哪一項來解決此問題？（選兩個。）

A. Add vanishing gradient to the model.

A. 在模型中加入消失梯度。

B. Perform data augmentation on the training data.

B. 對訓練資料執行資料增強。

C. Make the neural network architecture complex.

C. 使神經網路架構複雜化。

D. Use gradient checking in the model.

D. 在模型中使用梯度檢定。

E. Add L2 regularization to the model.

E. 在模型中加入L2 正規化。

問題# 30

When submitting Amazon SageMaker training jobs using one of the built-in algorithms, which common parameters MUST be specified? (Choose three.)

使用內建演算法之一提交Amazon SageMaker 訓練作業時，必須指定哪些常用參數？（選三個。）

A. The training channel identifying the location of training data on an Amazon S3 bucket.

A. 訓練通道，用於識別Amazon S3 儲存桶上訓練資料的位置。

B. The validation channel identifying the location of validation data on an Amazon S3 bucket.

B. 驗證通道，用於識別Amazon S3 儲存桶上驗證資料的位置。

C. The IAM role that Amazon SageMaker can assume to perform tasks on behalf of the users.

C. Amazon SageMaker 可以承擔代表使用者執行任務的IAM 角色。

D. Hyperparameters in a JSON array as documented for the algorithm used.

D. JSON 陣列中的超參數，如所用演算法所述。

E. The Amazon EC2 instance class specifying whether training will be run using CPU or GPU.

E. Amazon EC2 執行個體類，用於指定使用CPU 還是GPU 運行訓練。

F. The output path specifying where on an Amazon S3 bucket the trained model will persist.

F. 輸出路徑，用於指定訓練後的模型將在Amazon S3 儲存桶上的哪個位置保留。

問題# 31

A monitoring service generates 1 TB of scale metrics record data every minute. A Research team performs queries on this data using Amazon Athena. The queries run slowly due to the large volume of data, and the team requires better performance. How should the records be stored in Amazon S3 to improve query performance?

某監控服務每分鐘產生1TB 規模的指標記錄數據，某研究團隊使用Amazon Athena 查詢這些數據，由於數據量龐大，查詢運作緩慢，團隊需要更好的效能。如何將記錄儲存在Amazon S3 中以提高查詢效能？

A. CSV files.

A. CSV 檔案

B. Parquet files.

B. Parquet 檔案

C. Compressed JSON.

C. 壓縮JSON

D. RecordIO.

D. RecordIO。

問題# 32

Machine Learning Specialist is working with a media company to perform classification on popular articles from the company's website. The company is using random forests to classify how popular an article will be before it is published. A sample of the data being used is below. Given the dataset, the Specialist wants to convert the Day_Of_Week column to binary values. What technique should be used to convert this column to binary values?

機器學習專家正在與媒體公司合作，對公司網站上的熱門文章進行分類。該公司正在使用隨機森林對文章在發布之前的受歡迎程度進行分類。下面是正在使用的資料範例。給定資料集，專家希望將 Day_Of_Week 列轉換為二進位值。應該使用什麼技術將此列轉換為二進位值？

Article_Title	Author	Top_Keywords	Day_Of_Week	URL_of_Article	Page_Views
Building a Big Data Platform	Jane Doe	Big Data, Spark, Hadoop	Tuesday	http://examplecorp.com/data_platform.html	1300456
Getting Started with Deep Learning	John Doe	Deep Learning, Machine Learning, Spark	Tuesday	http://examplecorp.com/started_deep_learning.html	1230661
MXNet ML Guide	Jane Doe	Machine Learning, MXNet, Logistic Regression	Thursday	http://examplecorp.com/mxnet_guide.html	937291
Intro to NoSQL Databases	Mary Major	NoSQL, Operations, Database	Monday	http://examplecorp.com/nosql_intro_guide.html	407812

- A. Binarization.
- A. 二值化
- B. One-hot encoding.
- B. 獨熱編碼
- C. Tokenization.
- C. 標記化
- D. Normalization transformation.
- D. 規範化轉換。

A gaming company has launched an online game where people can start playing for free, but they need to pay if they choose to use certain features. The company needs to build an automated system to predict whether or not a new user will become a paid user within 1 year. The company has gathered a labeled dataset from 1 million users. The training dataset consists of 1,000 positive samples (from users who ended up paying within 1 year) and 999,000 negative samples (from users who did not use any paid features). Each data sample consists of 200 features including user age, device, location, and play patterns. Using this dataset for training, the Data Science team trained a random forest model that converged with over 99% accuracy on the training set. However, the prediction results on a test dataset were not satisfactory. Which of the following approaches should the Data Science team take to mitigate this issue? (Choose two.)

一家遊戲公司推出了一款線上遊戲，玩家可以免費開始玩，但如果選擇使用某些功能，則需要付費。該公司需要建立一個自動化系統來預測新用戶是否會在1年內成為付費用戶。該公司從100萬用戶收集了一個標籤的資料集。訓練資料集包含1,000個正樣本（來自1年內最終付費的使用者）和999,000個負樣本（來自未使用任何付費功能的使用者）。每個資料樣本包含200個特徵，包括使用者年齡、裝置、位置和遊戲模式。使用此資料集進行訓練，資料科學團隊訓練了一個隨機森林模型，該模型在訓練集上的收斂準確率超過99%。然而，測試資料集上的預測結果並不令人滿意。資料科學團隊應採取下列哪種方法來緩解此問題？（選兩個。）

A. Add more deep trees to the random forest to enable the model to learn more features.

A. 為隨機森林增加更多深樹，使模型學習到更多特徵。

B. Include a copy of the samples in the test dataset in the training dataset.

B. 在訓練資料集中包含測試資料集中樣本的副本。

C. Generate more positive samples by duplicating the positive samples and adding a small amount of noise to the duplicated data.

C. 透過複製正樣本並向重複資料添加少量雜訊來產生更多正樣本。

D. Change the cost function so that false negatives have a higher impact on the cost value than false positives.

D. 改變成本函數，使假陰性對成本值的影響大於假陽性。

E. Change the cost function so that false positives have a higher impact on the cost value than false negatives.

E. 改變成本函數，使假陽性對成本值的影響大於假陰性。

問題# 34

A Data Scientist is developing a machine learning model to predict future patient outcomes based on information collected about each patient and their treatment plans. The model should output a continuous value as its prediction. The data available includes labeled outcomes for a set of 4,000 patients. The study was conducted on a group of individuals over the age of 65 who have a particular disease that is known to worsen with age. Initial models have performed poorly. While reviewing the underlying data, the Data Scientist notices that, out of 4,000 patient observations, there are 450 where the patient age has been input as 0. The other features for these observations appear normal compared to the rest of the sample population. How should the Data Scientist correct this issue?

一位資料科學家正在開發一種機器學習模型，根據收集到的關於每位患者及其治療計劃的資訊來預

測未來的患者結果。此模型應輸出一個連續值作為其預測值。可用數據包括一組4,000 名患者的標記結果。這項研究是針對一組65 歲以上患有一種已知會隨著年齡增長而惡化的特定疾病的人進行的。初始模型表現不佳。在審查基礎資料時，資料科學家注意到，在4,000 個患者觀察結果中，有450 個患者的年齡輸入為0。與樣本人群的其餘部分相比，這些觀察結果的其他特徵看起來很正常。資料科學家應該如何修正這個問題？

A. Drop all records from the dataset where age has been set to 0.

A. 從資料集中刪除所有年齡設定為0 的記錄。

B. Replace the age field value for records with a value of 0 with the mean or median value from the dataset.

B. 將年齡為0 的記錄的年齡欄位值替換為資料集中的平均值或中位數。

C. Drop the age feature from the dataset and train the model using the rest of the features.

C. 從資料集中刪除年齡特徵，並使用其餘特徵訓練模型。

D. Use k-means clustering to handle missing features.

D. 使用k 均值聚類來處理缺失的特徵。

問題# 35

A Data Science team is designing a dataset repository where it will store a large amount of training data commonly used in its machine learning models. As DataScientists may create an arbitrary number of new datasets every day, the solution has to scale automatically and be cost-effective. Also, it must be possible to explore the data using SQL. Which storage scheme is MOST adapted to this scenario?

一個資料科學團隊正在設計一個資料集儲存庫，用於儲存機器學習模型中常用的大量訓練資料。由於資料科學家每天可能會建立任意數量的新資料集，因此此解決方案必須能夠自動擴展且具有成本效益。此外，還必須能夠使用SQL 探索資料。哪種儲存方案最適合這種情況？

A. Store datasets as files in Amazon S3.

A. 將資料集作為檔案儲存在Amazon S3 中。

B. Store datasets as files in an Amazon EBS volume attached to an Amazon EC2 instance.

B. 將資料集作為檔案儲存在附加到Amazon EC2 執行個體的Amazon EBS 磁碟區中。

C. Store datasets as tables in a multi-node Amazon Redshift cluster.

C. 將資料集儲存為表格儲存在多節點Amazon Redshift 叢集中。

D. Store datasets as global tables in Amazon DynamoDB.

D. 將資料集儲存為全域表儲存在Amazon DynamoDB 中。

問題# 36

A Machine Learning Specialist deployed a model that provides product recommendations on a company's website. Initially, the model was performing very well and resulted in customers buying more products on average. However, within the past few months, the Specialist has noticed that the effect of product recommendations has diminished and customers are starting to return to their original habits of spending less. The Specialist is unsure of what happened, as the model has not changed from its initial deployment over a year ago. Which method should the Specialist try to improve model performance?

一位機器學習專家在一家公司的網站上部署了一個提供產品推薦的模型。最初，該模型表現良好，平均而言，客戶購買的產品更多。然而，在過去的幾個月裡，這位專家注意到產品推薦的效果已經減弱，客戶開始恢復原來少花錢的習慣。這位專家不確定發生了什麼，因為該模型與一年多前首次部署時相比沒有變化。這位專家應該嘗試哪種方法來提升模型效能？

A. The model needs to be completely re-engineered because it is unable to handle product inventory changes.

A. 此模型需要完全重新設計，因為它無法處理產品庫存變化。

B. The model's hyperparameters should be periodically updated to prevent drift.

B. 此模型的超參數應定期更新以防止漂移。

C. The model should be periodically retrained from scratch using the original data while adding a regularization term to handle product inventory changes.

C. 此模型應定期使用原始資料從頭開始重新訓練，同時添加正規化項以處理產品庫存變化。

D. The model should be periodically retrained using the original training data plus new data as product inventory changes.

D. 此模型應定期使用原始訓練資料和新資料進行重新訓練，因為產品庫存會發生變化。

問題# 37

A Machine Learning Specialist working for an online fashion company wants to build a data ingestion solution for the company's Amazon S3-based data lake. The Specialist wants to create a set of ingestion mechanisms that will enable future capabilities comprised of: Real-time analytics Interactive analytics of historical data Clickstream analytics Product recommendations Which services should the Specialist use?

一家線上時尚公司的機器學習專家希望為該公司基於Amazon S3的資料湖建立資料擷取解決方案。專家希望創建一套提取機制，以實現以下未來功能：即時分析 歷史資料的互動式分析 點擊流分析 產品推薦專家應該使用哪些服務？

A. AWS Glue as the data catalog; Amazon Kinesis Data Streams and Amazon Kinesis Data Analytics for real-time data insights; Amazon Kinesis Data Firehose for delivery to Amazon ES for clickstream analytics; Amazon EMR to generate personalized product recommendations.

A. 使用AWS Glue 作為資料目錄；使用Amazon Kinesis Data Streams 和Amazon Kinesis Data Analytics 取得即時資料洞察；使用Amazon Kinesis Data Firehose 傳輸至Amazon ES 進行點擊流分析；使用Amazon EMR 產生個人化產品推薦。

B. Amazon Athena as the data catalog; Amazon Kinesis Data Streams and Amazon Kinesis Data Analytics for near-real-time data insights; Amazon Kinesis Data Firehose for clickstream analytics; AWS Glue to generate personalized product recommendations.

B. 使用Amazon Athena 作為資料目錄；使用Amazon Kinesis Data Streams 和Amazon Kinesis Data Analytics 獲得近乎即時的資料洞察；使用Amazon Kinesis Data Firehose 進行點擊流分析；使用AWS Glue 產生個人化產品推薦。

C. AWS Glue as the data catalog; Amazon Kinesis Data Streams and Amazon Kinesis Data Analytics for historical data insights; Amazon Kinesis Data Firehose for delivery to Amazon ES for clickstream analytics; Amazon EMR to generate personalized product recommendations.

C. 使用AWS Glue 作為資料目錄；使用Amazon Kinesis Data Streams 和Amazon Kinesis Data Analytics 取得歷史資料洞察；使用Amazon Kinesis Data Firehose 傳輸至Amazon ES 進行點擊流分析；使用Amazon EMR 產生個人化產品推薦。

D. Amazon Athena as the data catalog; Amazon Kinesis Data Streams and Amazon Kinesis Data Analytics for historical data insights; Amazon DynamoDB streams for clickstream analytics; AWS Glue to generate personalized product recommendations.

D. 使用Amazon Athena 作為資料目錄；使用Amazon Kinesis Data Streams 和Amazon Kinesis Data Analytics 獲得歷史資料洞察；使用Amazon DynamoDB streams 進行點擊流分析；使用AWS Glue 產生個人化產品推薦。

問題# 38

A company is observing low accuracy while training on the default built-in image classification algorithm in Amazon SageMaker. The Data Science team wants to use an Inception neural network architecture instead of a ResNet architecture. Which of the following will accomplish this? (Choose two.)

一家公司在使用Amazon SageMaker 中的預設內建影像分類演算法進行訓練時發現準確率較低。資料科學團隊希望使用Inception 神經網路架構而不是ResNet 架構。下列哪一項可以實現此目的？（選兩項。）

A. Customize the built-in image classification algorithm to use Inception and use this for model training.

A. 自訂內建影像分類演算法以使用Inception，並將其用於模型訓練。

B. Create a support case with the SageMaker team to change the default image classification algorithm to Inception.

B. 與SageMaker 團隊一起建立支援案例，將預設影像分類演算法變更為Inception。

C. Bundle a Docker container with TensorFlow Estimator loaded with an Inception network and use this for model training.

C. 將Docker 容器與裝有Inception 網路的TensorFlow Estimator 捆綁在一起，並將其用於模型訓練。

D. Use custom code in Amazon SageMaker with TensorFlow Estimator to load the model with an Inception network, and use this for model training.

D. 在Amazon SageMaker 中使用帶有TensorFlow Estimator 的自訂程式碼，將模型與Inception 網路一起加載，並將其用於模型訓練。

E. Download and apt-get install the inception network code into an Amazon EC2 instance and use this instance as a Jupyter notebook in Amazon SageMaker.

E. 下載並透過apt-get 安裝將inception 網路程式碼安裝到Amazon EC2 執行個體中，並將此執行個體用作Amazon SageMaker 中的Jupyter 筆記本。

問題# 39

A Machine Learning Specialist built an image classification deep learning model. However, the Specialist ran into an overfitting problem in which the training and testing accuracies were 99% and 75%, respectively. How should the Specialist address this issue and what is the reason behind it?

一位機器學習專家建立了一個影像分類深度學習模型。然而，專家遇到了過度擬合問題，訓練和測試準確率分別為99% 和75%。專家該如何解決這個問題，背後的原因是什麼？

A. The learning rate should be increased because the optimization process was trapped at a local minimum.

A. 應該增加學習率，因為最佳化過程陷入了局部最小值。

B. The dropout rate at the flatten layer should be increased because the model is not generalized enough.

B. 應該會增加平坦層的dropout 率，因為模型泛化能力不夠。

C. The dimensionality of dense layer next to the flatten layer should be increased because the model is not complex enough.

C. 應該會增加平坦層旁邊的密集層的維數，因為模型不夠複雜。

D. The epoch number should be increased because the optimization process was terminated before it reached the global minimum.

D. 應該會增加epoch 數，因為最佳化過程在達到全局最小值之前就終止了。

問題# 40

A Machine Learning team uses Amazon SageMaker to train an Apache MXNet handwritten digit classifier model using a research dataset. The team wants to receive a notification when the model is overfitting. Auditors want to view the Amazon SageMaker log activity report to ensure there are no unauthorized API calls. What should the Machine Learning team do to address the requirements with the least amount of code and fewest steps?

機器學習團隊使用Amazon SageMaker 訓練Apache MXNet 手寫數位分類器模型，並使用研究資料

集。團隊希望在模型過度擬合時收到通知。稽核人員希望查看Amazon SageMaker 日誌活動報告，以確保沒有未經授權的API 呼叫。機器學習團隊該怎麼做才能以最少的程式碼和最少的步驟來滿足要求？

A. Implement an AWS Lambda function to log Amazon SageMaker API calls to Amazon S3. Add code to push a custom metric to Amazon CloudWatch. Create an alarm in CloudWatch with Amazon SNS to receive a notification when the model is overfitting.

A. 實作AWS Lambda 函數以將Amazon SageMaker API 呼叫記錄到Amazon S3。新增程式碼以將自訂指標推送到Amazon CloudWatch。使用Amazon SNS 在CloudWatch 中建立警報，以便在模型過度擬合時接收通知。

B. Use AWS CloudTrail to log Amazon SageMaker API calls to Amazon S3. Add code to push a custom metric to Amazon CloudWatch. Create an alarm in CloudWatch with Amazon SNS to receive a notification when the model is overfitting.

B. 使用AWS CloudTrail 將Amazon SageMaker API 呼叫記錄到Amazon S3。新增程式碼以將自訂指標推送到Amazon CloudWatch。使用Amazon SNS 在CloudWatch 中建立警報，以便在模型過度擬合時接收通知。

C. Implement an AWS Lambda function to log Amazon SageMaker API calls to AWS CloudTrail. Add code to push a custom metric to Amazon CloudWatch. Create an alarm in CloudWatch with Amazon SNS to receive a notification when the model is overfitting.

C. 實作AWS Lambda 函數以將Amazon SageMaker API 呼叫記錄到AWS CloudTrail。新增程式碼以將自訂指標推送到Amazon CloudWatch。使用Amazon SNS 在CloudWatch 中建立警報，以便在模型過度擬合時接收通知。

D. Use AWS CloudTrail to log Amazon SageMaker API calls to Amazon S3. Set up Amazon SNS to receive a notification when the model is overfitting.

D. 使用AWS CloudTrail 將Amazon SageMaker API 呼叫記錄到Amazon S3。設定Amazon SNS 以在模型過度擬合時接收通知。

問題# 41

A Machine Learning Specialist is building a prediction model for a large number of features using linear models, such as linear regression and logistic regression. During exploratory data analysis, the Specialist observes that many features are highly correlated with each other. This may make the model unstable. What should be done to reduce the impact of having such a large number of features?

機器學習專家正在使用線性模型（例如線性迴歸和邏輯迴歸）為大量特徵建立預測模型。在探索性資料分析期間，專家觀察到許多特徵彼此高度相關。這可能會使模型不穩定。應該怎麼做才能減少如此大量特徵的影響？

A. Perform one-hot encoding on highly correlated features.

A. 對高度相關的特徵執行獨熱編碼。

B. Use matrix multiplication on highly correlated features.

B. 對高度相關的特徵使用矩陣乘法。

C. Create a new feature space using principal component analysis (PCA).

C. 使用主成分分析(PCA) 建立新的特徵空間。

D. Apply the Pearson correlation coefficient.

D. 應用皮爾遜相關係數。

問題# 42

A Machine Learning Specialist is implementing a full Bayesian network on a dataset that describes public transit in New York City. One of the random variables is discrete, and represents the number of minutes New Yorkers wait for a bus given that the buses cycle every 10 minutes, with a mean of 3 minutes. Which prior probability distribution should the ML Specialist use for this variable?

一位機器學習專家正在針對描述紐約市公共交通的資料集實施完整的貝葉斯網路。其中一個隨機變數是離散的，表示紐約人等待公車的分鐘數，假設公車每10分鐘一班，平均3分鐘一班。機器學習專家應該對這個變數使用哪一種先驗機率分佈？

A. Poisson distribution.

A. 泊松分佈。

B. Uniform distribution.

B. 均勻分佈。

C. Normal distribution.

C. 常態分佈。

D. Binomial distribution.

D. 二項分佈。

問題# 43

A Data Science team within a large company uses Amazon SageMaker notebooks to access data stored in Amazon S3 buckets. The IT Security team is concerned that internet-enabled notebook instances create a security vulnerability where malicious code running on the instances could compromise data privacy. The company mandates that all instances stay within a secured VPC with no internet access, and data communication traffic must stay within the AWS network. How should the Data Science team configure the notebook instance placement to meet these requirements?

一家大型公司的資料科學團隊使用Amazon SageMaker 筆記本存取儲存在Amazon S3 儲存桶中的資料。IT 安全團隊擔心支援網際網路的筆記型電腦實例會產生安全漏洞，在這些執行個體上執行的

惡意程式碼可能會危及資料隱私。該公司要求所有實例都位於沒有網路存取權限的安全VPC內，且資料通訊流量必須位於AWS網路內。資料科學團隊應如何配置筆記本實例位置以滿足這些要求？

A. Associate the Amazon SageMaker notebook with a private subnet in a VPC. Place the Amazon SageMaker endpoint and S3 buckets within the same VPC.

A. 將Amazon SageMaker 筆記本與VPC 中的私有子網路關聯。將Amazon SageMaker 終端節點和S3 儲存桶放在同一個VPC 內。

B. Associate the Amazon SageMaker notebook with a private subnet in a VPC. Use IAM policies to grant access to Amazon S3 and Amazon SageMaker.

B. 將Amazon SageMaker 筆記本與VPC 中的私有子網路關聯。使用IAM 策略授予對Amazon S3 和Amazon SageMaker 的存取權。

C. Associate the Amazon SageMaker notebook with a private subnet in a VPC. Ensure the VPC has S3 VPC endpoints and Amazon SageMaker VPC endpoints attached to it.

C. 將Amazon SageMaker 筆記本與VPC 中的私有子網路關聯。確保VPC 已連接S3 VPC 終端節點和Amazon SageMaker VPC 終端節點。

D. Associate the Amazon SageMaker notebook with a private subnet in a VPC. Ensure the VPC has a NAT gateway and an associated security group allowing only outbound connections to Amazon S3 and Amazon SageMaker.

D. 將Amazon SageMaker 筆記本與VPC 中的私有子網路關聯。確保VPC 具有NAT 閘道和關聯的安全性群組，僅允許與Amazon S3 和Amazon SageMaker 的出站連線。

問題# 44

A Machine Learning Specialist has created a deep learning neural network model that performs well on the training data but performs poorly on the test data. Which of the following methods should the Specialist consider using to correct this? (Choose three.)

機器學習專家創建了一個深度學習神經網路模型，在訓練資料上表現良好，但在測試資料上表現不佳。專家應該考慮使用下列哪一種方法來糾正這個問題？（選三個。）

A. Decrease regularization.

A. 減少正規化。

B. Increase regularization.

B. 增加正規化。

C. Increase dropout.

C. 增加dropout。

D. Decrease dropout.

D. 減少dropout。

E. Increase feature combinations.

E. 增加特徵組合。

F. Decrease feature combinations.

F. 減少特徵組合。

問題# 45

A Data Scientist needs to create a serverless ingestion and analytics solution for high-velocity, real-time streaming data. The ingestion process must buffer and convert incoming records from JSON to a query-optimized, columnar format without data loss. The output datastore must be highly available, and Analysts must be able to run SQL queries against the data and connect to existing business intelligence dashboards. Which solution should the Data Scientist build to satisfy the requirements?

資料科學家需要為高速即時串流資料創建無伺服器提取和分析解決方案。擷取過程必須緩衝傳入記錄並將其從JSON轉換為查詢最佳化的列式格式，且不會遺失資料。輸出資料儲存必須具有高可用性，分析師必須能夠針對資料執行SQL查詢並連接到現有的商業智慧儀表板。資料科學家應該建立哪種解決方案來滿足要求？

A. Create a schema in the AWS Glue Data Catalog of the incoming data format. Use an Amazon Kinesis Data Firehose delivery stream to stream the data and transform the data to Apache Parquet or ORC format using the AWS Glue Data Catalog before delivering to Amazon S3. Have the Analysts query the data directly from Amazon S3 using Amazon Athena, and connect to BI tools using the Athena Java Database Connectivity (JDBC) connector.

A. 在AWS Glue 資料目錄中建立傳入資料格式的架構。使用Amazon Kinesis Data Firehose 傳輸流來傳輸數據，並使用AWS Glue 數據目錄將數據轉換為Apache Parquet 或ORC 格式，然後再傳輸到Amazon S3。讓分析師使用Amazon Athena 直接從Amazon S3 查詢數據，並使用Athena Java 資料庫連接(JDBC) 連接器連接到BI 工具。

B. Write each JSON record to a staging location in Amazon S3. Use the S3 Put event to trigger an AWS Lambda function that transforms the data into Apache Parquet or ORC format and writes the data to a processed data location in Amazon S3. Have the Analysts query the data directly from Amazon S3 using Amazon Athena, and connect to BI tools using the Athena Java Database Connectivity (JDBC) connector.

B. 將每個JSON 記錄寫入Amazon S3 中的暫存位置。使用S3 Put 事件觸發AWS Lambda 函數，該函數將資料轉換為Apache Parquet 或ORC 格式，並將資料寫入Amazon S3 中的已處理資料位置。讓分析師使用Amazon Athena 直接從Amazon S3 查詢數據，並使用Athena Java 資料庫連接(JDBC) 連接器連接到BI 工具。

C. Write each JSON record to a staging location in Amazon S3. Use the S3 Put event to trigger an AWS Lambda function that transforms the data into Apache Parquet or ORC format and inserts it into an Amazon RDS PostgreSQL database. Have the Analysts query and run dashboards from the RDS database.

C. 將每個JSON 記錄寫入Amazon S3 中的暫存位置。使用S3 Put 事件觸發AWS Lambda 函數，將資料轉換為Apache Parquet 或ORC 格式，並插入Amazon RDS PostgreSQL 資料庫。讓分析師從RDS 資料庫查詢並執行儀表板。

D. Use Amazon Kinesis Data Analytics to ingest the streaming data and perform real-time SQL queries to convert the records to Apache Parquet before delivering to Amazon S3. Have the Analysts query the data directly from Amazon S3 using Amazon Athena and connect to BI tools using the Athena Java Database Connectivity (JDBC) connector.

D. 使用Amazon Kinesis Data Analytics 提取串流資料並執行即時SQL 查詢，將記錄轉換為Apache Parquet，然後再傳送到Amazon S3。讓分析師使用Amazon Athena 直接從Amazon S3 查詢數據，並使用Athena Java 資料庫連接(JDBC) 連接器連接到BI 工具。

問題# 46

An online reseller has a large, multi-column dataset with one column missing 30% of its data. A Machine Learning Specialist believes that certain columns in the dataset could be used to reconstruct the missing data. Which reconstruction approach should the Specialist use to preserve the integrity of the dataset?

一家線上經銷商擁有一個大型多列資料集，其中有一列缺失了30% 的資料。機器學習專家認為，資料集中的某些欄位可用於重建缺少的資料。專家應使用哪一種重建方法來保持資料集的完整性？

A. Listwise deletion.

A. 按清單刪除。

B. Last observation carried forward.

B. 結轉上次觀察值。

C. Multiple imputation.

C. 多重插補。

D. Mean substitution.

D. 均值替代。

問題# 47

A company is setting up an Amazon SageMaker environment. The corporate data security policy does not allow communication over the internet. How can the company enable the Amazon SageMaker service without enabling direct internet access to Amazon SageMaker notebook instances?

一家公司正在設定Amazon SageMaker 環境。公司資料安全策略不允許透過網際網路進行通訊。該公司如何在不啟用對Amazon SageMaker 筆記型電腦實例的直接網路存取的情況下啟用Amazon SageMaker 服務？

A. Create a NAT gateway within the corporate VPC.

A. 在企業VPC 內建立NAT 閘道。

B. Route Amazon SageMaker traffic through an on-premises network.

B. 透過本地網路路由Amazon SageMaker 流量。

C. Create Amazon SageMaker VPC interface endpoints within the corporate VPC.

C. 在企業VPC 內建立Amazon SageMaker VPC 介面終端節點。

D. Create VPC peering with Amazon VPC hosting Amazon SageMaker.

D. 與託管Amazon SageMaker 的Amazon VPC 建立VPC 對等連線。

問題# 48

A Machine Learning Specialist is training a model to identify the make and model of vehicles in images. The Specialist wants to use transfer learning and an existing model trained on images of general objects. The Specialist collated a large custom dataset of pictures containing different vehicle makes and models. What should the Specialist do to initialize the model to re-train it with the custom data?

機器學習專家正在訓練一個模型來識別影像中的車輛品牌和型號。專家希望使用遷移學習和針對一般物體影像訓練的現有模型。專家整理了一個包含不同車輛品牌和型號的大型自訂圖片資料集。專家應該如何初始化模型以使用自訂資料重新訓練它？

A. Initialize the model with random weights in all layers including the last fully connected layer.

A. 在所有層（包括最後一個全連接層）中使用隨機權重初始化模型。

B. Initialize the model with pre-trained weights in all layers and replace the last fully connected layer.

B. 在所有層中使用預先訓練的權重初始化模型，並取代最後一個全連接層。

C. Initialize the model with random weights in all layers and replace the last fully connected layer.

C. 在所有層中使用隨機權重初始化模型，並取代最後一個全連接層。

D. Initialize the model with pre-trained weights in all layers including the last fully connected layer.

D. 在所有層（包括最後一個全連接層）中使用預先訓練的權重初始化模型。

問題# 49

An office security agency conducted a successful pilot using 100 cameras installed at key locations within the main office. Images from the cameras were uploaded to Amazon S3 and tagged using Amazon Rekognition, and the results were stored in Amazon ES. The agency is now looking to expand the pilot into a full production system using thousands of video cameras in its office locations globally. The goal is to identify activities performed by non-employees in real time. Which solution should the agency consider?

一家辦公室安全機構進行了一次成功的試點，在總部關鍵位置安裝了100個攝影機。攝影機拍攝的影像上傳到Amazon S3並使用Amazon Rekognition進行標記，結果儲存在Amazon ES中。該機

構現在正考慮將試點擴展為完整的生產系統，使用全球各地辦公地點的數千台攝影機。目標是即時識別非員工執行的活動該機構應該考慮哪種解決方案？

A. Use a proxy server at each local office and for each camera, and stream the RTSP feed to a unique Amazon Kinesis Video Streams video stream. On each stream, use Amazon Rekognition Video and create a stream processor to detect faces from a collection of known employees, and alert when non-employees are detected.

A. 在每個本地辦事處和每個相機上使用代理伺服器，並將RTSP 來源串流傳輸到唯一的Amazon Kinesis Video Streams 視訊串流。在每個流上，使用Amazon Rekognition Video 並建立一個串流處理器來偵測已知員工集合中的臉孔，並在偵測到非員工時發出警報。

B. Use a proxy server at each local office and for each camera, and stream the RTSP feed to a unique Amazon Kinesis Video Streams video stream. On each stream, use Amazon Rekognition Image to detect faces from a collection of known employees and alert when non-employees are detected.

B. 在每個本地辦事處和每個攝影機上使用代理伺服器，並將RTSP 來源串流傳輸到唯一的Amazon Kinesis Video Streams 視訊串流。在每個流上，使用Amazon Rekognition Image 來偵測已知員工集合中的臉孔，並在偵測到非員工時發出警報。

C. Install AWS DeepLens cameras and use the DeepLens_Kinesis_Video module to stream video to Amazon Kinesis Video Streams for each camera. On each stream, use Amazon Rekognition Video and create a stream processor to detect faces from a collection on each stream, and alert when non-employees are detected.

C. 安裝AWS DeepLens 相機並使用DeepLens_Kinesis_Video 模組將視訊串流傳輸到每個相機的Amazon Kinesis Video Streams。在每個流上，使用Amazon Rekognition Video 並建立一個流處理器來偵測每個流上的集合中的臉孔，並在偵測到非員工時發出警報。

D. Install AWS DeepLens cameras and use the DeepLens_Kinesis_Video module to stream video to Amazon Kinesis Video Streams for each camera. On each stream, run an AWS Lambda function to capture image fragments and then call Amazon Rekognition Image to detect faces from a collection of known employees, and alert when non-employees are detected.

D. 安裝AWS DeepLens 相機並使用DeepLens_Kinesis_Video 模組將視訊串流傳輸到每個相機的Amazon Kinesis Video Streams。在每個流上，執行AWS Lambda 函數來捕捉影像片段，然後呼叫Amazon Rekognition Image 從一組已知員工中偵測臉部，並在偵測到非員工時發出警報。

問題# 50

A Marketing Manager at a pet insurance company plans to launch a targeted marketing campaign on social media to acquire new customers. Currently, the company has the following data in Amazon Aurora: Profiles for all past and existing customers Profiles for all past and existing insured pets Policy-level information Premiums received Claims paid What steps should be taken to implement a machine learning model to identify potential new customers on social media?

一家寵物保險公司的行銷經理計劃在社群媒體上進行有針對性的行銷活動，以吸引新客戶。目前，該公司在Amazon Aurora 中擁有以下數據：所有過去和現有客戶的資料 所有過去和現有投保寵物的資料 保單級別資訊 已收到的保費 已支付的索賠應採取哪些步驟來實施機器學習模型以在社群媒體上識別潛在的新客戶？

A. Use regression on customer profile data to understand key characteristics of consumer segments. Find similar profiles on social media.

A. 對客戶資料資料使用迴歸分析，了解消費者細分市場的關鍵特徵。在社群媒體上尋找類似的資料。

B. Use clustering on customer profile data to understand key characteristics of consumer segments. Find similar profiles on social media.

B. 對客戶資料資料使用聚類分析，了解消費者細分市場的關鍵特徵。在社群媒體上尋找類似的資料。

C. Use a recommendation engine on customer profile data to understand key characteristics of consumer segments. Find similar profiles on social media.

C. 對客戶資料資料使用推薦引擎，了解消費者細分市場的關鍵特徵。在社群媒體上尋找類似的資料。

D. Use a decision tree classifier engine on customer profile data to understand key characteristics of consumer segments. Find similar profiles on social media.

D. 對客戶資料資料使用決策樹分類器引擎，了解消費者細分市場的關鍵特徵。在社群媒體上尋找類似的資料。

問題# 51

A manufacturing company has a large set of labeled historical sales data. The manufacturer would like to predict how many units of a particular part should be produced each quarter. Which machine learning approach should be used to solve this problem?

一家製造公司擁有大量的標籤歷史銷售數據。製造商希望預測每季應生產多少個特定零件。應該使用哪種機器學習方法來解決這個問題？

A. Logistic regression.

A. 邏輯迴歸。

B. Random Cut Forest (RCF).

B. 隨機砍伐森林（RCF）。

C. Principal component analysis (PCA).

C. 主成分分析（PCA）。

D. Linear regression.

D. 線性迴歸。

問題# 52

A financial services company is building a robust serverless data lake on Amazon S3. The data lake should be flexible and meet the following requirements: Support querying old and new data on Amazon S3 through Amazon Athena and Amazon Redshift Spectrum. Support event-driven ETL pipelines Provide a quick and easy way to understand metadata Which approach meets these requirements?

一家金融服務公司正在Amazon S3 上建立一個強大的無伺服器資料湖。資料湖應具有靈活性並滿足以下要求：支援透過Amazon Athena 和Amazon Redshift Spectrum 查詢Amazon S3 上的舊資料和新資料。支援事件驅動的ETL 管道 提供一種快速簡便的方法來理解元資料哪種方法滿足這些要求？

A. Use an AWS Glue crawler to crawl S3 data, an AWS Lambda function to trigger an AWS Glue ETL job, and an AWS Glue Data catalog to search and discover metadata.

A. 使用AWS Glue 爬網程式爬取S3 數據，使用AWS Lambda 函數觸發AWS Glue ETL 作業，使用AWS Glue 資料目錄搜尋和發現元資料。

B. Use an AWS Glue crawler to crawl S3 data, an AWS Lambda function to trigger an AWS Batch job, and an external Apache Hive metastore to search and discover metadata.

B. 使用AWS Glue 爬網程式爬取S3 數據，使用AWS Lambda 函數觸發AWS Batch 作業，使用外部Apache Hive 元儲存搜尋和發現元資料。

C. Use an AWS Glue crawler to crawl S3 data, an Amazon CloudWatch alarm to trigger an AWS Batch job, and an AWS Glue Data Catalog to search and discover metadata.

C. 使用AWS Glue 爬網程序爬取S3 數據，使用Amazon CloudWatch 警報觸發AWS Batch 作業，使用AWS Glue 數據目錄搜尋和發現元數據。

D. Use an AWS Glue crawler to crawl S3 data, an Amazon CloudWatch alarm to trigger an AWS Glue ETL job, and an external Apache Hive metastore to search and discover metadata.

D. 使用AWS Glue 爬網程式爬取S3 數據，使用Amazon CloudWatch 警報觸發AWS Glue ETL 作業，使用外部Apache Hive 元儲存搜尋和發現元資料。

問題# 53

A company's Machine Learning Specialist needs to improve the training speed of a time-series forecasting model using TensorFlow. The training is currently implemented on a single-GPU machine and takes approximately 23 hours to complete. The training needs to be run daily. The model accuracy is acceptable, but the company anticipates a continuous increase in the size of the training data and a need to update the model on an hourly, rather than a daily, basis. The company also wants to minimize coding effort and infrastructure changes. What should the Machine Learning Specialist do to the training solution to allow it to scale for future demand?

一家公司的機器學習專家需要使用TensorFlow 來提高時間序列預測模型的訓練速度。訓練目前在單GPU 機器上實施，大約需要23 小時才能完成。訓練需要每天進行。模型準確度可以接受，但公司預計訓練資料量將持續增加，需要每小時更新一次模型，而不是每天更新一次。該公司還希望最大限度地減少編碼工作量和基礎設施變更。機器學習專家應該對訓練解決方案做些什麼，以使其能

夠擴展以滿足未來的需求？

A. Do not change the TensorFlow code. Change the machine to one with a more powerful GPU to speed up the training.

A. 不更改TensorFlow 程式碼。將機器換成GPU 更強大的機器，以加快訓練速度。

B. Change the TensorFlow code to implement a Horovod distributed framework supported by Amazon SageMaker. Parallelize the training to as many machines as needed to achieve the business goals.

B. 更改TensorFlow 程式碼，實作Amazon SageMaker 支援的Horovod 分散式框架。將訓練並行到盡可能多的機器，以實現業務目標。

C. Switch to using a built-in AWS SageMaker DeepAR model. Parallelize the training to as many machines as needed to achieve the business goals.

C. 改用內建AWS SageMaker DeepAR 模型。將訓練並行到盡可能多的機器，以實現業務目標。

D. Move the training to Amazon EMR and distribute the workload to as many machines as needed to achieve the business goals.

D. 將訓練移至Amazon EMR，並將工作負載分配到盡可能多的機器，以實現業務目標。

問題# 54

Which of the following metrics should a Machine Learning Specialist generally use to compare/evaluate machine learning classification models against each other?

機器學習專家通常應該使用下列哪一個指標來比較/評估機器學習分類模型？

A. Recall.

A. 召回率。

B. Misclassification rate.

B. 錯誤分類率。

C. Mean absolute percentage error (MAPE).

C. 平均絕對百分比誤差(MAPE)。

D. Area Under the ROC Curve (AUC).

D. ROC 曲線下面積(AUC)。

問題# 55

A company is running a machine learning prediction service that generates 100 TB of predictions every day. A Machine Learning Specialist must generate a visualization of the daily precision-recall curve from the predictions, and forward a read-only version to the Business team. Which solution requires the LEAST coding effort?

一家公司正在運行機器學習預測服務，該服務每天產生100 TB 的預測。機器學習專家必須根據預測產生每日精確召回率曲線的可視化，並將唯讀版本轉發給業務團隊。哪種解決方案需要最少的編碼工作量？

A. Run a daily Amazon EMR workflow to generate precision-recall data, and save the results in Amazon S3. Give the Business team read-only access to S3.

A. 執行每日Amazon EMR 工作流程以產生精確度-回想率數據，並將結果保存在Amazon S3 中。授予業務團隊對S3 的唯讀存取權限。

B. Generate daily precision-recall data in Amazon QuickSight, and publish the results in a dashboard shared with the Business team.

B. 在Amazon QuickSight 中產生每日精確度-召回率數據，並在與業務團隊共享的儀表板中發布結果。

C. Run a daily Amazon EMR workflow to generate precision-recall data, and save the results in Amazon S3. Visualize the arrays in Amazon QuickSight, and publish them in a dashboard shared with the Business team.

C. 執行每日Amazon EMR 工作流程以產生精確度-召回率數據，並將結果保存在Amazon S3 中。在Amazon QuickSight 中視覺化數組，並在與業務團隊共享的儀表板中發布它們。

D. Generate daily precision-recall data in Amazon ES, and publish the results in a dashboard shared with the Business team.

D. 在Amazon ES 中產生每日精確度-召回率數據，並在與業務團隊共享的儀表板中發布結果。

問題# 56

A Machine Learning Specialist is preparing data for training on Amazon SageMaker. The Specialist is using one of the SageMaker built-in algorithms for the training. The dataset is stored in CSV format and is transformed into a numpy.array, which appears to be negatively affecting the speed of the training. What should the Specialist do to optimize the data for training on SageMaker?

機器學習專家正在準備在Amazon SageMaker 上進行訓練的資料。專家正在使用SageMaker 內建演算法之一進行訓練。資料集以CSV 格式儲存並轉換為numpy.array，這似乎對訓練速度產生了負面影響。專家應該如何優化在SageMaker 上進行訓練的資料？

A. Use the SageMaker batch transform feature to transform the training data into a DataFrame.

A. 使用SageMaker 批次轉換功能將訓練資料轉換為DataFrame。

B. Use AWS Glue to compress the data into the Apache Parquet format.

B. 使用AWS Glue 將資料壓縮為Apache Parquet 格式。

C. Transform the dataset into the RecordIO protobuf format.

C. 將資料集轉換為RecordIO protobuf 格式。

D. Use the SageMaker hyperparameter optimization feature to automatically optimize the data.

D. 使用SageMaker 超參數最佳化功能自動最佳化資料。

問題# 57

A Machine Learning Specialist is required to build a supervised image-recognition model to identify a cat. The ML Specialist performs some tests and records the following results for a neural network-based image classifier: Total number of images available = 1,000 Test set images = 100 (constant test set) The ML Specialist notices that, in over 75% of the misclassified images, the cats were held upside down by their owners. Which techniques can be used by the ML Specialist to improve this specific test error?

需要一名機器學習專家來建立一個監督式影像辨識模型來識別貓。機器學習專家執行一些測試，並記錄基於神經網路的影像分類器的以下結果：可用影像總數= 1,000 測試集影像= 100（恆定測試集）機器學習專家注意到，在超過75% 的錯誤分類影像中，貓被主人倒抱。機器學習專家可以使用哪些技術來改進這個特定的測試錯誤？

A. Increase the training data by adding variation in rotation for training images.

A. 透過為訓練圖像添加旋轉變化來增加訓練資料。

B. Increase the number of epochs for model training.

B. 增加模型訓練的時期數。

C. Increase the number of layers for the neural network.

C. 增加神經網路的層數。

D. Increase the dropout rate for the second-to-last layer.

D. 增加倒數第二層的dropout 率。

問題# 58

A Machine Learning Specialist needs to be able to ingest streaming data and store it in Apache Parquet files for exploration and analysis. Which of the following services would both ingest and store this data in the correct format?

機器學習專家需要能夠獲取串流資料並將其儲存在Apache Parquet 檔案中以供探索和分析。下列哪一種服務可以以正確的格式取得和儲存這些資料？

A. AWS DMS.

A. AWS DMS。

B. Amazon Kinesis Data Streams.

B. Amazon Kinesis 資料流。

C. Amazon Kinesis Data Firehose.

C. Amazon Kinesis Data Firehose。

D. Amazon Kinesis Data Analytics.

D. Amazon Kinesis 數據分析。

問題# 59

A data scientist has explored and sanitized a dataset in preparation for the modeling phase of a supervised learning task. The statistical dispersion can vary widely between features, sometimes by several orders of magnitude. Before moving on to the modeling phase, the data scientist wants to ensure that the prediction performance on the production data is as accurate as possible. Which sequence of steps should the data scientist take to meet these requirements?

資料科學家已經探索並清理了資料集，為監督學習任務的建模階段做準備。統計離散度在不同特徵之間可能存在很大差異，有時甚至相差幾個數量級。在進入建模階段之前，資料科學家希望確保生產資料的預測效能盡可能準確。資料科學家應該採取哪些步驟來滿足這些要求？

A. Apply random sampling to the dataset. Then split the dataset into training, validation, and test sets.

A. 對資料集應用隨機抽樣。然後將資料集拆分為訓練集、驗證集和測試集。

B. Split the dataset into training, validation, and test sets. Then rescale the training set and apply the same scaling to the validation and test sets.

B. 將資料集拆分為訓練集、驗證集和測試集。然後重新縮放訓練集並將相同的縮放應用於驗證集和測試集。

C. Rescale the dataset. Then split the dataset into training, validation, and test sets.

C. 重新縮放資料集。然後將資料集拆分為訓練集、驗證集和測試集。

D. Split the dataset into training, validation, and test sets. Then rescale the training set, the validation set, and the test set independently.

D. 將資料集拆分為訓練集、驗證集和測試集。然後分別重新縮放訓練集、驗證集和測試集。

問題# 60

A Machine Learning Specialist is assigned a TensorFlow project using Amazon SageMaker for training, and needs to continue working for an extended period with no Wi-Fi access. Which approach should the Specialist use to continue working?

一位機器學習專家被分配了一個使用Amazon SageMaker 進行訓練的TensorFlow 項目，並且需要在沒有Wi-Fi 訪問的情況下長時間繼續工作。專家應該使用哪種方法來繼續工作？

A. Install Python 3 and boto3 on their laptop and continue the code development using that environment.

A. 在筆記型電腦上安裝Python 3 和boto3，並使用該環境繼續進程式碼開發。

B. Download the TensorFlow Docker container used in Amazon SageMaker from GitHub to their local environment, and use the Amazon SageMaker Python SDK to test the code.

B. 從GitHub 下載Amazon SageMaker 中使用的TensorFlow Docker 容器到本機環境，並使用Amazon SageMaker Python SDK 測試程式碼。

C. Download TensorFlow from tensorflow.org to emulate the TensorFlow kernel in the SageMaker environment.

C. 從下載TensorFlow tensorflow.org在SageMaker 環境中模擬TensorFlow 核心。

D. Download the SageMaker notebook to their local environment, then install Jupyter Notebooks on their laptop and continue the development in a local notebook.

D. 將SageMaker 筆記型電腦下載到他們的本機環境，然後在他們的筆記型電腦上安裝Jupyter Notebooks 並在本機筆記型電腦中繼續開發。

問題# 61

A Machine Learning Specialist is working with a large cybersecurity company that manages security events in real time for companies around the world. The cybersecurity company wants to design a solution that will allow it to use machine learning to score malicious events as anomalies on the data as it is being ingested. The company also wants to be able to save the results in its data lake for later processing and analysis. What is the MOST efficient way to accomplish these tasks?

一位機器學習專家正在與一家大型網路安全公司合作，該公司為世界各地的公司即時管理安全事件。這家網路安全公司希望設計一種解決方案，使其能夠使用機器學習在資料被提取時將惡意事件標記為異常。該公司還希望能夠將結果保存在其數據湖中以供以後處理和分析。完成這些任務最有效的方法是什麼？

A. Ingest the data using Amazon Kinesis Data Firehose, and use Amazon Kinesis Data Analytics Random Cut Forest (RCF) for anomaly detection. Then use Kinesis Data Firehose to stream the results to Amazon S3.

A. 使用Amazon Kinesis Data Firehose 擷取數據，並使用Amazon Kinesis Data Analytics Random Cut Forest (RCF) 進行異常檢測。然後使用Kinesis Data Firehose 將結果串流傳輸到Amazon S3。

B. Ingest the data into Apache Spark Streaming using Amazon EMR, and use Spark MLlib with k-means to perform anomaly detection. Then store the results in an Apache Hadoop Distributed File System (HDFS) using Amazon EMR with a replication factor of three as the data lake.

B. 使用Amazon EMR 將資料提取到Apache Spark Streaming 中，並使用帶有k-means 的Spark MLlib 執行異常檢測。然後使用Amazon EMR 將結果儲存在Apache Hadoop 分散式檔案系統(HDFS) 中，副本因子為3，作為資料湖。

C. Ingest the data and store it in Amazon S3. Use AWS Batch along with the AWS Deep Learning AMIs to train a k-means model using TensorFlow on the data in Amazon S3.

C. 提取資料並將其儲存在Amazon S3 中。使用AWS Batch 和AWS Deep Learning AMI 使用Tensor Flow 對Amazon S3 中的資料訓練k-means 模型。

D. Ingest the data and store it in Amazon S3. Have an AWS Glue job that is triggered on demand transform the new data. Then use the built-in Random Cut Forest (RCF) model within Amazon SageMaker to detect anomalies in the data.

D. 提取資料並將其儲存在Amazon S3 中。讓按需觸發的AWS Glue 作業轉換新資料。然後使用Amazon SageMaker 中內建的Random Cut Forest (RCF) 模型檢測資料中的異常。

問題# 62

A Data Scientist wants to gain real-time insights into a data stream of GZIP files. Which solution would allow the use of SQL to query the stream with the LEAST latency?

資料科學家希望即時了解GZIP 檔案的資料流。哪一種解決方案可以允許使用SQL 以最少的延遲查詢資料流？

A. Amazon Kinesis Data Analytics with an AWS Lambda function to transform the data.

A. 使用AWS Lambda 函數轉換資料的Amazon Kinesis Data Analytics。

B. AWS Glue with a custom ETL script to transform the data.

B. 使用自訂ETL 腳本轉換資料的AWS Glue。

C. An Amazon Kinesis Client Library to transform the data and save it to an Amazon ES cluster.

C. 使用Amazon Kinesis Client Library 轉換資料並將其儲存到Amazon ES 叢集。

D. Amazon Kinesis Data Firehose to transform the data and put it into an Amazon S3 bucket.

D. 使用Amazon Kinesis Data Firehose 轉換資料並將其放入Amazon S3 儲存桶。

問題# 63

A retail company intends to use machine learning to categorize new products. A labeled dataset of current products was provided to the Data Science team. The dataset includes 1,200 products. The labeled dataset has 15 features for each product such as title dimensions, weight, and price. Each product is labeled as belonging to one of six categories such as books, games, electronics, and movies. Which model should be used for categorizing new products using the provided dataset for training?

一家零售公司打算使用機器學習對新產品進行分類。數據科學團隊收到了當前產品的標籤資料集。此資料集包含1,200 種產品。標籤的資料集針對每種產品有15 個特徵，例如標題尺寸、重量和價格。每種產品都被標記為屬於書籍、遊戲、電子產品和電影等六個類別之一。使用提供的資料集進行訓練時，應該使用哪種模型對新產品進行分類？

A. An XGBoost model where the objective parameter is set to multi:softmax.

A. 一個目標參數設定為multi:softmax 的XGBoost 模型。

B. A deep convolutional neural network (CNN) with a softmax activation function for the last layer.

B. 一個深度卷積神經網路(CNN)，最後一層使用softmax 活化函數。

C. A regression forest where the number of trees is set equal to the number of product categories.

C. 一個回歸森林，其中樹的數量設定為與產品類別的數量相等。

D. A DeepAR forecasting model based on a recurrent neural network (RNN).

D. 一個基於循環神經網路(RNN) 的DeepAR 預測模型。

問題# 64

A Data Scientist is working on an application that performs sentiment analysis. The validation accuracy is poor, and the Data Scientist thinks that the cause may be a rich vocabulary and a low average frequency of words in the dataset. Which tool should be used to improve the validation accuracy?

一位資料科學家正在開發一款執行情緒分析的應用程式。驗證準確率很差，資料科學家認為原因可能是詞彙量豐富，資料集中單字的平均頻率較低。應該使用哪一種工具來提高驗證準確率？

A. Amazon Comprehend syntax analysis and entity detection.

A. Amazon Comprehend 語法分析和實體檢測。

B. Amazon SageMaker BlazingText cbow mode.

B. Amazon SageMaker BlazingText cbow 模式。

C. Natural Language Toolkit (NLTK) stemming and stop word removal.

C. 自然語言工具包(NLTK) 詞幹擷取與停用詞刪除。

D. Scikit-learn term frequency-inverse document frequency (TF-IDF) vectorizer.

問題# 65

Machine Learning Specialist is building a model to predict future employment rates based on a wide range of economic factors. While exploring the data, the Specialist notices that the magnitude of the input features vary greatly. The Specialist does not want variables with a larger magnitude to dominate the model. What should the Specialist do to prepare the data for model training?

機器學習專家正在建立一個模型，根據各種經濟因素預測未來的就業率。在探索資料時，專家注意到輸入特徵的量級差異很大。專家不希望量級較大的變數主導模型。專家應該如何準備模型訓練的資料？

A. Apply quantile binning to group the data into categorical bins to keep any relationships in the data by replacing the magnitude with distribution.

A. 應用分位數分箱將資料分組到分類箱中，透過以分佈取代幅度來保留資料中的任何關係。

B. Apply the Cartesian product transformation to create new combinations of fields that are independent of the magnitude.

B. 應用笛卡爾積變換來建立與幅度無關的新欄位組合。

C. Apply normalization to ensure each field will have a mean of 0 and a variance of 1 to remove any significant magnitude.

C. 應用規範化以確保每個欄位的平均值均為0，變異數為1，以消除任何顯著的振幅。

D. Apply the orthogonal sparse bigram (OSB) transformation to apply a fixed-size sliding window to generate new features of a similar magnitude.

D. 應用正交稀疏二元組(OSB) 變換來應用固定大小的滑動視窗來產生具有相似幅度的新特徵。

問題# 66

A Machine Learning Specialist must build out a process to query a dataset on Amazon S3 using Amazon Athena. The dataset contains more than 800,000 records stored as plaintext CSV files. Each record contains 200 columns and is approximately 1.5 MB in size. Most queries will span 5 to 10 columns only. How should the Machine Learning Specialist transform the dataset to minimize query runtime?

機器學習專家必須建立一個流程，使用Amazon Athena 查詢Amazon S3 上的資料集。此資料集包含800,000 多筆以純文字CSV 檔案形式儲存的記錄。每筆記錄包含200 列，大小約1.5 MB。大多數查詢僅跨越5 到10 列。機器學習專家應如何轉換資料集以最大限度地減少查詢運行時間？

A. Convert the records to Apache Parquet format.

A. 將記錄轉換為Apache Parquet 格式。

B. Convert the records to JSON format.

B. 將記錄轉換為JSON 格式。

C. Convert the records to GZIP CSV format.

C. 將記錄轉換為GZIP CSV 格式。

D. Convert the records to XML format.

D. 將記錄轉換為XML 格式。

問題# 67

A Machine Learning Specialist is developing a daily ETL workflow containing multiple ETL jobs. The workflow consists of the following processes: * Start the workflow as soon as data is uploaded to Amazon S3. * When all the datasets are available in Amazon S3, start an ETL job to join the uploaded datasets with multiple terabyte-sized datasets already stored in Amazon S3. * Store the results of joining datasets in Amazon S3. * If one of the jobs fails, send a notification to the Administrator. Which configuration will meet these requirements?

機器學習專家正在開發包含多個ETL作業的日常ETL工作流程。工作流程由下列流程組成：* 資料上傳至Amazon S3後立即啟動工作流程。* 當所有資料集在Amazon S3中可用時，啟動ETL作業以將上傳的資料集與已儲存在Amazon S3中的多個TB層級資料集合併。* 將合併資料集的結果儲存在Amazon S3中。* 如果其中一個作業失敗，則向管理員發送通知。哪種配置可以滿足這些要求？

A. Use AWS Lambda to trigger an AWS Step Functions workflow to wait for dataset uploads to complete in Amazon S3. Use AWS Glue to join the datasets. Use an Amazon CloudWatch alarm to send an SNS notification to the Administrator in the case of a failure.

A. 使用AWS Lambda 觸發AWS Step Functions 工作流程以等待資料集上傳在Amazon S3 完成。使用AWS Glue 加入資料集。使用Amazon CloudWatch 警報在發生故障時向管理員發送SNS 通知。

B. Develop the ETL workflow using AWS Lambda to start an Amazon SageMaker notebook instance. Use a lifecycle configuration script to join the datasets and persist the results in Amazon S3. Use an Amazon CloudWatch alarm to send an SNS notification to the Administrator in the case of a failure.

B. 使用AWS Lambda 開發ETL 工作流程以啟動Amazon SageMaker 筆電實例。使用生命週期配置腳本加入資料集並將結果儲存在Amazon S3 中。使用Amazon CloudWatch 警報在發生故障時向管理員發送SNS 通知。

C. Develop the ETL workflow using AWS Batch to trigger the start of ETL jobs when data is uploaded to Amazon S3. Use AWS Glue to join the datasets in Amazon S3. Use an Amazon CloudWatch alarm to send an SNS notification to the Administrator in the case of a failure.

C. 使用AWS Batch 開發ETL 工作流程以在資料上傳到Amazon S3 時觸發ETL 作業的啟動。使用AWS Glue 加入Amazon S3 中的資料集。使用Amazon CloudWatch 警報在發生故障時向管理員發送SNS 通知。

D. Use AWS Lambda to chain other Lambda functions to read and join the datasets in Amazon S3 as soon as the data is uploaded to Amazon S3. Use an Amazon CloudWatch alarm to send an SNS notification to the Administrator in the case of a failure.

D. 使用AWS Lambda 連結其他Lambda 函數以在資料上傳到Amazon S3 後立即讀取和加入Amazon S3 中的資料集。發生故障時，使用Amazon CloudWatch 警報向管理員發送SNS 通知。

問題# 68

An agency collects census information within a country to determine healthcare and social program needs by province and city. The census form collects responses for approximately 500 questions from each citizen. Which combination of algorithms would provide the appropriate insights? (Choose two.)

某機構收集某國的人口普查信息，以確定各省和城市的醫療保健和社會項目需求。人口普查表收集了每位公民對大約500 個問題的答案。哪一種演算法組合可以提供適當的見解？（選兩個。）

A. The factorization machines (FM) algorithm.

A. 因式分解機(FM) 演算法。

B. The Latent Dirichlet Allocation (LDA) algorithm.

B. 潛在狄利克雷分配(LDA) 演算法。

C. The principal component analysis (PCA) algorithm.

C. 主成分分析(PCA) 演算法。

D. The k-means algorithm.

D. k-means 演算法。

E. The Random Cut Forest (RCF) algorithm.

E. 隨機砍伐森林(RCF) 演算法。

問題# 69

A large consumer goods manufacturer has the following products on sale: * 34 different toothpaste variants * 48 different toothbrush variants * 43 different mouthwash variants The entire sales history of all these products is available in Amazon S3. Currently, the company is using custom-built autoregressive integrated moving average (ARIMA) models to forecast demand for these products. The company wants to predict the demand for a new product that will soon be launched. Which solution should a Machine Learning Specialist apply?

一家大型消費品製造商銷售以下產品：* 34 種不同的牙膏 * 48 種不同的牙刷 * 43 種不同的漱口水所有這些產品的完整銷售歷史記錄均可在Amazon S3 中找到。目前，該公司正在使用客製化的自回歸綜合移動平均(ARIMA) 模型來預測這些產品的需求。該公司希望預測即將推出的新產品的需求。機器學習專家應該應用哪一種解決方案？

A. Train a custom ARIMA model to forecast demand for the new product.

A. 訓練自訂ARIMA 模型來預測新產品的需求。

B. Train an Amazon SageMaker DeepAR algorithm to forecast demand for the new product.

B. 訓練Amazon SageMaker DeepAR 演算法來預測新產品的需求。

C. Train an Amazon SageMaker k-means clustering algorithm to forecast demand for the new product.

C. 訓練Amazon SageMaker k-means 聚類演算法來預測新產品的需求。

D. Train a custom XGBoost model to forecast demand for the new product.

D. 訓練自訂XGBoost 模型來預測新產品的需求。

問題# 70

A Machine Learning Specialist uploads a dataset to an Amazon S3 bucket protected with server-side encryption using AWS KMS. How should the ML Specialist define the Amazon SageMaker notebook instance so it can read the same dataset from Amazon S3?

機器學習專家使用AWS KMS 將資料集上傳到受伺服器端加密保護的Amazon S3 儲存桶。機器學習專家應如何定義Amazon SageMaker 筆記本實例，以便它可以從Amazon S3 讀取相同的資料集？

A. Define security group(s) to allow all HTTP inbound/outbound traffic and assign those security group(s) to the Amazon SageMaker notebook instance.

A. 定義安全群組以允許所有HTTP 入站/出站流量，並將這些安全群組指派給Amazon SageMaker 筆記本實例。

B. Configure the Amazon SageMaker notebook instance to have access to the VPC. Grant permission in the KMS key policy to the notebook's KMS role.

B. 設定Amazon SageMaker 筆記本實例以存取VPC。在KMS 金鑰策略中向筆記本的KMS 角色授予權限。

C. Assign an IAM role to the Amazon SageMaker notebook with S3 read access to the dataset. Grant permission in the KMS key policy to that role.

C. 為Amazon SageMaker 筆記本指派一個IAM 角色，並授予對資料集的S3 讀取存取權。在KMS 金鑰策略中向該角色授予權限。

D. Assign the same KMS key used to encrypt data in Amazon S3 to the Amazon SageMaker notebook instance.

D. 將用於加密Amazon S3 中資料的相同KMS 金鑰指派給Amazon SageMaker 筆記本實例。

問題# 71

A Data Scientist needs to migrate an existing on-premises ETL process to the cloud. The current process runs at regular time intervals and uses PySpark to combine and format multiple large data sources into a single consolidated output for downstream processing. The Data Scientist has been given the following requirements to the cloud solution: Combine multiple data sources. Reuse existing PySpark logic. Run the solution on the existing schedule. Minimize the number of servers that will need to be managed. Which architecture should the Data Scientist use to build this solution?

資料科學家需要將現有的本地ETL 流程遷移到雲端。目前流程定期運行，並使用PySpark 將多個大型資料來源合併並格式化為單一合併輸出，以供下游處理。資料科學家對雲端解決方案有以下要求：合併多個資料來源。重複使用現有的PySpark 邏輯。按照現有時間表運行解決方案。盡量減少需要管理的伺服器數量。資料科學家應該使用哪種架構來建立此解決方案？

A. Write the raw data to Amazon S3. Schedule an AWS Lambda function to submit a Spark step to a persistent Amazon EMR cluster based on the existing schedule. Use the existing PySpark logic to run the ETL job on the EMR cluster. Output the results to a €processed€ location in Amazon S3 that is accessible for downstream use.

A. 將原始資料寫入Amazon S3。根據現有計劃安排AWS Lambda 函數將Spark 步驟提交到持久性Amazon EMR 叢集。使用現有的PySpark 邏輯在EMR 叢集上執行ETL 作業。將結果輸出至Amazon S3 中可供下游使用的「已處理」位置。

B. Write the raw data to Amazon S3. Create an AWS Glue ETL job to perform the ETL processing against the input data. Write the ETL job in PySpark to leverage the existing logic. Create a new AWS Glue trigger to trigger the ETL job based on the existing schedule. Configure the output target of the ETL job to write to a €processed€ location in Amazon S3 that is accessible for downstream use.

B. 將原始資料寫入Amazon S3。建立AWS Glue ETL 作業以針對輸入資料執行ETL 處理。在PySpark 中編寫ETL 作業以利用現有邏輯。建立新的AWS Glue 觸發器以根據現有計劃觸發ETL 作業。配置ETL 作業的輸出目標以寫入Amazon S3 中可供下游使用的「已處理」位置。

C. Write the raw data to Amazon S3. Schedule an AWS Lambda function to run on the existing schedule and process the input data from Amazon S3. Write the Lambda logic in Python and implement the existing PySpark logic to perform the ETL process. Have the Lambda function output the results to a €processed€ location in Amazon S3 that is accessible for downstream use.

C. 將原始資料寫入Amazon S3。安排AWS Lambda 函數依現有計畫執行並處理來自Amazon S3 的輸入資料。用Python 編寫Lambda 邏輯並實作現有的PySpark 邏輯以執行ETL 流程。讓Lambda 函數將結果輸出到Amazon S3 中可供下游使用的「已處理」位置。

D. Use Amazon Kinesis Data Analytics to stream the input data and perform real-time SQL queries against the stream to carry out the required transformations within the stream. Deliver the output results to a €processed€ location in Amazon S3 that is accessible for downstream use.

D. 使用Amazon Kinesis Data Analytics 串流輸入資料並針對串流執行即時SQL 查詢以在串流中執行所需的轉換。將輸出結果傳遞至Amazon S3 中可供下游使用的「已處理」位置。

問題# 72

A Data Scientist is building a model to predict customer churn using a dataset of 100 continuous numerical features. The Marketing team has not provided any insight about which features are relevant for churn prediction. The Marketing team wants to interpret the model and see the direct impact of relevant features on the model outcome. While training a logistic regression model, the Data Scientist observes that there is a wide gap between the training and validation set accuracy. Which methods can the Data Scientist use to improve the model performance and satisfy the Marketing team's needs? (Choose two.)

一位資料科學家正在使用包含100 個連續數值特徵的資料集建立一個模型來預測客戶流失。行銷團隊尚未提供任何有關哪些特徵與客戶流失預測相關的見解。行銷團隊希望解釋該模型並了解相關特徵對模型結果的直接影響。在訓練邏輯迴歸模型時，資料科學家發現訓練集和驗證集準確率之間存在很大差距。資料科學家可以使用哪些方法來提高模型效能並滿足行銷團隊的需求？（選兩個。）

A. Add L1 regularization to the classifier.

A. 在分類器上新增L1 正規化。

B. Add features to the dataset.

B. 在資料集中新增特徵。

C. Perform recursive feature elimination.

C. 執行遞歸特徵消除。

D. Perform t-distributed stochastic neighbor embedding (t-SNE).

D. 執行t 分佈隨機鄰域嵌入(t-SNE)。

E. Perform linear discriminant analysis.

E. 執行線性判別分析。

問題# 73

An aircraft engine manufacturing company is measuring 200 performance metrics in a time-series. Engineers want to detect critical manufacturing defects in near- real time during testing. All of the data needs to be stored for offline analysis. What approach would be the MOST effective to perform near-real time defect detection?

一家飛機引擎製造公司正在按時間序列測量200 個性能指標。工程師希望在測試期間近乎即時地檢測關鍵製造缺陷。所有數據都需要儲存以供離線分析。哪一種方法最有效地執行近乎即時的缺陷檢測？

A. Use AWS IoT Analytics for ingestion, storage, and further analysis. Use Jupyter notebooks from within AWS IoT Analytics to carry out analysis for anomalies.

A. 使用AWS IoT Analytics 進行提取、儲存和進一步分析。使用AWS IoT Analytics 中的Jupyter 筆記本執行異常分析。

B. Use Amazon S3 for ingestion, storage, and further analysis. Use an Amazon EMR cluster to carry out Apache Spark ML k-means clustering to determine anomalies.

B. 使用Amazon S3 進行提取、儲存和進一步分析。使用Amazon EMR 叢集執行Apache Spark ML k-means 聚類以確定異常。

C. Use Amazon S3 for ingestion, storage, and further analysis. Use the Amazon SageMaker Random Cut Forest (RCF) algorithm to determine anomalies.

C. 使用Amazon S3 進行提取、儲存和進一步分析。使用Amazon SageMaker 隨機森林砍伐(RCF) 演算法確定異常。

D. Use Amazon Kinesis Data Firehose for ingestion and Amazon Kinesis Data Analytics Random Cut Forest (RCF) to perform anomaly detection. Use Kinesis Data Firehose to store data in Amazon S3 for further analysis.

D. 使用Amazon Kinesis Data Firehose 進行萃取並使用Amazon Kinesis Data Analytics 隨機森林砍伐(RCF) 執行異常檢測。使用Kinesis Data Firehose 將資料儲存在Amazon S3 中以進行進一步分析。

問題# 74

A Machine Learning team runs its own training algorithm on Amazon SageMaker. The training algorithm requires external assets. The team needs to submit both its own algorithm code and algorithm-specific parameters to Amazon SageMaker. What combination of services should the team use to build a custom algorithm in Amazon SageMaker? (Choose two.)

機器學習團隊在Amazon SageMaker 上運行自己的訓練演算法。訓練演算法需要外部資產。團隊需要向Amazon SageMaker 提交自己的演算法程式碼和演算法特定參數。團隊應使用哪些服務組合在Amazon SageMaker 中建立自訂演算法？（選兩個。）

A. AWS Secrets Manager.

A. AWS Secrets Manager。

B. AWS CodeStar.

B. AWS CodeStar。

C. Amazon ECR.

C. Amazon ECR。

D. Amazon ECS.

D. Amazon ECS。

E. Amazon S3.

E. Amazon S3。

問題# 75

A Machine Learning Specialist wants to determine the appropriate SageMakerVariantInvocationsPerInstance setting for an endpoint automatic scaling configuration. The Specialist has performed a load test on a single instance and determined that peak requests per second (RPS) without service degradation is about 20 RPS. As this is the first deployment, the Specialist intends to set the invocation safety factor to 0.5. Based on the stated parameters and given that the invocations per instance setting is measured on a per-minute basis, what should the Specialist set as the SageMakerVariantInvocationsPerInstance setting?

機器學習專家想要確定終端節點自動擴展配置的適當 SageMakerVariantInvocationsPerInstance 設定。專家已對單一實例執行了負載測試，並確定在不降低服務品質的情況下，每秒的峰值請求數(RPS) 約為20 RPS。由於這是首次部署，專家打算將呼叫安全係數設為0.5。根據所述參數，並假設每個實例的呼叫次數設定是按分鐘衡量的，專家應將 SageMakerVariantInvocationsPerInstance 設定為多少？

A. 10.

A. 10

B. 30.

B. 30

C. 600.

C. 600

D. 2,400.

D. 2,400.

問題# 76

A company uses a long short-term memory (LSTM) model to evaluate the risk factors of a particular energy sector. The model reviews multi-page text documents to analyze each sentence of the text and categorize it as either a potential risk or no risk. The model is not performing well, even though the Data Scientist has experimented with many different network structures and tuned the corresponding hyperparameters. Which approach will provide the MAXIMUM performance boost?

一家公司使用長短期記憶(LSTM) 模型來評估特定能源產業的風險因素。該模型審查多頁文本文檔，分析文本的每個句子並將其歸類為潛在風險或無風險。儘管資料科學家已經嘗試了許多不同的網路結構並調整了相應的超參數，但該模型的表現並不理想。哪種方法可以提供最大的效能提升？

A. Initialize the words by term frequency-inverse document frequency (TF-IDF) vectors pretrained on a large collection of news articles related to the energy sector.

A. 透過在大量與能源產業相關的新聞文章上進行預訓練的詞頻-逆文檔頻率(TF-IDF) 向量初始化單字。

B. Use gated recurrent units (GRUs) instead of LSTM and run the training process until the validation loss stops decreasing.

B. 使用門控循環單元(GRU) 代替LSTM，並運行訓練過程，直到驗證損失停止減少。

C. Reduce the learning rate and run the training process until the training loss stops decreasing.

C. 降低學習率並執行訓練過程，直到訓練損失停止減少。

D. Initialize the words by word2vec embeddings pretrained on a large collection of news articles related to the energy sector.

D. 透過在大量與能源產業相關的新聞文章上進行預先訓練的word2vec 嵌入初始化單字。

問題# 77

A Machine Learning Specialist needs to move and transform data in preparation for training. Some of the data needs to be processed in near-real time, and other data can be moved hourly. There are existing Amazon EMR MapReduce jobs to clean and feature engineering to perform on the data. Which of the following services can feed data to the MapReduce jobs? (Choose two.)

機器學習專家需要移動和轉換資料以準備進行訓練。部分數據需要近乎即時地處理，其他數據可以每小時移動一次。現有Amazon EMR MapReduce 作業需要清理資料並對資料執行特徵工程。下列哪一項服務可以將資料提供給MapReduce 作業？（選兩項。）

A. AWS DMS.

A.AWS DMS。

B. Amazon Kinesis.

B.Amazon Kinesis。

C. AWS Data Pipeline.

C.AWS Data Pipeline。

D. Amazon Athena.

D.Amazon Athena。

E. Amazon ES.

E.Amazon ES。

問題# 78

A Machine Learning Specialist previously trained a logistic regression model using scikit-learn on a local machine, and the Specialist now wants to deploy it to production for inference only. What steps should be taken to ensure Amazon SageMaker can host a model that was trained locally?

一位機器學習專家之前在本地機器上使用scikit-learn 訓練了一個邏輯回歸模型，現在該專家只想將其部署到生產環境中進行推理。應採取哪些步驟來確保Amazon SageMaker 可以託管在本地訓練的模型？

A. Build the Docker image with the inference code. Tag the Docker image with the registry hostname and upload it to Amazon ECR.

A. 使用推理程式碼建立Docker 映像。使用登錄主機名稱標記Docker 映像並將其上傳至Amazon ECR。

B. Serialize the trained model so the format is compressed for deployment. Tag the Docker image with the registry hostname and upload it to Amazon S3.

B. 序列化訓練後的模型，以便壓縮格式以便部署。使用登錄主機名稱標記Docker 映像並將其上傳至Amazon S3。

C. Serialize the trained model so the format is compressed for deployment. Build the image and upload it to Docker Hub.

C. 序列化訓練後的模型，以便壓縮格式以便部署。建置映像並將其上傳到Docker Hub。

D. Build the Docker image with the inference code. Configure Docker Hub and upload the image to Amazon ECR.

D. 使用推理程式碼建立Docker 映像。配置Docker Hub 並將映像上傳到Amazon ECR。

問題# 79

A trucking company is collecting live image data from its fleet of trucks across the globe. The data is growing rapidly and approximately 100 GB of new data is generated every day. The company wants to explore machine learning uses cases while ensuring the data is only accessible to specific IAM users. Which storage option provides the most processing flexibility and will allow access control with IAM?

一家卡車運輸公司正在從遍布全球的卡車車隊收集即時影像數據。數據成長迅速，每天約產生100 GB 的新數據。該公司希望探索機器學習用例，同時確保只有特定的IAM 用戶才能存取資料。哪種儲存選項提供最大的處理彈性並允許使用IAM 進行存取控制？

A. Use a database, such as Amazon DynamoDB, to store the images, and set the IAM policies to restrict access to only the desired IAM users.

A. 使用資料庫（例如Amazon DynamoDB）來儲存映像，並設定IAM 策略以將存取權限限制為僅所需的IAM 使用者。

B. Use an Amazon S3-backed data lake to store the raw images, and set up the permissions using bucket policies.

B. 使用Amazon S3 支援的資料湖來儲存原始影像，並使用儲存桶策略設定權限。

C. Setup up Amazon EMR with Hadoop Distributed File System (HDFS) to store the files, and restrict access to the EMR instances using IAM policies.

C. 使用Hadoop 分散式檔案系統(HDFS) 設定Amazon EMR 來儲存文件，並使用IAM 策略限制對EMR 實例的存取。

D. Configure Amazon EFS with IAM policies to make the data available to Amazon EC2 instances owned by the IAM users.

D. 使用IAM 策略配置Amazon EFS，以使資料可供IAM 使用者擁有的Amazon EC2 執行個體使用。

問題# 80

A credit card company wants to build a credit scoring model to help predict whether a new credit card applicant will default on a credit card payment. The company has collected data from a large number of sources with thousands of raw attributes. Early experiments to train a classification model revealed that many attributes are highly correlated, the large number of features slows down the training speed significantly, and that there are some overfitting issues. The Data Scientist on this project would like to speed up the model training time without losing a lot of information from the original dataset. Which feature engineering technique should the Data Scientist use to meet the objectives?

一家信用卡公司希望建立一個信用評分模型，以幫助預測新的信用卡申請人是否會拖欠信用卡付款。該公司從大量來源收集了數千個原始屬性的數據。訓練分類模型的早期實驗表明，許多屬性高度相關，大量特徵會顯著降低訓練速度，並且存在一些過度擬合問題。該計畫的資料科學家希望加快模型訓練時間，而不會失去原始資料集中的大量資訊。資料科學家應該使用哪種特徵工程技術來實現目標？

A. Run self-correlation on all features and remove highly correlated features.

A. 對所有特徵執行自相關並刪除高度相關的特徵。

B. Normalize all numerical values to be between 0 and 1.

B. 將所有數值標準化為0 到1 之間。

C. Use an autoencoder or principal component analysis (PCA) to replace original features with new features.

C. 使用自動編碼器或主成分分析(PCA) 將原始特徵替換為新特徵。

D. Cluster raw data using k-means and use sample data from each cluster to build a new dataset.

D. 使用k 平均值對原始資料進行聚類，並使用每個聚類中的樣本資料建立新資料集。

問題# 81

A Data Scientist is training a multilayer perception (MLP) on a dataset with multiple classes. The target class of interest is unique compared to the other classes within the dataset, but it does not achieve an acceptable recall metric. The Data Scientist has already tried varying the number and size of the MLP's hidden layers, which has not significantly improved the results. A solution to improve recall must be implemented as quickly as possible. Which techniques should be used to meet these requirements?

一位資料科學家正在一個具有多個類別的資料集上訓練多層感知器(MLP)。目標類別與資料集內的其他類別相比是獨一無二的，但它沒有達到可接受的召回率指標。資料科學家已經嘗試改變MLP隱藏層的數量和大小，但結果並沒有顯著改善。必須盡快實施提高召回率的解決方案。應該使用哪些技術來滿足這些要求？

A. Gather more data using Amazon Mechanical Turk and then retrain.

A. 使用Amazon Mechanical Turk 收集更多數據，然後重新訓練。

B. Train an anomaly detection model instead of an MLP.

B. 訓練異常檢測模型而不是MLP。

C. Train an XGBoost model instead of an MLP.

C. 訓練XGBoost 模型而不是MLP。

D. Add class weights to the MLP's loss function and then retrain.

D. 將類別權重加入到MLP 的損失函數，然後重新訓練。

問題# 82

A Machine Learning Specialist works for a credit card processing company and needs to predict which transactions may be fraudulent in near-real time. Specifically, the Specialist must train a model that returns the probability that a given transaction may be fraudulent. How should the Specialist frame this business problem?

機器學習專家在信用卡處理公司工作，需要近乎即時地預測哪些交易可能是詐欺行為。具體來說，專家必須訓練一個模型，該模型返回給定交易可能是欺詐行為的機率。專家應該如何建構這個業務問題？

A. Streaming classification.

A. 流式分類。

B. Binary classification.

B. 二元分類。

C. Multi-category classification.

C. 多類別分類。

D. Regression classification.

D. 迴歸分類。

問題# 83

A real estate company wants to create a machine learning model for predicting housing prices based on a historical dataset. The dataset contains 32 features. Which model will meet the business requirement?

一家房地產公司希望創建一個基於歷史資料集預測房價的機器學習模型。此資料集包含32 個特徵。哪種模型可以滿足業務需求？

A. Logistic regression.

A. 邏輯迴歸。

B. Linear regression.

B. 線性迴歸。

C. K-means.

C. K 均值。

D. Principal component analysis (PCA).

D. 主成分分析(PCA)。

問題# 84

A Machine Learning Specialist is applying a linear least squares regression model to a dataset with 1,000 records and 50 features. Prior to training, the ML Specialist notices that two features are perfectly linearly dependent. Why could this be an issue for the linear least squares regression model?

機器學習專家正在將線性最小平方法迴歸模型應用於包含1,000 筆記錄和50 個特徵的資料集。在訓練之前，ML Specialist 注意到兩個特質完全線性相關。為什麼這會成為線性最小平方法迴歸模型的問題？

A. It could cause the backpropagation algorithm to fail during training.

A. 它可能導致反向傳播演算法在訓練期間失敗。

B. It could create a singular matrix during optimization, which fails to define a unique solution.

B. 它可能會在最佳化期間創建一個奇異矩陣，從而無法定義唯一的解決方案。

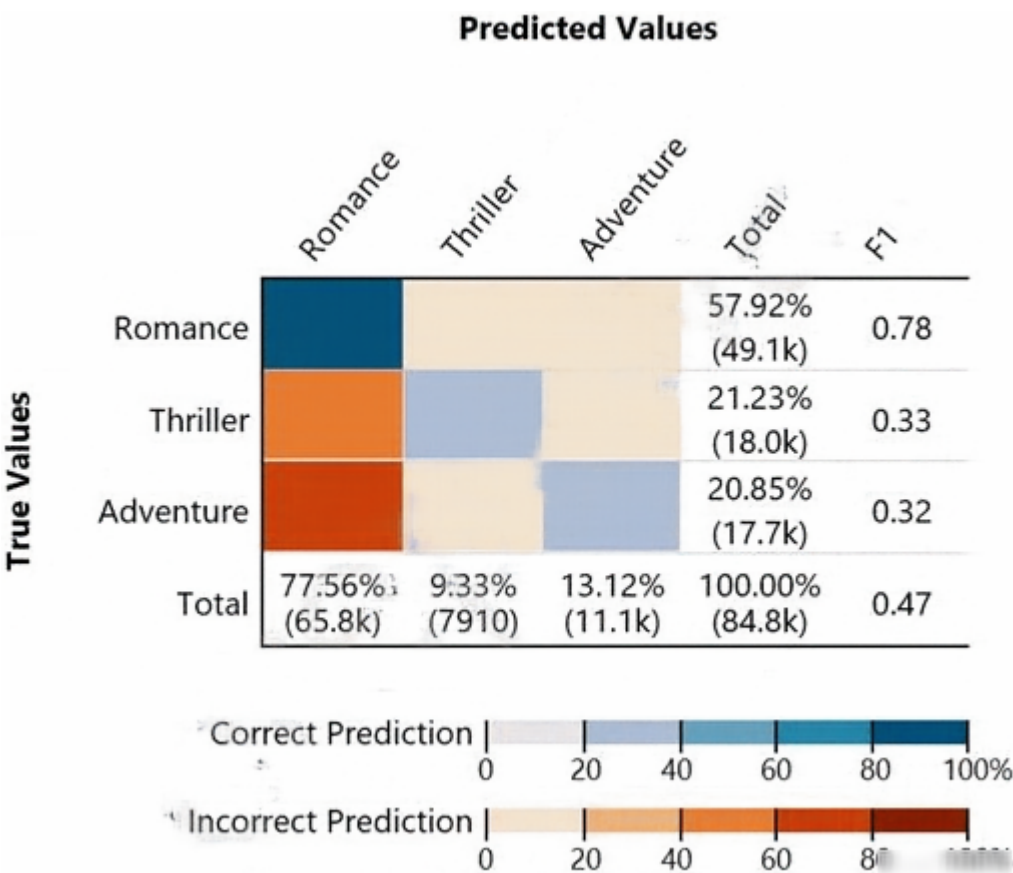
C. It could modify the loss function during optimization, causing it to fail during training.

- C. 它可能會在最佳化期間修改損失函數，導致其在訓練期間失敗。
- D. It could introduce non-linear dependencies within the data, which could invalidate the linear assumptions of the model.
- D. 它可能會在資料中引入非線性依賴關係，從而導致模型的線性假設無效。

問題# 85

Given the following confusion matrix for a movie classification model, what is the true class frequency for Romance and the predicted class frequency forAdventure?

給定以下電影分類模型的混淆矩陣，浪漫的真实類別頻率和冒險的預測類別頻率是多少？



- A. The true class frequency for Romance is 77.56% and the predicted class frequency for Adventure is 20.85%.
- A. 浪漫類的真實類頻為77.56%，而冒險類的預測類頻為20.85%。
- B. The true class frequency for Romance is 57.92% and the predicted class frequency for Adventure is 13.12%.
- B. 浪漫類的真實類頻為57.92%，而冒險類的預測類頻為13.12%。
- C. The true class frequency for Romance is 0.78 and the predicted class frequency for Adventure is (0.47-0.32).
- C. 浪漫類的真實類頻為0.78，而冒險類的預測類頻為(0.47-0.32)。

D. The true class frequency for Romance is 77.56% — 0.78 and the predicted class frequency for Adventure is 20.85% — 0.32.

D. 浪漫類的真實類頻為77.56% — 0.78，而冒險類的預測類頻為20.85% — 0.32。

問題# 86

A Machine Learning Specialist wants to bring a custom algorithm to Amazon SageMaker. The Specialist implements the algorithm in a Docker container supported by Amazon SageMaker. How should the Specialist package the Docker container so that Amazon SageMaker can launch the training correctly?

機器學習專家希望將自訂演算法引入Amazon SageMaker。專家在Amazon SageMaker 支援的Docker 容器中實作該演算法。專家應如何打包Docker 容器，以便Amazon SageMaker 可以正確啟動訓練？

A. Modify the bash_profile file in the container and add a bash command to start the training program.

A. 修改容器中的bash_profile 文件，新增啟動訓練程式的bash 指令。

B. Use CMD config in the Dockerfile to add the training program as a CMD of the image.

B. 使用Dockerfile 中的CMD config 將訓練程式新增為映像的CMD。

C. Configure the training program as an ENTRYPOINT named train.

C. 將訓練程式配置為名為train 的ENTRYPOINT。

D. Copy the training program to directory /opt/ml/train.

D. 將訓練程序複製到目錄/opt/ml/train。

問題# 87

A Data Scientist needs to analyze employment data. The dataset contains approximately 10 million observations on people across 10 different features. During the preliminary analysis, the Data Scientist notices that income and age distributions are not normal. While income levels shows a right skew as expected, with fewer individuals having a higher income, the age distribution also shows a right skew, with fewer older individuals participating in the workforce. Which feature transformations can the Data Scientist apply to fix the incorrectly skewed data? (Choose two.)

數據科學家需要分析就業數據。此資料集包含約1000 萬個關於10 個不同特徵的人員觀察值。在初步分析期間，資料科學家注意到收入和年齡分佈不正常。雖然收入水準如預期的呈現右偏態，收入較高的人數較少，但年齡分佈也呈現右偏態，參與勞動力的老年人較少。資料科學家可以應用哪些特徵轉換來修復錯誤傾斜的資料？（選兩個。）

A. Cross-validation.

A. 交叉驗證

B. Numerical value binning.

B. 數值分箱

C. High-degree polynomial transformation.

C. 高次多項式變換

D. Logarithmic transformation.

D. 對數轉換

E. One hot encoding.

E. 獨熱編碼。

問題# 88

A web-based company wants to improve its conversion rate on its landing page. Using a large historical dataset of customer visits, the company has repeatedly trained a multi-class deep learning network algorithm on Amazon SageMaker. However, there is an overfitting problem: training data shows 90% accuracy in predictions, while test data shows 70% accuracy only. The company needs to boost the generalization of its model before deploying it into production to maximize conversions of visits to purchases. Which action is recommended to provide the HIGHEST accuracy model for the company's test and validation data?

一家網路公司希望提高其登陸頁面的轉換率。該公司使用大量客戶存取歷史資料集，在Amazon SageMaker 上反覆訓練多類深度學習網路演算法。然而，有過度擬合問題：訓練資料顯示預測準確率為90%，而測試資料顯示準確率僅70%。該公司需要在將模型部署到生產環境之前提高其泛化能力，以最大限度地提高訪問購買的轉換率。建議採取哪種措施來為公司的測試和驗證數據提供最準確的模型？

A. Increase the randomization of training data in the mini-batches used in training.

A. 增加訓練中所使用的小批量中訓練資料的隨機化。

B. Allocate a higher proportion of the overall data to the training dataset.

B. 將更高比例的整體資料分配給訓練資料集。

C. Apply L1 or L2 regularization and dropouts to the training.

C. 對訓練應用L1 或L2 正規化和dropouts。

D. Reduce the number of layers and units (or neurons) from the deep learning network.

D. 減少深度學習網路中的層數和單元數（或神經元數）。

問題# 89

A Machine Learning Specialist is given a structured dataset on the shopping habits of a company's customer base. The dataset contains thousands of columns of data and hundreds of numerical columns for each customer. The Specialist wants to identify whether there are natural groupings for these columns across all customers and visualize the results as quickly as possible. What approach should the Specialist take to accomplish these tasks?

機器學習專家獲得了一個關於公司客戶群購物習慣的結構化資料集。該資料集包含數千列資料和每個客戶的數百個數值列。專家希望確定這些列在所有客戶中是否存在自然分組，並儘快將結果視覺化。專家應該採取什麼方法來完成這些任務？

A. Embed the numerical features using the t-distributed stochastic neighbor embedding (t-SNE) algorithm and create a scatter plot.

A. 使用t 分佈隨機鄰域嵌入(t-SNE) 演算法嵌入數值特徵並建立散佈圖。

B. Run k-means using the Euclidean distance measure for different values of k and create an elbow plot.

B. 使用歐幾里德距離測量對不同的k 值運行k 均值並建立肘形圖。

C. Embed the numerical features using the t-distributed stochastic neighbor embedding (t-SNE) algorithm and create a line graph.

C. 使用t 分佈隨機鄰域嵌入(t-SNE) 演算法嵌入數值特徵並建立折線圖。

D. Run k-means using the Euclidean distance measure for different values of k and create box plots for each numerical column within each cluster.

D. 使用歐幾里德距離測量對不同的k 值運行k 均值並為每個聚類內的每個數值列建立箱線圖。

問題# 90

A Machine Learning Specialist is planning to create a long-running Amazon EMR cluster. The EMR cluster will have 1 master node, 10 core nodes, and 20 task nodes. To save on costs, the Specialist will use Spot Instances in the EMR cluster. Which nodes should the Specialist launch on Spot Instances?

一位機器學習專家計劃創建一個長期運行的Amazon EMR 叢集。EMR 叢集將有1 個主節點、10 個核心節點和20 個任務節點。為了節省成本，專家將在EMR 叢集中使用Spot 實例。專家應該在Spot 實例上啟動哪些節點？

A. Master node.

A. 主節點。

B. Any of the core nodes.

B. 任意核心節點。

C. Any of the task nodes.

C. 任意任務節點。

D. Both core and task nodes.

D. 核心節點和任務節點。

問題# 91

A manufacturer of car engines collects data from cars as they are being driven. The data collected includes timestamp, engine temperature, rotations per minute(RPM), and other sensor readings. The company wants to predict when an engine is going to have a problem, so it can notify drivers in advance to get engine maintenance. The engine data is loaded into a data lake for training. Which is the MOST suitable predictive model that can be deployed into production?

一家汽車引擎製造商在汽車行駛過程中收集數據。收集的數據包括時間戳記、引擎溫度、每分鐘轉速(RPM) 和其他感測器讀數。該公司希望預測引擎何時會出現問題，以便提前通知駕駛員進行引擎維護。發動機數據被加載到數據湖中進行訓練。哪種預測模型最適合部署到生產中？

A. Add labels over time to indicate which engine faults occur at what time in the future to turn this into a supervised learning problem. Use a recurrent neural network (RNN) to train the model to recognize when an engine might need maintenance for a certain fault.

A. 隨著時間的推移添加標籤，以指示哪些引擎故障在未來的什麼時候發生，從而將其轉變為監督學習問題。使用循環神經網路(RNN) 訓練模型以識別引擎何時可能需要維護某個故障。

B. This data requires an unsupervised learning algorithm. Use Amazon SageMaker k-means to cluster the data.

B. 此數據需要無監督學習演算法。使用Amazon SageMaker k-means 對資料進行聚類。

C. Add labels over time to indicate which engine faults occur at what time in the future to turn this into a supervised learning problem. Use a convolutional neural network (CNN) to train the model to recognize when an engine might need maintenance for a certain fault.

C. 隨著時間的推移添加標籤，以指示哪些引擎故障在未來的什麼時候發生，從而將其轉變為監督學習問題。使用卷積神經網路(CNN) 訓練模型以識別引擎何時可能需要維護某個故障。

D. This data is already formulated as a time series. Use Amazon SageMaker seq2seq to model the time series.

D. 此數據已製定為時間序列。使用Amazon SageMaker seq2seq 對時間序列進行建模。

問題# 92

A company wants to predict the sale prices of houses based on available historical sales data. The target variable in the company's dataset is the sale price. The features include parameters such as the lot size, living area measurements, non-living area measurements, number of bedrooms, number of bathrooms, year built, and postal code. The company wants to use multi-variable linear regression to predict house sale prices. Which step should a machine learning specialist take to remove features that are irrelevant for the analysis and reduce the model's complexity?

一家公司希望根據可用的歷史銷售數據預測房屋的銷售價格。該公司資料集中的目標變數是銷售價格。這些特徵包括地塊大小、居住面積測量、非居住面積測量、臥室數量、浴室數量、建造年份和郵遞區號等參數。該公司希望使用多變量線性迴歸來預測房屋銷售價格。機器學習專家應該採取哪

些步驟來刪除與分析無關的特徵並降低模型的複雜性？

A. Plot a histogram of the features and compute their standard deviation. Remove features with high variance.

A. 繪製特徵的直方圖並計算其標準差。刪除方差大的特徵。

B. Plot a histogram of the features and compute their standard deviation. Remove features with low variance.

B. 繪製特徵的直方圖並計算其標準差。刪除方差低的特徵。

C. Build a heatmap showing the correlation of the dataset against itself. Remove features with low mutual correlation scores.

C. 建立熱圖以顯示資料集與自身的相關性。刪除相互相關分數低的特徵。

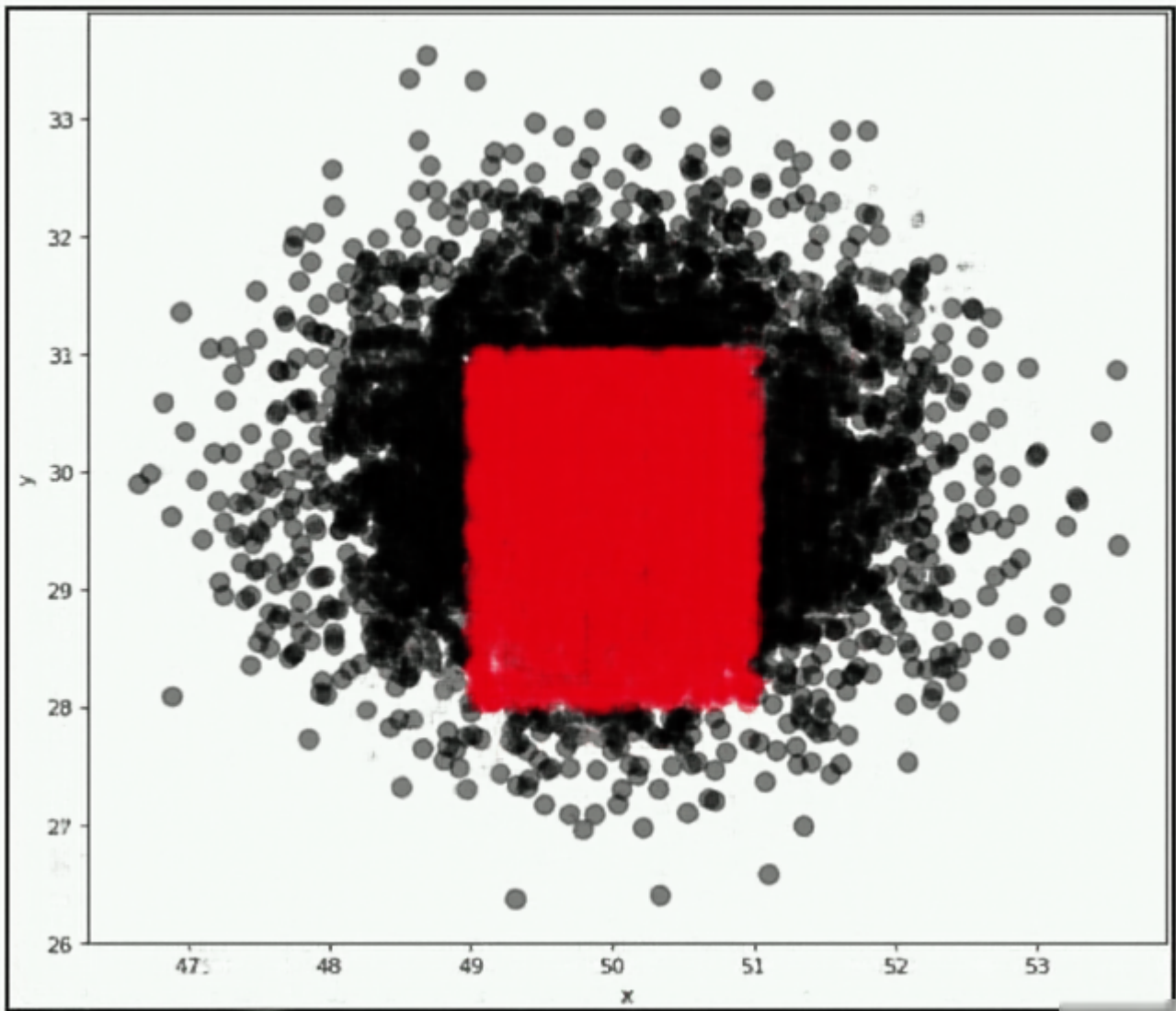
D. Run a correlation check of all features against the target variable. Remove features with low target variable correlation scores.

D. 針對目標變數執行所有特徵的相關性檢查。刪除目標變數相關性分數低的特徵。

問題# 93

A company wants to classify user behavior as either fraudulent or normal. Based on internal research, a machine learning specialist will build a binary classifier based on two features: age of account, denoted by x , and transaction month, denoted by y . The class distributions are illustrated in the provided figure. The positive class is portrayed in red, while the negative class is portrayed in black. Which model would have the HIGHEST accuracy?

一家公司希望將用戶行為分類為詐欺行為或正常行為。根據內部研究，機器學習專家將基於兩個特徵建立二元分類器：帳戶年齡（以 x 表示）和交易月份（以 y 表示）。類別分佈如提供的圖所示。正類用紅色表示，負類用黑色表示。哪個模型的準確率最高？



A. Linear support vector machine (SVM).

A. 線性支援向量機(SVM)。

B. Decision tree.

B. 決策樹。

C. Support vector machine (SVM) with a radial basis function kernel.

C. 具有徑向基底函數核的支援向量機(SVM)。

D. Single perceptron with a Tanh activation function.

D. 具有Tanh 激活函數的單一感知器。

A health care company is planning to use neural networks to classify their X-ray images into normal and abnormal classes. The labeled data is divided into a training set of 1,000 images and a test set of 200 images. The initial training of a neural network model with 50 hidden layers yielded 99% accuracy on the training set, but only 55% accuracy on the test set. What changes should the Specialist consider to solve this issue? (Choose three.)

一家醫療保健公司計劃使用神經網路將X光影像分為正常和異常類別。標記資料分為1,000張圖像的訓練集和200張圖像的測試集。具有50個隱藏層的神經網路模型的初始訓練在訓練集上的準確率為99%，但在測試集上的準確率僅為55%。專家應考慮進行哪些更改來解決這個問題？（選三個。）

A. Choose a higher number of layers.

A. 選擇較高的層數。

B. Choose a lower number of layers.

B. 選擇較低的層數。

C. Choose a smaller learning rate.

C. 選擇較小的學習率。

D. Enable dropout.

D. 啟用dropout。

E. Include all the images from the test set in the training set.

E. 將測試集中的所有圖像包含在訓練集中。

F. Enable early stopping.

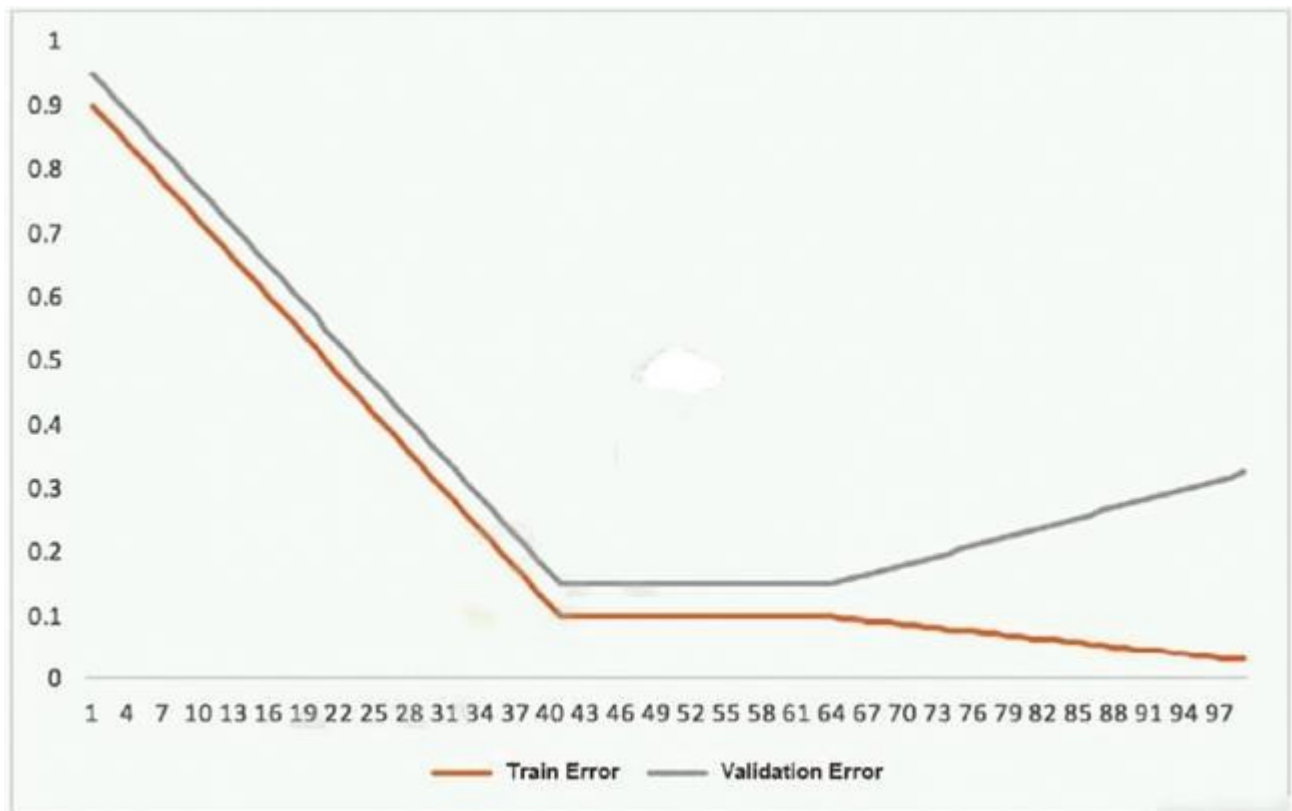
F. 啟用早期停止。

問題# 95

This graph shows the training and validation loss against the epochs for a neural network. The network being trained is as follows: Two dense layers, one output neuron 100 neurons in each layer 100 epochs Random initialization of weights Which technique can be used to improve model performance in terms of accuracy in the validation set?

此圖顯示了神經網路在各個時期的訓練和驗證損失。正在訓練的網路如下：兩個密集層，一個輸出神經元 每層100個神經元 100個時期隨機初始化權重 哪種技術可用於提高模型在驗證集準確率方面的性能？

▪



A. Early stopping.

A. 提前停止。

B. Random initialization of weights with appropriate seed.

B. 使用適當的種子隨機初始化權重。

C. Increasing the number of epochs.

C. 增加迭代次數。

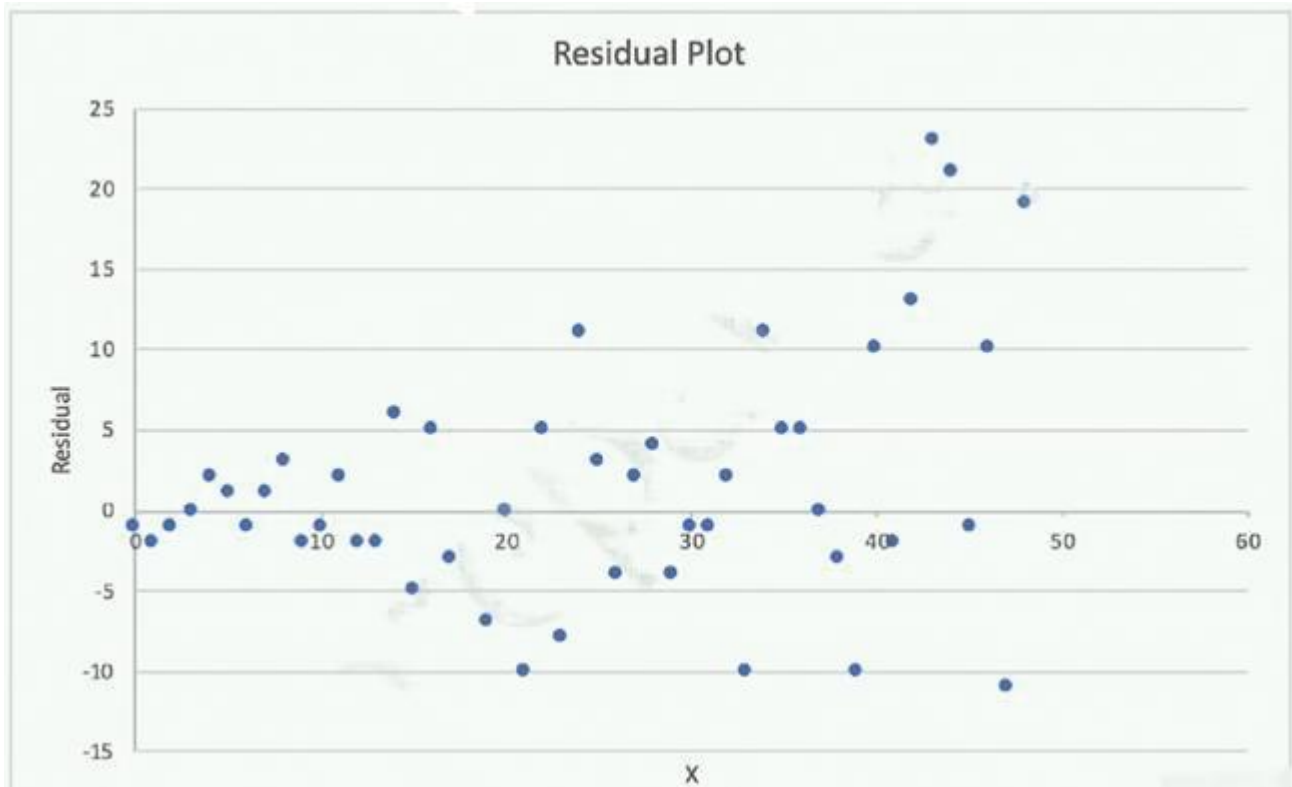
D. Adding another layer with the 100 neurons.

D. 增加另一層，其中包含100 個神經元。

問題# 96

A Machine Learning Specialist is attempting to build a linear regression model. Given the displayed residual plot only, what is the MOST likely problem with the model?

機器學習專家正在嘗試建立線性迴歸模型。僅給出顯示的殘差圖，模型最可能的問題是什麼？



A. Linear regression is inappropriate. The residuals do not have constant variance.

A. 線性迴歸不合適。殘差不具有恆定變異數。

B. Linear regression is inappropriate. The underlying data has outliers.

B. 線性迴歸不合適。基礎數據有異常值。

C. Linear regression is appropriate. The residuals have a zero mean.

C. 線性迴歸合適。殘差具有零均值。

D. Linear regression is appropriate. The residuals have constant variance.

D. 線性迴歸合適。殘差具有恆定方差。

問題# 97

A large company has developed a BI application that generates reports and dashboards using data collected from various operational metrics. The company wants to provide executives with an enhanced experience so they can use natural language to get data from the reports. The company wants the executives to be able ask questions using written and spoken interfaces. Which combination of services can be used to build this conversational interface? (Choose three.)

一家大型公司開發了一款BI 應用程式，該應用程式使用從各種營運指標收集的數據來產生報告和儀表板。該公司希望為高階主管提供增強的體驗，以便他們可以使用自然語言從報告中獲取數據。該公司希望高階主管能夠使用書面和口頭介面提出問題。可以使用哪種服務組合來建立此對話式介面？（選三個。）

A. Alexa for Business.

A. Alexa for Business。

B. Amazon Connect.

B. Amazon Connect。

C. Amazon Lex.

C. Amazon Lex。

D. Amazon Polly.

D. Amazon Polly。

E. Amazon Comprehend.

E. Amazon Comprehend。

F. Amazon Transcribe.

F. Amazon Transcribe。

問題# 98

A machine learning specialist works for a fruit processing company and needs to build a system that categorizes apples into three types. The specialist has collected a dataset that contains 150 images for each type of apple and applied transfer learning on a neural network that was pretrained on ImageNet with this dataset. The company requires at least 85% accuracy to make use of the model. After an exhaustive grid search, the optimal hyperparameters produced the following: 68% accuracy on the training set 67% accuracy on the validation set. What can the machine learning specialist do to improve the system's accuracy?

一家水果加工公司的機器學習專家需要建造一個將蘋果分為三種類型的系統。該專家收集了一個包含每種蘋果150張圖像的資料集，並在一個使用該資料集在ImageNet上預先訓練的神經網路上應用了遷移學習。該公司要求至少85%的準確率才能使用該模型。經過詳盡的網格搜尋後，最佳超參數產生了以下結果：訓練集準確率為68% 驗證集準確率為67%。機器學習專家可以做些什麼來提高系統的準確率？

A. Upload the model to an Amazon SageMaker notebook instance and use the Amazon SageMaker HPO feature to optimize the model's hyperparameters.

A. 將模型上傳至Amazon SageMaker 筆電實例並使用Amazon SageMaker HPO 功能最佳化模型的超參數。

B. Add more data to the training set and retrain the model using transfer learning to reduce the bias.

B. 在訓練集中添加更多資料並使用遷移學習重新訓練模型以減少偏差。

C. Use a neural network model with more layers that are pretrained on ImageNet and apply transfer learning to increase the variance.

C. 使用在ImageNet 上預先訓練的具有更多層的神經網路模型並應用遷移學習來增加方差。

D. Train a new model using the current neural network architecture.

D. 使用目前的神經網路架構訓練新模型。

問題# 99

A company uses camera images of the tops of items displayed on store shelves to determine which items were removed and which ones still remain. After several hours of data labeling, the company has a total of 1,000 hand-labeled images covering 10 distinct items. The training results were poor. Which machine learning approach fulfills the company's long-term needs?

一家公司使用相機拍攝的貨架上陳列商品頂部的圖像來確定哪些商品被移除，哪些商品仍然保留。經過幾個小時的數據標記，該公司總共擁有1,000 張手工標記的圖像，涵蓋10 個不同的商品。訓練結果很差。哪一種機器學習方法可以滿足公司的長期需求？

A. Convert the images to grayscale and retrain the model.

A. 將影像轉換為灰階並重新訓練模型。

B. Reduce the number of distinct items from 10 to 2, build the model, and iterate.

B. 將不同項目的數量從10 個減少到2 個，建立模型並進行迭代。

C. Attach different colored labels to each item, take the images again, and build the model.

C. 為每個項目附加不同顏色的標籤，再次拍攝影像並建立模型。

D. Augment training data for each item using image variants like inversions and translations, build the model, and iterate.

D. 使用反轉和平移等影像變體增強每個項目的訓練數據，建立模型並進行迭代。

問題# 100

A Data Scientist is developing a binary classifier to predict whether a patient has a particular disease on a series of test results. The Data Scientist has data on 400 patients randomly selected from the population. The disease is seen in 3% of the population. Which cross-validation strategy should the Data Scientist adopt?

一位資料科學家正在開發一個二元分類器，根據一系列測試結果預測患者是否患有某種疾病。資料科學家擁有從人群中隨機選擇的400 名患者的數據。3% 的人口患有這種疾病。資料科學家應該採用哪種交叉驗證策略？

A. A k-fold cross-validation strategy with $k=5$.

A. $k=5$ 的 k 倍交叉驗證策略。

B. A stratified k-fold cross-validation strategy with $k=5$.

B. $k=5$ 的分層 k 倍交叉驗證策略。

C. A k-fold cross-validation strategy with $k=5$ and 3 repeats.

C. $k=5$ 且重複3次的 k 倍交叉驗證策略。

D. An 80/20 stratified split between training and validation.

D. 訓練與驗證之間的80/20分層分割。

問題# 101

A technology startup is using complex deep neural networks and GPU compute to recommend the company's products to its existing customers based upon each customer's habits and interactions. The solution currently pulls each dataset from an Amazon S3 bucket before loading the data into a TensorFlow model pulled from the company's Git repository that runs locally. This job then runs for several hours while continually outputting its progress to the same S3 bucket. The job can be paused, restarted, and continued at any time in the event of a failure, and is run from a central queue. Senior managers are concerned about the complexity of the solution's resource management and the costs involved in repeating the process regularly. They ask for the workload to be automated so it runs once a week, starting Monday and completing by the close of business Friday. Which architecture should be used to scale the solution at the lowest cost?

一家科技新創公司正在使用複雜的深度神經網路和GPU運算，根據每個客戶的習慣和互動向現有客戶推薦公司的產品。該解決方案目前從Amazon S3儲存桶中提取每個資料集，然後將資料載入到從本地運行的公司Git儲存庫中提取的TensorFlow模型中。然後，此作業運行數小時，同時不斷將其進度輸出到同一個S3儲存桶。如果發生故障，作業可以隨時暫停、重新啟動和繼續，並且從中央佇列執行。高階經理擔心解決方案資源管理的複雜性以及定期重複流程所涉及的成本。他們要求工作負載自動化，以便每週運行一次，從週一開始，到週五下班前完成。應該使用哪種架構以最低成本擴展解決方案？

A. Implement the solution using AWS Deep Learning Containers and run the container as a job using AWS Batch on a GPU-compatible Spot Instance.

A. 使用AWS Deep Learning Containers實作解決方案，並使用AWS Batch在GPU相容的Spot實例上將容器作為作業運行。

B. Implement the solution using a low-cost GPU-compatible Amazon EC2 instance and use the AWS Instance Scheduler to schedule the task.

B. 使用低成本的GPU相容Amazon EC2執行個體實作解決方案，並使用AWS Instance Scheduler安排任務。

C. Implement the solution using AWS Deep Learning Containers, run the workload using AWS Fargate running on Spot Instances, and then schedule the task using the built-in task scheduler.

C. 使用AWS Deep Learning Containers 實施解決方案，使用在Spot 執行個體上執行的AWS Fargate 執行工作負載，然後使用內建任務排程程式安排任務。

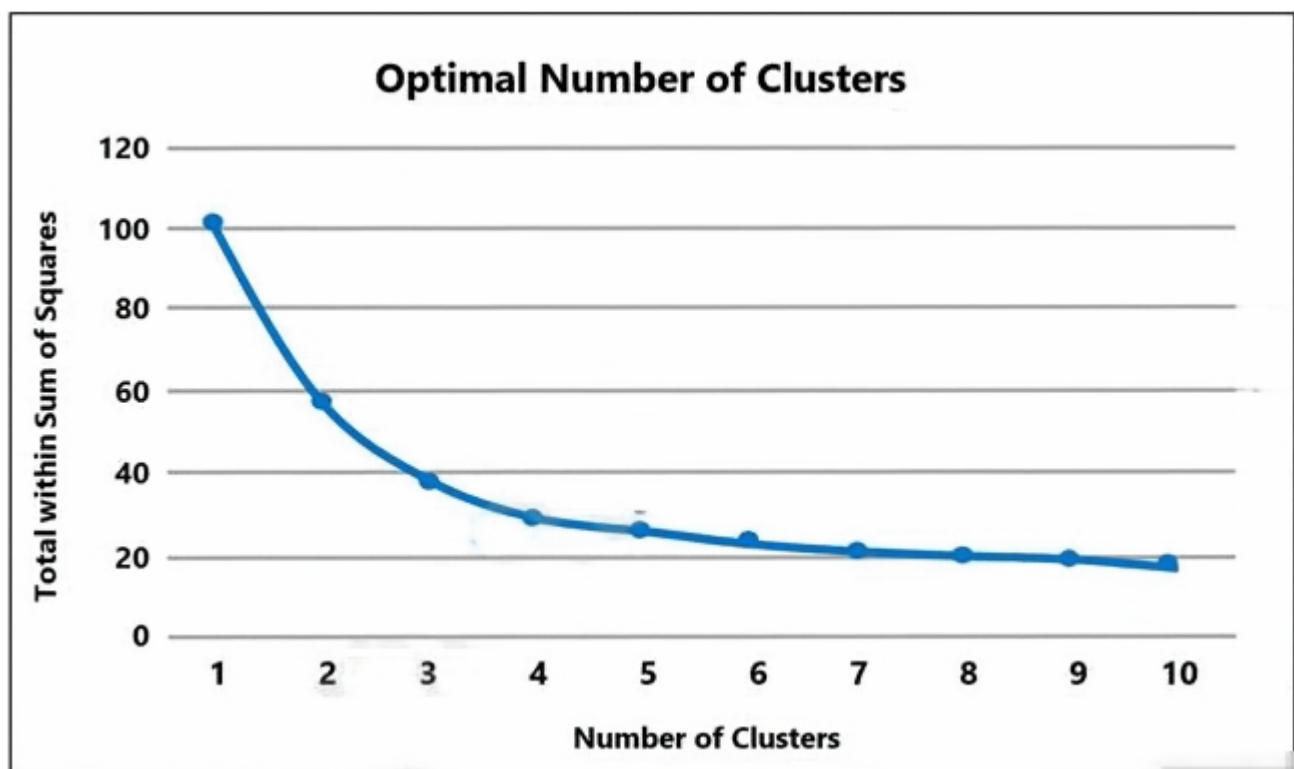
D. Implement the solution using Amazon ECS running on Spot Instances and schedule the task using the ECS service scheduler.

D. 使用在Spot 實例上執行的Amazon ECS 實施解決方案，並使用ECS 服務計劃程式安排任務。

問題# 102

A Machine Learning Specialist prepared the following graph displaying the results of k-means for $k = [1..10]$: Considering the graph, what is a reasonable selection for the optimal choice of k ?

一位機器學習專家準備了以下圖表，展示了 $k = [1..10]$ 的 k 均值結果：考慮該圖，最佳 k 選擇的合理選擇是什麼？



A. 1.

A. 1

B. 4.

B. 4

C. 7.

C. 7

D. 10.

D. 10

問題# 103

A media company with a very large archive of unlabeled images, text, audio, and video footage wishes to index its assets to allow rapid identification of relevant content by the Research team. The company wants to use machine learning to accelerate the efforts of its in-house researchers who have limited machine learning expertise. Which is the FASTEST route to index the assets?

一家媒體公司擁有大量未標記的圖像、文字、音訊和視訊片段檔案，希望對其資產進行索引，以便研究團隊快速識別相關內容。該公司希望使用機器學習來加速其內部研究人員的工作，因為這些研究人員的機器學習專業知識有限。哪一種是索引資產最快的方法？

A. Use Amazon Rekognition, Amazon Comprehend, and Amazon Transcribe to tag data into distinct categories/classes.

A. 使用Amazon Rekognition、Amazon Comprehend 和Amazon Transcribe 將資料標記為不同的類別/類別。

B. Create a set of Amazon Mechanical Turk Human Intelligence Tasks to label all footage.

B. 創建一組Amazon Mechanical Turk 人類智慧任務來標記所有鏡頭。

C. Use Amazon Transcribe to convert speech to text. Use the Amazon SageMaker Neural Topic Model (NTM) and Object Detection algorithms to tag data into distinct categories/classes.

C. 使用Amazon Transcribe 將語音轉換為文字。使用Amazon SageMaker 神經主題模型(NTM) 和物件偵測演算法將資料標記為不同的類別/類別。

D. Use the AWS Deep Learning AMI and Amazon EC2 GPU instances to create custom models for audio transcription and topic modeling, and use object detection to tag data into distinct categories/classes.

D. 使用AWS Deep Learning AMI 和Amazon EC2 GPU 執行個體為音訊轉錄和主題建模建立自訂模型，並使用物件偵測將資料標記為不同的類別/類別。

問題# 104

A Machine Learning Specialist is working for an online retailer that wants to run analytics on every customer visit, processed through a machine learning pipeline. The data needs to be ingested by Amazon Kinesis Data Streams at up to 100 transactions per second, and the JSON data blob is 100 KB in size. What is the MINIMUM number of shards in Kinesis Data Streams the Specialist should use to successfully ingest this data?

一位機器學習專家正在為一家線上零售商工作，該公司希望對每次客戶訪問進行分析，並透過機器學習管道進行處理。資料需要由Amazon Kinesis Data Streams 以每秒最多100 個事務的速度擷取，JSON 資料blob 大小為100 KB。專家應使用Kinesis Data Streams 中的最少分片數是多少才能成功提取這些資料？

A. 1 shards.

A. 1 個碎片。

B. 10 shards.

B. 10 個碎片。

C. 100 shards.

C. 100 個碎片。

D. 1,000 shards.

D. 1,000 個碎片。

問題# 105

A Machine Learning Specialist is deciding between building a naive Bayesian model or a full Bayesian network for a classification problem. The Specialist computes the Pearson correlation coefficients between each feature and finds that their absolute values range between 0.1 to 0.95. Which model describes the underlying data in this situation?

機器學習專家正在決定是建立樸素貝葉斯模型還是完整貝葉斯網路來解決分類問題。專家計算每個特徵之間的皮爾森相關係數，發現它們的絕對值介於0.1 到0.95 之間。在這種情況下，哪種模型可以描述基礎資料？

A. A naive Bayesian model, since the features are all conditionally independent.

A. 一個樸素貝葉斯模型，因為所有特徵都是條件獨立的。

B. A full Bayesian network, since the features are all conditionally independent.

B. 一個完整的貝葉斯網路，因為所有特徵都是條件獨立的。

C. A naive Bayesian model, since some of the features are statistically dependent.

C. 一個樸素貝葉斯模型，因為一些特徵在統計上是相關的。

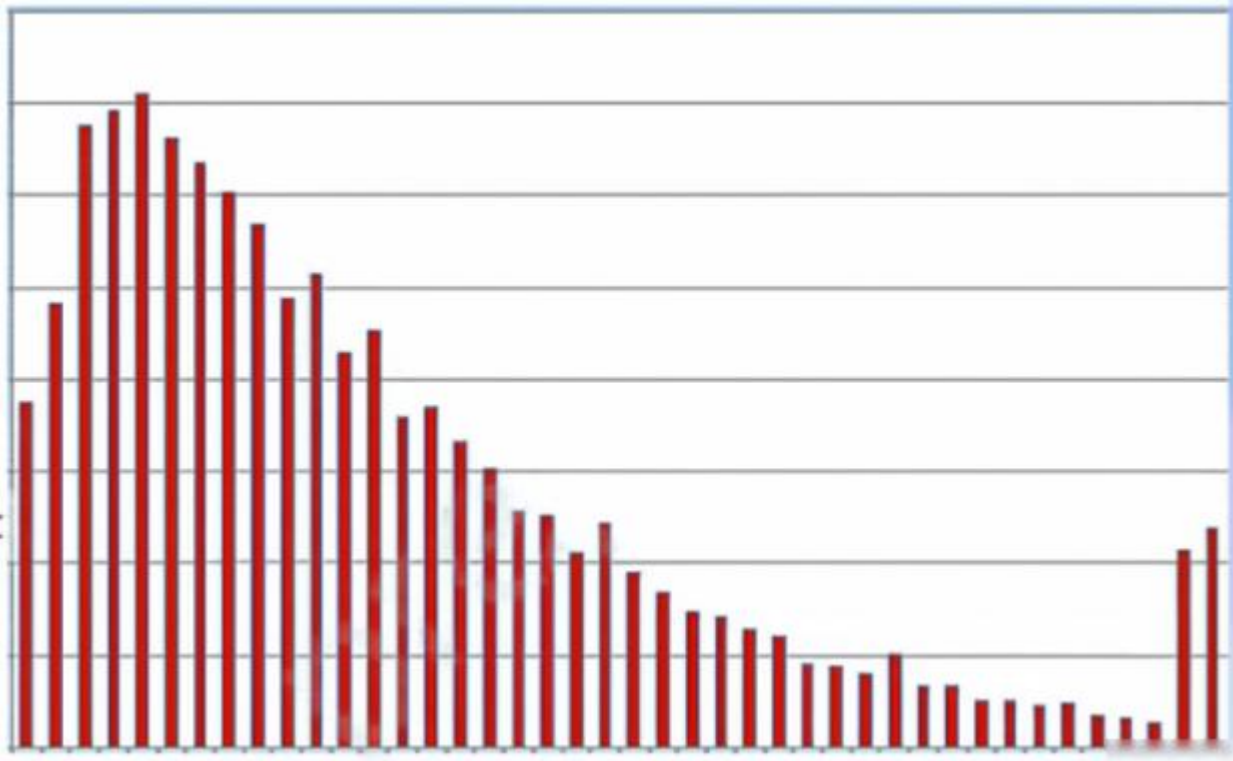
D. A full Bayesian network, since some of the features are statistically dependent.

D. 一個完整的貝葉斯網路，因為一些特徵在統計上是相關的。

問題# 106

A Data Scientist is building a linear regression model and will use resulting p-values to evaluate the statistical significance of each coefficient. Upon inspection of the dataset, the Data Scientist discovers that most of the features are normally distributed. The plot of one feature in the dataset is shown in the graphic. What transformation should the Data Scientist apply to satisfy the statistical assumptions of the linear regression model?

資料科學家正在建立線性迴歸模型，並將使用所得的p 值來評估每個係數的統計顯著性。在檢查資料集後，資料科學家發現大多數特徵呈現常態分佈。圖中顯示了資料集中一個特徵的圖。資料科學家應該應用哪一種轉換來滿足線性迴歸模型的統計假設？



A. Exponential transformation.

A. 指數變換。

B. Logarithmic transformation.

B. 對數轉換。

C. Polynomial transformation.

C. 多項式變換。

D. Sinusoidal transformation.

D. 正弦變換。

A Machine Learning Specialist is assigned to a Fraud Detection team and must tune an XGBoost model, which is working appropriately for test data. However, with unknown data, it is not working as expected. The existing parameters are provided as follows. Which parameter tuning guidelines should the Specialist follow to avoid overfitting?

機器學習專家被分配到詐欺偵測團隊，必須調整XGBoost 模型，該模型適用於測試資料。但是，對於未知數據，它無法按預期工作。現有參數如下。專家應遵循哪些參數調整指南以避免過度擬合？

```
param = {  
    'eta': 0.05, # the training step for each iteration  
    'silent': 1, # logging mode - quiet  
    'n_estimators': 2000,  
    'max_depth': 30,  
    'min_child_weight': 3,  
    'gamma': 0,  
    'subsample': 0.8,  
    'objective': 'multi:softprob', # error evaluation for multiclass training  
    'num_class': 201} # the number of classes that exist in this dataset  
num_round = 60 # the number of training iterations
```

A. Increase the max_depth parameter value.

A. 增加max_depth 參數值。

B. Lower the max_depth parameter value.

B. 降低max_depth 參數值。

C. Update the objective to binary:logistic.

C. 將目標更新為binary:logistic。

D. Lower the min_child_weight parameter value.

D. 降低min_child_weight 參數值。

問題# 108

A data scientist is developing a pipeline to ingest streaming web traffic data. The data scientist needs to implement a process to identify unusual web traffic patterns as part of the pipeline. The patterns will be used downstream for alerting and incident response. The data scientist has access to unlabeled historic data to use, if needed. The solution needs to do the following: Calculate an anomaly score for each web traffic entry. Adapt unusual event identification to changing web patterns over time. Which approach should the data scientist implement to meet these requirements?

一位資料科學家正在開發一個管道來提取串流網路流量資料。資料科學家需要實施一個流程來識別異常的網路流量模式作為管道的一部分。這些模式將在下游用於警報和事件響應。如果需要，資料科學家可以存取未標記的歷史資料以供使用。解決方案需要執行以下操作：計算每個網路流量條目的異常分數。使異常事件識別適應隨時間變化的網路模式。資料科學家應該實施哪種方法來滿足這些要求？

▪

A. Use historic web traffic data to train an anomaly detection model using the Amazon SageMaker Random Cut Forest (RCF) built-in model. Use an Amazon Kinesis Data Stream to process the incoming web traffic data. Attach a preprocessing AWS Lambda function to perform data enrichment by calling the RCF model to calculate the anomaly score for each record.

A. 使用歷史網路流量數據，透過Amazon SageMaker 隨機森林(RCF) 內建模型訓練異常偵測模型。使用Amazon Kinesis 資料流處理傳入的網路流量資料。附加預處理AWS Lambda 函數，透過呼叫RCF 模型計算每筆記錄的異常分數來執行資料豐富。

B. Use historic web traffic data to train an anomaly detection model using the Amazon SageMaker built-in XGBoost model. Use an Amazon Kinesis Data Stream to process the incoming web traffic data. Attach a preprocessing AWS Lambda function to perform data enrichment by calling the XGBoost model to calculate the anomaly score for each record.

B. 使用歷史網路流量數據，透過Amazon SageMaker 內建XGBoost 模型訓練異常偵測模型。使用Amazon Kinesis 資料流處理傳入的網路流量資料。附加預處理AWS Lambda 函數，透過呼叫XGBoost 模型計算每筆記錄的異常分數來執行資料豐富。

C. Collect the streaming data using Amazon Kinesis Data Firehose. Map the delivery stream as an input source for Amazon Kinesis Data Analytics. Write a SQL query to run in real time against the streaming data with the k-Nearest Neighbors (kNN) SQL extension to calculate anomaly scores for each record using a tumbling window.

C. 使用Amazon Kinesis Data Firehose 收集流資料。將傳輸流對應為Amazon Kinesis Data Analytics 的輸入源。編寫SQL 查詢以使用k-Nearest Neighbors (kNN) SQL 擴充功能即時針對串流資料運行，以使用滾動視窗計算每筆記錄的異常分數。

D. Collect the streaming data using Amazon Kinesis Data Firehose. Map the delivery stream as an input source for Amazon Kinesis Data Analytics. Write a SQL query to run in real time against the streaming data with the Amazon Random Cut Forest (RCF) SQL extension to calculate anomaly scores for each record using a sliding window.

D. 使用Amazon Kinesis Data Firehose 收集流資料。將交付流對應為Amazon Kinesis Data Analytics 的輸入來源。編寫SQL 查詢以使用Amazon Random Cut Forest (RCF) SQL 擴充功能針對串流資料即時執行，以使用滑動視窗計算每筆記錄的異常分數。

問題# 109

A Data Scientist received a set of insurance records, each consisting of a record ID, the final outcome among 200 categories, and the date of the final outcome. Some partial information on claim contents is also provided, but only for a few of the 200 categories. For each outcome category, there are hundreds of records distributed over the past 3 years. The Data Scientist wants to predict how many claims to expect in each category from month to month, a few months in advance. What type of machine learning model should be used?

一位資料科學家收到了一組保險記錄，每筆記錄都包含一個記錄ID、200 個類別中的最終結果以及最終結果的日期。還提供了索賠內容的部分信息，但僅針對200 個類別中的幾個。對於每個結果類別，都有數百筆記錄分佈在過去3 年。資料科學家想要提前幾個月預測每個類別每月預計有多少索賠。應該使用哪種類型的機器學習模型？

A. Classification month-to-month using supervised learning of the 200 categories based on claim contents.

A. 使用基於索賠內容的200 個類別的監督學習進行逐月分類。

B. Reinforcement learning using claim IDs and timestamps where the agent will identify how many claims in each category to expect from month to month.

B. 使用索賠ID 和時間戳進行強化學習，其中代理商將確定每個類別中預計每月有多少索賠。

C. Forecasting using claim IDs and timestamps to identify how many claims in each category to expect from month to month.

C. 使用索賠ID 和時間戳進行預測，以確定每個類別中預計每月有多少索賠。

D. Classification with supervised learning of the categories for which partial information on claim contents is provided, and forecasting using claim IDs and timestamps for all other categories.

D. 使用監督學習對提供索賠內容部分資訊的類別進行分類，並使用索賠ID 和時間戳對所有其他類別進行預測。

問題# 110

A company that promotes healthy sleep patterns by providing cloud-connected devices currently hosts a sleep tracking application on AWS. The application collects device usage information from device users. The company's Data Science team is building a machine learning model to predict if and when a user will stop utilizing the company's devices. Predictions from this model are used by a downstream application that determines the best approach for contacting users. The Data Science team is building multiple versions of the machine learning model to evaluate each version against the company's business goals. To measure long-term effectiveness, the team wants to run multiple versions of the model in parallel for long periods of time, with the ability to control the portion of inferences served by the models. Which solution satisfies these requirements with MINIMAL effort?

一家透過提供雲端連接設備來促進健康睡眠模式的公司目前在AWS 上託管了一個睡眠追蹤應用程式。該應用程式從設備用戶收集設備使用資訊。該公司的數據科學團隊正在建立一個機器學習模型，以預測用戶是否以及何時會停止使用該公司的設備。下游應用程式使用此模型的預測來確定聯繫用戶的最佳方法。數據科學團隊正在建立機器學習模型的多個版本，以根據公司的業務目標評估每個版本。為了衡量長期有效性，團隊希望長時間並行運行多個版本的模型，並能夠控制模型提供的推理部分。哪一種解決方案可以以最少的努力滿足這些要求？

A. Build and host multiple models in Amazon SageMaker. Create multiple Amazon SageMaker endpoints, one for each model. Programmatically control invoking different models for inference at the application layer.

A. 在Amazon SageMaker 中建立和託管多個模型。建立多個Amazon SageMaker 終端節點，每個模型一個。以程式方式控制在應用程式層呼叫不同的模型進行推理。

B. Build and host multiple models in Amazon SageMaker. Create an Amazon SageMaker endpoint configuration with multiple production variants. Programmatically control the portion of the inferences served by the multiple models by updating the endpoint configuration.

B. 在Amazon SageMaker 中建立和託管多個模型。建立具有多個生產變體的Amazon SageMaker 終端節點配置。透過更新終端節點配置，以程式方式控制多個模型提供的推理部分。

C. Build and host multiple models in Amazon SageMaker Neo to take into account different types of medical devices. Programmatically control which model is invoked for inference based on the medical device type.

C. 在Amazon SageMaker Neo 中建立和託管多個模型，以考慮不同類型的醫療設備。以程式方式控制根據醫療設備類型呼叫哪個模型進行推理。

D. Build and host multiple models in Amazon SageMaker. Create a single endpoint that accesses multiple models. Use Amazon SageMaker batch transform to control invoking the different models through the single endpoint.

D. 在Amazon SageMaker 中建立和託管多個模型。建立可存取多個模型的單一終端節點。使用Amazon SageMaker 批次轉換來控制透過單一終端節點呼叫不同的模型。

問題# 111

An agricultural company is interested in using machine learning to detect specific types of weeds in a 100-acre grassland field. Currently, the company uses tractor-mounted cameras to capture multiple images of the field as 10 × 10 grids. The company also has a large training dataset that consists of annotated images of popular weed classes like broadleaf and non-broadleaf docks. The company wants to build a weed detection model that will detect specific types of weeds and the location of each type within the field. Once the model is ready, it will be hosted on Amazon SageMaker endpoints. The model will perform real-time inferencing using the images captured by the cameras. Which approach should a Machine Learning Specialist take to obtain accurate predictions?

一家農業公司有興趣使用機器學習來檢測100 英畝草地中的特定類型的雜草。目前，該公司使用拖拉機安裝的攝影機以10 和10 的網格形式捕捉田地的多個影像。該公司還擁有一個大型訓練資料集，其中包含常見雜草類別（如闊葉和非闊葉雜草）的註釋圖像。該公司希望建立一個雜草檢測模型，以檢測田地內特定類型的雜草及其位置。模型準備就緒後，將託管在Amazon SageMaker 終端節點上。該模型將使用攝影機捕獲的圖像執行即時推理。機器學習專家應該採用哪一種方法來獲得準確的預測？

A. Prepare the images in RecordIO format and upload them to Amazon S3. Use Amazon SageMaker to train, test, and validate the model using an image classification algorithm to categorize images into various weed classes.

A. 準備RecordIO 格式的映像並將其上傳至Amazon S3。使用Amazon SageMaker 透過影像分類演算法訓練、測試和驗證模型，將影像分類為各種雜草類別。

B. Prepare the images in Apache Parquet format and upload them to Amazon S3. Use Amazon SageMaker to train, test, and validate the model using an object- detection single-shot multibox detector (SSD) algorithm.

B. 準備Apache Parquet 格式的映像並將其上傳到Amazon S3。使用Amazon SageMaker 透過物件偵測單次多框偵測器(SSD) 演算法訓練、測試和驗證模型。

C. Prepare the images in RecordIO format and upload them to Amazon S3. Use Amazon SageMaker to train, test, and validate the model using an object- detection single-shot multibox detector (SSD) algorithm.

C. 準備RecordIO 格式的圖像並將其上傳到Amazon S3。使用Amazon SageMaker 透過物件偵測單次多框偵測器(SSD) 演算法訓練、測試和驗證模型。

D. Prepare the images in Apache Parquet format and upload them to Amazon S3. Use Amazon SageMaker to train, test, and validate the model using an image classification algorithm to categorize images into various weed classes.

D. 準備Apache Parquet 格式的映像並將其上傳到Amazon S3。使用Amazon SageMaker 透過影像分類演算法訓練、測試和驗證模型，將影像分類為各種雜草類別。

問題# 112

A manufacturer is operating a large number of factories with a complex supply chain relationship where unexpected downtime of a machine can cause production to stop at several factories. A data scientist wants to analyze sensor data from the factories to identify equipment in need of preemptive maintenance and then dispatch a service team to prevent unplanned downtime. The sensor readings from a single machine can include up to 200 data points including temperatures, voltages, vibrations, RPMs, and pressure readings. To collect this sensor data, the manufacturer deployed Wi-Fi and LANs across the factories. Even though many factory locations do not have reliable or high-speed internet connectivity, the manufacturer would like to maintain near-real-time inference capabilities. Which deployment architecture for the model will address these business requirements?

一家製造商經營大量工廠，這些工廠的供應鏈關係複雜，一台機器的意外停機可能會導致多家工廠停產。一位資料科學家想要分析工廠的感測器數據，以確定需要預先維護的設備，然後派遣服務團隊以防止意外停機。單一機器的感測器讀數最多可包含200個數據點，包括溫度、電壓、振動、轉速和壓力讀數。為了收集這些感測器數據，製造商在工廠內部署了Wi-Fi和LAN。儘管許多工廠沒有可靠或高速的互聯網連接，但製造商希望保持近乎即時的推理能力。哪種模型部署架構可以滿足這些業務需求？

A. Deploy the model in Amazon SageMaker. Run sensor data through this model to predict which machines need maintenance.

A. 在Amazon SageMaker 中部署模型。透過此模型運行感測器資料以預測哪些機器需要維護。

B. Deploy the model on AWS IoT Greengrass in each factory. Run sensor data through this model to infer which machines need maintenance.

B. 在每個工廠的AWS IoT Greengrass 上部署模型。透過此模型運行感測器資料以推斷哪些機器需要維護。

C. Deploy the model to an Amazon SageMaker batch transformation job. Generate inferences in a daily batch report to identify machines that need maintenance.

C. 將模型部署到Amazon SageMaker 批次轉換作業。在每日批次報告中產生推理以識別需要維護的機器。

D. Deploy the model in Amazon SageMaker and use an IoT rule to write data to an Amazon DynamoDB table. Consume a DynamoDB stream from the table with an AWS Lambda function to invoke the endpoint.

D. 在Amazon SageMaker 中部署模型並使用IoT 規則將資料寫入Amazon DynamoDB 表。使用AWS Lambda 函數從表中使用DynamoDB 流來呼叫終端節點。

問題# 113

A Machine Learning Specialist is designing a scalable data storage solution for Amazon SageMaker. There is an existing TensorFlow-based model implemented as a train.py script that relies on static training data that is currently stored as TFRecords. Which method of providing training data to Amazon SageMaker would meet the business requirements with the LEAST development overhead?

機器學習專家正在為Amazon SageMaker 設計可擴展的資料儲存解決方案。目前有一個基於TensorFlow 的模型，該模型以train.py 腳本的形式實現，該腳本依賴目前儲存為TFRecords 的靜態訓練資料。哪種向Amazon SageMaker 提供訓練資料的方法可以滿足業務需求且開發開銷最少？

A. Use Amazon SageMaker script mode and use train.py unchanged. Point the Amazon SageMaker training invocation to the local path of the data without reformatting the training data.

A. 使用Amazon SageMaker 腳本模式並使用不變的train.py。將Amazon SageMaker 訓練呼叫指向資料的本機路徑，而無需重新格式化訓練資料。

B. Use Amazon SageMaker script mode and use train.py unchanged. Put the TFRecord data into an Amazon S3 bucket. Point the Amazon SageMaker training invocation to the S3 bucket without reformatting the training data.

B. 使用Amazon SageMaker 腳本模式並使用不變的train.py。將TFRecord 資料放入Amazon S3 儲存桶中。將Amazon SageMaker 訓練呼叫指向S3 儲存桶，而無需重新格式化訓練資料。

C. Rewrite the train.py script to add a section that converts TFRecords to protobuf and ingests the protobuf data instead of TFRecords.

C. 重寫train.py 腳本以新增將TFRecords 轉換為protobuf 並提取protobuf 資料（而不是TFRecords）的部分。

D. Prepare the data in the format accepted by Amazon SageMaker. Use AWS Glue or AWS Lambda to reformat and store the data in an Amazon S3 bucket.

D. 以Amazon SageMaker 接受的格式準備資料。使用AWS Glue 或AWS Lambda 重新格式化並將資料儲存在Amazon S3 儲存桶中。

問題# 114

The chief editor for a product catalog wants the research and development team to build a machine learning system that can be used to detect whether or not individuals in a collection of images are wearing the company's retail brand. The team has a set of training data. Which machine learning algorithm should the researchers use that BEST meets their requirements?

產品目錄的主編希望研發團隊建立一個機器學習系統，該系統可用於偵測影像集合中的個人是否穿著該公司的零售品牌服裝。該團隊有一組訓練資料。研究人員應該使用哪種機器學習演算法才能最好地滿足他們的要求？

A. Latent Dirichlet Allocation (LDA).

A. 潛在狄利克雷分佈（LDA）。

B. Recurrent neural network (RNN).

B. 循環神經網路 (RNN) 。

C. K-means.

C. K-均值。

D. Convolutional neural network (CNN).

D. 卷積神經網路 (CNN) 。

問題# 115

A retail company is using Amazon Personalize to provide personalized product recommendations for its customers during a marketing campaign. The company sees a significant increase in sales of recommended items to existing customers immediately after deploying a new solution version, but these sales decrease a short time after deployment. Only historical data from before the marketing campaign is available for training. How should a data scientist adjust the solution?

一家零售公司正在使用Amazon Personalize 在行銷活動期間為客戶提供個人化產品推薦。部署新版本的解決方案後，該公司立即發現向現有客戶推薦的商品銷售量大幅增加，但部署後不久銷售就下降了。只有行銷活動之前的歷史資料可用於訓練。資料科學家該如何調整解決方案？

A. Use the event tracker in Amazon Personalize to include real-time user interactions.

A. 使用Amazon Personalize 中的事件追蹤器來包含即時使用者互動。

B. Add user metadata and use the HRNN-Metadata recipe in Amazon Personalize.

B. 新增使用者元資料並在Amazon Personalize 中使用HRNN-Metadata 配方。

C. Implement a new solution using the built-in factorization machines (FM) algorithm in Amazon SageMaker.

C. 使用Amazon SageMaker 中的內建分解機(FM) 演算法實作新解決方案。

D. Add event type and event value fields to the interactions dataset in Amazon Personalize.

D. 將事件類型和事件值欄位新增至Amazon Personalize 中的互動資料集。

問題# 116

A machine learning (ML) specialist wants to secure calls to the Amazon SageMaker Service API. The specialist has configured Amazon VPC with a VPC interface endpoint for the Amazon SageMaker Service API and is attempting to secure traffic from specific sets of instances and IAM users. The VPC is configured with a single public subnet. Which combination of steps should the ML specialist take to secure the traffic? (Choose two.)

機器學習(ML) 專家希望保護對Amazon SageMaker 服務API 的呼叫。該專家已為Amazon VPC 配置了用於Amazon SageMaker 服務API 的VPC 介面終端節點，並嘗試保護來自特定實例集和IAM 使

用者的流量。VPC 配置了一個公共子網路。ML 專家應採取哪些步驟組合來保護流量？（選兩個。）

A. Add a VPC endpoint policy to allow access to the IAM users.

A. 新增VPC 終端節點策略以允許IAM 使用者存取。

B. Modify the users' IAM policy to allow access to Amazon SageMaker Service API calls only.

B. 修改使用者的IAM 策略以僅允許存取Amazon SageMaker 服務API 呼叫。

C. Modify the security group on the endpoint network interface to restrict access to the instances.

C. 修改終端節點網路介面上的安全群組以限制對實例的存取。

D. Modify the ACL on the endpoint network interface to restrict access to the instances.

D. 修改終端節點網路介面上的ACL 以限制對實例的存取。

E. Add a SageMaker Runtime VPC endpoint interface to the VPC.

E. 為VPC 新增SageMaker Runtime VPC 終端節點介面。

問題# 117

An e commerce company wants to launch a new cloud-based product recommendation feature for its web application. Due to data localization regulations, any sensitive data must not leave its on-premises data center, and the product recommendation model must be trained and tested using nonsensitive data only. Data transfer to the cloud must use IPsec. The web application is hosted on premises with a PostgreSQL database that contains all the data. The company wants the data to be uploaded securely to Amazon S3 each day for model retraining. How should a machine learning specialist meet these requirements?

一家電子商務公司希望為其Web 應用程式推出一項新的基於雲端的产品推薦功能。根據資料本地化法規，任何敏感資料都不得離開其本地資料中心，且產品推薦模型必須僅使用非敏感資料進行訓練和測試。將資料傳輸到雲端必須使用IPsec。Web 應用程式託管在本地，其中包含所有資料的PostgreSQL 資料庫。該公司希望每天將資料安全地上傳到Amazon S3 進行模型再訓練。機器學習專家應如何滿足這些要求？

A. Create an AWS Glue job to connect to the PostgreSQL DB instance. Ingest tables without sensitive data through an AWS Site-to-Site VPN connection directly into Amazon S3.

A. 建立AWS Glue 作業以連線到PostgreSQL DB 實例。透過AWS Site-to-Site VPN 連線將不包含敏感資料的表直接提取到Amazon S3 中。

B. Create an AWS Glue job to connect to the PostgreSQL DB instance. Ingest all data through an AWS Site-to-Site VPN connection into Amazon S3 while removing sensitive data using a PySpark job.

B. 建立AWS Glue 作業以連線到PostgreSQL DB 實例。透過AWS Site-to-Site VPN 連線將所有資料提取到Amazon S3 中，同時使用PySpark 作業刪除敏感資料。

C. Use AWS Database Migration Service (AWS DMS) with table mapping to select PostgreSQL tables with no sensitive data through an SSL connection. Replicate data directly into Amazon S3.

C. 使用具有表格映射的AWS Database Migration Service (AWS DMS) 透過SSL 連線選擇不包含敏感資料的PostgreSQL 表。將資料直接複製到Amazon S3。

D. Use PostgreSQL logical replication to replicate all data to PostgreSQL in Amazon EC2 through AWS Direct Connect with a VPN connection. Use AWS Glue to move data from Amazon EC2 to Amazon S3.

D. 使用PostgreSQL 邏輯複製透過帶有VPN 連線的AWS Direct Connect 將所有資料複製到Amazon EC2 中的PostgreSQL。使用AWS Glue 將資料從Amazon EC2 移至Amazon S3。

問題# 118

A logistics company needs a forecast model to predict next month's inventory requirements for a single item in 10 warehouses. A machine learning specialist uses Amazon Forecast to develop a forecast model from 3 years of monthly data. There is no missing data. The specialist selects the DeepAR+ algorithm to train a predictor. The predictor means absolute percentage error (MAPE) is much larger than the MAPE produced by the current human forecasters. Which changes to the CreatePredictor API call could improve the MAPE? (Choose two.)

一家物流公司需要一個預測模型來預測10 個倉庫中單一商品下個月的庫存需求。一位機器學習專家使用Amazon Forecast 根據3 年的月度數據開發了預測模型。沒有缺失資料。專家選擇DeepAR+ 演算法來訓練預測器。預測器平均絕對百分比誤差(MAPE) 比目前人工預測器產生的MAPE 大得多。對CreatePredictor API 呼叫進行哪些變更可以改善MAPE? (選兩個。)

A. Set PerformAutoML to true.

A. 將PerformAutoML 設定為true。

B. Set ForecastHorizon to 4.

B. 將ForecastHorizon 設定為4。

C. Set ForecastFrequency to W for weekly.

C. 將ForecastFrequency 設定為W (每週)。

D. Set PerformHPO to true.

D. 將PerformHPO 設定為true。

E. Set FeaturizationMethodName to filling.

E. 將FeaturizationMethodName 設定為填滿。

問題# 119

A data scientist wants to use Amazon Forecast to build a forecasting model for inventory demand for a retail company. The company has provided a dataset of historic inventory demand for its products as a.csv file stored in an Amazon S3 bucket. The table below shows a sample of the dataset.How should the data scientist transform the data?

一位資料科學家想要使用Amazon Forecast 為一家零售公司建立庫存需求預測模型。該公司已提供其產品歷史庫存需求資料集，以.csv 檔案形式儲存在Amazon S3 儲存桶中。下表顯示了資料集的範例。資料科學家應如何轉換資料？

timestamp	item_id	demand	category	lead_time
2019-12-14	uni_000736	120	hardware	90
2020-01-31	uni_003429	98	hardware	30
2020-03-04	uni_000211	234	accessories	10

A. Use ETL jobs in AWS Glue to separate the dataset into a target time series dataset and an item metadata dataset. Upload both datasets as.csv files to Amazon S3.

A. 使用AWS Glue 中的ETL 作業將資料集分成目標時間序列資料集和項目元資料集。將兩個資料集作為.csv 檔案上傳到Amazon S3。

B. Use a Jupyter notebook in Amazon SageMaker to separate the dataset into a related time series dataset and an item metadata dataset. Upload both datasets as tables in Amazon Aurora.

B. 使用Amazon SageMaker 中的Jupyter 筆記本將資料集分割成相關時間序列資料集和項目元資料集。將兩個資料集作為表格上傳到Amazon Aurora。

C. Use AWS Batch jobs to separate the dataset into a target time series dataset, a related time series dataset, and an item metadata dataset. Upload them directly to Forecast from a local machine.

C. 使用AWS Batch 作業將資料集分成目標時間序列資料集、相關時間序列資料集和專案元資料集。將它們從本機直接上傳到Forecast。

D. Use a Jupyter notebook in Amazon SageMaker to transform the data into the optimized protobuf recordIO format. Upload the dataset in this format to Amazon S3.

D. 使用Amazon SageMaker 中的Jupyter 筆記本將資料轉換為優化的protobuf recordIO 格式。將此格式的資料集上傳到Amazon S3。

問題# 120

A machine learning specialist is running an Amazon SageMaker endpoint using the built-in object detection algorithm on a P3 instance for real-time predictions in a company's production application. When evaluating the model's resource utilization, the specialist notices that the model is using only a fraction of the GPU.Which architecture changes would ensure that provisioned resources are being utilized effectively?

一位機器學習專家正在使用內建物件偵測演算法在P3 實例上執行Amazon SageMaker 終端節點，以便在該公司的生產應用程式中進行即時預測。在評估模型的資源利用率時，專家注意到模型僅使用了GPU 的一小部分。哪些架構變化可以確保有效利用預先配置的資源？

A. Redeploy the model as a batch transform job on an M5 instance.

A. 在M5 實例上將模型作為批次轉換作業重新部署。

B. Redeploy the model on an M5 instance. Attach Amazon Elastic Inference to the instance.

B. 在M5 實例上重新部署模型。將Amazon Elastic Inference 附加到該實例。

C. Redeploy the model on a P3dn instance.

C. 在P3dn 實例上重新部署模型。

D. Deploy the model onto an Amazon Elastic Container Service (Amazon ECS) cluster using a P3 instance.

D. 使用P3 實例將模型部署到Amazon Elastic Container Service (Amazon ECS) 叢集上。

問題# 121

A data scientist uses an Amazon SageMaker notebook instance to conduct data exploration and analysis. This requires certain Python packages that are not natively available on Amazon SageMaker to be installed on the notebook instance. How can a machine learning specialist ensure that required packages are automatically available on the notebook instance for the data scientist to use?

資料科學家使用Amazon SageMaker 筆記本實例進行資料探索和分析。這需要在筆記本實例上安裝某些Amazon SageMaker 上沒有的Python 軟體包。機器學習專家如何確保所需的軟體套件在筆記本實例上自動可用，以供資料科學家使用？

A. Install AWS Systems Manager Agent on the underlying Amazon EC2 instance and use Systems Manager Automation to execute the package installation commands.

A. 在底層Amazon EC2 執行個體上安裝AWS Systems Manager 代理，並使用Systems Manager Automation 執行軟體包安裝指令。

B. Create a Jupyter notebook file (.ipynb) with cells containing the package installation commands to execute and place the file under the /etc/init directory of each Amazon SageMaker notebook instance.

B. 建立一個Jupyter 筆記本檔案(.ipynb)，其中的儲存格包含要執行的軟體包安裝命令，並將該檔案放在每個Amazon SageMaker 筆記本實例的/etc/init 目錄下。

C. Use the conda package manager from within the Jupyter notebook console to apply the necessary conda packages to the default kernel of the notebook.

C. 使用Jupyter 筆記本控制台中的conda 套件管理器將必要的conda 套件應用於筆記本的預設核心。

D. Create an Amazon SageMaker lifecycle configuration with package installation commands and assign the lifecycle configuration to the notebook instance.

D. 使用軟體包安裝指令建立Amazon SageMaker 生命週期配置，並將生命週期配置指派給筆記本實例。

問題# 122

A data scientist needs to identify fraudulent user accounts for a company's ecommerce platform. The company wants the ability to determine if a newly created account is associated with a previously known fraudulent user. The data scientist is using AWS Glue to cleanse the company's application logs during ingestion. Which strategy will allow the data scientist to identify fraudulent accounts?

一位資料科學家需要識別公司電子商務平台的詐欺性使用者帳戶。該公司希望能夠確定新建立的帳戶是否與先前已知的詐騙用戶相關聯。資料科學家正在使用AWS Glue 在提取資料期間清理公司的應用程式日誌。哪種策略可以讓資料科學家識別詐欺性帳戶？

A. Execute the built-in FindDuplicates Amazon Athena query.

A. 執行內建的FindDuplicates Amazon Athena 查詢。

B. Create a FindMatches machine learning transform in AWS Glue.

B. 在AWS Glue 中建立FindMatches 機器學習轉換。

C. Create an AWS Glue crawler to infer duplicate accounts in the source data.

C. 建立AWS Glue 爬蟲程式以推斷來源資料中的重複帳戶。

D. Search for duplicate accounts in the AWS Glue Data Catalog.

D. 在AWS Glue 資料目錄中搜尋重複帳戶。

問題# 123

A Data Scientist is developing a machine learning model to classify whether a financial transaction is fraudulent. The labeled data available for training consists of 100,000 non-fraudulent observations and 1,000 fraudulent observations. The Data Scientist applies the XGBoost algorithm to the data, resulting in the following confusion matrix when the trained model is applied to a previously unseen validation dataset. The accuracy of the model is 99.1%, but the Data Scientist needs to reduce the number of false negatives. Which combination of steps should the Data Scientist take to reduce the number of false negative predictions by the model? (Choose two.)

一位資料科學家正在開發一個機器學習模型來對金融交易是否為詐欺進行分類。可用於訓練的標記資料包括100,000 個非詐欺性觀察和1,000 個詐欺性觀察。資料科學家將XGBoost 演算法應用於數據，當將訓練模型應用於先前未見過的驗證資料集時，會產生以下混淆矩陣。此模型的準確率為99.1%，但資料科學家需要減少假陰性的數量。資料科學家應採取哪些步驟組合來減少模型的假陰性預測數量？（選擇兩個。）

	Predicted 0	Predicted 1
Actual 0	99,966	34
Actual 1	877	123

A. Change the XGBoost eval_metric parameter to optimize based on Root Mean Square Error (RMSE).

A. 更改XGBoost eval_metric 參數以基於均方根誤差（RMSE）進行最佳化。

B. Increase the XGBoost scale_pos_weight parameter to adjust the balance of positive and negative weights.

B. 增加XGBoost scale_pos_weight 參數來調整正負權重的平衡。

C. Increase the XGBoost max_depth parameter because the model is currently underfitting the data.

C. 增加XGBoost max_depth 參數，因為模型目前對資料擬合不足。

D. Change the XGBoost eval_metric parameter to optimize based on Area Under the ROC Curve (AUC).

D. 更改XGBoost eval_metric 參數以基於ROC 曲線下面積（AUC）進行最佳化。

E. Decrease the XGBoost max_depth parameter because the model is currently overfitting the data.

E. 減少XGBoost max_depth 參數，因為模型目前對資料擬合過度。

問題# 124

A data scientist has developed a machine learning translation model for English to Japanese by using Amazon SageMaker's built-in seq2seq algorithm with 500,000 aligned sentence pairs. While testing with sample sentences, the data scientist finds that the translation quality is reasonable for an example as short as five words. However, the quality becomes unacceptable if the sentence is 100 words long. Which action will resolve the problem?

一位資料科學家使用Amazon SageMaker 內建的seq2seq 演算法開發了一個英語到日語的機器學習翻譯模型，該模型包含500,000 個對齊的句子對。在使用範例句子進行測試時，資料科學家發現，對於短至五個單字的範例，翻譯品質還算合理。但是，如果句子長度為100 個單詞，品質就變得不可接受。哪種操作可以解決問題？

A. Change preprocessing to use n-grams.

A. 更改預處理以使用n-gram。

B. Add more nodes to the recurrent neural network (RNN) than the largest sentence's word count.

B. 在循環神經網路(RNN) 中加入比最大句子的字數更多的節點。

C. Adjust hyperparameters related to the attention mechanism.

C. 調整與注意力機制相關的超參數。

D. Choose a different weight initialization type.

D. 選擇不同的權重初始化類型。

問題# 125

A financial company is trying to detect credit card fraud. The company observed that, on average, 2% of credit card transactions were fraudulent. A data scientist trained a classifier on a year's worth of credit card transactions data. The model needs to identify the fraudulent transactions (positives) from the regular ones(negatives). The company's goal is to accurately capture as many positives as possible.Which metrics should the data scientist use to optimize the model? (Choose two.)

一家金融公司正在嘗試偵測信用卡詐欺行為。該公司發現，平均有2% 的信用卡交易是詐騙的。一位資料科學家根據一年的信用卡交易資料訓練了一個分類器。此模型需要從常規交易（負數）中辨識出詐欺交易（正數）。該公司的目標是盡可能準確地捕捉到盡可能多的正數。資料科學家應該使用哪些指標來優化模型？（選兩個。）

A. Specificity.

A. 特異性。

B. False positive rate.

B. 假陽性率。

C. Accuracy.

C. 準確度。

D. Area under the precision-recall curve.

D. 精確度-召回率曲線下面積。

E. True positive rate.

E. 真陽性率。

問題# 126

A machine learning specialist is developing a proof of concept for government users whose primary concern is security. The specialist is using AmazonSageMaker to train a convolutional neural network (CNN) model for a photo classifier application. The specialist wants to protect the data so that it cannot be accessed and transferred to a remote host by malicious code accidentally installed on the training container.Which action will provide the MOST secure protection?

一位機器學習專家正在為主要關注安全性的政府使用者開發概念驗證。該專家正在使用AmazonSageMaker 為照片分類器應用程式訓練卷積神經網路(CNN) 模型。該專家希望保護數據，以防止意外安裝在訓練容器上的惡意程式碼存取資料並將其傳輸到遠端主機。哪種操作可以提供最安全的保護？

A. Remove Amazon S3 access permissions from the SageMaker execution role.

A. 從SageMaker 執行角色中刪除Amazon S3 存取權限。

B. Encrypt the weights of the CNN model.

B. 加密CNN 模型的權重。

C. Encrypt the training and validation dataset.

C. 加密訓練和驗證資料集。

D. Enable network isolation for training jobs.

D. 為訓練作業啟用網路隔離。

問題# 127

A medical imaging company wants to train a computer vision model to detect areas of concern on patients' CT scans. The company has a large collection of unlabeled CT scans that are linked to each patient and stored in an Amazon S3 bucket. The scans must be accessible to authorized users only. A machine learning engineer needs to build a labeling pipeline. Which set of steps should the engineer take to build the labeling pipeline with the LEAST effort?

一家醫學影像公司希望訓練電腦視覺模型來檢測患者CT 掃描中值得關注的區域。該公司擁有大量未標記的CT 掃描，這些掃描與每位患者相關聯並儲存在Amazon S3 儲存桶中。掃描必須僅供授權使用者存取。機器學習工程師需要建立標記管道。工程師應採取哪一組步驟以最少的努力建構標記管道？

A. Create a workforce with AWS Identity and Access Management (IAM). Build a labeling tool on Amazon EC2 Queue images for labeling by using Amazon Simple Queue Service (Amazon SQS). Write the labeling instructions.

A. 使用AWS Identity and Access Management (IAM) 建立員工團隊。使用Amazon Simple Queue Service (Amazon SQS) 在Amazon EC2 佇列影像上建立標記工具以進行標記。編寫標記說明。

B. Create an Amazon Mechanical Turk workforce and manifest file. Create a labeling job by using the built-in image classification task type in Amazon SageMaker Ground Truth. Write the labeling instructions.

B. 創建Amazon Mechanical Turk 員工隊伍和清單文件。使用Amazon SageMaker Ground Truth 中內建的影像分類任務類型建立標記作業。編寫標記說明。

C. Create a private workforce and manifest file. Create a labeling job by using the built-in bounding box task type in Amazon SageMaker Ground Truth. Write the labeling instructions.

C. 建立私有員工隊伍和清單檔案。使用Amazon SageMaker Ground Truth 中的內建邊界框任務類型建立標記作業。編寫標記說明。

D. Create a workforce with Amazon Cognito. Build a labeling web application with AWS Amplify. Build a labeling workflow backend using AWS Lambda. Write the labeling instructions.

D. 使用Amazon Cognito 創建員工團隊。使用AWS Amplify 建立標記Web 應用程式。使用AWS Lambda 建置標記工作流程後端。編寫標記說明。

問題# 128

A company is using Amazon Textract to extract textual data from thousands of scanned text-heavy legal documents daily. The company uses this information to process loan applications automatically. Some of the documents fail business validation and are returned to human reviewers, who investigate the errors. This activity increases the time to process the loan applications. What should the company do to reduce the processing time of loan applications?

一家公司每天使用Amazon Textract 從數千份掃描的文本繁多的法律文件中提取文字資料。該公司使用這些資訊自動處理貸款申請。有些文件未通過業務驗證，並被退回給人工審核人員，由他們調查錯誤。此活動增加了處理貸款申請的時間。該公司該怎麼做才能減少貸款申請的處理時間？

A. Configure Amazon Textract to route low-confidence predictions to Amazon SageMaker Ground Truth. Perform a manual review on those words before performing a business validation.

A. 設定Amazon Textract 以將低置信度預測路由到Amazon SageMaker Ground Truth。在執行業務驗證之前對這些字詞進行手動審核。

B. Use an Amazon Textract synchronous operation instead of an asynchronous operation.

B. 使用Amazon Textract 同步操作而不是非同步操作。

C. Configure Amazon Textract to route low-confidence predictions to Amazon Augmented AI (Amazon A2I). Perform a manual review on those words before performing a business validation.

C. 配置Amazon Textract 將低置信度預測路由到Amazon Augmented AI (Amazon A2I)。在執行業務驗證之前對這些字詞進行手動審核。

D. Use Amazon Rekognition's feature to detect text in an image to extract the data from scanned images. Use this information to process the loan applications.

D. 使用Amazon Rekognition 的功能檢測影像中的文字以從掃描影像中提取資料。使用此資訊來處理貸款申請。

問題# 129

A company ingests machine learning (ML) data from web advertising clicks into an Amazon S3 data lake. Click data is added to an Amazon Kinesis data stream by using the Kinesis Producer Library (KPL). The data is loaded into the S3 data lake from the data stream by using an Amazon Kinesis Data Firehose delivery stream. As the data volume increases, an ML specialist notices that the rate of data ingested into Amazon S3 is relatively constant. There also is an increasing backlog of data for Kinesis Data Streams and Kinesis Data Firehose to ingest. Which next step is MOST likely to improve the data ingestion rate into Amazon S3?

一家公司將網路廣告點擊的機器學習(ML) 資料提取到Amazon S3 資料湖中。使用Kinesis Producer Library (KPL) 將點擊資料新增至Amazon Kinesis 資料流。使用Amazon Kinesis Data Firehose 傳輸流將資料從資料流載入到S3 資料湖。隨著資料量的增加，ML 專家注意到提取到Amazon S3 的資料速率相對穩定。Kinesis Data Streams 和Kinesis Data Firehose 需要擷取的資料積壓也在增加。下一步哪一步最有可能提高Amazon S3 的資料提取率？

A. Increase the number of S3 prefixes for the delivery stream to write to.

A. 增加傳輸流要寫入的S3 前綴數量。

B. Decrease the retention period for the data stream.

B. 減少資料流的保留期。

C. Increase the number of shards for the data stream.

C. 增加資料流的分片數量。

D. Add more consumers using the Kinesis Client Library (KCL).

D. 使用Kinesis 用戶端庫(KCL) 新增更多用戶。

問題# 130

A data scientist must build a custom recommendation model in Amazon SageMaker for an online retail company. Due to the nature of the company's products, customers buy only 4-5 products every 5-10 years. So, the company relies on a steady stream of new customers. When a new customer signs up, the company collects data on the customer's preferences. Below is a sample of the data available to the data scientist. How should the data scientist split the dataset into a training and test set for this use case?

資料科學家必須為一家線上零售公司在Amazon SageMaker 中建立自訂推薦模型。由於公司產品的性質，客戶每5-10 年僅購買4-5 種產品。因此，該公司依賴穩定的新客戶流。當新客戶註冊時，公司會收集有關客戶偏好的資料。以下是資料科學家可用的資料範例。資料科學家應如何將資料集拆分為此用例的訓練集和測試集？

timestamp	user_id	product_id	preference_1	...	preference_10
2020-03-04	90	25	0.234	...	0.374
2020-03-04	90	61	0	...	0.374
2020-02-21	203	56	1	...	0.098

A. Shuffle all interaction data. Split off the last 10% of the interaction data for the test set.

A. 對所有互動資料進行打亂。分離出最後10% 的交互資料用於測試集。

B. Identify the most recent 10% of interactions for each user. Split off these interactions for the test set.

B. 決定每個使用者最近的10% 的互動。分離出這些交互作用用於測試集。

C. Identify the 10% of users with the least interaction data. Split off all interaction data from these users for the test set.

C. 決定交互資料最少的10% 的使用者。分離出這些使用者的所有互動資料用於測試集。

D. Randomly select 10% of the users. Split off all interaction data from these users for the test set.

D. 隨機選擇10% 的用戶。分離出這些使用者的所有互動資料用於測試集。

問題# 131

A financial services company wants to adopt Amazon SageMaker as its default data science environment. The company's data scientists run machine learning(ML) models on confidential financial data. The company is worried about data egress and wants an ML engineer to secure the environment. Which mechanisms can the ML engineer use to control data egress from SageMaker? (Choose three.)

一家金融服務公司希望採用Amazon SageMaker 作為其預設資料科學環境。該公司的資料科學家在機密財務資料上運行機器學習(ML) 模型。該公司擔心資料流出，希望ML 工程師保護環境。ML 工程師可以使用哪些機制來控制來自SageMaker 的資料流出？（選三個。）

A. Connect to SageMaker by using a VPC interface endpoint powered by AWS PrivateLink.

A. 使用由AWS PrivateLink 提供支援的VPC 介面終端節點連接到SageMaker。

B. Use SCPs to restrict access to SageMaker.

B. 使用SCP 限制對SageMaker 的存取。

C. Disable root access on the SageMaker notebook instances.

C. 停用SageMaker 筆記本實例上的根存取權。

D. Enable network isolation for training jobs and models.

D. 為訓練作業和模型啟用網路隔離。

E. Restrict notebook presigned URLs to specific IPs used by the company.

E. 將筆記本預簽名URL 限制為公司使用的特定IP。

F. Protect data with encryption at rest and in transit. Use AWS Key Management Service (AWS KMS) to manage encryption keys.

F. 使用加密保護靜態和傳輸中的資料。使用AWS Key Management Service (AWS KMS) 管理加密金鑰。

問題# 133

A company is converting a large number of unstructured paper receipts into images. The company wants to create a model based on natural language processing(NLP) to find relevant entities such as date, location, and notes, as well as some custom entities such as receipt numbers. The company is using optical character recognition (OCR) to extract text for data labeling. However, documents are in different structures and formats, and the company is facing challenges with setting up the manual workflows for each document type. Additionally, the company trained a named entity recognition (NER) model for custom entity detection using a small sample size. This model has a very low confidence score and will require retraining with a large dataset. Which solution for text extraction and entity detection will require the LEAST amount of effort?

一家公司正在將大量非結構化紙質收據轉換為圖像。該公司希望創建一個基於自然語言處理(NLP) 的模型來尋找相關實體（例如日期、地點和註釋）以及一些自訂實體（例如收據編號）。該公司正在使用光學字元辨識(OCR) 提取文字以進行資料標記。但是，文件的結構和格式各不相同，該公司

在為每種文件類型設定手動工作流程方面面臨挑戰。此外，該公司使用小樣本量訓練了一個命名實體識別(NER) 模型以進行自訂實體檢測。此模型的置信度得分非常低，需要使用大型資料集重新訓練。哪種文字擷取和實體偵測解決方案所需的工作量最少？

A. Extract text from receipt images by using Amazon Textract. Use the Amazon SageMaker BlazingText algorithm to train on the text for entities and custom entities.

A. 使用Amazon Textract 從收據圖片中擷取文字。使用Amazon SageMaker BlazingText 演算法對實體和自訂實體的文字進行訓練。

B. Extract text from receipt images by using a deep learning OCR model from the AWS Marketplace. Use the NER deep learning model to extract entities.

B. 使用AWS Marketplace 中的深度學習OCR 模型從收據圖像中提取文字。使用NER 深度學習模型擷取實體。

C. Extract text from receipt images by using Amazon Textract. Use Amazon Comprehend for entity detection, and use Amazon Comprehend custom entity recognition for custom entity detection.

C. 使用Amazon Textract 從收據圖片中提取文字。使用Amazon Comprehend 進行實體偵測，使用Amazon Comprehend 自訂實體識別進行自訂實體偵測。

D. Extract text from receipt images by using a deep learning OCR model from the AWS Marketplace. Use Amazon Comprehend for entity detection, and use Amazon Comprehend custom entity recognition for custom entity detection.

D. 使用AWS Marketplace 中的深度學習OCR 模型從收據圖像中提取文字。使用Amazon Comprehend 進行實體偵測，使用Amazon Comprehend 自訂實體識別進行自訂實體偵測。

問題# 134

A company is building a predictive maintenance model based on machine learning (ML). The data is stored in a fully private Amazon S3 bucket that is encrypted at rest with AWS Key Management Service (AWS KMS) CMKs. An ML specialist must run data preprocessing by using an Amazon SageMaker Processing job that is triggered from code in an Amazon SageMaker notebook. The job should read data from Amazon S3, process it, and upload it back to the same S3 bucket. The preprocessing code is stored in a container image in Amazon Elastic Container Registry (Amazon ECR). The ML specialist needs to grant permissions to ensure a smooth data preprocessing workflow. Which set of actions should the ML specialist take to meet these requirements?

一家公司正在建立基於機器學習(ML) 的預測性維護模型。資料儲存在完全私有的Amazon S3 儲存桶中，該儲存桶使用AWS Key Management Service (AWS KMS) CMK 進行靜態加密。ML 專家必須使用Amazon SageMaker 筆記本中的程式碼觸發的Amazon SageMaker 處理作業來執行資料預處理。該作業應從Amazon S3 讀取資料、對其進行處理並將其上傳回同一S3 儲存桶。預處理程式碼儲存在Amazon Elastic Container Registry (Amazon ECR) 中的容器映像中。ML 專家需要授予權限以確保資料預處理工作流程順暢。ML 專家應採取哪一組操作來滿足這些要求？

A. Create an IAM role that has permissions to create Amazon SageMaker Processing jobs, S3 read and write access to the relevant S3 bucket, and appropriate KMS and ECR permissions. Attach the role to the SageMaker notebook instance. Create an Amazon SageMaker Processing job from the notebook.

A. 建立一個IAM 角色，該角色具有建立Amazon SageMaker 處理作業的權限、對相關S3 儲存桶的

S3 讀寫存取權限以及適當的KMS 和ECR 權限。將角色附加到SageMaker 筆記本實例。從筆記本建立一個Amazon SageMaker 處理作業。

B. Create an IAM role that has permissions to create Amazon SageMaker Processing jobs. Attach the role to the SageMaker notebook instance. Create an Amazon SageMaker Processing job with an IAM role that has read and write permissions to the relevant S3 bucket, and appropriate KMS and ECR permissions.

B. 建立一個IAM 角色，該角色具有建立Amazon SageMaker 處理作業的權限。將角色附加到Sage Maker 筆記本實例。使用IAM 角色建立Amazon SageMaker 處理作業，該角色具有相關S3 儲存桶的讀取和寫入權限以及適當的KMS 和ECR 權限。

C. Create an IAM role that has permissions to create Amazon SageMaker Processing jobs and to access Amazon ECR. Attach the role to the SageMaker notebook instance. Set up both an S3 endpoint and a KMS endpoint in the default VPC. Create Amazon SageMaker Processing jobs from the notebook.

C. 建立一個IAM 角色，該角色具有建立Amazon SageMaker 處理作業和存取Amazon ECR 的權限。將角色附加到SageMaker 筆記本實例。在預設VPC 中設定S3 終端節點和KMS 終端節點。從筆記本建立Amazon SageMaker 處理作業。

D. Create an IAM role that has permissions to create Amazon SageMaker Processing jobs. Attach the role to the SageMaker notebook instance. Set up an S3 endpoint in the default VPC. Create Amazon SageMaker Processing jobs with the access key and secret key of the IAM user with appropriate KMS and ECR permissions.

D. 建立一個IAM 角色，該角色具有建立Amazon SageMaker 處理作業的權限。將角色附加到Sage Maker 筆記本實例。在預設VPC 中設定S3 端點。使用具有適當KMS 和ECR 權限的IAM 使用者的存取金鑰和金鑰來建立Amazon SageMaker 處理作業。

問題# 135

A data scientist has been running an Amazon SageMaker notebook instance for a few weeks. During this time, a new version of Jupyter Notebook was released along with additional software updates. The security team mandates that all running SageMaker notebook instances use the latest security and software updates provided by SageMaker. How can the data scientist meet this requirements?

一位資料科學家已經運行Amazon SageMaker 筆記本實例幾週了。在此期間，Jupyter Notebook 發布了新版本以及其他軟體更新。安全團隊要求所有正在執行的SageMaker 筆記型電腦實例都使用SageMaker 提供的最新安全性和軟體更新。數據科學家如何滿足這項要求？

A. Call the CreateNotebookInstanceLifecycleConfig API operation.

A. 呼叫CreateNotebookInstanceLifecycleConfig API 操作。

B. Create a new SageMaker notebook instance and mount the Amazon Elastic Block Store (Amazon EBS) volume from the original instance.

B. 建立新的SageMaker 筆記本實例並從原始實例掛載Amazon Elastic Block Store (Amazon EBS) 磁碟區。

C. Stop and then restart the SageMaker notebook instance.

C. 停止然後重新啟動SageMaker 筆記本實例。

D. Call the UpdateNotebookInstanceLifecycleConfig API operation.

D. 呼叫UpdateNotebookInstanceLifecycleConfig API 作業。

問題# 136

A library is developing an automatic book-borrowing system that uses Amazon Rekognition. Images of library members' faces are stored in an Amazon S3 bucket. When members borrow books, the Amazon Rekognition CompareFaces API operation compares real faces against the stored faces in Amazon S3. The library needs to improve security by making sure that images are encrypted at rest. Also, when the images are used with Amazon Rekognition, they need to be encrypted in transit. The library also must ensure that the images are not used to improve Amazon Rekognition as a service. How should a machine learning specialist architect the solution to satisfy these requirements?

一家圖書館正在開發使用Amazon Rekognition 的自動圖書借閱系統。圖書館會員的臉部圖像儲存在Amazon S3 儲存桶中。當會員借閱書籍時，Amazon Rekognition CompareFaces API 作業會將真實臉部與Amazon S3 中儲存的臉部進行比較。圖書館需要透過確保圖像在靜止時加密來提高安全性。此外，當圖像與Amazon Rekognition 一起使用時，它們需要在傳輸過程中加密。圖書館還必須確保圖像不用於改進Amazon Rekognition 服務。機器學習專家應如何設計解決方案以滿足這些要求？

A. Enable server-side encryption on the S3 bucket. Submit an AWS Support ticket to opt out of allowing images to be used for improving the service, and follow the process provided by AWS Support.

A. 在S3 儲存桶上啟用伺服器端加密。提交AWS Support 工單以選擇不允許使用圖像來改進服務，並按照AWS Support 提供的流程進行操作。

B. Switch to using an Amazon Rekognition collection to store the images. Use the IndexFaces and SearchFacesByImage API operations instead of the CompareFaces API operation.

B. 切換到使用Amazon Rekognition 集合來儲存映像。使用IndexFaces 和SearchFacesByImage API 操作，而不是CompareFaces API 操作。

C. Switch to using the AWS GovCloud (US) Region for Amazon S3 to store images and for Amazon Rekognition to compare faces. Set up a VPN connection and only call the Amazon Rekognition API operations through the VPN.

C. 切換到使用AWS GovCloud (US) 區域，讓Amazon S3 儲存影像，讓Amazon Rekognition 比較人臉。設定VPN 連接，並且僅透過VPN 呼叫Amazon Rekognition API 操作。

D. Enable client-side encryption on the S3 bucket. Set up a VPN connection and only call the Amazon Rekognition API operations through the VPN.

D. 在S3 儲存桶上啟用客戶端加密。設定VPN 連接，並且僅透過VPN 呼叫Amazon Rekognition API 操作。

問題# 137

A company is building a line-counting application for use in a quick-service restaurant. The company wants to use video cameras pointed at the line of customers at a given register to measure how many people are in line and deliver notifications to managers if the line grows too long. The restaurant locations have limited bandwidth for connections to external services and cannot accommodate multiple video streams without impacting other operations. Which solution should a machine learning specialist implement to meet these requirements?

一家公司正在開發一款用於快餐店的排隊計數應用程式。該公司希望使用攝影機對準收銀台的顧客隊伍，以測量排隊人數，並在隊伍過長時向經理發送通知。餐廳的頻寬有限，無法連接外部服務，無法在不影響其他操作的情況下容納多個視訊串流。機器學習專家應該實施哪種解決方案來滿足這些要求？

A. Install cameras compatible with Amazon Kinesis Video Streams to stream the data to AWS over the restaurant's existing internet connection. Write an AWS Lambda function to take an image and send it to Amazon Rekognition to count the number of faces in the image. Send an Amazon Simple Notification Service (Amazon SNS) notification if the line is too long.

A. 安裝與Amazon Kinesis Video Streams 相容的攝像頭，透過餐廳現有的網路連線將資料傳輸到AWS。編寫AWS Lambda 函數來拍攝圖像並將其發送到Amazon Rekognition 以計算圖像中的人臉數量。如果隊伍太長，則發送Amazon Simple Notification Service (Amazon SNS) 通知。

B. Deploy AWS DeepLens cameras in the restaurant to capture video. Enable Amazon Rekognition on the AWS DeepLens device, and use it to trigger a local AWS Lambda function when a person is recognized. Use the Lambda function to send an Amazon Simple Notification Service (Amazon SNS) notification if the line is too long.

B. 在餐廳部署AWS DeepLens 攝影機來捕捉影片。在AWS DeepLens 裝置上啟用Amazon Rekognition，並在識別出人員時使用它來觸發本機AWS Lambda 函數。如果隊伍太長，則使用Lambda 函數傳送Amazon Simple Notification Service (Amazon SNS) 通知。

C. Build a custom model in Amazon SageMaker to recognize the number of people in an image. Install cameras compatible with Amazon Kinesis Video Streams in the restaurant. Write an AWS Lambda function to take an image. Use the SageMaker endpoint to call the model to count people. Send an Amazon Simple Notification Service (Amazon SNS) notification if the line is too long.

C. 在Amazon SageMaker 中建立自訂模型以識別影像中的人數。在餐廳安裝與Amazon Kinesis Video Streams 相容的攝影機。編寫AWS Lambda 函數來拍攝影像。使用SageMaker 終端節點呼叫模型來計算人數。如果隊伍太長，則發送Amazon Simple Notification Service (Amazon SNS) 通知。

D. Build a custom model in Amazon SageMaker to recognize the number of people in an image. Deploy AWS DeepLens cameras in the restaurant. Deploy the model to the cameras. Deploy an AWS Lambda function to the cameras to use the model to count people and send an Amazon Simple Notification Service (Amazon SNS) notification if the line is too long.

D. 在Amazon SageMaker 中建立自訂模型以識別影像中的人數。在餐廳部署AWS DeepLens 攝影機。將模型部署到攝影機。將AWS Lambda 函數部署到攝影機以使用該模型來計算人數，如果排隊太長，則發送Amazon Simple Notification Service (Amazon SNS) 通知。

問題# 138

A company has set up and deployed its machine learning (ML) model into production with an endpoint using Amazon SageMaker hosting services. The ML team has configured automatic scaling for its SageMaker instances to support workload changes. During testing, the team notices that additional instances are being launched before the new instances are ready. This behavior needs to change as soon as possible. How can the ML team solve this issue?

一家公司已使用Amazon SageMaker 託管服務設定並部署了其機器學習(ML) 模型，並將其部署到生產環境中。ML 團隊已為其SageMaker 執行個體配置了自動擴展，以支援工作負載變更。在測試期間，團隊注意到在新實例準備就緒之前，其他實例正在啟動。這種行為需要盡快改變。ML 團隊如何解決這個問題？

A. Decrease the cooldown period for the scale-in activity. Increase the configured maximum capacity of instances.

A. 減少縮減活動的冷卻時間。增加配置的最大實例容量。

B. Replace the current endpoint with a multi-model endpoint using SageMaker.

B. 使用SageMaker 將目前終端節點替換為多模型終端節點。

C. Set up Amazon API Gateway and AWS Lambda to trigger the SageMaker inference endpoint.

C. 設定Amazon API Gateway 和AWS Lambda 以觸發SageMaker 推理終端節點。

D. Increase the cooldown period for the scale-out activity.

D. 增加擴展活動的冷卻時間。

問題# 139

A telecommunications company is developing a mobile app for its customers. The company is using an Amazon SageMaker hosted endpoint for machine learning model inferences. Developers want to introduce a new version of the model for a limited number of users who subscribed to a preview feature of the app. After the new version of the model is tested as a preview, developers will evaluate its accuracy. If a new version of the model has better accuracy, developers need to be able to gradually release the new version for all users over a fixed period of time. How can the company implement the testing model with the LEAST amount of operational overhead?

一家電信公司正在為其客戶開發一款行動應用程式。該公司正在使用Amazon SageMaker 託管終端節點進行機器學習模型推理。開發人員希望向訂閱了該應用程式預覽功能的有限數量的用戶推出新版本的模型。在新版本的模型作為預覽版進行測試後，開發人員將評估其準確性。如果新版本的模型具有更好的準確性，開發人員需要能夠在固定的時間內逐步向所有使用者發布新版本。公司如何以最少的營運開銷實施測試模型？

A. Update the ProductionVariant data type with the new version of the model by using the CreateEndpointConfig operation with the InitialVariantWeight parameter set to 0. Specify the TargetVariant parameter for InvokeEndpoint calls for users who subscribed to the preview feature. When the new version of the model is ready for release, gradually increase InitialVariantWeight until all users have the updated version.

A. 使用CreateEndpointConfig 操作並將InitialVariantWeight 參數設為0，使用新版本的模型更新ProductionVariant 資料類型。為訂閱預覽功能的使用者的InvokeEndpoint 呼叫指定TargetVariant 參數。當新版本的模型準備發佈時，逐漸增加InitialVariantWeight，直到所有使用者都擁有更

新的版本。

B. Configure two SageMaker hosted endpoints that serve the different versions of the model. Create an Application Load Balancer (ALB) to route traffic to both endpoints based on the TargetVariant query string parameter. Reconfigure the app to send the TargetVariant query string parameter for users who subscribed to the preview feature. When the new version of the model is ready for release, change the ALB's routing algorithm to weighted until all users have the updated version.

B. 配置兩個SageMaker 託管終端節點，為不同版本的模型提供服務。建立一個應用程式負載平衡器(ALB)，以根據TargetVariant 查詢字串參數將流量路由到兩個終端節點。重新配置應用程式以向訂閱預覽功能的使用者發送TargetVariant 查詢字串參數。當新版本的模型準備發佈時，將ALB 的路由演算法變更為加權，直到所有使用者都擁有更新的版本。

C. Update the DesiredWeightsAndCapacity data type with the new version of the model by using the UpdateEndpointWeightsAndCapacities operation with the DesiredWeight parameter set to 0. Specify the TargetVariant parameter for InvokeEndpoint calls for users who subscribed to the preview feature. When the new version of the model is ready for release, gradually increase DesiredWeight until all users have the updated version.

C. 使用UpdateEndpointWeightsAndCapacities 作業並將DesiredWeight 參數設為0，並使用新版本的模型更新DesiredWeightsAndCapacity 資料類型。為訂閱預覽功能的使用者的InvokeEndpoint 呼叫指定TargetVariant 參數。當新版本的模型準備好發佈時，逐漸增加DesiredWeight，直到所有使用者都擁有更新的版本。

D. Configure two SageMaker hosted endpoints that serve the different versions of the model. Create an Amazon Route 53 record that is configured with a simple routing policy and that points to the current version of the model. Configure the mobile app to use the endpoint URL for users who subscribed to the preview feature and to use the Route 53 record for other users. When the new version of the model is ready for release, add a new model version endpoint to Route 53, and switch the policy to weighted until all users have the updated version.

D. 配置兩個SageMaker 託管終端節點，為不同版本的模型提供服務。建立一條Amazon Route 53 記錄，該記錄配置了簡單的路由策略，並指向目前版本的模型。配置行動應用程式以使用終端節點 URL 來對待訂閱預覽功能的用戶，並使用Route 53 記錄來對待其他用戶。當新版本的模型準備好發佈時，請向Route 53 新增新的模型版本終端節點，並將策略切換為加權，直到所有使用者都擁有更新的版本。

問題# 140

A company offers an online shopping service to its customers. The company wants to enhance the site's security by requesting additional information when customers access the site from locations that are different from their normal location. The company wants to update the process to call a machine learning (ML) model to determine when additional information should be requested. The company has several terabytes of data from its existing ecommerce web servers containing the source IP addresses for each request made to the web server. For authenticated requests, the records also contain the login name of the requesting user. Which approach should an ML specialist take to implement the new security feature in the web application?

一家公司為客戶提供線上購物服務。該公司希望透過在客戶從非正常位置訪問網站時請求更多資訊來增強網站的安全性。該公司希望更新流程以調用機器學習(ML) 模型來確定何時應請求更多資訊。該公司擁有來自其現有電子商務Web 伺服器的數TB 數據，其中包含對Web 伺服器發出的每個請求的來源IP 位址。對於經過身份驗證的請求，記錄還包含請求使用者的登入名稱。ML 專家應採用哪

種方法來在Web 應用程式中實現新的安全功能？

A. Use Amazon SageMaker Ground Truth to label each record as either a successful or failed access attempt. Use Amazon SageMaker to train a binary classification model using the factorization machines (FM) algorithm.

A. 使用Amazon SageMaker Ground Truth 將每筆記錄標記為成功或失敗的存取嘗試。使用Amazon SageMaker 使用因式分解機(FM) 演算法訓練二元分類模型。

B. Use Amazon SageMaker to train a model using the IP Insights algorithm. Schedule updates and retraining of the model using new log data nightly.

B. 使用Amazon SageMaker 使用IP Insights 演算法訓練模型。安排每晚使用新日誌資料更新和重新訓練模型。

C. Use Amazon SageMaker Ground Truth to label each record as either a successful or failed access attempt. Use Amazon SageMaker to train a binary classification model using the IP Insights algorithm.

C. 使用Amazon SageMaker Ground Truth 將每筆記錄標記為成功或失敗的存取嘗試。使用Amazon SageMaker 使用IP Insights 演算法訓練二元分類模型。

D. Use Amazon SageMaker to train a model using the Object2Vec algorithm. Schedule updates and retraining of the model using new log data nightly.

D. 使用Amazon SageMaker 使用Object2Vec 演算法訓練模型。安排每晚使用新日誌資料更新和重新訓練模型。

問題# 141

A retail company wants to combine its customer orders with the product description data from its product catalog. The structure and format of the records in each dataset is different. A data analyst tried to use a spreadsheet to combine the datasets, but the effort resulted in duplicate records and records that were not properly combined. The company needs a solution that it can use to combine similar records from the two datasets and remove any duplicates. Which solution will meet these requirements?

一家零售公司希望將其客戶訂單與其產品目錄中的產品描述資料結合。每個資料集中的記錄結構和格式都不同。一位資料分析師嘗試使用電子表格來合併資料集，但這項努力導致記錄重複以及記錄合併不正確。該公司需要一個解決方案來合併兩個資料集中的相似記錄並刪除任何重複項。哪種解決方案可以滿足這些要求？

A. Use an AWS Lambda function to process the data. Use two arrays to compare equal strings in the fields from the two datasets and remove any duplicates.

A. 使用AWS Lambda 函數處理資料。使用兩個陣列比較兩個資料集中欄位的相等字串並刪除任何重複項。

B. Create AWS Glue crawlers for reading and populating the AWS Glue Data Catalog. Call the AWS Glue SearchTables API operation to perform a fuzzy- matching search on the two datasets, and cleanse the data accordingly.

B. 建立AWS Glue 爬蟲以讀取和填充AWS Glue 資料目錄。呼叫AWS Glue SearchTables API 操作

對兩個資料集執行模糊匹配搜索，並相應地清理資料。

C. Create AWS Glue crawlers for reading and populating the AWS Glue Data Catalog. Use the FindMatches transform to cleanse the data.

C. 建立AWS Glue 爬蟲以讀取和填充AWS Glue 資料目錄。使用FindMatches 轉換清理資料。

D. Create an AWS Lake Formation custom transform. Run a transformation for matching products from the Lake Formation console to cleanse the data automatically.

D. 建立AWS Lake Formation 自訂轉換。從Lake Formation 控制台執行轉換以符合產品以自動清理資料。

問題# 142

A company provisions Amazon SageMaker notebook instances for its data science team and creates Amazon VPC interface endpoints to ensure communication between the VPC and the notebook instances. All connections to the Amazon SageMaker API are contained entirely and securely using the AWS network. However, the data science team realizes that individuals outside the VPC can still connect to the notebook instances across the internet. Which set of actions should the data science team take to fix the issue?

一家公司為其資料科學團隊配置了Amazon SageMaker 筆記本實例，並創建了Amazon VPC 介面終端節點，以確保VPC 與筆記本實例之間的通訊。所有與Amazon SageMaker API 的連線都完全安全地包含在AWS 網路中。但是，資料科學團隊意識到VPC 以外的個人仍然可以透過網路連接到筆記型電腦實例。數據科學團隊應該採取哪些措施來解決這個問題？

A. Modify the notebook instances' security group to allow traffic only from the CIDR ranges of the VPC. Apply this security group to all of the notebook instances' VPC interfaces.

A. 修改筆記本執行個體的安全性群組以僅允許來自VPC 的CIDR 範圍的流量。將此安全性群組套用至所有筆記本執行個體的VPC 介面。

B. Create an IAM policy that allows the sagemaker:CreatePresignedNotebookInstanceUrl and sagemaker:DescribeNotebookInstance actions from only the VPC endpoints. Apply this policy to all IAM users, groups, and roles used to access the notebook instances.

B. 建立一個IAM 策略，僅允許來自VPC 終端節點的sagemaker:CreatePresignedNotebookInstanceUrl 和sagemaker:DescribeNotebookInstance 操作。將此策略套用至用於存取筆記本實例的所有IAM 使用者、群組和角色。

C. Add a NAT gateway to the VPC. Convert all of the subnets where the Amazon SageMaker notebook instances are hosted to private subnets. Stop and start all of the notebook instances to reassign only private IP addresses.

C. 向VPC 新增NAT 閘道。將託管Amazon SageMaker 筆記型電腦執行個體的所有子網路轉換為私有子網路。停止並啟動所有筆記本實例以僅重新指派私有IP 位址。

D. Change the network ACL of the subnet the notebook is hosted in to restrict access to anyone outside the VPC.

D. 更改筆記本所在子網路的網路ACL，以限制VPC 以外任何人的存取。

問題# 143

A company will use Amazon SageMaker to train and host a machine learning (ML) model for a marketing campaign. The majority of data is sensitive customer data. The data must be encrypted at rest. The company wants AWS to maintain the root of trust for the master keys and wants encryption key usage to be logged. Which implementation will meet these requirements?

一家公司將使用Amazon SageMaker 來訓練和託管行銷活動的機器學習(ML) 模型。大多數數據都是敏感的客戶資料。資料必須在靜止時加密。該公司希望AWS 維護主金鑰的信任根，並希望記錄加密金鑰的使用情況。哪種實施可以滿足這些要求？

A. Use encryption keys that are stored in AWS Cloud HSM to encrypt the ML data volumes, and to encrypt the model artifacts and data in Amazon S3.

A. 使用儲存在AWS Cloud HSM 中的加密金鑰加密ML 資料卷，並加密Amazon S3 中的模型構件和資料。

B. Use SageMaker built-in transient keys to encrypt the ML data volumes. Enable default encryption for new Amazon Elastic Block Store (Amazon EBS) volumes.

B. 使用SageMaker 內建臨時金鑰加密ML 資料卷。為新的Amazon Elastic Block Store (Amazon EBS) 磁碟區啟用預設加密。

C. Use customer managed keys in AWS Key Management Service (AWS KMS) to encrypt the ML data volumes, and to encrypt the model artifacts and data in Amazon S3.

C. 使用AWS Key Management Service (AWS KMS) 中的客戶管理金鑰加密ML 資料卷，並加密Amazon S3 中的模型構件和資料。

D. Use AWS Security Token Service (AWS STS) to create temporary tokens to encrypt the ML storage volumes, and to encrypt the model artifacts and data in Amazon S3.

D. 使用AWS Security Token Service (AWS STS) 建立臨時令牌以加密ML 儲存卷，並加密Amazon S3 中的模型構件和資料。

問題# 144

A machine learning specialist stores IoT soil sensor data in Amazon DynamoDB table and stores weather event data as JSON files in Amazon S3. The dataset in DynamoDB is 10 GB in size and the dataset in Amazon S3 is 5 GB in size. The specialist wants to train a model on this data to help predict soil moisture levels as a function of weather events using Amazon SageMaker. Which solution will accomplish the necessary transformation to train the Amazon SageMaker model with the LEAST amount of administrative overhead?

機器學習專家將物聯網土壤感測器資料儲存在Amazon DynamoDB 表中，並將天氣事件資料作為JSON 檔案儲存在Amazon S3 中。DynamoDB 中的資料集大小為10 GB，Amazon S3 中的資料集大小為5 GB。專家希望使用Amazon SageMaker 根據這些數據訓練一個模型，以幫助根據天氣事件預測土壤濕度水平。哪種解決方案可以以最少的管理開銷完成訓練Amazon SageMaker 模型所需的轉換？

A. Launch an Amazon EMR cluster. Create an Apache Hive external table for the DynamoDB table and S3 data. Join the Hive tables and write the results out to Amazon S3.

A. 啟動Amazon EMR 叢集。為DynamoDB 資料表和S3 資料建立Apache Hive 外部表。連接Hive 表並將結果寫入Amazon S3。

B. Crawl the data using AWS Glue crawlers. Write an AWS Glue ETL job that merges the two tables and writes the output to an Amazon Redshift cluster.

B. 使用AWS Glue 爬蟲程式爬取資料。寫一個AWS Glue ETL 作業，合併兩個表格並將輸出寫入Amazon Redshift 叢集。

C. Enable Amazon DynamoDB Streams on the sensor table. Write an AWS Lambda function that consumes the stream and appends the results to the existing weather files in Amazon S3.

C. 在感測器表上啟用Amazon DynamoDB Streams。編寫一個AWS Lambda 函數，使用該流並將結果附加到Amazon S3 中現有的天氣檔案。

D. Crawl the data using AWS Glue crawlers. Write an AWS Glue ETL job that merges the two tables and writes the output in CSV format to Amazon S3.

D. 使用AWS Glue 爬蟲程式爬取資料。寫一個AWS Glue ETL 作業，合併兩個表並將輸出以CSV 格式寫入Amazon S3。

問題# 145

A company sells thousands of products on a public website and wants to automatically identify products with potential durability problems. The company has 1,000 reviews with date, star rating, review text, review summary, and customer email fields, but many reviews are incomplete and have empty fields. Each review has already been labeled with the correct durability result. A machine learning specialist must train a model to identify reviews expressing concerns over product durability. The first model needs to be trained and ready to review in 2 days. What is the MOST direct approach to solve this problem within 2 days?

一家公司在公共網站上銷售數千種產品，並希望自動識別可能存在耐久性問題的產品。該公司有1,000 條評論，其中包含日期、星級、評論文字、評論摘要和客戶電子郵件字段，但許多評論不完整且有空白字段。每則評論都已貼上正確的耐久性結果標籤。機器學習專家必須訓練一個模型來識別表達對產品耐久性擔憂的評論。第一個模型需要在2 天內完成訓練並準備好進行審查。在2 天內解決這個問題的最直接方法是什麼？

A. Train a custom classifier by using Amazon Comprehend.

A. 使用Amazon Comprehend 訓練自訂分類器。

B. Build a recurrent neural network (RNN) in Amazon SageMaker by using Gluon and Apache MXNet.

B. 使用Gluon 和Apache MXNet 在Amazon SageMaker 中建構循環神經網路(RNN)。

C. Train a built-in BlazingText model using Word2Vec mode in Amazon SageMaker.

C. 使用Amazon SageMaker 中的Word2Vec 模式訓練內建BlazingText 模型。

D. Use a built-in seq2seq model in Amazon SageMaker.

D. 使用Amazon SageMaker 內建的seq2seq 模型。

問題# 146

A company that runs an online library is implementing a chatbot using Amazon Lex to provide book recommendations based on category. This intent is fulfilled by an AWS Lambda function that queries an Amazon DynamoDB table for a list of book titles, given a particular category. For testing, there are only three categories implemented as the custom slot types: "comedy," "adventure," and "documentary." A machine learning (ML) specialist notices that sometimes the request cannot be fulfilled because Amazon Lex cannot understand the category spoken by users with utterances such as "funny," "fun," and "humor." The ML specialist needs to fix the problem without changing the Lambda code or data in DynamoDB. How should the ML specialist fix the problem?

一家經營線上圖書館的公司正在使用Amazon Lex 實現聊天機器人，以根據類別提供書籍推薦。此意圖由AWS Lambda 函數實現，該函數查詢Amazon DynamoDB 表中給定特定類別的書籍標題清單。為了進行測試，只有三個類別實現為自訂槽類型：「喜劇」、「冒險」和「紀錄片」。機器學習(ML) 專家注意到，有時無法滿足請求，因為Amazon Lex 無法理解使用者所說的類別，例如「有趣」、「有趣」和「幽默」。ML 專家需要在不更改Lambda 程式碼或DynamoDB 中的資料的情況下解決問題。ML 專家該如何解決問題？

A. Add the unrecognized words in the enumeration values list as new values in the slot type.

A. 將無法辨識的單字作為新值加入到枚舉值清單中。

B. Create a new custom slot type, add the unrecognized words to this slot type as enumeration values, and use this slot type for the slot.

B. 建立新的自訂插槽類型，將無法辨識的單字作為枚舉值新增至此插槽類型，並將此插槽類型用於插槽。

C. Use the AMAZON.SearchQuery built-in slot types for custom searches in the database.

C. 使用AMAZON.SearchQuery 內建插槽類型在資料庫中進行自訂搜尋。

D. Add the unrecognized words as synonyms in the custom slot type.

D. 將無法辨識的單字作為同義詞加入自訂插槽類型。

問題# 147

A manufacturing company uses machine learning (ML) models to detect quality issues. The models use images that are taken of the company's product at the end of each production step. The company has thousands of machines at the production site that generate one image per second on average. The company ran a successful pilot with a single manufacturing machine. For the pilot, ML specialists used an industrial PC that ran AWS IoT Greengrass with a long-running AWS Lambda function that uploaded the images to Amazon S3. The uploaded images invoked a Lambda function that was written in Python to perform inference by using an Amazon SageMaker endpoint that ran a custom model. The inference results were forwarded back to a web service that was hosted at the production site to prevent faulty products from being shipped. The company scaled the solution out to all manufacturing machines by installing similarly configured industrial PCs on each production machine. However, latency for predictions increased beyond acceptable limits. Analysis shows that the internet connection is at its capacity limit. How can the company resolve this issue MOST cost-effectively?

一家製造公司使用機器學習(ML)模型來檢測品質問題。這些模型使用在每個生產步驟結束時拍攝的公司產品影像。該公司在生產現場擁有數千台機器，平均每秒鐘產生一張影像。該公司使用一台製造機器進行了成功的試點。在試點中，ML專家使用運行AWS IoT Greengrass的工業PC和長期運行的AWS Lambda函數將映像上傳到Amazon S3。上傳的映像呼叫用Python編寫的Lambda函數，透過使用運行自訂模型的Amazon SageMaker終端節點執行推理。推理結果被轉發回託管在生產現場的Web服務，以防止運送有缺陷的產品。該公司透過在每台生產機器上安裝類似配置的工業PC，將解決方案擴展到所有製造機器。然而，預測的延遲超出了可接受的限度。分析顯示網路連線已達到其容量極限。公司如何以最具成本效益的方式解決這個問題？

A. Set up a 10 Gbps AWS Direct Connect connection between the production site and the nearest AWS Region. Use the Direct Connect connection to upload the images. Increase the size of the instances and the number of instances that are used by the SageMaker endpoint.

A. 在生產站點和最近的AWS區域之間建立10 Gbps AWS Direct Connect連線。使用Direct Connect連線上傳影像。增加實例的大小和SageMaker終端節點所使用的實例數量。

B. Extend the long-running Lambda function that runs on AWS IoT Greengrass to compress the images and upload the compressed files to Amazon S3. Decompress the files by using a separate Lambda function that invokes the existing Lambda function to run the inference pipeline.

B. 擴展在AWS IoT Greengrass上運行的長時間運行的Lambda函數以壓縮映像並將壓縮檔案上傳到Amazon S3。使用單獨的Lambda函數解壓縮文件，該函數呼叫現有的Lambda函數來運行推理管道。

C. Use auto scaling for SageMaker. Set up an AWS Direct Connect connection between the production site and the nearest AWS Region. Use the Direct Connect connection to upload the images.

C. 對SageMaker使用自動擴充。在生產站點和最近的AWS區域之間建立AWS Direct Connect連線。使用Direct Connect連線上傳影像。

D. Deploy the Lambda function and the ML models onto the AWS IoT Greengrass core that is running on the industrial PCs that are installed on each machine. Extend the long-running Lambda function that runs on AWS IoT Greengrass to invoke the Lambda function with the captured images and run the inference on the edge component that forwards the results directly to the web service.

D. 將Lambda函數和ML模型部署到在每台機器上安裝的工業PC上運行的AWS IoT Greengrass核心上。擴展在AWS IoT Greengrass上運行的長時間運行的Lambda函數以使用捕獲的圖像調用Lambda函數，並在將結果直接轉發到Web服務的邊緣組件上運行推理。

問題# 148

A data scientist is using an Amazon SageMaker notebook instance and needs to securely access data stored in a specific Amazon S3 bucket. How should the data scientist accomplish this?

一位資料科學家正在使用Amazon SageMaker 筆記本實例，並且需要安全地存取儲存在特定Amazon S3 儲存桶中的資料。數據科學家應如何實現這一目標？

A. Add an S3 bucket policy allowing GetObject, PutObject, and ListBucket permissions to the Amazon SageMaker notebook ARN as principal.

A. 新增一個S3 儲存桶策略，允許將GetObject、PutObject 和ListBucket 權限作為主體授予Amazon SageMaker 筆記本ARN。

B. Encrypt the objects in the S3 bucket with a custom AWS Key Management Service (AWS KMS) key that only the notebook owner has access to.

B. 使用只有筆記本擁有者才有權存取的自訂AWS Key Management Service (AWS KMS) 金鑰加密S3 儲存桶中的物件。

C. Attach the policy to the IAM role associated with the notebook that allows GetObject, PutObject, and ListBucket operations to the specific S3 bucket.

C. 將策略附加到與筆記本相關的IAM 角色，允許對特定S3 儲存桶執行GetObject、PutObject 和ListBucket 操作。

D. Use a script in a lifecycle configuration to configure the AWS CLI on the instance with an access key ID and secret.

D. 使用生命週期配置中的腳本，使用存取金鑰ID 和金鑰在實例上設定AWS CLI。

問題# 149

A company is launching a new product and needs to build a mechanism to monitor comments about the company and its new product on social media. The company needs to be able to evaluate the sentiment expressed in social media posts, and visualize trends and configure alarms based on various thresholds. The company needs to implement this solution quickly, and wants to minimize the infrastructure and data science resources needed to evaluate the messages. The company already has a solution in place to collect posts and store them within an Amazon S3 bucket. What services should the data science team use to deliver this solution?

一家公司正在推出一款新產品，需要建立一種機制來監控社群媒體上有關該公司及其新產品的評論。該公司需要能夠評估社交媒體貼文中表達的情緒，並視覺化趨勢並根據各種閾值配置警報。該公司需要快速實施此解決方案，並希望最大限度地減少評估訊息所需的基礎設施和數據科學資源。該公司已經有一個解決方案來收集帖子並將其存儲在Amazon S3 存儲桶中。數據科學團隊應該使用哪些服務來提供此解決方案？

A. Train a model in Amazon SageMaker by using the BlazingText algorithm to detect sentiment in the corpus of social media posts. Expose an endpoint that can be called by AWS Lambda. Trigger a Lambda function when posts are added to the S3 bucket to invoke the endpoint and record the sentiment in an Amazon DynamoDB table and in a custom Amazon CloudWatch metric. Use CloudWatch alarms to notify analysts of trends.

A. 使用BlazingText 演算法在Amazon SageMaker 中訓練模型，以偵測社群媒體貼文語料庫中的情緒。公開可由AWS Lambda 呼叫的終端節點。當貼文新增至S3 儲存桶時觸發Lambda 函數以呼叫終端節點並在Amazon DynamoDB 表和自訂Amazon CloudWatch 指標中記錄情緒。使用CloudWatch 警報通知分析師趨勢。

B. Train a model in Amazon SageMaker by using the semantic segmentation algorithm to model the semantic content in the corpus of social media posts. Expose an endpoint that can be called by AWS Lambda. Trigger a Lambda function when objects are added to the S3 bucket to invoke the endpoint and record the sentiment in an Amazon DynamoDB table. Schedule a second Lambda function to query recently added records and send an Amazon Simple Notification Service (Amazon SNS) notification to notify analysts of trends.

B. 使用語義分割演算法在Amazon SageMaker 中訓練模型，以對社群媒體貼文語料庫中的語意內容進行建模。公開可由AWS Lambda 呼叫的終端節點。當物件新增至S3 儲存桶時觸發Lambda 函數以呼叫終端節點並在Amazon DynamoDB 表中記錄情緒。安排第二個Lambda 函數以查詢最近新增的記錄並發送Amazon Simple Notification Service (Amazon SNS) 通知以通知分析師趨勢。

C. Trigger an AWS Lambda function when social media posts are added to the S3 bucket. Call Amazon Comprehend for each post to capture the sentiment in the message and record the sentiment in an Amazon DynamoDB table. Schedule a second Lambda function to query recently added records and send an Amazon Simple Notification Service (Amazon SNS) notification to notify analysts of trends.

C. 當社群媒體貼文新增至S3 儲存桶時觸發AWS Lambda 函數。為每個貼文呼叫Amazon Comprehend 以捕捉訊息中的情緒並將情緒記錄在Amazon DynamoDB 表中。安排第二個Lambda 函數來查詢最近新增的記錄並發送Amazon Simple Notification Service (Amazon SNS) 通知以通知分析師趨勢。

D. Trigger an AWS Lambda function when social media posts are added to the S3 bucket. Call Amazon Comprehend for each post to capture the sentiment in the message and record the sentiment in a custom Amazon CloudWatch metric and in S3. Use CloudWatch alarms to notify analysts of trends.

D. 當社群媒體貼文新增至S3 儲存桶時觸發AWS Lambda 函數。為每個貼文呼叫Amazon Comprehend 以捕捉訊息中的情緒並將情緒記錄在自訂Amazon CloudWatch 指標和S3 中。使用CloudWatch 警報通知分析師趨勢。

A bank wants to launch a low-rate credit promotion. The bank is located in a town that recently experienced economic hardship. Only some of the bank's customers were affected by the crisis, so the bank's credit team must identify which customers to target with the promotion. However, the credit team wants to make sure that loyal customers' full credit history is considered when the decision is made. The bank's data science team developed a model that classifies account transactions and understands credit eligibility. The data science team used the XGBoost algorithm to train the model. The team used 7 years of bank transaction historical data for training and hyperparameter tuning over the course of several days. The accuracy of the model is sufficient, but the credit team is struggling to explain accurately why the model denies credit to some customers. The credit team has almost no skill in data science. What should the data science team do to address this issue in the MOST operationally efficient manner?

一家銀行希望推出低利率信貸促銷活動。該銀行位於一個最近遭遇經濟困難的城鎮。只有部分銀行客戶受到危機的影響，因此銀行的信貸團隊必須確定促銷活動的目標客戶。但是，信貸團隊希望確保在做出決定時考慮忠實客戶的完整信用歷史。銀行的數據科學團隊開發了一個模型，該模型可以對帳戶交易進行分類並了解信貸資格。資料科學團隊使用XGBoost 演算法來訓練此模型。該團隊使用了7年的銀行交易歷史數據，在幾天內進行了訓練和超參數調整。該模型的準確性足夠高，但信貸團隊很難準確解釋為什麼模型拒絕向某些客戶提供信貸。信貸團隊幾乎沒有數據科學技能。數據科學團隊該怎麼做才能以最有效率的方式解決這個問題？

A. Use Amazon SageMaker Studio to rebuild the model. Create a notebook that uses the XGBoost training container to perform model training. Deploy the model at an endpoint. Enable Amazon SageMaker Model Monitor to store inferences. Use the inferences to create Shapley values that help explain model behavior. Create a chart that shows features and SHapley Additive exPlanations (SHAP) values to explain to the credit team how the features affect the model outcomes.

A. 使用Amazon SageMaker Studio 重建模型。建立一個使用XGBoost 訓練容器執行模型訓練的筆記本。在終端部署模型。啟用Amazon SageMaker 模型監控器來儲存推理。使用推理建立有助於解釋模型行為的Shapley 值。建立一個顯示特徵和SHapley Additive exPlanations (SHAP) 值的圖表，以向信用團隊解釋這些特徵如何影響模型結果。

B. Use Amazon SageMaker Studio to rebuild the model. Create a notebook that uses the XGBoost training container to perform model training. Activate Amazon SageMaker Debugger, and configure it to calculate and collect Shapley values. Create a chart that shows features and SHapley Additive exPlanations (SHAP) values to explain to the credit team how the features affect the model outcomes.

B. 使用Amazon SageMaker Studio 重建模型。建立一個使用XGBoost 訓練容器執行模型訓練的筆記本。啟動Amazon SageMaker Debugger，並將其配置為計算和收集Shapley 值。建立一個顯示特徵和SHapley Additive exPlanations (SHAP) 值的圖表，以向信用團隊解釋這些特徵如何影響模型結果。

C. Create an Amazon SageMaker notebook instance. Use the notebook instance and the XGBoost library to locally retrain the model. Use the plot_importance() method in the Python XGBoost interface to create a feature importance chart. Use that chart to explain to the credit team how the features affect the model outcomes.

C. 建立一個Amazon SageMaker 筆記本實例。使用筆記本實例和XGBoost 函式庫在本地重新訓練模型。使用Python XGBoost 介面中的plot_importance() 方法建立特徵重要性圖表。使用該圖表向信用團隊解釋這些特徵如何影響模型結果。

D. Use Amazon SageMaker Studio to rebuild the model. Create a notebook that uses the XGBoost training container to perform model training. Deploy the model at an endpoint. Use Amazon SageMaker Processing to post-analyze the model and create a feature importance explainability chart automatically for the credit team.

D. 使用Amazon SageMaker Studio 重建模型。建立一個使用XGBoost 訓練容器執行模型訓練的筆記本。在終端節點部署模型。使用Amazon SageMaker Processing 對模型進行後期分析，並為信用團隊自動建立特徵重要性可解釋性圖表。

問題# 151

A data science team is planning to build a natural language processing (NLP) application. The application's text preprocessing stage will include part-of-speech tagging and key phrase extraction. The preprocessed text will be input to a custom classification algorithm that the data science team has already written and trained using Apache MXNet. Which solution can the team build MOST quickly to meet these requirements?

一個數據科學團隊計劃建立一個自然語言處理(NLP) 應用程式。該應用程式的文本預處理階段將包括詞性標記和關鍵階段提取。預處理後的文字將輸入到資料科學團隊已經使用Apache MXNet 編寫和訓練的自訂分類演算法。團隊可以最快速地建立哪種解決方案來滿足這些要求？

A. Use Amazon Comprehend for the part-of-speech tagging, key phrase extraction, and classification tasks.

A. 使用Amazon Comprehend 進行詞性標記、關鍵階段擷取與分類任務。

B. Use an NLP library in Amazon SageMaker for the part-of-speech tagging. Use Amazon Comprehend for the key phrase extraction. Use AWS Deep Learning Containers with Amazon SageMaker to build the custom classifier.

B. 使用Amazon SageMaker 中的NLP 庫進行詞性標記。使用Amazon Comprehend 進行關鍵階段擷取。將AWS Deep Learning Containers 與Amazon SageMaker 結合使用以建立自訂分類器。

C. Use Amazon Comprehend for the part-of-speech tagging and key phrase extraction tasks. Use Amazon SageMaker built-in Latent Dirichlet Allocation (LDA) algorithm to build the custom classifier.

C. 使用Amazon Comprehend 進行詞性標記和關鍵階段擷取任務。使用Amazon SageMaker 內建的潛在狄利克雷分配(LDA) 演算法建立自訂分類器。

D. Use Amazon Comprehend for the part-of-speech tagging and key phrase extraction tasks. Use AWS Deep Learning Containers with Amazon SageMaker to build the custom classifier.

D. 使用Amazon Comprehend 進行詞性標記和關鍵階段擷取任務。將AWS Deep Learning Containers 與Amazon SageMaker 結合使用以建立自訂分類器。

問題# 152

A machine learning (ML) specialist must develop a classification model for a financial services company. A domain expert provides the dataset, which is tabular with 10,000 rows and 1,020 features. During exploratory data analysis, the specialist finds no missing values and a small percentage of duplicate rows. There are correlation scores of > 0.9 for 200 feature pairs. The mean value of each feature is similar to its 50th percentile. Which feature engineering strategy should the ML specialist use with Amazon SageMaker?

機器學習(ML) 專家必須為金融服務公司開發分類模型。領域專家提供了資料集，該資料集是表格形式，包含10,000 行和1,020 個特徵。在探索性資料分析期間，專家發現沒有缺失值，但重複行的比例很小。200 個特徵對的相關性得分 > 0.9 。每個特徵的平均值與其第50 個百分位數相似。ML 專家應該將哪種特徵工程策略與Amazon SageMaker 結合使用？

A. Apply dimensionality reduction by using the principal component analysis (PCA) algorithm.

A. 使用主成分分析(PCA) 演算法進行降維。

B. Drop the features with low correlation scores by using a Jupyter notebook.

B. 使用Jupyter 筆記本刪除相關性分數低的特徵。

C. Apply anomaly detection by using the Random Cut Forest (RCF) algorithm.

C. 使用隨機砍伐森林(RCF) 演算法應用異常檢測。

D. Concatenate the features with high correlation scores by using a Jupyter notebook.

D. 使用Jupyter 筆記本連結相關性分數高的特徵。

問題# 153

A manufacturing company asks its machine learning specialist to develop a model that classifies defective parts into one of eight defect types. The company has provided roughly 100,000 images per defect type for training. During the initial training of the image classification model, the specialist notices that the validation accuracy is 80%, while the training accuracy is 90%. It is known that human-level performance for this type of image classification is around 90%. What should the specialist consider to fix this issue?

一家製造公司要求其機器學習專家開發一種模型，將缺陷部件歸類為八種缺陷類型之一。該公司為每種缺陷類型提供了大約100,000 張圖像用於訓練。在影像分類模型的初始訓練期間，專家注意到驗證準確率為80%，而訓練準確率為90%。眾所周知，這種類型的圖像分類的人類層級表現約為90%。專家應該考慮什麼來解決這個問題？

A. A longer training time.

A. 延長訓練時間。

B. Making the network larger.

B. 擴大網路。

C. Using a different optimizer.

C. 使用不同的優化器。

D. Using some form of regularization.

D. 使用某種形式的正規化。

問題# 154

A machine learning specialist needs to analyze comments on a news website with users across the globe. The specialist must find the most discussed topics in the comments that are in either English or Spanish. What steps could be used to accomplish this task? (Choose two.)

機器學習專家需要分析一個擁有全球用戶的新聞網站上的評論。專家必須在評論中找到討論最多的英語或西班牙語主題。可以使用哪些步驟來完成此任務？（選兩個。）

A. Use an Amazon SageMaker BlazingText algorithm to find the topics independently from language. Proceed with the analysis.

A. 使用Amazon SageMaker BlazingText 演算法獨立於語言尋找主題。繼續分析。

B. Use an Amazon SageMaker seq2seq algorithm to translate from Spanish to English, if necessary. Use a SageMaker Latent Dirichlet Allocation (LDA) algorithm to find the topics.

B. 使用Amazon SageMaker seq2seq 演算法將西班牙文翻譯成英文（如有需要）。使用SageMaker 潛在狄利克雷分配(LDA) 演算法找出主題。

C. Use Amazon Translate to translate from Spanish to English, if necessary. Use Amazon Comprehend topic modeling to find the topics.

C. 使用Amazon Translate 將西班牙文翻譯成英文（如有必要）。使用Amazon Comprehend 主題建模找出主題。

D. Use Amazon Translate to translate from Spanish to English, if necessary. Use Amazon Lex to extract topics from the content.

D. 使用Amazon Translate 將西班牙文翻譯成英文（如有必要）。使用Amazon Lex 從內容中提取主題。

E. Use Amazon Translate to translate from Spanish to English, if necessary. Use Amazon SageMaker Neural Topic Model (NTM) to find the topics.

E. 使用Amazon Translate 將西班牙文翻譯成英文（如有必要）。使用Amazon SageMaker 神經主題模型(NTM) 尋找主題。

問題# 155

A machine learning (ML) specialist is administering a production Amazon SageMaker endpoint with model monitoring configured. Amazon SageMaker ModelMonitor detects violations on the SageMaker endpoint, so the ML specialist retrains the model with the latest dataset. This dataset is statistically representative of the current production traffic. The ML specialist notices that even after deploying the new SageMaker model and running the first monitoring job, the SageMaker endpoint still has violations. What should the ML specialist do to resolve the violations?

機器學習(ML)專家正在管理配置了模型監控的生產Amazon SageMaker 終端節點。Amazon SageMaker ModelMonitor 偵測到SageMaker 終端節點上的違規行為，因此ML專家使用最新資料集重新訓練模型。此資料集在統計上代表了當前生產流量。ML專家注意到，即使部署新的SageMaker模型並執行第一個監控作業後，SageMaker 終端節點仍然有違規行為。ML專家該怎麼做才能解決違規行為？

A. Manually trigger the monitoring job to re-evaluate the SageMaker endpoint traffic sample.

A. 手動觸發監控作業以重新評估SageMaker 端點流量樣本。

B. Run the Model Monitor baseline job again on the new training set. Configure Model Monitor to use the new baseline.

B. 在新的訓練集上再次執行模型監控基線作業。配置模型監控以使用新基準。

C. Delete the endpoint and recreate it with the original configuration.

C. 刪除端點並使用原始配置重新建立它。

D. Retrain the model again by using a combination of the original training set and the new training set.

D. 使用原始訓練集和新訓練集的組合再次重新訓練模型。

問題# 156

A company supplies wholesale clothing to thousands of retail stores. A data scientist must create a model that predicts the daily sales volume for each item for each store. The data scientist discovers that more than half of the stores have been in business for less than 6 months. Sales data is highly consistent from week to week. Daily data from the database has been aggregated weekly, and weeks with no sales are omitted from the current dataset. Five years (100 MB) of sales data is available in Amazon S3. Which factors will adversely impact the performance of the forecast model to be developed, and which actions should the data scientist take to mitigate them? (Choose two.)

一家公司為數千家零售店提供批發服裝。資料科學家必須建立一個模型來預測每家商店每種商品的每日銷售量。資料科學家發現，超過一半的商店開業時間不到6個月。每週的銷售數據高度一致。資料庫中的每日資料已按週匯總，當前資料集中省略了沒有銷售的周數。Amazon S3中提供了五年（100 MB）的銷售數據。哪些因素會對要開發的預測模型的表現產生不利影響，而資料科學家應採取哪些措施來緩解這些影響？（選兩個。）

A. Detecting seasonality for the majority of stores will be an issue. Request categorical data to relate new stores with similar stores that have more historical data.

A. 檢測大多數商店的季節性將是一個問題。請求分類資料以將新商店與具有更多歷史資料的類似商店聯繫起來。

B. The sales data does not have enough variance. Request external sales data from other industries to improve the model's ability to generalize.

B. 銷售資料沒有足夠的變異數。請求來自其他行業的外部銷售資料以提高模型的泛化能力。

C. Sales data is aggregated by week. Request daily sales data from the source database to enable building a daily model.

C. 銷售資料按週匯總。從來源資料庫請求每日銷售資料以建立每日模型。

D. The sales data is missing zero entries for item sales. Request that item sales data from the source database include zero entries to enable building the model.

D. 銷售資料缺少零個商品銷售條目。請求來源資料庫中的商品銷售資料包含零個條目以啟用建置模型。

E. Only 100 MB of sales data is available in Amazon S3. Request 10 years of sales data, which would provide 200 MB of training data for the model.

E. Amazon S3 中只有100 MB 的銷售資料可用。請求10 年的銷售數據，這將為模型提供200 MB 的訓練數據。

問題# 157

An ecommerce company is automating the categorization of its products based on images. A data scientist has trained a computer vision model using the Amazon SageMaker image classification algorithm. The images for each product are classified according to specific product lines. The accuracy of the model is too low when categorizing new products. All of the product images have the same dimensions and are stored within an Amazon S3 bucket. The company wants to improve the model so it can be used for new products as soon as possible. Which steps would improve the accuracy of the solution? (Choose three.)

一家電子商務公司正在根據圖像自動對其產品進行分類。一位資料科學家使用Amazon SageMaker 影像分類演算法訓練了一個電腦視覺模型。每種產品的圖像都根據特定的產品線進行分類。在對新產品進行分類時，模型的準確性太低。所有產品圖像的尺寸都相同，並儲存在Amazon S3 儲存桶中。該公司希望改進該模型，以便盡快將其用於新產品。哪些步驟可以提高解決方案的準確性？（選三個。）

A. Use the SageMaker semantic segmentation algorithm to train a new model to achieve improved accuracy.

A. 使用SageMaker 語意分割演算法訓練新模型，以提高準確率。

B. Use the Amazon Rekognition DetectLabels API to classify the products in the dataset.

B. 使用Amazon Rekognition DetectLabels API 對資料集中的產品進行分類。

C. Augment the images in the dataset. Use open source libraries to crop, resize, flip, rotate, and adjust the brightness and contrast of the images.

C. 增強資料集中的影像。使用開源庫裁剪、調整大小、翻轉、旋轉以及調整影像的亮度和對比度。

D. Use a SageMaker notebook to implement the normalization of pixels and scaling of the images. Store the new dataset in Amazon S3.

D. 使用SageMaker 筆記型電腦實現像素的規範化和影像的縮放。將新資料集儲存在Amazon S3 中。

E. Use Amazon Rekognition Custom Labels to train a new model.

E. 使用Amazon Rekognition 自訂標籤訓練新模型。

F. Check whether there are class imbalances in the product categories, and apply oversampling or undersampling as required. Store the new dataset in Amazon S3.

F. 檢查產品類別中是否存在類別不平衡，並根據需要應用過採樣或欠採樣。將新資料集儲存在Amazon S3 中。

問題# 158

A data scientist is training a text classification model by using the Amazon SageMaker built-in BlazingText algorithm. There are 5 classes in the dataset, with 300 samples for category A, 292 samples for category B, 240 samples for category C, 258 samples for category D, and 310 samples for category E. The data scientist shuffles the data and splits off 10% for testing. After training the model, the data scientist generates confusion matrices for the training and test sets. What could the data scientist conclude from these results?

一位資料科學家正在使用Amazon SageMaker 內建的BlazingText 演算法訓練文字分類模型。資料集中有5 個類別，其中A 類有300 個樣本，B 類有292 個樣本，C 類有240 個樣本，D 類有258 個樣本，E 類有310 個樣本。資料科學家對資料進行打亂，並分離出10% 用於測試。在訓練模型後，資料科學家為訓練集和測試集產生混淆矩陣。數據科學家可以從這些結果中得到什麼結論？

Training data confusion matrix

		Predicted class					Total
		A	B	C	D	E	
True class	A	270	0	0	0	0	270
	B	1	260	0	0	2	263
	C	0	0	111	100	5	216
	D	4	3	132	92	1	232
	E	0	0	2	3	274	279
	Total	275	263	245	195	282	1260

Test data confusion matrix

		Predicted class					Total
		A	B	C	D	E	
True class	A	9	1	0	0	0	10
	B	2	25	0	2	0	29
	C	10	2	11	10	1	34
	D	1	0	12	14	0	27
	E	9	1	4	1	25	40
	Total	31	29	27	27	26	140

A. Classes C and D are too similar.

A. C 類和D 類太相似。

B. The dataset is too small for holdout cross-validation.

B. 資料集太小，無法進行保留交叉驗證。

C. The data distribution is skewed.

C. 數據分佈不均。

D. The model is overfitting for classes B and E.

D. 此模型過度擬合B 類和E 類。

問題# 159

A company that manufactures mobile devices wants to determine and calibrate the appropriate sales price for its devices. The company is collecting the relevant data and is determining data features that it can use to train machine learning (ML) models. There are more than 1,000 features, and the company wants to determine the primary features that contribute to the sales price. Which techniques should the company use for feature selection? (Choose three.)

一家生產行動裝置的公司希望確定並調整其設備的合適銷售價格。該公司正在收集相關數據，並確定可用於訓練機器學習(ML)模型的數據特徵。有1,000多個特徵，該公司希望確定影響銷售價格的主要特徵。該公司應使用哪些技術進行特徵選擇？（選三個。）

A. Data scaling with standardization and normalization.

A. 透過標準化和標準化進行資料縮放。

B. Correlation plot with heat maps.

B. 有熱圖的相關圖。

C. Data binning.

C. 數據分箱。

D. Univariate selection.

D. 單變量選擇。

E. Feature importance with a tree-based classifier.

E. 使用基於樹的分類器來確定特徵重要性。

F. Data augmentation.

F. 數據增強。

問題# 160

A power company wants to forecast future energy consumption for its customers in residential properties and commercial business properties. Historical power consumption data for the last 10 years is available. A team of data scientists who performed the initial data analysis and feature selection will include the historical power consumption data and data such as weather, number of individuals on the property, and public holidays. The data scientists are using Amazon Forecast to generate the forecasts. Which algorithm in Forecast should the data scientists use to meet these requirements?

一家電力公司希望預測其住宅物業和商業物業客戶的未來能源消耗。過去10年的歷史用電量資料可用。執行初步數據分析和特徵選擇的數據科學家團隊將包括歷史用電量數據以及天氣、物業上個人數量和公共假期等數據。資料科學家正在使用Amazon Forecast來產生預測。資料科學家應該使用Forecast中的哪一種演算法來滿足這些要求？

A. Autoregressive Integrated Moving Average (AIRMA).

A. 自迴歸綜合移動平均線 (AIRMA) 。

B. Exponential Smoothing (ETS).

B. 指數平滑法 (ETS) 。

C. Convolutional Neural Network - Quantile Regression (CNN-QR).

C. 卷積神經網路- 分位數迴歸 (CNN-QR) 。

D. Prophet.

D. 先知。

問題# 161

A company wants to use automatic speech recognition (ASR) to transcribe messages that are less than 60 seconds long from a voicemail-style application. The company requires the correct identification of 200 unique product names, some of which have unique spellings or pronunciations. The company has 4,000 words of Amazon SageMaker Ground Truth voicemail transcripts it can use to customize the chosen ASR model. The company needs to ensure that everyone can update their customizations multiple times each hour. Which approach will maximize transcription accuracy during the development phase?

一家公司希望使用自動語音辨識(ASR)轉錄語音郵件式應用程式中少於60秒的訊息。該公司需要正確識別200個獨特的產品名稱，其中一些具有獨特的拼寫或發音。該公司擁有4,000個單字的Amazon SageMaker Ground Truth語音郵件記錄，可用於自訂所選的ASR模型。該公司需要確保每個人每小時都可以多次更新他們的自訂內容。哪一種方法可以在開發階段最大限度地提高轉錄準確性？

A. Use a voice-driven Amazon Lex bot to perform the ASR customization. Create customer slots within the bot that specifically identify each of the required product names. Use the Amazon Lex synonym mechanism to provide additional variations of each product name as mis-transcriptions are identified in development.

A. 使用語音驅動的Amazon Lex機器人執行ASR自訂。在機器人內創建客戶槽，專門標識每個所需的產品名稱。使用Amazon Lex同義詞機制提供每個產品名稱的其他變體，因為在開發過程中會發現錯誤轉錄。

B. Use Amazon Transcribe to perform the ASR customization. Analyze the word confidence scores in the transcript, and automatically create or update a custom vocabulary file with any word that has a confidence score below an acceptable threshold value. Use this updated custom vocabulary file in all future transcription tasks.

B. 使用Amazon Transcribe執行ASR自訂。分析轉錄中的單字置信度分數，並使用任何置信度分數低於可接受閾值的單字自動建立或更新自訂詞彙表檔案。在所有未來的轉錄任務中使用此更新的自訂詞彙表檔案。

C. Create a custom vocabulary file containing each product name with phonetic pronunciations, and use it with Amazon Transcribe to perform the ASR customization. Analyze the transcripts and manually update the custom vocabulary file to include updated or additional entries for those names that are not being correctly identified.

C. 建立一個包含每個產品名稱及其語音發音的自訂詞彙表文件，並將其與Amazon Transcribe一起使用以執行ASR自訂。分析轉錄並手動更新自訂詞彙表文件，以包含那些未正確識別的名稱的更新或附加條目。

D. Use the audio transcripts to create a training dataset and build an Amazon Transcribe custom language model. Analyze the transcripts and update the training dataset with a manually corrected version of transcripts where product names are not being transcribed correctly. Create an updated custom language model.

D. 使用音訊轉錄建立訓練資料集並建立Amazon Transcribe自訂語言模型。分析轉錄並使用手動更正的轉錄版本更新訓練資料集，其中產品名稱未正確轉錄。建立更新的自訂語言模型。

問題# 162

A company is building a demand forecasting model based on machine learning (ML). In the development stage, an ML specialist uses an Amazon SageMaker notebook to perform feature engineering during work hours that consumes low amounts of CPU and memory resources. A data engineer uses the same notebook to perform data preprocessing once a day on average that requires very high memory and completes in only 2 hours. The data preprocessing is not configured to use GPU. All the processes are running well on an ml.m5.4xlarge notebook instance. The company receives an AWS Budgets alert that the billing for this month exceeds the allocated budget. Which solution will result in the MOST cost savings?

一家公司正在建立基於機器學習(ML)的需求預測模型。在開發階段，ML專家使用Amazon SageMaker 筆電在工作時間執行特徵工程，消耗少量CPU和記憶體資源。資料工程師使用同一筆記本平均每天執行一次資料預處理，需要非常高的內存，並且只需2小時即可完成。資料預處理未配置為使用GPU。所有流程都在ml.m5.4xlarge 筆記本實例上運作良好。該公司收到AWS 預算警報，提示本月的帳單超出分配的預算。哪種解決方案將帶來最大的成本節省？

A. Change the notebook instance type to a memory optimized instance with the same vCPU number as the ml.m5.4xlarge instance has. Stop the notebook when it is not in use. Run both data preprocessing and feature engineering development on that instance.

A. 將筆記本實例類型變更為記憶體最佳化實例，其vCPU數量與ml.m5.4xlarge 實例相同。在不使用時停止筆記本。在該實例上運行資料預處理和功能工程開發。

B. Keep the notebook instance type and size the same. Stop the notebook when it is not in use. Run data preprocessing on a P3 instance type with the same memory as the ml.m5.4xlarge instance by using Amazon SageMaker Processing.

B. 保持筆記本實例類型和大小相同。在不使用時停止筆記本。使用Amazon SageMaker Processing 在具有與ml.m5.4xlarge 實例相同記憶體的P3 實例類型上執行資料預處理。

C. Change the notebook instance type to a smaller general purpose instance. Stop the notebook when it is not in use. Run data preprocessing on an ml.r5 instance with the same memory size as the ml.m5.4xlarge instance by using Amazon SageMaker Processing.

C. 將筆記本實例類型變更為較小的通用實例。在不使用時停止筆記本。使用Amazon SageMaker Processing 在具有與ml.m5.4xlarge 實例相同記憶體大小的ml.r5 實例上執行資料預處理。

D. Change the notebook instance type to a smaller general purpose instance. Stop the notebook when it is not in use. Run data preprocessing on an R5 instance with the same memory size as the ml.m5.4xlarge instance by using the Reserved Instance option.

D. 將筆記本實例類型變更為較小的通用實例。在不使用時停止筆記本。使用預留實例選項在具有與ml.m5.4xlarge 實例相同記憶體大小的R5 實例上執行資料預處理。

問題# 163

A machine learning specialist is developing a regression model to predict rental rates from rental listings. A variable named Wall_Color represents the most prominent exterior wall color of the property. The following is the sample data, excluding all other variables: The specialist chose a model that needs numerical input data. Which feature engineering approaches should the specialist use to allow the regression model to learn from the Wall_Color data? (Choose two.)

一位機器學習專家正在開發一個迴歸模型，以根據租賃清單預測租金。名為Wall_Color的變數表示

房產最突出的外牆顏色。以下是樣本數據，不包括所有其他變數：專家選擇了一個需要數字輸入數據的模型。專家應該使用哪些特徵工程方法讓迴歸模型從Wall_Color 資料中學習？（選擇兩個。）

Property_ID	Wall_Color
1000	Red
1001	White
1002	Green

A. Apply integer transformation and set Red = 1, White = 5, and Green = 10.

A. 套用整數轉換並設定紅色= 1、白色= 5 和綠色= 10。

B. Add new columns that store one-hot representation of colors.

B. 新增用於儲存顏色獨熱表示的新欄位。

C. Replace the color name string by its length.

C. 用顏色名稱字串的長度替換顏色名稱字串。

D. Create three columns to encode the color in RGB format.

D. 建立三列以RGB 格式對顏色進行編碼。

E. Replace each color name by its training set frequency.

E. 用每個顏色名稱的訓練集頻率取代每個顏色名稱。

問題# 164

A data scientist is working on a public sector project for an urban traffic system. While studying the traffic patterns, it is clear to the data scientist that the traffic behavior at each light is correlated, subject to a small stochastic error term. The data scientist must model the traffic behavior to analyze the traffic patterns and reduce congestion. How will the data scientist MOST effectively model the problem?

一位資料科學家正在為城市交通系統進行公共部門計畫。在研究交通模式時，資料科學家清楚知道，每個交通燈處的交通行為都是相關的，但會受到一個小的隨機誤差項的影響。資料科學家必須對交通行為進行建模，以分析交通模式並減少擁塞。資料科學家將如何最有效地對問題進行建模？

A. The data scientist should obtain a correlated equilibrium policy by formulating this problem as a multi-agent reinforcement learning problem.

A. 資料科學家應該將此問題表述為多智能體強化學習問題來獲得相關平衡策略。

B. The data scientist should obtain the optimal equilibrium policy by formulating this problem as a single-agent reinforcement learning problem.

B. 資料科學家應該透過將此問題表述為單智能體強化學習問題來獲得最佳平衡策略。

C. Rather than finding an equilibrium policy, the data scientist should obtain accurate predictors of traffic flow by using historical data through a supervised learning approach.

C. 資料科學家不應該尋找平衡策略，而應該透過監督學習方法使用歷史資料來獲得準確的交通流預測指標。

D. Rather than finding an equilibrium policy, the data scientist should obtain accurate predictors of traffic flow by using unlabeled simulated data representing the new traffic patterns in the city and applying an unsupervised learning approach.

D. 資料科學家不應該尋找平衡策略，而應該透過使用代表城市新交通模式的未標記模擬資料並應用無監督學習方法來獲得準確的交通流預測指標。

問題# 165

A data scientist is using the Amazon SageMaker Neural Topic Model (NTM) algorithm to build a model that recommends tags from blog posts. The raw blog post data is stored in an Amazon S3 bucket in JSON format. During model evaluation, the data scientist discovered that the model recommends certain stopwords such as "a," "an," and "the" as tags to certain blog posts, along with a few rare words that are present only in certain blog entries. After a few iterations of tag review with the content team, the data scientist notices that the rare words are unusual but feasible. The data scientist also must ensure that the tag recommendations of the generated model do not include the stopwords. What should the data scientist do to meet these requirements?

一位資料科學家正在使用Amazon SageMaker 神經主題模型(NTM) 演算法建立一個模型，用於推薦部落格文章中的標籤。原始部落格文章資料以JSON 格式儲存在Amazon S3 儲存桶中。在模型評估期間，資料科學家發現該模型推薦某些停用詞（例如「a」、「an」和「the」）作為某些部落格文章的標籤，以及一些僅出現在某些部落格文章中的罕見詞。在與內容團隊進行幾次標籤審查後，資料科學家注意到這些罕見字詞雖然不常見，但確實可行。資料科學家還必須確保產生的模型的標籤建議不包含停用詞。資料科學家該怎麼做才能滿足這些要求？

A. Use the Amazon Comprehend entity recognition API operations. Remove the detected words from the blog post data. Replace the blog post data source in the S3 bucket.

A. 使用Amazon Comprehend 實體辨識API 作業。從部落格文章資料中刪除偵測到的單字。替換S3 儲存桶中的部落格文章資料來源。

B. Run the SageMaker built-in principal component analysis (PCA) algorithm with the blog post data from the S3 bucket as the data source. Replace the blog post data in the S3 bucket with the results of the training job.

B. 使用S3 儲存桶中的部落格文章資料作為資料來源，執行SageMaker 內建主成分分析(PCA) 演算法。以訓練作業的結果取代S3 儲存桶中的部落格文章資料。

C. Use the SageMaker built-in Object Detection algorithm instead of the NTM algorithm for the training job to process the blog post data.

C. 使用SageMaker 內建物件偵測演算法（而不是訓練作業的NTM 演算法）來處理部落格文章資料。

D. Remove the stopwords from the blog post data by using the CountVectorizer function in the scikit-learn library. Replace the blog post data in the S3 bucket with the results of the vectorizer.

D. 使用scikit-learn 函式庫中的CountVectorizer 函數從部落格文章資料中刪除停用詞。用向量化器的結果取代S3 儲存桶中的部落格文章資料。

問題# 166

A company wants to create a data repository in the AWS Cloud for machine learning (ML) projects. The company wants to use AWS to perform complete ML lifecycles and wants to use Amazon S3 for the data storage. All of the company's data currently resides on premises and is 40 GB in size. The company wants a solution that can transfer and automatically update data between the on-premises object storage and Amazon S3. The solution must support encryption, scheduling, monitoring, and data integrity validation. Which solution meets these requirements?

一家公司希望在AWS 雲端為機器學習(ML) 專案建立一個資料儲存庫。該公司希望使用AWS 執行完整的ML 生命週期，並希望使用Amazon S3 進行資料儲存。該公司的所有資料目前都駐留在本地，大小為40 GB之本地物件儲存和Amazon S3。該公司需要一個解決方案，可以在本地物件儲存和Amazon S3之間傳輸和自動更新資料。該解決方案必須支援加密、調度、監控和資料完整性驗證。哪種解決方案符合這些要求？

A. Use the S3 sync command to compare the source S3 bucket and the destination S3 bucket. Determine which source files do not exist in the destination S3 bucket and which source files were modified.

A. 使用S3 sync 指令比較來源S3 儲存桶和目標S3 儲存桶。確定目標S3 儲存桶中不存在哪些來源檔案以及哪些來源檔案已被修改。

B. Use AWS Transfer for FTPS to transfer the files from the on-premises storage to Amazon S3.

B. 使用AWS Transfer for FTPS 將檔案從本機儲存傳輸到Amazon S3。

C. Use AWS DataSync to make an initial copy of the entire dataset. Schedule subsequent incremental transfers of changing data until the final cutover from on premises to AWS.

C. 使用AWS DataSync 製作整個資料集的初始副本。安排後續增量傳輸變更數據，直到最終從本地切換到AWS。

D. Use S3 Batch Operations to pull data periodically from the on-premises storage. Enable S3 Versioning on the S3 bucket to protect against accidental overwrites.

D. 使用S3 批次操作定期從本地儲存中提取資料。在S3 儲存桶上啟用S3 版本控制以防止意外覆蓋。

問題# 167

A company has video feeds and images of a subway train station. The company wants to create a deep learning model that will alert the station manager if any passenger crosses the yellow safety line when there is no train in the station. The alert will be based on the video feeds. The company wants the model to detect the yellow line, the passengers who cross the yellow line, and the trains in the video feeds. This task requires labeling. The video data must remain confidential. A data scientist creates a bounding box to label the sample data and uses an object detection model. However, the object detection model cannot clearly demarcate the yellow line, the passengers who cross the yellow line, and the trains. Which labeling approach will help the company improve this model?

一家公司擁有地鐵站的影片和圖像。該公司希望創建一個深度學習模型，當車站中沒有火車時，如果有乘客越過黃色安全線，該模型將向車站經理發出警報。警報將基於視訊來源。該公司希望該模型能夠偵測視訊來源中的黃線、越過黃線的乘客和火車。此任務需要標記。視訊資料必須保密。資料科學家創建了一個邊界框來標記樣本資料並使用物件偵測模型。但是，物件偵測模型無法清楚地劃分黃線、越過黃線的乘客和火車。哪一種標記方法可以幫助公司改進模型？

A. Use Amazon Rekognition Custom Labels to label the dataset and create a custom Amazon Rekognition object detection model. Create a private workforce. Use Amazon Augmented AI (Amazon A2I) to review the low-confidence predictions and retrain the custom Amazon Rekognition model.

A. 使用Amazon Rekognition 自訂標籤標記資料集並建立自訂Amazon Rekognition 物件偵測模型。建立私人員工。使用Amazon Augmented AI (Amazon A2I) 檢視低置信度預測並重新訓練自訂Amazon Rekognition 模型。

B. Use an Amazon SageMaker Ground Truth object detection labeling task. Use Amazon Mechanical Turk as the labeling workforce.

B. 使用Amazon SageMaker Ground Truth 物件偵測標記任務。使用Amazon Mechanical Turk 作為標記員工。

C. Use Amazon Rekognition Custom Labels to label the dataset and create a custom Amazon Rekognition object detection model. Create a workforce with a third-party AWS Marketplace vendor. Use Amazon Augmented AI (Amazon A2I) to review the low-confidence predictions and retrain the custom Amazon Rekognition model.

C. 使用Amazon Rekognition 自訂標籤標記資料集並建立自訂Amazon Rekognition 物件偵測模型。與第三方AWS Marketplace 供應商一起創建員工。使用Amazon Augmented AI (Amazon A2I) 檢視低置信度預測並重新訓練自訂Amazon Rekognition 模型。

D. Use an Amazon SageMaker Ground Truth semantic segmentation labeling task. Use a private workforce as the labeling workforce.

D. 使用Amazon SageMaker Ground Truth 語意分割標記任務。使用私人員工作為標記員工。

問題# 168

A data engineer at a bank is evaluating a new tabular dataset that includes customer data. The data engineer will use the customer data to create a new model to predict customer behavior. After creating a correlation matrix for the variables, the data engineer notices that many of the 100 features are highly correlated with each other. Which steps should the data engineer take to address this issue? (Choose two.)

銀行的一名資料工程師正在評估包含客戶資料的新表格資料集。資料工程師將使用客戶資料建立一

個新模型來預測客戶行為。在為變數建立相關矩陣後，資料工程師注意到100 個特徵中的許多特徵彼此高度相關。資料工程師應採取哪些步驟來解決這個問題？（選兩個。）

A. Use a linear-based algorithm to train the model.

A. 使用基於線性的演算法來訓練模型。

B. Apply principal component analysis (PCA).

B. 應用主成分分析（PCA）。

C. Remove a portion of highly correlated features from the dataset.

C. 從資料集中刪除一部分高度相關的特徵。

D. Apply min-max feature scaling to the dataset.

D. 對資料集應用最小-最大特徵縮放。

E. Apply one-hot encoding category-based variables.

E. 應用獨熱編碼基於類別的變數。

問題# 169

A company is building a new version of a recommendation engine. Machine learning (ML) specialists need to keep adding new data from users to improve personalized recommendations. The ML specialists gather data from the users' interactions on the platform and from sources such as external websites and social media. The pipeline cleans, transforms, enriches, and compresses terabytes of data daily, and this data is stored in Amazon S3. A set of Python scripts was coded to do the job and is stored in a large Amazon EC2 instance. The whole process takes more than 20 hours to finish, with each script taking at least an hour. The company wants to move the scripts out of Amazon EC2 into a more managed solution that will eliminate the need to maintain servers. Which approach will address all of these requirements with the LEAST development effort?

一家公司正在建立新版本的推薦引擎。機器學習(ML) 專家需要不斷添加來自使用者的新資料以改善個人化推薦。ML 專家從使用者在平台上的互動以及外部網站和社群媒體等來源收集資料。管道每天清理、轉換、豐富和壓縮數TB 的數據，並將這些數據儲存在Amazon S3 中。一組Python 腳本被編碼來完成這項工作，並儲存在一個大型Amazon EC2 實例中。整個過程需要20 多個小時才能完成，每個腳本至少需要一個小時。該公司希望將腳本從Amazon EC2 移出到更託管的解決方案中，因此無需維護伺服器。哪種方法可以以最少的開發工作量來滿足所有這些要求？

A. Load the data into an Amazon Redshift cluster. Execute the pipeline by using SQL. Store the results in Amazon S3.

A. 將資料載入到Amazon Redshift 叢集中。使用SQL 執行管道。將結果儲存在Amazon S3 中。

B. Load the data into Amazon DynamoDB. Convert the scripts to an AWS Lambda function. Execute the pipeline by triggering Lambda executions. Store the results in Amazon S3.

B. 將資料載入到Amazon DynamoDB 中。將腳本轉換為AWS Lambda 函數。透過觸發Lambda 執行來執行管道。將結果儲存在Amazon S3 中。

C. Create an AWS Glue job. Convert the scripts to PySpark. Execute the pipeline. Store the results in Amazon S3.

C. 建立AWS Glue 作業。將腳本轉換為PySpark。執行管道。將結果儲存在Amazon S3 中。

D. Create a set of individual AWS Lambda functions to execute each of the scripts. Build a step function by using the AWS Step Functions Data Science SDK. Store the results in Amazon S3.

D. 建立一組單獨的AWS Lambda 函數來執行每個腳本。使用AWS Step Functions Data Science SDK 建立步驟函數。將結果儲存在Amazon S3 中。

問題# 170

A retail company is selling products through a global online marketplace. The company wants to use machine learning (ML) to analyze customer feedback and identify specific areas for improvement. A developer has built a tool that collects customer reviews from the online marketplace and stores them in an Amazon S3 bucket. This process yields a dataset of 40 reviews. A data scientist building the ML models must identify additional sources of data to increase the size of the dataset. Which data sources should the data scientist use to augment the dataset of reviews? (Choose three.)

一家零售公司正在透過全球線上市場銷售產品。該公司希望使用機器學習(ML) 來分析客戶回饋並確定需要改進的特定領域。開發人員建立了一個工具，用於從線上市場收集客戶評論並將其儲存在Amazon S3 儲存桶中。此過程產生了一個包含40 則評論的資料集。建構ML 模型的資料科學家必須確定其他資料來源以增加資料集的大小。資料科學家應該使用哪些資料來源來擴充評論資料集？（選三個。）

A. Emails exchanged by customers and the company's customer service agents.

A. 客戶與公司客服人員往來的電子郵件。

B. Social media posts containing the name of the company or its products.

B. 包含公司或產品名稱的社群媒體貼文。

C. A publicly available collection of news articles.

C. 公開的新聞文章集合。

D. A publicly available collection of customer reviews.

D. 公開的顧客評論集合。

E. Product sales revenue figures for the company.

E. 公司的產品銷售收入資料。

F. Instruction manuals for the company's products.

F. 公司產品的使用說明書。

問題# 171

A machine learning (ML) specialist wants to create a data preparation job that uses a PySpark script with complex window aggregation operations to create data for training and testing. The ML specialist needs to evaluate the impact of the number of features and the sample count on model performance. Which approach should the ML specialist use to determine the ideal data transformations for the model?

一位機器學習(ML)專家想要建立一個資料準備作業，該作業使用帶有複雜視窗聚合操作的PySpark腳本來建立用於訓練和測試的資料。ML專家需要評估特徵數量和樣本數量對模型效能的影響。ML專家應該使用哪種方法來確定模型的理想資料轉換？

A. Add an Amazon SageMaker Debugger hook to the script to capture key metrics. Run the script as an AWS Glue job.

A. 在腳本中加入Amazon SageMaker Debugger 掛鉤以捕捉關鍵指標。將腳本作為AWS Glue 作業運行。

B. Add an Amazon SageMaker Experiments tracker to the script to capture key metrics. Run the script as an AWS Glue job.

B. 為腳本新增Amazon SageMaker Experiments 追蹤器以擷取關鍵指標。將腳本作為AWS Glue 作業運行。

C. Add an Amazon SageMaker Debugger hook to the script to capture key parameters. Run the script as a SageMaker processing job.

C. 在腳本中新增Amazon SageMaker Debugger 掛鉤以擷取關鍵參數。將腳本作為SageMaker 處理作業運行。

D. Add an Amazon SageMaker Experiments tracker to the script to capture key parameters. Run the script as a SageMaker processing job.

D. 在腳本中新增Amazon SageMaker Experiments 追蹤器以擷取關鍵參數。將腳本作為SageMaker 處理作業運行。

問題# 172

A data scientist has a dataset of machine part images stored in Amazon Elastic File System (Amazon EFS). The data scientist needs to use Amazon SageMaker to create and train an image classification machine learning model based on this dataset. Because of budget and time constraints, management wants the data scientist to create and train a model with the least number of steps and integration work required. How should the data scientist meet these requirements?

一位資料科學家在Amazon Elastic File System (Amazon EFS) 中儲存了一個機器零件影像資料集。資料科學家需要使用Amazon SageMaker 根據此資料集建立和訓練一個影像分類機器學習模型。由於預算和時間限制，管理層希望資料科學家以最少的步驟和整合工作來創建和訓練一個模型。資料科學家應該如何滿足這些要求？

A. Mount the EFS file system to a SageMaker notebook and run a script that copies the data to an Amazon FSx for Lustre file system. Run the SageMaker training job with the FSx for Lustre file system as the data source.

A. 將EFS 檔案系統掛載到SageMaker 筆記本並執行將資料複製到Amazon FSx for Lustre 檔案系統的腳本。使用FSx for Lustre 檔案系統作為資料來源執行SageMaker 訓練作業。

B. Launch a transient Amazon EMR cluster. Configure steps to mount the EFS file system and copy the data to an Amazon S3 bucket by using S3DistCp. Run the SageMaker training job with Amazon S3 as the data source.

B. 啟動臨時Amazon EMR 叢集。設定使用S3DistCp 掛載EFS 檔案系統並將資料複製到Amazon S3 儲存桶的步驟。使用Amazon S3 作為資料來源執行SageMaker 訓練作業。

C. Mount the EFS file system to an Amazon EC2 instance and use the AWS CLI to copy the data to an Amazon S3 bucket. Run the SageMaker training job with Amazon S3 as the data source.

C. 將EFS 檔案系統掛載到Amazon EC2 執行個體並使用AWS CLI 將資料複製到Amazon S3 儲存桶。使用Amazon S3 作為資料來源執行SageMaker 訓練作業。

D. Run a SageMaker training job with an EFS file system as the data source.

D. 使用EFS 檔案系統作為資料來源來執行SageMaker 訓練作業。

問題# 173

A retail company uses a machine learning (ML) model for daily sales forecasting. The company's brand manager reports that the model has provided inaccurate results for the past 3 weeks. At the end of each day, an AWS Glue job consolidates the input data that is used for the forecasting with the actual daily sales data and the predictions of the model. The AWS Glue job stores the data in Amazon S3. The company's ML team is using an Amazon SageMaker Studio notebook to gain an understanding about the source of the model's inaccuracies. What should the ML team do on the SageMaker Studio notebook to visualize the model's degradation MOST accurately?

一家零售公司使用機器學習(ML) 模型進行每日銷售預測。該公司的品牌經理報告稱，該模型在過去3 週內提供的結果不準確。每天結束時，AWS Glue 作業都會將用於預測的輸入資料與實際每日銷售資料和模型的預測進行整合。AWS Glue 作業將資料儲存在Amazon S3 中。該公司的ML 團隊正在使用Amazon SageMaker Studio 筆記本來了解模型不準確的根源。ML 團隊應該在SageMaker Studio 筆記本上做什麼才能最準確地視覺化模型的退化？

A. Create a histogram of the daily sales over the last 3 weeks. In addition, create a histogram of the daily sales from before that period.

A. 建立過去3 週的每日銷售額直方圖。此外，建立該期間之前的每日銷售額直方圖。

B. Create a histogram of the model errors over the last 3 weeks. In addition, create a histogram of the model errors from before that period.

B. 建立過去3 週的模型誤差直方圖。此外，建立該期間先前的模型誤差直方圖。

C. Create a line chart with the weekly mean absolute error (MAE) of the model.

C. 建立模型每週平均絕對誤差(MAE) 的折線圖。

D. Create a scatter plot of daily sales versus model error for the last 3 weeks. In addition, create a scatter plot of daily sales versus model error from before that period.

D. 建立過去3 週的每日銷售額與模型誤差的散佈圖。此外，建立該期間之前的每日銷售額與模型誤差的散佈圖。

問題# 174

An ecommerce company sends a weekly email newsletter to all of its customers. Management has hired a team of writers to create additional targeted content. A data scientist needs to identify five customer segments based on age, income, and location. The customers' current segmentation is unknown. The data scientist previously built an XGBoost model to predict the likelihood of a customer responding to an email based on age, income, and location. Why does the XGBoost model NOT meet the current requirements, and how can this be fixed?

一家電子商務公司每周向其所有客戶發送一封電子郵件簡報。管理層已聘請了一支寫作團隊來創建額外的目標內容。資料科學家需要根據年齡、收入和位置來確定五個客戶細分群體。客戶目前的細分情況未知。資料科學家先前建立了一個XGBoost 模型來預測客戶根據年齡、收入和位置回覆電子郵件的可能性。為什麼XGBoost 模型不能滿足目前要求？如何解決這個問題？

A. The XGBoost model provides a true/false binary output. Apply principal component analysis (PCA) with five feature dimensions to predict a segment.

A. XGBoost 模型提供真/假二進位輸出。應用具有五個特徵維度的主成分分析(PCA) 來預測一個片段。

B. The XGBoost model provides a true/false binary output. Increase the number of classes the XGBoost model predicts to five classes to predict a segment.

B. XGBoost 模型提供真/假二進位輸出。將XGBoost 模型預測的類別數增加到五個類別來預測一個片段。

C. The XGBoost model is a supervised machine learning algorithm. Train a k-Nearest-Neighbors (kNN) model with $K = 5$ on the same dataset to predict a segment.

C. XGBoost 模型是一種監督機器學習演算法。在同一資料集上訓練 $K = 5$ 的k-最近鄰(kNN) 模型來預測一個片段。

D. The XGBoost model is a supervised machine learning algorithm. Train a k-means model with $K = 5$ on the same dataset to predict a segment.

D. XGBoost 模型是一種監督機器學習演算法。在同一資料集上訓練 $K = 5$ 的k-均值模型來預測一個片段。

問題# 175

A global financial company is using machine learning to automate its loan approval process. The company has a dataset of customer information. The dataset contains some categorical fields, such as customer location by city and housing status. The dataset also includes financial fields in different units, such as account balances in US dollars and monthly interest in US cents. The company's data scientists are using a gradient boosting regression model to infer the credit score for each customer. The model has a training accuracy of 99% and a testing accuracy of 75%. The data scientists want to improve the model's testing accuracy. Which process will improve the testing accuracy the MOST?

一家全球金融公司正在使用機器學習來自動化其貸款審批流程。該公司擁有一個客戶資訊資料集。該資料集包含一些分類字段，例如按城市劃分的客戶位置和住房狀況。該資料集還包括不同單位的財務字段，例如以美元計的帳戶餘額和以美分計的每月利息。該公司的資料科學家正在使用梯度提升迴歸模型來推斷每個客戶的信用評分。此模型的訓練準確率為99%，測試準確率為75%。數據科學家希望提高模型的測試準確度。哪個過程將最大程度地提高測試準確率？

A. Use a one-hot encoder for the categorical fields in the dataset. Perform standardization on the financial fields in the dataset. Apply L1 regularization to the data.

A. 對資料集中的分類欄位使用獨熱編碼器。對資料集中的金融欄位執行標準化。對數據應用L1 正規化。

B. Use tokenization of the categorical fields in the dataset. Perform binning on the financial fields in the dataset. Remove the outliers in the data by using the z- score.

B. 對資料集中的分類欄位使用標記化。對資料集中的金融欄位執行分箱。使用z 分數刪除資料中的異常值。

C. Use a label encoder for the categorical fields in the dataset. Perform L1 regularization on the financial fields in the dataset. Apply L2 regularization to the data.

C. 對資料集中的分類欄位使用標籤編碼器。對資料集中的金融欄位執行L1 正規化。對資料應用L2 正規化。

D. Use a logarithm transformation on the categorical fields in the dataset. Perform binning on the financial fields in the dataset. Use imputation to populate missing values in the dataset.

D. 對資料集中的分類欄位使用對數轉換。對資料集中的金融欄位執行分箱。使用插補來填入資料集中的缺失值。

問題# 176

A machine learning (ML) specialist needs to extract embedding vectors from a text series. The goal is to provide a ready-to-ingest feature space for a data scientist to develop downstream ML predictive models. The text consists of curated sentences in English. Many sentences use similar words but in different contexts. There are questions and answers among the sentences, and the embedding space must differentiate between them. Which options can produce the required embedding vectors that capture word context and sequential QA information? (Choose two.)

機器學習(ML) 專家需要從一系列文本中提取嵌入向量。目標是為資料科學家提供隨時可用的特徵空間，以開發下游ML 預測模型。文字由精選的英文句子組成。許多句子使用相似的單詞，但上下文不同。句子中存在問題和答案，嵌入空間必須區分它們。哪些選項可以產生所需的嵌入向量，以捕

獲單字上下文和順序QA 資訊？（選兩個。）

A. Amazon SageMaker seq2seq algorithm.

A. Amazon SageMaker seq2seq 演算法

B. Amazon SageMaker BlazingText algorithm in Skip-gram mode.

B. Skip-gram 模式下的Amazon SageMaker BlazingText 演算法。

C. Amazon SageMaker Object2Vec algorithm.

C. Amazon SageMaker Object2Vec 演算法

D. Amazon SageMaker BlazingText algorithm in continuous bag-of-words (CBOW) mode.

D. 連續詞袋(CBOW) 模式下的Amazon SageMaker BlazingText 演算法。

E. Combination of the Amazon SageMaker BlazingText algorithm in Batch Skip-gram mode with a custom recurrent neural network (RNN).

E. Batch Skip-gram 模式下的Amazon SageMaker BlazingText 演算法與自訂循環神經網路(RNN) 的組合。

問題# 177

A retail company wants to update its customer support system. The company wants to implement automatic routing of customer claims to different queues to prioritize the claims by category. Currently, an operator manually performs the category assignment and routing. After the operator classifies and routes the claim, the company stores the claim's record in a central database. The claim's record includes the claim's category. The company has no data science team or experience in the field of machine learning (ML). The company's small development team needs a solution that requires no ML expertise. Which solution meets these requirements?

一家零售公司希望更新其客戶支援系統。該公司希望實現將客戶索賠自動路由到不同隊列的功能，以便按類別對索賠進行優先排序。目前，操作員手動執行類別分配和路由。操作員對索賠進行分類和路由後，公司將索賠記錄儲存在中央資料庫中。索賠記錄包括索賠類別。該公司沒有數據科學團隊或機器學習(ML) 領域的經驗。該公司的小型開發團隊需要一個不需要ML 專業知識的解決方案。哪種解決方案可以滿足這些要求？

A. Export the database to a.csv file with two columns: claim_label and claim_text. Use the Amazon SageMaker Object2Vec algorithm and the.csv file to train a model. Use SageMaker to deploy the model to an inference endpoint. Develop a service in the application to use the inference endpoint to process incoming claims, predict the labels, and route the claims to the appropriate queue.

A. 將資料庫匯出到包含兩列的.csv 檔案：claim_label 和claim_text。使用Amazon SageMaker Object2Vec 演算法和.csv 檔案來訓練模型。使用SageMaker 將模型部署到推理終端節點。在應用程式中開發一項服務，以使用推理終端節點處理傳入的聲明、預測標籤並將聲明路由到適當的佇列。

B. Export the database to a.csv file with one column: claim_text. Use the Amazon SageMaker Latent Dirichlet Allocation (LDA) algorithm and the.csv file to train a model. Use the LDA algorithm to detect labels automatically. Use SageMaker to deploy the model to an inference endpoint. Develop a service in the application to use the inference endpoint to process incoming claims, predict the labels, and route the claims to the appropriate queue.

B. 將資料庫匯出到包含一列的.csv 檔案：claim_text。使用Amazon SageMaker Latent Dirichlet Allocation (LDA) 演算法和.csv 檔案來訓練模型。使用LDA 演算法自動偵測標籤。使用SageMaker 將模型部署到推理終端節點。在應用程式中開發一項服務，以使用推理終端節點處理傳入的聲明、預測標籤並將聲明路由到適當的佇列。

C. Use Amazon Textract to process the database and automatically detect two columns: claim_label and claim_text. Use Amazon Comprehend custom classification and the extracted information to train the custom classifier. Develop a service in the application to use the Amazon Comprehend API to process incoming claims, predict the labels, and route the claims to the appropriate queue.

C. 使用Amazon Textract 處理資料庫並自動偵測兩個欄位：claim_label 和claim_text。使用Amazon Comprehend 自訂分類和提取的資訊來訓練自訂分類器。在應用程式中開發一項服務，以使用Amazon Comprehend API 處理傳入的索賠、預測標籤並將索賠路由到適當的佇列。

D. Export the database to a.csv file with two columns: claim_label and claim_text. Use Amazon Comprehend custom classification and the.csv file to train the custom classifier. Develop a service in the application to use the Amazon Comprehend API to process incoming claims, predict the labels, and route the claims to the appropriate queue.

D. 將資料庫匯出到包含兩列的.csv 檔案：claim_label 和claim_text。使用Amazon Comprehend 自訂分類和.csv 檔案來訓練自訂分類器。在應用程式中開發一項服務，以使用Amazon Comprehend API 處理傳入的索賠、預測標籤並將索賠路由到適當的佇列。

問題# 178

A machine learning (ML) specialist is using Amazon SageMaker hyperparameter optimization (HPO) to improve a model's accuracy. The learning rate parameter is specified in the following HPO configuration: During the results analysis, the ML specialist determines that most of the training jobs had a learning rate between 0.01 and 0.1. The best result had a learning rate of less than 0.01. Training jobs need to run regularly over a changing dataset. The ML specialist needs to find a tuning mechanism that uses different learning rates more evenly from the provided range between MinValue and MaxValue. Which solution provides the MOST accurate result?

機器學習(ML) 專家正在使用Amazon SageMaker 超參數最佳化(HPO) 來提高模型的準確性。學習率參數在以下HPO 配置中指定：在結果分析期間，ML 專家確定大多數訓練作業的學習率在0.01 到0.1 之間。最佳結果的學習率小於0.01。訓練作業需要在不斷變化的資料集上定期運行。ML 專家需要找到一種調整機制，從提供的MinValue 和MaxValue 之間的範圍內更均勻地使用不同的學習率。哪一種解決方案提供最準確的結果？

```
{  
    "Name": "learning_rate",  
    "MaxValue" : "0.0001",  
    "MinValue": "0.1"  
}
```

A. Modify the HPO configuration as follows: Select the most accurate hyperparameter configuration form this HPO job.

A. 依下列方式修改HPO 設定：從此HPO 作業中選擇最精確的超參數配置。

```
{
  "Name": "learning_rate",
  "MaxValue" : "0.0001",
  "MinValue": "0.1"
  "ScalingType": "ReverseLogarithmic"
}
```

B. Run three different HPO jobs that use different learning rates form the following intervals for MinValue and MaxValue while using the same number of training jobs for each HPO job: [0.01, 0.1] [0.001, 0.01] [0.0001, 0.001] Select the most accurate hyperparameter configuration form these three HPO jobs.

B. 執行三個不同的HPO 作業，這些作業對MinValue 和MaxValue 使用以下間隔中的不同學習率，同時對每個HPO 作業使用相同數量的訓練作業： [0.01, 0.1] [0.001, 0.01] [0.0001, 0.001]從這三個HPO作業中選擇最準確的超參數配置。

```
{
  "Name": "learning_rate",
  "MaxValue" : "0.0001",
  "MinValue": "0.1"
  "ScalingType": "ReverseLogarithmic"
}
```

C. Modify the HPO configuration as follows: Select the most accurate hyperparameter configuration form this training job.

C. 依下列方式修改HPO 配置：從此訓練作業中選擇最精確的超參數配置。

```
{
  "Name": "learning_rate",
  "MaxValue" : "0.0001",
  "MinValue": "0.1"
  "ScalingType": "Logarithmic"
}
```

D. Run three different HPO jobs that use different learning rates form the following intervals for MinValue and MaxValue. Divide the number of training jobs for each HPO job by three: [0.01, 0.1] [0.001, 0.01] [0.0001, 0.001] Select the most accurate hyperparameter configuration form these three HPO jobs.

D. 執行三個不同的HPO 作業，這些作業對MinValue 和MaxValue 使用以下間隔中的不同學習率。將每個HPO 作業的訓練作業數量除以三： [0.01, 0.1] [0.001, 0.01] [0.0001, 0.001] 從這三個HPO 作業中選擇最準確的超參數配置。

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A manufacturing company wants to use machine learning (ML) to automate quality control in its facilities. The facilities are in remote locations and have limited internet connectivity. The company has 20 TB of training data that consists of labeled images of defective product parts. The training data is in the corporate on-premises data center. The company will use this data to train a model for real-time defect detection in new parts as the parts move on a conveyor belt in the facilities. The company needs a solution that minimizes costs for compute infrastructure and that maximizes the scalability of resources for training. The solution also must facilitate the company's use of an ML model in the low-connectivity environments. Which solution will meet these requirements?

一家製造公司希望使用機器學習(ML)來自動化其設施的品質控制。這些設施位於偏遠地區，網路連線有限。該公司擁有20 TB的訓練數據，其中包括有缺陷產品零件的標記圖像。訓練資料位於公司內部資料中心。當零件在設施中的傳送帶上移動時，公司將使用這些資料來訓練模型，以即時檢測新零件中的缺陷。該公司需要一種解決方案，以最大限度地降低計算基礎設施的成本，並最大限度地提高訓練資源的可擴展性。此解決方案還必須方便公司在低連接環境中使用ML模型。哪種解決方案可以滿足這些要求？

A. Move the training data to an Amazon S3 bucket. Train and evaluate the model by using Amazon SageMaker. Optimize the model by using SageMaker Neo. Deploy the model on a SageMaker hosting services endpoint.

A. 將訓練資料移至Amazon S3 儲存桶。使用Amazon SageMaker 訓練和評估模型。使用SageMaker Neo 最佳化模型。在SageMaker 託管服務終端節點上部署模型。

B. Train and evaluate the model on premises. Upload the model to an Amazon S3 bucket. Deploy the model on an Amazon SageMaker hosting services endpoint.

B. 在本地訓練和評估模型。將模型上傳到Amazon S3 儲存桶。在Amazon SageMaker 託管服務終端節點上部署模型。

C. Move the training data to an Amazon S3 bucket. Train and evaluate the model by using Amazon SageMaker. Optimize the model by using SageMaker Neo. Set up an edge device in the manufacturing facilities with AWS IoT Greengrass. Deploy the model on the edge device.

C. 將訓練資料移至Amazon S3 儲存桶。使用Amazon SageMaker 訓練和評估模型。使用SageMaker Neo 最佳化模型。使用AWS IoT Greengrass 在製造設施中設置邊緣設備。在邊緣設備上部署模型。

D. Train the model on premises. Upload the model to an Amazon S3 bucket. Set up an edge device in the manufacturing facilities with AWS IoT Greengrass. Deploy the model on the edge device.

D. 在本地訓練模型。將模型上傳到Amazon S3 儲存桶。使用AWS IoT Greengrass 在製造設施中設置邊緣設備。在邊緣設備上部署模型。

A company has an ecommerce website with a product recommendation engine built in TensorFlow. The recommendation engine endpoint is hosted by Amazon SageMaker. Three compute-optimized instances support the expected peak load of the website. Response times on the product recommendation page are increasing at the beginning of each month. Some users are encountering errors. The website receives the majority of its traffic between 8 AM and 6 PM on weekdays in a single time zone. Which of the following options are the MOST effective in solving the issue while keeping costs to a minimum? (Choose two.)

一家公司擁有一個電子商務網站，該網站內建了TensorFlow的產品推薦引擎。推薦引擎終端由Amazon SageMaker託管。三個計算優化實例支援該網站的預期峰值負載。產品推薦頁面的回應時間在每個月初都在增加。有些用戶遇到錯誤。該網站的大部分流量來自工作日上午8點至下午6點之間的單一時區。下列哪個選項最能有效解決問題，同時將成本降至最低？（選兩個。）

A. Configure the endpoint to use Amazon Elastic Inference (EI) accelerators.

A. 配置終端節點以使用Amazon Elastic Inference (EI) 加速器。

B. Create a new endpoint configuration with two production variants.

B. 建立具有兩個生產變體的新終端節點配置。

C. Configure the endpoint to automatically scale with the `InvocationsPerInstance` metric.

C. 配置終端節點以使用`InvocationsPerInstance` 指標自動擴充。

D. Deploy a second instance pool to support a blue/green deployment of models.

D. 部署第二個實例池以支援模型的藍/綠部署。

E. Reconfigure the endpoint to use burstable instances.

E. 重新配置終端節點以使用可突發實例。

問題# 181

A real-estate company is launching a new product that predicts the prices of new houses. The historical data for the properties and prices is stored in.csv format in an Amazon S3 bucket. The data has a header, some categorical fields, and some missing values. The company's data scientists have used Python with a common open-source library to fill the missing values with zeros. The data scientists have dropped all of the categorical fields and have trained a model by using the open-source linear regression algorithm with the default parameters. The accuracy of the predictions with the current model is below 50%. The company wants to improve the model performance and launch the new product as soon as possible. Which solution will meet these requirements with the LEAST operational overhead?

一家房地產公司正在推出一款預測新房價的新產品。房產和價格的歷史資料以.csv 格式儲存在Amazon S3 儲存桶中。資料有一個標題、一些分類欄位和一些缺失值。該公司的資料科學家使用Python 和通用開源函式庫將缺失值填入零。資料科學家刪除了所有分類字段，並使用開源線性回歸演算法和預設參數訓練了一個模型。目前模型的預測準確率低於50%。該公司希望提高模型性能並儘快推出新產品。哪種解決方案能夠以最少的營運開銷滿足這些要求？

A. Create a service-linked role for Amazon Elastic Container Service (Amazon ECS) with access to the S3 bucket. Create an ECS cluster that is based on an AWS Deep Learning Containers image. Write the code to perform the feature engineering. Train a logistic regression model for predicting the price, pointing to the bucket with the dataset. Wait for the training job to complete. Perform the inferences.

A. 為Amazon Elastic Container Service (Amazon ECS) 建立服務相關角色，並允許其存取S3 儲存桶。建立基於AWS Deep Learning Containers 映像的ECS 叢集。編寫程式碼以執行特徵工程。訓練邏輯迴歸模型以預測價格，並指向包含資料集的儲存桶。等待訓練作業完成。執行推理。

B. Create an Amazon SageMaker notebook with a new IAM role that is associated with the notebook. Pull the dataset from the S3 bucket. Explore different combinations of feature engineering transformations, regression algorithms, and hyperparameters. Compare all the results in the notebook, and deploy the most accurate configuration in an endpoint for predictions.

B. 使用與筆記本相關的新IAM 角色建立Amazon SageMaker 筆記本。從S3 儲存桶中提取資料集。探索特徵工程轉換、迴歸演算法和超參數的不同組合。比較筆記本中的所有結果，並在終端節點中部署最準確的配置以進行預測。

C. Create an IAM role with access to Amazon S3, Amazon SageMaker, and AWS Lambda. Create a training job with the SageMaker built-in XGBoost model pointing to the bucket with the dataset. Specify the price as the target feature. Wait for the job to complete. Load the model artifact to a Lambda function for inference on prices of new houses.

C. 建立可以存取Amazon S3、Amazon SageMaker 和AWS Lambda 的IAM 角色。使用SageMaker 內建XGBoost 模型建立訓練作業，並指向包含資料集的儲存桶。將價格指定為目標特徵。等待作業完成。將模型工件載入到Lambda 函數中，以推斷新房價。

D. Create an IAM role for Amazon SageMaker with access to the S3 bucket. Create a SageMaker AutoML job with SageMaker Autopilot pointing to the bucket with the dataset. Specify the price as the target attribute. Wait for the job to complete. Deploy the best model for predictions.

D. 為Amazon SageMaker 建立IAM 角色，並存取S3 儲存桶。使用SageMaker Autopilot 建立一個指向包含資料集的儲存桶的SageMaker AutoML 作業。將價格指定為目標屬性。等待作業完成。部署最佳模型進行預測。

問題# 182

A data scientist is reviewing customer comments about a company's products. The data scientist needs to present an initial exploratory analysis by using charts and a word cloud. The data scientist must use feature engineering techniques to prepare this analysis before starting a natural language processing (NLP) model. Which combination of feature engineering techniques should the data scientist use to meet these requirements? (Choose two.)

一位資料科學家正在查看客戶對某家公司產品的評論。資料科學家需要使用圖表和詞雲來呈現初步探索性分析。在開始自然語言處理(NLP) 模型之前，資料科學家必須使用特徵工程技術來準備此分析。資料科學家應該使用哪種特徵工程技術組合來滿足這些要求？（選兩個。）

A. Named entity recognition.

A. 命名實體辨識

B. Coreference.

B. 共指

C. Stemming.

C. 詞幹擷取

D. Term frequency-inverse document frequency (TF-IDF).

D. 詞頻-逆文檔頻率 (TF-IDF)

E. Sentiment analysis.

E. 情緒分析。

問題# 183

A data scientist is evaluating a GluonTS on Amazon SageMaker DeepAR model. The evaluation metrics on the test set indicate that the coverage score is 0.489 and 0.889 at the 0.5 and 0.9 quantiles, respectively. What can the data scientist reasonably conclude about the distributional forecast related to the test set?

一位資料科學家正在評估Amazon SageMaker DeepAR 模型上的GluonTS。測試集上的評估指標表明，在0.5 和0.9 分位數處，覆蓋率得分分別為0.489 和0.889。資料科學家可以合理地得出與測試集相關的分佈預測的什麼結論？

A. The coverage scores indicate that the distributional forecast is poorly calibrated. These scores should be approximately equal to each other at all quantiles.

A. 覆蓋率分數表示分佈預測校準不佳。這些分數在所有分位數上應大致相等。

B. The coverage scores indicate that the distributional forecast is poorly calibrated. These scores should peak at the median and be lower at the tails.

B. 覆蓋率分數表示分佈預測校準不佳。這些分數應在中位數達到峰值，在尾部較低。

C. The coverage scores indicate that the distributional forecast is correctly calibrated. These scores should always fall below the quantile itself.

C. 覆蓋率分數表示分佈預測校準正確。這些分數應始終低於分位數本身。

D. The coverage scores indicate that the distributional forecast is correctly calibrated. These scores should be approximately equal to the quantile itself.

D. 覆蓋率分數表示分佈預測校準正確。這些分數應大致等於分位數本身。

問題# 184

An energy company has wind turbines, weather stations, and solar panels that generate telemetry data. The company wants to perform predictive maintenance on these devices. The devices are in various locations and have unstable internet connectivity. A team of data scientists is using the telemetry data to perform machine learning (ML) to conduct anomaly detection and predict maintenance before the devices start to deteriorate. The team needs a scalable, secure, high-velocity data ingestion mechanism. The team has decided to use Amazon S3 as the data storage location. Which approach meets these requirements?

一家能源公司擁有風力渦輪機、氣象站和太陽能板，它們會產生遙測數據。該公司希望對這些設備進行預測性維護。這些設備位於不同位置，網路連線不穩定。一組資料科學家正在使用遙測資料進行機器學習(ML)，以在設備開始惡化之前進行異常檢測並預測維護。該團隊需要一種可擴展、安全、高速的資料提取機制。該團隊已決定使用Amazon S3 作為資料儲存位置。哪一種方法可以滿足這些要求？

A. Ingest the data by using an HTTP API call to a web server that is hosted on Amazon EC2. Set up EC2 instances in an Auto Scaling configuration behind an Elastic Load Balancer to load the data into Amazon S3.

A. 使用託管在Amazon EC2 上的Web 伺服器的HTTP API 呼叫來提取資料。在Elastic Load Balancer 後面的Auto Scaling 配置中設定EC2 實例，以將資料載入到Amazon S3 中。

B. Ingest the data over Message Queuing Telemetry Transport (MQTT) to AWS IoT Core. Set up a rule in AWS IoT Core to use Amazon Kinesis Data Firehose to send data to an Amazon Kinesis data stream that is configured to write to an S3 bucket.

B. 透過訊息佇列遙測傳輸(MQTT) 將資料提取到AWS IoT Core。在AWS IoT Core 中設定規則以使用Amazon Kinesis Data Firehose 將資料傳送至配置為寫入S3 儲存桶的Amazon Kinesis 資料流。

C. Ingest the data over Message Queuing Telemetry Transport (MQTT) to AWS IoT Core. Set up a rule in AWS IoT Core to direct all MQTT data to an Amazon Kinesis Data Firehose delivery stream that is configured to write to an S3 bucket.

C. 透過訊息佇列遙測傳輸(MQTT) 將資料提取到AWS IoT Core。在AWS IoT Core 中設定規則以將所有MQTT 資料定向到配置為寫入S3 儲存桶的Amazon Kinesis Data Firehose 傳輸流。

D. Ingest the data over Message Queuing Telemetry Transport (MQTT) to Amazon Kinesis data stream that is configured to write to an S3 bucket.

D. 透過訊息佇列遙測傳輸(MQTT) 將資料提取到配置為寫入S3 儲存桶的Amazon Kinesis 資料流。

問題# 185

A retail company collects customer comments about its products from social media, the company website, and customer call logs. A team of data scientists and engineers wants to find common topics and determine which products the customers are referring to in their comments. The team is using natural language processing (NLP) to build a model to help with this classification. Each product can be classified into multiple categories that the company defines. These categories are related but are not mutually exclusive. For example, if there is mention of "Sample Yogurt" in the document of customer comments, then "Sample Yogurt" should be classified as "yogurt," "snack," and "dairy product." The team is using Amazon Comprehend to train the model and must complete the project as soon as possible. Which functionality of Amazon Comprehend should the team use to meet these requirements?

一家零售公司從社交媒體、公司網站和客戶通話記錄中收集客戶對其產品的評論。由資料科學家和工程師組成的團隊希望找到共同的主題，並確定客戶在評論中提到的產品。該團隊正在使用自然語言處理(NLP) 建立一個模型來幫助進行這種分類。每種產品都可以歸入公司定義的多個類別。這些類別是相關的，但並不互相排斥。例如，如果在客戶評論文件中提到“樣品優格”，那麼“樣品優格”應該歸類為“優格”、“零食”和“乳製品”。該團隊正在使用Amazon Comprehend 來訓練模型，並且必須盡快完成專案。該團隊應該使用Amazon Comprehend 的哪些功能來滿足這些要求？

A. Custom classification with multi-class mode.

A. 多類別模式的自訂分類。

B. Custom classification with multi-label mode.

B. 多標籤模式的自訂分類。

C. Custom entity recognition.

C. 自訂實體識別。

D. Built-in models.

D. 內建模型。

問題# 186

A data engineer is using AWS Glue to create optimized, secure datasets in Amazon S3. The data science team wants the ability to access the ETL scripts directly from Amazon SageMaker notebooks within a VPC. After this setup is complete, the data science team wants the ability to run the AWS Glue job and invoke the SageMaker training job. Which combination of steps should the data engineer take to meet these requirements? (Choose three.)

資料工程師正在使用AWS Glue 在Amazon S3 中建立最佳化的安全資料集。資料科學團隊希望能夠直接從VPC 中的Amazon SageMaker 筆記本存取ETL 腳本。完成此設定後，資料科學團隊希望能夠執行AWS Glue 作業並呼叫SageMaker 訓練作業。資料工程師應採取哪些步驟組合來滿足這些要求？（選三個。）

A. Create a SageMaker development endpoint in the data science team's VPC.

A. 在資料科學團隊的VPC 中建立SageMaker 開發終端節點。

B. Create an AWS Glue development endpoint in the data science team's VPC.

B. 在資料科學團隊的VPC 中建立AWS Glue 開發終端節點。

C. Create SageMaker notebooks by using the AWS Glue development endpoint.

C. 使用AWS Glue 開發終端節點建立SageMaker 筆記型電腦。

D. Create SageMaker notebooks by using the SageMaker console.

D. 使用SageMaker 控制台建立SageMaker 筆記本。

E. Attach a decryption policy to the SageMaker notebooks.

E. 將解密策略附加到SageMaker 筆記本。

F. Create an IAM policy and an IAM role for the SageMaker notebooks.

F. 為SageMaker 筆記本建立IAM 策略和IAM 角色。

問題# 187

A data engineer needs to provide a team of data scientists with the appropriate dataset to run machine learning training jobs. The data will be stored in Amazon S3. The data engineer is obtaining the data from an Amazon Redshift database and is using join queries to extract a single tabular dataset. A portion of the schema is as follows: TransactionTimestamp (Timestamp) CardName (Varchar) CardNo (Varchar). The data engineer must provide the data so that any row with a CardNo value of NULL is removed. Also, the TransactionTimestamp column must be separated into a TransactionDate column and a TransactionTime column. Finally, the CardName column must be renamed to NameOnCard. The data will be extracted on a monthly basis and will be loaded into an S3 bucket. The solution must minimize the effort that is needed to set up infrastructure for the ingestion and transformation. The solution also must be automated and must minimize the load on the Amazon Redshift cluster. Which solution meets these requirements?

資料工程師需要為資料科學家團隊提供適當的資料集來執行機器學習訓練作業。資料將儲存在Amazon S3 中。資料工程師正在從Amazon Redshift 資料庫獲取數據，並使用連接查詢來提取單一表格資料集。部分架構如下：TransactionTimestamp (Timestamp) CardName (Varchar) CardNo (Varchar)。資料工程師必須提供數據，以便刪除CardNo 值為NULL 的任何行。此外，TransactionTimestamp 欄位必須分為TransactionDate 欄位和TransactionTime 欄位。最後，CardName 欄位必須重新命名為NameOnCard。資料將按月提取，並載入到S3 儲存桶中。此解決方案必須最大限度地減少設定用於提取和轉換的基礎設施所需的工作量。此解決方案還必須是自動化的，並且必須最大限度地減少Amazon Redshift 叢集上的負載。哪種解決方案滿足這些要求？

A. Set up an Amazon EMR cluster. Create an Apache Spark job to read the data from the Amazon Redshift cluster and transform the data. Load the data into the S3 bucket. Schedule the job to run monthly.

A. 設定Amazon EMR 叢集。建立Apache Spark 作業以從Amazon Redshift 叢集讀取資料並轉換資料。將資料載入到S3 儲存桶中。安排作業每月運行一次。

B. Set up an Amazon EC2 instance with a SQL client tool, such as SQL Workbench/J, to query the data from the Amazon Redshift cluster directly. Export the resulting dataset into a file. Upload the file into the S3 bucket. Perform these tasks monthly.

B. 使用SQL 用戶端工具（例如SQL Workbench/J）設定Amazon EC2 實例，以直接從Amazon Redshift 叢集查詢資料。將結果資料集匯出到文件中。將檔案上傳到S3 儲存桶。每月執行這些任務。

C. Set up an AWS Glue job that has the Amazon Redshift cluster as the source and the S3 bucket as the destination. Use the built-in transforms Filter, Map, and RenameField to perform the required transformations. Schedule the job to run monthly.

C. 設定AWS Glue 作業，將Amazon Redshift 叢集作為來源，將S3 儲存桶作為目標。使用內建轉換Filter、Map 和RenameField 執行所需的轉換。安排作業每月運行一次。

D. Use Amazon Redshift Spectrum to run a query that writes the data directly to the S3 bucket. Create an AWS Lambda function to run the query monthly.

D. 使用Amazon Redshift Spectrum 執行將資料直接寫入S3 儲存桶的查詢。建立AWS Lambda 函數以每月執行查詢。

問題# 188

A machine learning (ML) specialist wants to bring a custom training algorithm to Amazon SageMaker. The ML specialist implements the algorithm in a Docker container that is supported by SageMaker. How should the ML specialist package the Docker container so that SageMaker can launch the training correctly?

一位機器學習(ML) 專家想要將自訂訓練演算法引入Amazon SageMaker。ML 專家在SageMaker 支援的Docker 容器中實作該演算法。ML 專家應如何打包Docker 容器，以便SageMaker 能夠正確啟動訓練？

A. Specify the server argument in the ENTRYPOINT instruction in the Dockerfile.

A. 在Dockerfile 中的ENTRYPOINT 指令中指定伺服器參數。

B. Specify the training program in the ENTRYPOINT instruction in the Dockerfile.

B. 在Dockerfile 中的ENTRYPOINT 指令中指定訓練程式。

C. Include the path to the training data in the docker build command when packaging the container.

C. 在打包容器時，在docker build 指令中包含訓練資料的路徑。

D. Use a COPY instruction in the Dockerfile to copy the training program to the /opt/ml/train directory.

D. 使用Dockerfile 中的COPY 指令將訓練程式複製到/opt/ml/train 目錄。

問題# 189

An ecommerce company wants to use machine learning (ML) to monitor fraudulent transactions on its website. The company is using Amazon SageMaker to research, train, deploy, and monitor the ML models. The historical transactions data is in a.csv file that is stored in Amazon S3. The data contains features such as the user's IP address, navigation time, average time on each page, and the number of clicks for each session. There is no label in the data to indicate if a transaction is anomalous. Which models should the company use in combination to detect anomalous transactions? (Choose two.)

一家電子商務公司希望使用機器學習(ML) 來監控其網站上的詐騙交易。該公司正在使用Amazon SageMaker 研究、訓練、部署和監控ML 模型。歷史交易資料位於儲存在Amazon S3 中的.csv 檔案中。資料包含使用者IP 位址、導航時間、每個頁面的平均時間以及每個會話的點擊次數等特徵。數據中沒有標籤來指示交易是否異常。公司應該結合使用哪些模型來偵測異常交易？（選兩個。）

A. IP Insights.

A. IP 洞察

B. K-nearest neighbors (k-NN).

B. K-最近鄰(k-NN)。

C. Linear learner with a logistic function.

C. 具有邏輯函數的線性學習器。

D. Random Cut Forest (RCF).

D. 隨機砍伐森林(RCF)。

E. XGBoost.

E. XGBoost。

問題# 190

A healthcare company is using an Amazon SageMaker notebook instance to develop machine learning (ML) models. The company's data scientists will need to be able to access datasets stored in Amazon S3 to train the models. Due to regulatory requirements, access to the data from instances and services used for training must not be transmitted over the internet. Which combination of steps should an ML specialist take to provide this access? (Choose two.)

一家醫療保健公司正在使用Amazon SageMaker 筆記型電腦實例來開發機器學習(ML) 模型。該公司的資料科學家需要能夠存取儲存在Amazon S3 中的資料集來訓練模型。由於監管要求，用於訓練的實例和服務的資料存取不得透過網際網路傳輸。 ML 專家應採取哪些步驟組合來提供此存取權限？（選兩個。）

A. Configure the SageMaker notebook instance to be launched with a VPC attached and internet access disabled.

A. 設定要啟動的SageMaker 筆記型電腦實例，連接VPC 並停用Internet 存取。

B. Create and configure a VPN tunnel between SageMaker and Amazon S3.

B. 在SageMaker 和Amazon S3 之間建立並設定VPN 隧道。

C. Create and configure an S3 VPC endpoint Attach it to the VPC.

C. 建立並設定S3 VPC 終端節點並將其連接到VPC。

D. Create an S3 bucket policy that allows traffic from the VPC and denies traffic from the internet.

D. 建立允許來自VPC 的流量並拒絕來自Internet 的流量的S3 儲存桶策略。

E. Deploy AWS Transit Gateway Attach the S3 bucket and the SageMaker instance to the gateway.

E. 部署AWS Transit Gateway 將S3 儲存桶和SageMaker 執行個體連接到網關。

問題# 191

A machine learning (ML) specialist at a retail company is forecasting sales for one of the company's stores. The ML specialist is using data from the past 10 years. The company has provided a dataset that includes the total amount of money in sales each day for the store. Approximately 5% of the days are missing sales data. The ML specialist builds a simple forecasting model with the dataset and discovers that the model performs poorly. The performance is poor around the time of seasonal events, when the model consistently predicts sales figures that are too low or too high. Which actions should the ML specialist take to try to improve the model's performance? (Choose two.)

一家零售公司的機器學習(ML)專家正在預測該公司一家商店的銷售額。ML專家使用的是過去10年的資料。該公司提供了一個數據集，其中包括該商店每天的銷售總額。大約有5%的天數缺少銷售數據。ML專家使用該資料集建立了一個簡單的預測模型，發現該模型表現不佳。在季節性活動期間，模型的表現很差，此時模型的預測銷售數字始終過低或過高。ML專家應採取哪些措施來嘗試提高模型的效能？（選兩個。）

A. Add information about the store's sales periods to the dataset.

A. 將有關商店銷售期的資訊新增至資料集。

B. Aggregate sales figures from stores in the same proximity.

B. 匯總相同距離內商店的銷售資料。

C. Apply smoothing to correct for seasonal variation.

C. 應用平滑處理以校正季節性變化。

D. Change the forecast frequency from daily to weekly.

D. 將預測頻率從每日更改為每週。

E. Replace missing values in the dataset by using linear interpolation.

E. 使用線性內插法取代資料集中的缺失值。

問題# 192

A newspaper publisher has a table of customer data that consists of several numerical and categorical features, such as age and education history, as well as subscription status. The company wants to build a targeted marketing model for predicting the subscription status based on the table data. Which Amazon SageMaker built-in algorithm should be used to model the targeted marketing?

一家報紙出版商有一張客戶資料表，其中包含多個數字和分類特徵，例如年齡和教育歷史以及訂閱狀態。該公司希望根據表格數據建立一個有針對性的行銷模型，以預測訂閱狀態。應該使用哪種Amazon SageMaker內建演算法來建模有針對性的行銷？

A. Random Cut Forest (RCF).

A. 隨機砍伐森林(RCF)。

B. XGBoost.

B. XGBoost。

C. Neural Topic Model (NTM).

C. 神經主題模型(NTM)。

D. DeepAR forecasting.

D. DeepAR 預測。

問題# 193

A company will use Amazon SageMaker to train and host a machine learning model for a marketing campaign. The data must be encrypted at rest. Most of the data is sensitive customer data. The company wants AWS to maintain the root of trust for the encryption keys and wants key usage to be logged. Which solution will meet these requirements with the LEAST operational overhead?

一家公司將使用Amazon SageMaker 來訓練和託管行銷活動的機器學習模型。資料必須在靜止時加密。大多數數據都是敏感的客戶資料。該公司希望AWS 維護加密金鑰的信任根，並希望記錄金鑰使用情況。哪種解決方案可以以最少的營運開銷滿足這些要求？

A. Use AWS Security Token Service (AWS STS) to create temporary tokens to encrypt the storage volumes for all SageMaker instances and to encrypt the model artifacts and data in Amazon S3.

A. 使用AWS Security Token Service (AWS STS) 建立臨時令牌，以加密所有SageMaker 實例的儲存磁碟區並加密Amazon S3 中的模型工件和資料。

B. Use customer managed keys in AWS Key Management Service (AWS KMS) to encrypt the storage volumes for all SageMaker instances and to encrypt the model artifacts and data in Amazon S3.

B. 使用AWS Key Management Service (AWS KMS) 中的客戶管理金鑰加密所有SageMaker 執行個體的儲存磁碟區並加密Amazon S3 中的模型工件和資料。

C. Use encryption keys stored in AWS CloudHSM to encrypt the storage volumes for all SageMaker instances and to encrypt the model artifacts and data in Amazon S3.

C. 使用儲存在AWS CloudHSM 中的加密金鑰加密所有SageMaker 執行個體的儲存磁碟區並加密Amazon S3 中的模型工件和資料。

D. Use SageMaker built-in transient keys to encrypt the storage volumes for all SageMaker instances. Enable default encryption for new Amazon Elastic Block Store (Amazon EBS) volumes.

D. 使用SageMaker 內建臨時金鑰加密所有SageMaker 實例的儲存磁碟區。啟用預設加密的新Amazon Elastic Block Store (Amazon EBS) 磁碟區。

問題# 194

A data scientist is working on a model to predict a company's required inventory stock levels. All historical data is stored in.csv files in the company's data lake on Amazon S3. The dataset consists of approximately 500 GB of data. The data scientist wants to use SQL to explore the data before training the model. The company wants to minimize costs. Which option meets these requirements with the LEAST operational overhead?

一位資料科學家正在開發一個模型來預測公司所需的庫存水準。所有歷史資料都以.csv 檔案的形式儲存在Amazon S3 上的公司資料湖中。資料集包含大約500 GB 的資料。資料科學家希望在訓練模型之前使用SQL 探索資料。該公司希望最大限度地降低成本。哪種選擇可以以最少的營運開銷滿足這些要求？

A. Create an Amazon EMR cluster. Create external tables in the Apache Hive metastore, referencing the data that is stored in the S3 bucket. Explore the data from the Hive console.

A. 建立Amazon EMR 群集。在Apache Hive 元儲存中建立外部表，引用儲存在S3 儲存桶中的資料。從Hive 控制台探索數據。

B. Use AWS Glue to crawl the S3 bucket and create tables in the AWS Glue Data Catalog. Use Amazon Athena to explore the data.

B. 使用AWS Glue 抓取S3 儲存桶並在AWS Glue 資料目錄中建立表格。使用Amazon Athena 探索數據。

C. Create an Amazon Redshift cluster. Use the COPY command to ingest the data from Amazon S3. Explore the data from the Amazon Redshift query editor GUI.

C. 建立Amazon Redshift 叢集。使用COPY 指令從Amazon S3 中擷取資料。從Amazon Redshift 查詢編輯器GUI 探索資料。

D. Create an Amazon Redshift cluster. Create external tables in an external schema, referencing the S3 bucket that contains the data. Explore the data from the Amazon Redshift query editor GUI.

D. 建立Amazon Redshift 叢集。在外部架構中建立外部表，引用包含資料的S3 儲存桶。從Amazon Redshift 查詢編輯器GUI 探索資料。

問題# 195

A geospatial analysis company processes thousands of new satellite images each day to produce vessel detection data for commercial shipping. The company stores the training data in Amazon S3. The training data incrementally increases in size with new images each day. The company has configured an Amazon SageMaker training job to use a single ml.p2.xlarge instance with File input mode to train the built-in Object Detection algorithm. The training process was successful last month but is now failing because of a lack of storage. Aside from the addition of training data, nothing has changed in the model training process. A machine learning (ML) specialist needs to change the training configuration to fix the problem. The solution must optimize performance and must minimize the cost of training. Which solution will meet these requirements?

一家地理空間分析公司每天處理數千張新的衛星影像，以產生用於商業航運的船舶偵測資料。該公

司將訓練資料儲存在Amazon S3 中。訓練資料的大小會隨著每天的新影像而逐漸增加。該公司已配置Amazon SageMaker 訓練作業，以使用具有檔案輸入模式的單一ml.p2.xlarge 實例來訓練內建的物件偵測演算法。訓練過程上個月成功了，但現在由於缺乏儲存空間而失敗了。除了增加訓練資料外，模型訓練過程沒有任何變化。機器學習(ML) 專家需要更改訓練配置來解決問題。解決方案必須優化效能，並且必須最大限度地降低訓練成本。哪種解決方案可以滿足這些要求？

A. Modify the training configuration to use two ml.p2.xlarge instances.

A. 修改訓練配置以使用兩個ml.p2.xlarge 實例。

B. Modify the training configuration to use Pipe input mode.

B. 修改訓練配置以使用管路輸入模式。

C. Modify the training configuration to use a single ml.p3.2xlarge instance.

C. 修改訓練配置以使用單一ml.p3.2xlarge 實例。

D. Modify the training configuration to use Amazon Elastic File System (Amazon EFS) instead of Amazon S3 to store the input training data.

D. 修改訓練配置以使用Amazon Elastic File System (Amazon EFS) 而不是Amazon S3 來儲存輸入訓練資料。

問題# 196

A company is using Amazon SageMaker to build a machine learning (ML) model to predict customer churn based on customer call transcripts. Audio files from customer calls are located in an on-premises VoIP system that has petabytes of recorded calls. The on-premises infrastructure has high-velocity networking and connects to the company's AWS infrastructure through a VPN connection over a 100 Mbps connection. The company has an algorithm for transcribing customer calls that requires GPUs for inference. The company wants to store these transcriptions in an Amazon S3 bucket in the AWS Cloud for model development. Which solution should an ML specialist use to deliver the transcriptions to the S3 bucket as quickly as possible?

一家公司正在使用Amazon SageMaker 建立機器學習(ML) 模型，以根據客戶通話記錄預測客戶流失。客戶通話的音訊檔案位於本地VoIP 系統中，該系統具有PB 級的通話記錄。本地基礎設施具有高速網絡，並透過100 Mbps 的VPN 連接連接到公司的AWS 基礎設施。該公司有一種轉錄客戶通話的演算法，需要GPU 進行推理。該公司希望將這些轉錄儲存在AWS 雲端中的Amazon S3 儲存桶中以進行模型開發。ML 專家應該使用哪種解決方案盡快將轉錄內容傳送到S3 儲存桶？

A. Order and use an AWS Snowball Edge Compute Optimized device with an NVIDIA Tesla module to run the transcription algorithm. Use AWS DataSync to send the resulting transcriptions to the transcription S3 bucket.

A. 訂購並使用具有NVIDIA Tesla 模組的AWS Snowball Edge Compute Optimized 設備來運行轉錄演算法。使用AWS DataSync 將產生的轉錄傳送到轉錄S3 儲存桶。

B. Order and use an AWS Snowcone device with Amazon EC2 Inf1 instances to run the transcription algorithm. Use AWS DataSync to send the resulting transcriptions to the transcription S3 bucket.

B. 訂購並使用帶有Amazon EC2 Inf1 執行個體的AWS Snowcone 設備來運行轉錄演算法。使用AW

S DataSync 將產生的轉錄傳送到轉錄S3 儲存桶。

C. Order and use AWS Outposts to run the transcription algorithm on GPU-based Amazon EC2 instances. Store the resulting transcriptions in the transcription S3 bucket.

C. 訂購並使用AWS Outposts 在基於GPU 的Amazon EC2 執行個體上執行轉錄演算法。將產生的轉錄儲存在轉錄S3 儲存桶中。

D. Use AWS DataSync to ingest the audio files to Amazon S3. Create an AWS Lambda function to run the transcription algorithm on the audio files when they are uploaded to Amazon S3. Configure the function to write the resulting transcriptions to the transcription S3 bucket.

D. 使用AWS DataSync 將音訊檔案提取到Amazon S3。建立AWS Lambda 函數以在將音訊檔案上傳到Amazon S3 時在其上執行轉錄演算法。配置此函數以將產生的轉錄寫入轉錄S3 儲存桶。

問題# 197

A company has a podcast platform that has thousands of users. The company has implemented an anomaly detection algorithm to detect low podcast engagement based on a 10-minute running window of user events such as listening, pausing, and exiting the podcast. A machine learning (ML) specialist is designing the data ingestion of these events with the knowledge that the event payload needs some small transformations before inference. How should the ML specialist design the data ingestion to meet these requirements with the LEAST operational overhead?

一家公司擁有一個擁有數千名用戶的播客平台。該公司實施了一種異常檢測演算法，根據10 分鐘的用戶事件運行視窗（例如收聽、暫停和退出播客）來檢測播客參與度低的情況。機器學習(ML) 專家正在設計這些事件的資料擷取，並知道事件負載在推理之前需要進行一些小的轉換。ML 專家應如何設計資料擷取以滿足這些要求，同時將營運開銷降到最低？

A. Ingest event data by using a GraphQLAPI in AWS AppSync. Store the data in an Amazon DynamoDB table. Use DynamoDB Streams to call an AWS Lambda function to transform the most recent 10 minutes of data before inference.

A. 使用AWS AppSync 中的GraphQLAPI 擷取事件資料。將資料儲存在Amazon DynamoDB 表中。使用DynamoDB Streams 呼叫AWS Lambda 函數在推理之前轉換最近10 分鐘的資料。

B. Ingest event data by using Amazon Kinesis Data Streams. Store the data in Amazon S3 by using Amazon Kinesis Data Firehose. Use AWS Glue to transform the most recent 10 minutes of data before inference.

B. 使用Amazon Kinesis Data Streams 擷取事件資料。使用Amazon Kinesis Data Firehose 將資料儲存在Amazon S3 中。使用AWS Glue 在推理之前轉換最近10 分鐘的資料。

C. Ingest event data by using Amazon Kinesis Data Streams. Use an Amazon Kinesis Data Analytics for Apache Flink application to transform the most recent 10 minutes of data before inference.

C. 使用Amazon Kinesis Data Streams 擷取事件資料。使用Amazon Kinesis Data Analytics for Apache Flink 應用程式在推理之前轉換最近10 分鐘的資料。

D. Ingest event data by using Amazon Managed Streaming for Apache Kafka (Amazon MSK). Use an AWS Lambda function to transform the most recent 10 minutes of data before inference.

D. 使用Amazon Managed Streaming for Apache Kafka (Amazon MSK) 擷取事件資料。使用AWS

問題# 198

A company wants to predict the classification of documents that are created from an application. New documents are saved to an Amazon S3 bucket every 3 seconds. The company has developed three versions of a machine learning (ML) model within Amazon SageMaker to classify document text. The company wants to deploy these three versions to predict the classification of each document. Which approach will meet these requirements with the LEAST operational overhead?

一家公司希望預測從應用程式建立的文件的分類。每3 秒將新文件儲存到Amazon S3 儲存桶中。該公司在Amazon SageMaker 中開發了三種版本的機器學習(ML) 模型來對文件文字進行分類。該公司希望部署這三個版本來預測每個文件的分類。哪種方法可以以最少的營運開銷滿足這些要求？

A. Configure an S3 event notification that invokes an AWS Lambda function when new documents are created. Configure the Lambda function to create three SageMaker batch transform jobs, one batch transform job for each model for each document.

A. 設定一個S3 事件通知，在建立新文件時呼叫AWS Lambda 函數。配置Lambda 函數以建立三個SageMaker 批次轉換作業，每個文件的每個模型一個批次轉換作業。

B. Deploy all the models to a single SageMaker endpoint. Treat each model as a production variant. Configure an S3 event notification that invokes an AWS Lambda function when new documents are created. Configure the Lambda function to call each production variant and return the results of each model.

B. 將所有模型部署到單一SageMaker 終端節點。將每個模型視為生產變體。配置一個S3 事件通知，在建立新文件時呼叫AWS Lambda 函數。配置Lambda 函數以呼叫每個生產變體並傳回每個模型的結果。

C. Deploy each model to its own SageMaker endpoint. Configure an S3 event notification that invokes an AWS Lambda function when new documents are created. Configure the Lambda function to call each endpoint and return the results of each model.

C. 將每個模型部署到自己的SageMaker 終端節點配置一個S3 事件通知，在建立新文件時呼叫AWS Lambda 函數。配置Lambda 函數以呼叫每個終端節點並傳回每個模型的結果。

D. Deploy each model to its own SageMaker endpoint. Create three AWS Lambda functions. Configure each Lambda function to call a different endpoint and return the results. Configure three S3 event notifications to invoke the Lambda functions when new documents are created.

D. 將每個模型部署到自己的SageMaker 終端節點。建立三個AWS Lambda 函數。配置每個Lambda 函數以呼叫不同的終端節點並傳回結果。配置三個S3 事件通知以在建立新文件時呼叫Lambda 函數。

問題# 199

A manufacturing company needs to identify returned smartphones that have been damaged by moisture. The company has an automated process that produces 2,000 diagnostic values for each phone. The database contains more than five million phone evaluations. The evaluation process is consistent, and there are no missing values in the data. A machine learning (ML) specialist has trained an Amazon SageMaker linear learner ML model to classify phones as moisture damaged or not moisture damaged by using all available features. The model's F1 score is 0.6. Which changes in model training would MOST likely improve the model's F1 score? (Choose two.)

一家製造公司需要識別退回的受潮智慧型手機。該公司有一個自動化流程，可以為每支手機產生2,000個診斷值。資料庫包含超過五百萬個手機評估。評估過程是一致的，資料中沒有缺失值。機器學習(ML)專家訓練了一個Amazon SageMaker 線性學習器ML模型，使用所有可用特徵將手機分類為受潮損壞或未受潮損壞。此模型的F1得分為0.6。模型訓練中的哪些變化最有可能提高模型的F1得分？（選兩個。）

A. Continue to use the SageMaker linear learner algorithm. Reduce the number of features with the SageMaker principal component analysis (PCA) algorithm.

A. 繼續使用SageMaker 線性學習器演算法。使用SageMaker 主成分分析(PCA) 演算法減少特徵數量。

B. Continue to use the SageMaker linear learner algorithm. Reduce the number of features with the scikit-learn multi-dimensional scaling (MDS) algorithm.

B. 繼續使用SageMaker 線性學習器演算法。使用scikit-learn 多維縮放(MDS) 演算法減少特徵數量。

C. Continue to use the SageMaker linear learner algorithm. Set the predictor type to regressor.

C. 繼續使用SageMaker 線性學習器演算法。將預測器類型設定為回歸器。

D. Use the SageMaker k-means algorithm with k of less than 1,000 to train the model.

D. 使用k 小於1,000 的SageMaker k-means 演算法來訓練模型。

E. Use the SageMaker k-nearest neighbors (k-NN) algorithm. Set a dimension reduction target of less than 1,000 to train the model.

E. 使用SageMaker k-最近鄰(k-NN) 演算法。設定小於1,000 的降維目標來訓練模型。

A company is building a machine learning (ML) model to classify images of plants. An ML specialist has trained the model using the Amazon SageMaker built-in Image Classification algorithm. The model is hosted using a SageMaker endpoint on an ml.m5.xlarge instance for real-time inference. When used by researchers in the field, the inference has greater latency than is acceptable. The latency gets worse when multiple researchers perform inference at the same time on their devices. Using Amazon CloudWatch metrics, the ML specialist notices that the ModelLatency metric shows a high value and is responsible for most of the response latency. The ML specialist needs to fix the performance issue so that researchers can experience less latency when performing inference from their devices. Which action should the ML specialist take to meet this requirement?

一家公司正在建立一個機器學習(ML)模型來對植物影像進行分類。ML專家使用Amazon SageMaker內建的影像分類演算法訓練了該模型。此模型使用ml.m5.xlarge實例上的SageMaker終端節點託管，用於即時推理。當研究人員在現場使用時，推理的延遲大於可接受的延遲。當多個研究人員同時在其設備上執行推理時，延遲會變得更糟。使用Amazon CloudWatch指標，ML專家注意到ModelLatency指標顯示高值且是大部分回應延遲的原因。ML專家需要修復效能問題，以便研究人員從其裝置執行推理時可以減少延遲。ML專家應採取哪些措施來滿足此要求？

A. Change the endpoint instance to an ml.t3 burstable instance with the same vCPU number as the ml.m5.xlarge instance has.

A. 將終端節點實例變更為ml.t3可突增實例，其vCPU數量與ml.m5.xlarge實例相同。

B. Attach an Amazon Elastic Inference ml.eia2.medium accelerator to the endpoint instance.

B. 將Amazon Elastic Inference ml.eia2.medium加速器附加到終端節點實例。

C. Enable Amazon SageMaker Autopilot to automatically tune performance of the model.

C. 啟用Amazon SageMaker Autopilot以自動調整模型的效能。

D. Change the endpoint instance to use a memory optimized ML instance.

D. 將終端節點實例變更為使用記憶體最佳化的ML實例。

問題# 201

An automotive company is using computer vision in its autonomous cars. The company has trained its models successfully by using transfer learning from a convolutional neural network (CNN). The models are trained with PyTorch through the use of the Amazon SageMaker SDK. The company wants to reduce the time that is required for performing inferences, given the low latency that is required for self-driving. Which solution should the company use to evaluate and improve the performance of the models?

一家汽車公司正在其自動駕駛汽車中使用電腦視覺。該公司已使用卷積神經網路(CNN)的遷移學習成功訓練了其模型。這些模型透過使用Amazon SageMaker SDK使用PyTorch進行訓練。考慮到自動駕駛所需的低延遲，該公司希望減少執行推理所需的時間。該公司應該使用哪種解決方案來評估和改進模型的效能？

A. Use Amazon CloudWatch algorithm metrics for visibility into the SageMaker training weights, gradients, biases, and activation outputs. Compute the filter ranks based on this information. Apply pruning to remove the low-ranking filters. Set the new weights. Run a new training job with the pruned model.

A. 使用Amazon CloudWatch 演算法指標查看SageMaker 訓練權重、梯度、偏差和啟動輸出。根據此資訊計算過濾器排名。應用修剪以刪除排名較低的過濾器。設定新權重。使用修剪後的模型執行新的訓練作業。

B. Use SageMaker Debugger for visibility into the training weights, gradients, biases, and activation outputs. Adjust the model hyperparameters, and look for lower inference times. Run a new training job.

B. 使用SageMaker Debugger 查看訓練權重、梯度、偏差和啟動輸出。調整模型超參數，並尋找更低的推理時間。運行新的訓練作業。

C. Use SageMaker Debugger for visibility into the training weights, gradients, biases, and activation outputs. Compute the filter ranks based on this information. Apply pruning to remove the low-ranking filters. Set the new weights. Run a new training job with the pruned model.

C. 使用SageMaker Debugger 查看訓練權重、梯度、偏差和啟動輸出。根據此資訊計算過濾器排名。應用修剪以刪除排名較低的過濾器。設定新權重。使用修剪後的模型執行新的訓練作業。

D. Use SageMaker Model Monitor for visibility into the ModelLatency metric and OverheadLatency metric of the model after the model is deployed. Adjust the model hyperparameters, and look for lower inference times. Run a new training job.

D. 使用SageMaker Model Monitor 查看部署模型後的ModelLatency 指標和OverheadLatency 指標。調整模型超參數，並尋找更低的推理時間。運行一項新的訓練工作。

問題# 202

A company's machine learning (ML) specialist is designing a scalable data storage solution for Amazon SageMaker. The company has an existing TensorFlow-based model that uses a train.py script. The model relies on static training data that is currently stored in TFRecord format. What should the ML specialist do to provide the training data to SageMaker with the LEAST development overhead?

一家公司的機器學習(ML) 專家正在為Amazon SageMaker 設計可擴展的資料儲存解決方案。該公司有一個基於TensorFlow 的現有模型，該模型使用train.py 腳本。此模型依賴目前以TFRecord 格式儲存的靜態訓練資料。ML 專家應該怎麼做才能以最少的開發開銷向SageMaker 提供訓練資料？

A. Put the TFRecord data into an Amazon S3 bucket. Use AWS Glue or AWS Lambda to reformat the data to protobuf format and store the data in a second S3 bucket. Point the SageMaker training invocation to the second S3 bucket.

A. 將TFRecord 資料放入Amazon S3 儲存桶。使用AWS Glue 或AWS Lambda 將資料重新格式化為protobuf 格式，並將資料儲存在第二個S3 儲存桶中。將SageMaker 訓練呼叫指向第二個S3 儲存桶。

B. Rewrite the train.py script to add a section that converts TFRecord data to protobuf format. Point the SageMaker training invocation to the local path of the data. Ingest the protobuf data instead of the TFRecord data.

B. 重寫train.py 腳本以新增將TFRecord 資料轉換為protobuf 格式的部分。將SageMaker 訓練呼

叫指向資料的本機路徑。提取protobuf 資料而不是TFRecord 資料。

C. Use SageMaker script mode, and use train.py unchanged. Point the SageMaker training invocation to the local path of the data without reformatting the training data.

C. 使用SageMaker 腳本模式，並使用不變的train.py。將SageMaker 訓練呼叫指向資料的本機路徑，而無需重新格式化訓練資料。

D. Use SageMaker script mode, and use train.py unchanged. Put the TFRecord data into an Amazon S3 bucket. Point the SageMaker training invocation to the S3 bucket without reformatting the training data.

D. 使用SageMaker 腳本模式，並使用不變的train.py。將TFRecord 資料放入Amazon S3 儲存桶。將SageMaker 訓練呼叫指向S3 儲存桶，而無需重新格式化訓練資料。

問題# 203

An ecommerce company wants to train a large image classification model with 10,000 classes. The company runs multiple model training iterations and needs to minimize operational overhead and cost. The company also needs to avoid loss of work and model retraining. Which solution will meet these requirements?

一家電子商務公司想要訓練一個包含10,000 個類別的大型影像分類模型。該公司運行多個模型訓練迭代，並需要盡量減少營運開銷和成本。該公司還需要避免工作損失和模型重新訓練。哪種解決方案可以滿足這些要求？

A. Create the training jobs as AWS Batch jobs that use Amazon EC2 Spot Instances in a managed compute environment.

A. 在託管運算環境中將訓練作業建立為使用Amazon EC2 Spot 執行個體的AWS Batch 作業。

B. Use Amazon EC2 Spot Instances to run the training jobs. Use a Spot Instance interruption notice to save a snapshot of the model to Amazon S3 before an instance is terminated.

B. 使用Amazon EC2 Spot 執行個體執行訓練作業。在執行個體終止之前，使用Spot 執行個體中斷通知將模型快照儲存到Amazon S3。

C. Use AWS Lambda to run the training jobs. Save model weights to Amazon S3.

C. 使用AWS Lambda 執行訓練作業。將模型權重保存到Amazon S3。

D. Use managed spot training in Amazon SageMaker. Launch the training jobs with checkpointing enabled.

D. 在Amazon SageMaker 中使用託管Spot 訓練。在啟用檢查點的情況下啟動訓練作業。

問題# 204

A retail company uses a machine learning (ML) model for daily sales forecasting. The model has provided inaccurate results for the past 3 weeks. At the end of each day, an AWS Glue job consolidates the input data that is used for the forecasting with the actual daily sales data and the predictions of the model. The AWS Glue job stores the data in Amazon S3. The company's ML team determines that the inaccuracies are occurring because of a change in the value distributions of the model features. The ML team must implement a solution that will detect when this type of change occurs in the future. Which solution will meet these requirements with the LEAST amount of operational overhead?

一家零售公司使用機器學習(ML) 模型進行每日銷售預測。此模型在過去3 週內提供的結果不準確。每天結束時，AWS Glue 作業都會將用於預測的輸入資料與實際每日銷售資料和模型的預測進行整合。AWS Glue 作業將資料儲存在Amazon S3 中。該公司的ML 團隊確定不準確的原因是模型特徵的值分佈發生了變化。ML 團隊必須實施解決方案，以偵測未來何時發生此類變更。哪種解決方案能夠以最少的營運開銷滿足這些要求？

A. Use Amazon SageMaker Model Monitor to create a data quality baseline. Confirm that the `emit_metrics` option is set to Enabled in the baseline constraints file. Set up an Amazon CloudWatch alarm for the metric.

A. 使用Amazon SageMaker 模型監視器建立資料品質基準。確認基線約束檔案中的`emit_metrics` 選項設定為Enabled。為指標設定Amazon CloudWatch 警報。

B. Use Amazon SageMaker Model Monitor to create a model quality baseline. Confirm that the `emit_metrics` option is set to Enabled in the baseline constraints file. Set up an Amazon CloudWatch alarm for the metric.

B. 使用Amazon SageMaker 模型監視器建立模型品質基準。確認基線約束檔案中的`emit_metrics` 選項設定為Enabled。為指標設定Amazon CloudWatch 警報。

C. Use Amazon SageMaker Debugger to create rules to capture feature values. Set up an Amazon CloudWatch alarm for the rules.

C. 使用Amazon SageMaker 偵錯器建立規則以擷取特徵值為規則設定Amazon CloudWatch 警報。

D. Use Amazon CloudWatch to monitor Amazon SageMaker endpoints. Analyze logs in Amazon CloudWatch Logs to check for data drift.

D. 使用Amazon CloudWatch 監控Amazon SageMaker 終端節點。分析Amazon CloudWatch Logs 中的日誌以檢查資料漂移。

問題# 205

A machine learning (ML) specialist has prepared and used a custom container image with Amazon SageMaker to train an image classification model. The ML specialist is performing hyperparameter optimization (HPO) with this custom container image to produce a higher quality image classifier. The ML specialist needs to determine whether HPO with the SageMaker built-in image classification algorithm will produce a better model than the model produced by HPO with the custom container image. All ML experiments and HPO jobs must be invoked from scripts inside SageMaker Studio notebooks. How can the ML specialist meet these requirements in the LEAST amount of time?

機器學習(ML) 專家已準備並使用自訂容器映像和Amazon SageMaker 來訓練影像分類模型。ML 專家正在使用此自訂容器映像執行超參數最佳化(HPO)，以產生更高品質的映像分類器。ML 專家需要確定使用SageMaker 內建影像分類演算法的HPO 是否會產生比使用自訂容器映像的HPO 產生

的模型更好的模型。所有ML 實驗和HPO 作業都必須從SageMaker Studio 筆記本中的腳本呼叫。ML 專家如何在最短的時間內滿足這些要求？

A. Prepare a custom HPO script that runs multiple training jobs in SageMaker Studio in local mode to tune the model of the custom container image. Use the automatic model tuning capability of SageMaker with early stopping enabled to tune the model of the built-in image classification algorithm. Select the model with the best objective metric value.

A. 準備一個自訂的HPO 腳本，在SageMaker Studio 中以本機模式執行多個訓練作業，以調整自訂容器鏡像的模型。使用SageMaker 的自動模型調整功能並啟用早期停止來調整內建鏡像分類演算法的模型。選擇具有最佳客觀指標值的模型。

B. Use SageMaker Autopilot to tune the model of the custom container image. Use the automatic model tuning capability of SageMaker with early stopping enabled to tune the model of the built-in image classification algorithm. Compare the objective metric values of the resulting models of the SageMaker AutopilotAutoML job and the automatic model tuning job. Select the model with the best objective metric value.

B. 使用SageMaker Autopilot 調整自訂容器鏡像的模型。使用SageMaker 的自動模型調整功能並啟用早期停止來調整內建鏡像分類演算法的模型。比較SageMaker AutopilotAutoML 作業和自動模型調整作業的結果模型的客觀指標值。選擇具有最佳客觀指標值的模型。

C. Use SageMaker Experiments to run and manage multiple training jobs and tune the model of the custom container image. Use the automatic model tuning capability of SageMaker to tune the model of the built-in image classification algorithm. Select the model with the best objective metric value.

C. 使用SageMaker Experiments 運行和管理多個訓練作業並調整自訂容器鏡像的模型。使用SageMaker 的自動模型調整功能調整內建鏡像分類演算法的模型。選擇具有最佳客觀指標值的模型。

D. Use the automatic model tuning capability of SageMaker to tune the models of the custom container image and the built-in image classification algorithm at the same time. Select the model with the best objective metric value.

D. 利用SageMaker 的自動模型調優能力，同時對自訂容器鏡像和內建鏡像分類演算法的模型進行調優，選擇目標指標值最優的模型。

問題# 206

A company wants to deliver digital car management services to its customers. The company plans to analyze data to predict the likelihood of users changing cars. The company has 10 TB of data that is stored in an Amazon Redshift cluster. The company's data engineering team is using Amazon SageMaker Studio for data analysis and model development. Only a subset of the data is relevant for developing the machine learning models. The data engineering team needs a secure and cost-effective way to export the data to a data repository in Amazon S3 for model development. Which solutions will meet these requirements? (Choose two.)

一家公司希望向客戶提供數位汽車管理服務。該公司計劃分析數據以預測用戶換車的可能性。該公司有10 TB 的資料儲存在Amazon Redshift 叢集中。該公司的資料工程團隊正在使用Amazon SageMaker Studio 進行資料分析和模型開發。只有一部分數據與開發機器學習模型相關。資料工程團隊需要一種安全且經濟高效的方式將資料匯出到Amazon S3 中的資料儲存庫以進行模型開發。哪些解決方案可以滿足這些要求？（選兩個。）

A. Launch multiple medium-sized instances in a distributed SageMaker Processing job. Use the prebuilt Docker images for Apache Spark to query and plot the relevant data and to export the relevant data from Amazon Redshift to Amazon S3.

A. 在分散式SageMaker 處理作業中啟動多個中型實例。使用Apache Spark 的預先建置Docker 映像查詢和繪製相關數據，並將相關數據從Amazon Redshift 匯出到Amazon S3。

B. Launch multiple medium-sized notebook instances with a PySpark kernel in distributed mode. Download the data from Amazon Redshift to the notebook cluster. Query and plot the relevant data. Export the relevant data from the notebook cluster to Amazon S3.

B. 在分散式模式下使用PySpark 核心啟動多個中型筆記本實例。將資料從Amazon Redshift 下載到筆電叢集。查詢和繪製相關資料。將相關數據從筆記本集群匯出到Amazon S3。

C. Use AWS Secrets Manager to store the Amazon Redshift credentials. From a SageMaker Studio notebook, use the stored credentials to connect to Amazon Redshift with a Python adapter. Use the Python client to query the relevant data and to export the relevant data from Amazon Redshift to Amazon S3.

C. 使用AWS Secrets Manager 儲存Amazon Redshift 憑證。從SageMaker Studio 筆記本中，使用儲存的憑證透過Python 適配器連接到Amazon Redshift。使用Python 用戶端查詢相關數據，並將相關數據從Amazon Redshift 匯出到Amazon S3。

D. Use AWS Secrets Manager to store the Amazon Redshift credentials. Launch a SageMaker extra-large notebook instance with block storage that is slightly larger than 10 TB. Use the stored credentials to connect to Amazon Redshift with a Python adapter. Download, query, and plot the relevant data. Export the relevant data from the local notebook drive to Amazon S3.

D. 使用AWS Secrets Manager 儲存Amazon Redshift 憑證。啟動具有略大於10 TB 的區塊儲存的SageMaker 超大型筆記本實例。使用儲存的憑證透過Python 適配器連接到Amazon Redshift。下載、查詢和繪製相關資料。將相關資料從本機筆記本磁碟機匯出到Amazon S3。

E. Use SageMaker Data Wrangler to query and plot the relevant data and to export the relevant data from Amazon Redshift to Amazon S3.

E. 使用SageMaker Data Wrangler 查詢和繪製相關數據，並將相關數據從Amazon Redshift 匯出到Amazon S3。

問題# 207

A company is building an application that can predict spam email messages based on email text. The company can generate a few thousand human-labeled datasets that contain a list of email messages and a label of "spam" or "not spam" for each email message. A machine learning (ML) specialist wants to use transfer learning with a Bidirectional Encoder Representations from Transformers (BERT) model that is trained on English Wikipedia text data. What should the ML specialist do to initialize the model to fine-tune the model with the custom data?

一家公司正在開發一款應用程式，可以根據電子郵件文字預測垃圾郵件。該公司可以產生數千個個人工標記的資料集，其中包含電子郵件清單以及每封電子郵件的「垃圾郵件」或「非垃圾郵件」標籤。一位機器學習(ML) 專家希望使用遷移學習和基於英語維基百科文字資料訓練的Transformers 雙向編碼器表示(BERT) 模型。ML 專家應該如何初始化模型，以便使用自訂資料對模型進行微調？

A. Initialize the model with pretrained weights in all layers except the last fully connected layer.

A. 在所有層中，除了最後一個全連接層之外，使用預先訓練的權重初始化模型。

B. Initialize the model with pretrained weights in all layers. Stack a classifier on top of the first output position. Train the classifier with the labeled data.

B. 在所有層中，使用預先訓練的權重初始化模型。在第一個輸出位置之上堆疊一個分類器。使用標記資料訓練分類器。

C. Initialize the model with random weights in all layers. Replace the last fully connected layer with a classifier. Train the classifier with the labeled data.

C. 在所有層中使用隨機權重初始化模型。用分類器取代最後一個全連接層。使用標記資料訓練分類器。

D. Initialize the model with pretrained weights in all layers. Replace the last fully connected layer with a classifier. Train the classifier with the labeled data.

D. 在所有層中，使用預先訓練的權重初始化模型。用分類器取代最後一個全連接層。使用標記資料訓練分類器。

問題# 208

A company is using a legacy telephony platform and has several years remaining on its contract. The company wants to move to AWS and wants to implement the following machine learning features:

- calls based on the transcript
- calls

indication and scoring of that sentiment Which AWS solution will meet these requirements with the LEAST amount of custom model training?

一家公司正在使用傳統電話平台，合約還有幾年到期。該公司希望遷移到AWS，並希望實現以下機器學習功能：

- 多種語言的通話轉錄
- 根據轉錄對通話進行分類
- 檢測通話中的主要客戶問題
- 對轉錄的每一行進行客戶情緒分析，並對該情緒進行正面或負面的指示和評分

哪種AWS 解決方案能夠以最少的自訂模型訓練量滿足這些要求？

A. Use Amazon Transcribe to process audio calls to produce transcripts, categorize calls, and detect issues. Use Amazon Comprehend to analyze sentiment.

A. 使用Amazon Transcribe 處理音訊通話以產生記錄、對通話進行分類並偵測問題。使用Amazon Comprehend 分析情緒。

B. Use Amazon Transcribe to process audio calls to produce transcripts. Use Amazon Comprehend to categorize calls, detect issues, and analyze sentiment.

B. 使用Amazon Transcribe 處理音訊呼叫以產生記錄。使用Amazon Comprehend 對通話進行分類、偵測問題並分析情緒。

C. Use Contact Lens for Amazon Connect to process audio calls to produce transcripts, categorize calls, detect issues, and analyze sentiment.

C. 使用Contact Lens for Amazon Connect 處理音訊通話以產生記錄、對通話進行分類、偵測問題並分析情緒。

D. Use Contact Lens for Amazon Connect to process audio calls to produce transcripts. Use Amazon Comprehend to categorize calls, detect issues, and analyze sentiment.

D. 使用Contact Lens for Amazon Connect 處理音訊呼叫以產生記錄。使用Amazon Comprehend 對通話進行分類、偵測問題並分析情緒。

問題# 209

A finance company needs to forecast the price of a commodity. The company has compiled a dataset of historical daily prices. A data scientist must train various forecasting models on 80% of the dataset and must validate the efficacy of those models on the remaining 20% of the dataset. How should the data scientist split the dataset into a training dataset and a validation dataset to compare model performance?

一家金融公司需要預測某種商品的價格。該公司編制了一份歷史每日價格資料集。資料科學家必須對80%的資料集訓練各種預測模型，並且必須在剩餘20%的資料集上驗證這些模型的有效性。資料科學家應如何將資料集拆分為訓練資料集和驗證資料集，以比較模型效能？

A. Pick a date so that 80% of the data points precede the date. Assign that group of data points as the training dataset. Assign all the remaining data points to the validation dataset.

A. 選擇一個日期，使得80%的資料點出現在該日期之前。將該組資料點指定為訓練資料集。將所有剩餘的數據點分配給驗證數據集。

B. Pick a date so that 80% of the data points occur after the date. Assign that group of data points as the training dataset. Assign all the remaining data points to the validation dataset.

B. 選擇一個日期，使得80%的資料點出現在該日期之後。將該組資料點指定為訓練資料集。將所有剩餘的數據點分配給驗證數據集。

C. Starting from the earliest date in the dataset, pick eight data points for the training dataset and two data points for the validation dataset. Repeat this stratified sampling until no data points remain.

C. 從資料集中最早的日期開始，為訓練資料集挑選八個資料點，為驗證資料集挑選兩個資料點。重複此分層抽樣，直到沒有資料點剩餘。

D. Sample data points randomly without replacement so that 80% of the data points are in the training dataset. Assign all the remaining data points to the validation dataset.

D. 隨機抽樣資料點而不進行替換，使得80%的資料點位於訓練資料集中。將所有剩餘的數據點分配給驗證數據集。

問題# 210

A retail company wants to build a recommendation system for the company's website. The system needs to provide recommendations for existing users and needs to base those recommendations on each user's past browsing history. The system also must filter out any items that the user previously purchased. Which solution will meet these requirements with the LEAST development effort?

一家零售公司希望為公司的網站建立一個推薦系統。該系統需要為現有用戶提供推薦，並且需要根據每個用戶過去的瀏覽歷史記錄提供這些推薦。該系統還必須過濾掉用戶先前購買的任何商品。哪一種解決方案能夠以最少的開發工作量來滿足這些要求？

A. Train a model by using a user-based collaborative filtering algorithm on Amazon SageMaker. Host the model on a SageMaker real-time endpoint. Configure an Amazon API Gateway API and an AWS Lambda function to handle real-time inference requests that the web application sends. Exclude the items that the user previously purchased from the results before sending the results back to the web application.

A. 使用基於使用者的協同過濾演算法在Amazon SageMaker 上訓練模型。將模型託管在SageMaker 即時終端節點上。設定Amazon API Gateway API 和AWS Lambda 函數以處理Web 應用程式傳送的即時推理請求。在將結果傳回Web 應用程式之前，從結果中排除使用者先前購買的商品。

B. Use an Amazon Personalize PERSONALIZED_RANKING recipe to train a model. Create a real-time filter to exclude items that the user previously purchased. Create and deploy a campaign on Amazon Personalize. Use the GetPersonalizedRanking API operation to get the real-time recommendations.

B. 使用Amazon Personalize PERSONALIZED_RANKING 配方訓練模型。建立即時過濾器以排除用戶先前購買的商品。在Amazon Personalize 上建立並部署活動。使用GetPersonalizedRanking API 操作取得即時推薦。

C. Use an Amazon Personalize USER_PERSONALIZATION recipe to train a model. Create a real-time filter to exclude items that the user previously purchased. Create and deploy a campaign on Amazon Personalize. Use the GetRecommendations API operation to get the real-time recommendations.

C. 使用Amazon Personalize USER_PERSONALIZATION 配方訓練模型。建立即時過濾器以排除用戶先前購買的商品。在Amazon Personalize 上建立並部署活動。使用GetRecommendations API 操作取得即時推薦。

D. Train a neural collaborative filtering model on Amazon SageMaker by using GPU instances. Host the model on a SageMaker real-time endpoint. Configure an Amazon API Gateway API and an AWS Lambda function to handle real-time inference requests that the web application sends. Exclude the items that the user previously purchased from the results before sending the results back to the web application.

D. 使用GPU 實例在Amazon SageMaker 上訓練神經協同過濾模型。在SageMaker 即時終端節點上託管模型。設定Amazon API Gateway API 和AWS Lambda 函數來處理Web 應用程式傳送的即時推理請求。在將結果傳回Web 應用程式之前，從結果中排除使用者先前購買的商品。

問題# 211

A bank wants to use a machine learning (ML) model to predict if users will default on credit card payments. The training data consists of 30,000 labeled records and is evenly balanced between two categories. For the model, an ML specialist selects the Amazon SageMaker built-in XGBoost algorithm and configures a SageMaker automatic hyperparameter optimization job with the Bayesian method. The ML specialist uses the validation accuracy as the objective metric. When the bank implements the solution with this model, the prediction accuracy is 75%. The bank has given the ML specialist 1 day to improve the model in production. Which approach is the FASTEST way to improve the model's accuracy?

一家銀行希望使用機器學習(ML)模型來預測用戶是否會拖欠信用卡付款。訓練資料由30,000個標籤的記錄組成，並且在兩個類別之間均衡。對於該模型，ML專家選擇Amazon SageMaker內建的XGBoost演算法，並使用貝葉斯方法配置SageMaker自動超參數最佳化作業。ML專家使用驗證準確度作為客觀指標。當銀行使用此模型實施解決方案時，預測準確度為75%。銀行給了ML專家1天時間來改進生產中的模型。哪一種方法是提高模型準確性的最快方法？

A. Run a SageMaker incremental training based on the best candidate from the current model's tuning job. Monitor the same metric that was used as the objective metric in the previous tuning, and look for improvements.

A. 根據目前模型調整作業中最佳的候選運行SageMaker增量訓練。監控與上一次調整中用作目標指標的相同指標，並尋找改進。

B. Set the Area Under the ROC Curve (AUC) as the objective metric for a new SageMaker automatic hyperparameter tuning job. Use the same maximum training jobs parameter that was used in the previous tuning job.

B. 將ROC曲線下面積(AUC)設定為新SageMaker自動超參數調整作業的目標指標。使用與上一次調整作業中使用的相同最大訓練作業參數。

C. Run a SageMaker warm start hyperparameter tuning job based on the current model's tuning job. Use the same objective metric that was used in the previous tuning.

C. 根據目前模型調整作業執行SageMaker熱啟動超參數調整作業。使用與上一次調整中使用的相同目標指標。

D. Set the F1 score as the objective metric for a new SageMaker automatic hyperparameter tuning job. Double the maximum training jobs parameter that was used in the previous tuning job.

D. 將F1分數設定為新SageMaker自動超參數調整作業的目標指標。將上一次調整作業中使用的最大訓練作業參數加倍。

問題# 212

A data scientist has 20 TB of data in CSV format in an Amazon S3 bucket. The data scientist needs to convert the data to Apache Parquet format. How can the data scientist convert the file format with the LEAST amount of effort?

一位資料科學家在Amazon S3儲存桶中擁有20 TB的CSV格式資料。資料科學家需要將資料轉換為Apache Parquet格式。資料科學家如何以最少的精力轉換文件格式？

A. Use an AWS Glue crawler to convert the file format.

A. 使用AWS Glue爬蟲程式轉換檔案格式。

B. Write a script to convert the file format. Run the script as an AWS Glue job.

B. 編寫腳本來轉換檔案格式。將腳本作為AWS Glue 作業運行。

C. Write a script to convert the file format. Run the script on an Amazon EMR cluster.

C. 編寫腳本來轉換檔案格式。在Amazon EMR 叢集上執行腳本。

D. Write a script to convert the file format. Run the script in an Amazon SageMaker notebook.

D. 編寫腳本來轉換檔案格式。在Amazon SageMaker 筆記本中執行腳本。

問題# 213

A company is building a pipeline that periodically retrains its machine learning (ML) models by using new streaming data from devices. The company's data engineering team wants to build a data ingestion system that has high throughput, durable storage, and scalability. The company can tolerate up to 5 minutes of latency for data ingestion. The company needs a solution that can apply basic data transformation during the ingestion process. Which solution will meet these requirements with the MOST operational efficiency?

一家公司正在建立一個管道，使用來自設備的新流資料定期重新訓練其機器學習(ML) 模型。該公司的資料工程團隊希望建立一個具有高吞吐量、持久性儲存和可擴展性的資料提取系統。該公司可以容忍長達5 分鐘的資料擷取延遲。該公司需要一個可以在提取過程中應用基本資料轉換的解決方案。哪種解決方案能夠以最高的營運效率滿足這些要求？

A. Configure the devices to send streaming data to an Amazon Kinesis data stream. Configure an Amazon Kinesis Data Firehose delivery stream to automatically consume the Kinesis data stream, transform the data with an AWS Lambda function, and save the output into an Amazon S3 bucket.

A. 配置設備以將流資料傳送至Amazon Kinesis 資料流。配置Amazon Kinesis Data Firehose 傳輸流以自動使用Kinesis 資料流，使用AWS Lambda 函數轉換數據，並將輸出儲存至Amazon S3 儲存桶中。

B. Configure the devices to send streaming data to an Amazon S3 bucket. Configure an AWS Lambda function that is invoked by S3 event notifications to transform the data and load the data into an Amazon Kinesis data stream. Configure an Amazon Kinesis Data Firehose delivery stream to automatically consume the Kinesis data stream and load the output back into the S3 bucket.

B. 配置設備以將流資料傳送至Amazon S3 儲存桶。配置由S3 事件通知呼叫的AWS Lambda 函數以轉換資料並將資料載入到Amazon Kinesis 資料流中。配置Amazon Kinesis Data Firehose 傳輸流以自動使用Kinesis 資料流並將輸出載入回S3 儲存桶中。

C. Configure the devices to send streaming data to an Amazon S3 bucket. Configure an AWS Glue job that is invoked by S3 event notifications to read the data, transform the data, and load the output into a new S3 bucket.

C. 配置設備以將流資料傳送至Amazon S3 儲存桶。配置由S3 事件通知呼叫的AWS Glue 作業以讀取資料、轉換資料並將輸出載入到新的S3 儲存桶中。

D. Configure the devices to send streaming data to an Amazon Kinesis Data Firehose delivery stream. Configure an AWS Glue job that connects to the delivery stream to transform the data and load the output into an Amazon S3 bucket.

D. 配置設備以將流資料傳送至Amazon Kinesis Data Firehose 傳輸流。配置連接到傳輸流的AWS Glue 作業以轉換資料並將輸出載入到Amazon S3 儲存桶中。

問題# 214

A retail company is ingesting purchasing records from its network of 20,000 stores to Amazon S3 by using Amazon Kinesis Data Firehose. The company uses a small, server-based application in each store to send the data to AWS over the internet. The company uses this data to train a machine learning model that is retrained each day. The company's data science team has identified existing attributes on these records that could be combined to create an improved model. Which change will create the required transformed records with the LEAST operational overhead?

一家零售公司正在使用Amazon Kinesis Data Firehose 將來自其20,000 家商店網路的購買記錄匯入到Amazon S3。該公司在每個商店中使用一個小型的基於伺服器的應用程式透過網路將資料發送到AWS。該公司使用這些數據來訓練一個機器學習模型，該模型每天都會重新訓練。該公司的數據科學團隊已經確定了這些記錄上的現有屬性，這些屬性可以組合起來以創建改進的模型。哪項變更將以最少的營運開銷建立所需的轉換記錄？

A. Create an AWS Lambda function that can transform the incoming records. Enable data transformation on the ingestion Kinesis Data Firehose delivery stream. Use the Lambda function as the invocation target.

A. 建立一個可以轉換傳入記錄的AWS Lambda 函數。在擷取Kinesis Data Firehose 傳輸流上啟用資料轉換。使用Lambda 函數作為呼叫目標。

B. Deploy an Amazon EMR cluster that runs Apache Spark and includes the transformation logic. Use Amazon EventBridge (Amazon CloudWatch Events) to schedule an AWS Lambda function to launch the cluster each day and transform the records that accumulate in Amazon S3. Deliver the transformed records to Amazon S3.

B. 部署執行Apache Spark 並包含轉換邏輯的Amazon EMR 叢集。使用Amazon EventBridge (Amazon CloudWatch Events) 安排AWS Lambda 函數每天啟動叢集並轉換在Amazon S3 中累積的記錄。將轉換後的記錄傳送到Amazon S3。

C. Deploy an Amazon S3 File Gateway in the stores. Update the in-store software to deliver data to the S3 File Gateway. Use a scheduled daily AWS Glue job to transform the data that the S3 File Gateway delivers to Amazon S3.

C. 在商店中部署Amazon S3 檔案閘道。更新店內軟體以將資料傳送到S3 檔案閘道。使用計劃的每日AWS Glue 作業轉換S3 檔案閘道傳送到Amazon S3 的資料。

D. Launch a fleet of Amazon EC2 instances that include the transformation logic. Configure the EC2 instances with a daily cron job to transform the records that accumulate in Amazon S3. Deliver the transformed records to Amazon S3.

D. 啟動包含轉換邏輯的Amazon EC2 執行個體佇列。使用每日cron 作業配置EC2 實例以轉換在Amazon S3 中累積的記錄。將轉換後的記錄傳送到Amazon S3。

問題# 215

A sports broadcasting company is planning to introduce subtitles in multiple languages for a live broadcast. The commentary is in English. The company needs the transcriptions to appear on screen in French or Spanish, depending on the broadcasting country. The transcriptions must be able to capture domain-specific terminology, names, and locations based on the commentary context. The company needs a solution that can support options to provide tuning data. Which combination of AWS services and features will meet these requirements with the LEAST operational overhead? (Choose two.)

一家體育廣播公司計劃在直播中引入多種語言的字幕。解說詞為英語。該公司需要轉錄文字以法語或西班牙語顯示在螢幕上，具體取決於廣播國家。轉錄文本必須能夠根據解說語境捕捉特定領域的術語、名稱和位置。該公司需要一個可以支援提供調整數據選項的解決方案。哪種AWS 服務和功能組合可以滿足這些要求且營運開銷最少？（選兩個。）

A. Amazon Transcribe with custom vocabularies.

A. 具有自訂詞彙的Amazon Transcribe。

B. Amazon Transcribe with custom language models.

B. 自訂語言模型的Amazon Transcribe。

C. Amazon SageMaker Seq2Seq.

C. Amazon SageMaker Seq2Seq。

D. Amazon SageMaker with Hugging Face Speech2Text.

D. 搭配Hugging Face Speech2Text 的Amazon SageMaker。

E. Amazon Translate.

E. Amazon Translate。

問題# 216

A data scientist at a retail company is forecasting sales for a product over the next 3 months. After preliminary analysis, the data scientist identifies that sales are seasonal and that holidays affect sales. The data scientist also determines that sales of the product are correlated with sales of other products in the same category. The data scientist needs to train a sales forecasting model that incorporates this information. Which solution will meet this requirement with the LEAST development effort?

一家零售公司的資料科學家正在預測未來3 個月內某款產品的銷售情況。經過初步分析，資料科學家發現銷售具有季節性，假日會影響銷售。資料科學家也確定該產品的銷售與同一類別的其他產品的銷售相關。資料科學家需要訓練一個包含此資訊的銷售預測模型。哪種解決方案能夠以最少的開發工作量滿足此要求？

A. Use Amazon Forecast with Holidays featurization and the built-in autoregressive integrated moving average (ARIMA) algorithm to train the model.

- A. 使用具有假期特徵的Amazon Forecast 和內建的自回歸綜合移動平均(ARIMA) 演算法來訓練模型。
- B. Use Amazon Forecast with Holidays featurization and the built-in DeepAR+ algorithm to train the model.
B. 使用具有假期特徵的Amazon Forecast 和內建的DeepAR+ 演算法來訓練模型。
- C. Use Amazon SageMaker Processing to enrich the data with holiday information. Train the model by using the SageMaker DeepAR built-in algorithm.
- C. 使用Amazon SageMaker Processing 以假期資訊豐富資料。使用SageMaker DeepAR 內建演算法訓練模型。
- D. Use Amazon SageMaker Processing to enrich the data with holiday information. Train the model by using the Gluon Time Series (GluonTS) toolkit.
- D. 使用Amazon SageMaker Processing 以假期資訊豐富資料。使用Gluon Time Series (GluonTS) 工具包訓練模型。
-

問題# 217

A company is building a predictive maintenance model for its warehouse equipment. The model must predict the probability of failure of all machines in the warehouse. The company has collected 10,000 event samples within 3 months. The event samples include 100 failure cases that are evenly distributed across 50 different machine types. How should the company prepare the data for the model to improve the model's accuracy?

一家公司正在為其倉庫設備建立預測性維護模型。此模型必須預測倉庫中所有機器發生故障的機率。該公司在3個月內收集了10,000個事件樣本。事件樣本包括100個故障案例，這些故障案例均勻分佈在50種不同的機器類型中。該公司應如何為模型準備數據以提高模型的準確性？

- A. Adjust the class weight to account for each machine type.
- A. 調整類別權重以適應每種機器類型。
- B. Oversample the failure cases by using the Synthetic Minority Oversampling Technique (SMOTE).
B. 使用合成少數過採樣技術(SMOTE) 對故障案例進行過採樣。
- C. Undersample the non-failure events. Stratify the non-failure events by machine type.
- C. 對非故障事件進行欠採樣。依機器類型對非故障事件進行分層。
- D. Undersample the non-failure events by using the Synthetic Minority Oversampling Technique (SMOTE).
D. 使用合成少數過採樣技術(SMOTE) 對非故障事件進行欠採樣。
-

問題# 218

A company stores its documents in Amazon S3 with no predefined product categories. A data scientist needs to build a machine learning model to categorize the documents for all the company's products. Which solution will meet these requirements with the MOST operational efficiency?

一家公司將其文件儲存在Amazon S3 中，沒有預先定義的產品類別。資料科學家需要建立一個機器學習模型來對公司所有產品的文檔進行分類。哪種解決方案能夠以最高的營運效率滿足這些要求？

A. Build a custom clustering model. Create a Dockerfile and build a Docker image. Register the Docker image in Amazon Elastic Container Registry (Amazon ECR). Use the custom image in Amazon SageMaker to generate a trained model.

A. 建立自訂叢集模型。建立Dockerfile 並建立Docker 映像。在Amazon Elastic Container Registry (Amazon ECR) 中註冊Docker 映像。在Amazon SageMaker 中使用自訂映像來產生經過訓練的模型。

B. Tokenize the data and transform the data into tabular data. Train an Amazon SageMaker k-means model to generate the product categories.

B. 將資料標記並將資料轉換為表格資料。訓練Amazon SageMaker k-means 模型以產生產品類別。

C. Train an Amazon SageMaker Neural Topic Model (NTM) model to generate the product categories.

C. 訓練Amazon SageMaker 神經主題模型(NTM) 模型以產生產品類別。

D. Train an Amazon SageMaker Blazing Text model to generate the product categories.

D. 訓練Amazon SageMaker Blazing Text 模型以產生產品類別。

問題# 219

A sports analytics company is providing services at a marathon. Each runner in the marathon will have their race ID printed as text on the front of their shirt. The company needs to extract race IDs from images of the runners. Which solution will meet these requirements with the LEAST operational overhead?

一家體育分析公司正在為馬拉松賽事提供服務。馬拉松賽事的每位參賽者都將在襯衫正面以文字形式印上他們的比賽ID。該公司需要從參賽者的圖像中提取比賽ID。哪種解決方案能夠以最少的營運開銷滿足這些要求？

A. Use Amazon Rekognition.

A. 使用Amazon Rekognition。

B. Use a custom convolutional neural network (CNN).

B. 使用自訂卷積神經網路(CNN)。

C. Use the Amazon SageMaker Object Detection algorithm.

C. 使用Amazon SageMaker 物件偵測演算法。

D. Use Amazon Lookout for Vision.

D. 使用Amazon Lookout for Vision。

問題# 220

A manufacturing company wants to monitor its devices for anomalous behavior. A data scientist has trained an Amazon SageMaker scikit-learn model that classifies a device as normal or anomalous based on its 4-day telemetry. The 4-day telemetry of each device is collected in a separate file and is placed in an Amazon S3 bucket once every hour. The total time to run the model across the telemetry for all devices is 5 minutes. What is the MOST cost-effective solution for the company to use to run the model across the telemetry for all the devices?

一家製造公司希望監控其設備是否有異常行為。一位資料科學家訓練了一個Amazon SageMaker scikit-learn 模型，該模型根據設備的4 天遙測資料將設備分類為正常或異常。每台裝置的4 天遙測資料都收集在一個單獨的檔案中，每小時放置在Amazon S3 儲存桶中一次。在所有設備的遙測資料上運行模型的總時間為5 分鐘。對於公司來說，在所有設備的遙測資料上運行模型最具成本效益的解決方案是什麼？

A. SageMaker Batch Transform.

A. SageMaker 批次轉換

B. SageMaker Asynchronous Inference.

B. SageMaker 非同步推理

C. SageMaker Processing.

C. SageMaker 處理

D. A SageMaker multi-container endpoint.

D. SageMaker 多容器端點。

問題# 221

A company wants to segment a large group of customers into subgroups based on shared characteristics. The company's data scientist is planning to use the Amazon SageMaker built-in k-means clustering algorithm for this task. The data scientist needs to determine the optimal number of subgroups (k) to use. Which data visualization approach will MOST accurately determine the optimal value of k?

一家公司希望根據共同特徵將一大群客戶細分為子組。該公司的資料科學家計劃使用Amazon SageMaker 內建的k-means 聚類演算法來完成這項任務。資料科學家需要確定要使用的最佳子組數量(k)。哪種資料視覺化方法可以最準確地確定k 的最佳值？

A. Calculate the principal component analysis (PCA) components. Run the k-means clustering algorithm for a range of k by using only the first two PCA components. For each value of k, create a scatter plot with a different color for each cluster. The optimal value of k is the value where the clusters start to look reasonably separated.

A. 計算主成分分析(PCA) 成分。僅使用前兩個PCA 成分對一系列k 運行k 均值聚類演算法。對於每個k 值，為每個聚類建立一個不同顏色的散佈圖。 k 的最佳值是聚類開始看起來合理分離的值。

B. Calculate the principal component analysis (PCA) components. Create a line plot of the number of components against the explained variance. The optimal value of k is the number of PCA components after which the curve starts decreasing in a linear fashion.

B. 計算主成分分析(PCA) 成分。建立成分數量與解釋變異數的線圖。 k 的最佳值是曲線開始以線性方式下降的PCA 成分數量。

C. Create a t-distributed stochastic neighbor embedding (t-SNE) plot for a range of perplexity values. The optimal value of k is the value of perplexity, where the clusters start to look reasonably separated.

C. 為一系列困惑度值建立t 分佈隨機鄰域嵌入(t-SNE) 圖。 k 的最佳值是聚類開始看起來合理分離的困惑度值。

D. Run the k-means clustering algorithm for a range of k. For each value of k, calculate the sum of squared errors (SSE). Plot a line chart of the SSE for each value of k. The optimal value of k is the point after which the curve starts decreasing in a linear fashion.

D. 對一系列k 運行k 均值聚類演算法。對於每個k 值，計算平方誤差和(SSE)。為每個k 值繪製SSE 的線圖。 k 的最佳值是曲線開始以線性方式下降的點。

問題# 222

A data scientist at a financial services company used Amazon SageMaker to train and deploy a model that predicts loan defaults. The model analyzes new loan applications and predicts the risk of loan default. To train the model, the data scientist manually extracted loan data from a database. The data scientist performed the model training and deployment steps in a Jupyter notebook that is hosted on SageMaker Studio notebooks. The model's prediction accuracy is decreasing over time. Which combination of steps is the MOST operationally efficient way for the data scientist to maintain the model's accuracy? (Choose two.)

一家金融服務公司的資料科學家使用Amazon SageMaker 訓練和部署一個預測貸款違約的模型。此模型分析新的貸款申請並預測貸款違約的風險。為了訓練模型，資料科學家手動從資料庫中提取貸款資料。資料科學家在SageMaker Studio 筆記本上託管的Jupyter 筆記本中執行模型訓練和部署步驟。該模型的預測準確性隨著時間的推移而下降。哪一種步驟組合是資料科學家保持模型準確性最有效的方法？（選兩個。）

A. Use SageMaker Pipelines to create an automated workflow that extracts fresh data, trains the model, and deploys a new version of the model.

A. 使用SageMaker Pipelines 建立自動化工作流程，擷取新資料、訓練模型並部署新版本的模型。

B. Configure SageMaker Model Monitor with an accuracy threshold to check for model drift. Initiate an Amazon CloudWatch alarm when the threshold is exceeded. Connect the workflow in SageMaker Pipelines with the CloudWatch alarm to automatically initiate retraining.

B. 使用準確度閾值配置SageMaker Model Monitor 以檢查模型漂移。超過閾值時啟動Amazon CloudWatch 警報。將SageMaker Pipelines 中的工作流程與CloudWatch 警報連接起來以自動啟動重新訓練。

C. Store the model predictions in Amazon S3. Create a daily SageMaker Processing job that reads the predictions from Amazon S3, checks for changes in model prediction accuracy, and sends an email notification if a significant change is detected.

C. 將模型預測儲存在Amazon S3 中。建立一個每日SageMaker Processing 作業，從Amazon S3 讀取預測，檢查模型預測準確性的變化，並在偵測到重大變化時發送電子郵件通知。

D. Rerun the steps in the Jupyter notebook that is hosted on SageMaker Studio notebooks to retrain the model and redeploy a new version of the model.

D. 重新運行託管在SageMaker Studio 筆記本上的Jupyter 筆記本中的步驟以重新訓練模型並重新部署新版本的模型。

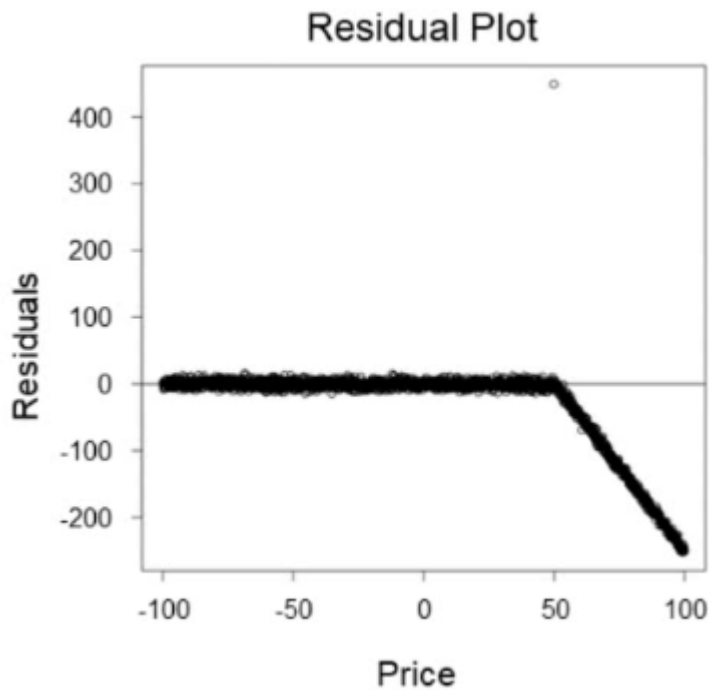
E. Export the training and deployment code from the SageMaker Studio notebooks into a Python script. Package the script into an Amazon Elastic Container Service (Amazon ECS) task that an AWS Lambda function can initiate.

E. 將SageMaker Studio 筆記本中的訓練和部署程式碼匯出到Python 腳本。將腳本打包到AWS Lambda 函數可以啟動的Amazon Elastic Container Service (Amazon ECS) 任務中。

問題# 223

A retail company wants to create a system that can predict sales based on the price of an item. A machine learning (ML) engineer built an initial linear model that resulted in the following residual plot: Which actions should the ML engineer take to improve the accuracy of the predictions in the next phase of model building? (Choose three.)

一家零售公司希望創建一個可以根據商品價格預測銷售額的系統。一位機器學習(ML) 工程師建立了一個初始線性模型，得出以下殘差圖：ML 工程師應採取哪些措施來提高模型建構下一階段的預測準確性？（選擇三項。）



A. Downsample the data uniformly to reduce the amount of data.

A. 統一對資料進行下採樣以減少資料量。

B. Create two different models for different sections of the data.

B. 為資料的不同部分建立兩個不同的模型。

C. Downsample the data in sections where $\text{Price} < 50$.

C. 對 $\text{Price} < 50$ 的部分資料進行下採樣。

D. Offset the input data by a constant value where $\text{Price} > 50$.

D. 將 $\text{Price} > 50$ 的部分資料偏移一個常數值。

E. Examine the input data, and apply non-linear data transformations where appropriate.

E. 檢查輸入數據，並在適當的情況下應用非線性數據轉換。

F. Use a non-linear model instead of a linear model.

F. 使用非線性模型取代線性模型。

問題# 224

A data scientist at a food production company wants to use an Amazon SageMaker built-in model to classify different vegetables. The current dataset has many features. The company wants to save on memory costs when the data scientist trains and deploys the model. The company also wants to be able to find similar data points for each test data point. Which algorithm will meet these requirements?

一家食品生產公司的資料科學家希望使用Amazon SageMaker 內建模型對不同的蔬菜進行分類。

目前資料集具有許多特徵。該公司希望在資料科學家訓練和部署模型時節省記憶體成本。該公司還希望能夠為每個測試數據點找到相似的數據點。哪種演算法可以滿足這些要求？

A. K-nearest neighbors (k-NN) with dimension reduction.

A. 具有降維功能的K 最近鄰(k-NN)。

B. Linear learner with early stopping.

B. 具有早期停止功能的線性學習器。

C. K-means.

C. K 均值。

D. Principal component analysis (PCA) with the algorithm mode set to random.

D. 演算法模式設定為隨機的主成分分析(PCA)。

問題# 225

A data scientist is training a large PyTorch model by using Amazon SageMaker. It takes 10 hours on average to train the model on GPU instances. The data scientist suspects that training is not converging and that resource utilization is not optimal. What should the data scientist do to identify and address training issues with the LEAST development effort?

一位資料科學家正在使用Amazon SageMaker 訓練大型PyTorch 模型。在GPU 實例上訓練模型平均需要10 小時。資料科學家懷疑訓練沒有收斂，資源利用率也不是最優的。資料科學家應該怎麼做才能以最少的開發工作量來識別和解決訓練問題？

A. Use CPU utilization metrics that are captured in Amazon CloudWatch. Configure a CloudWatch alarm to stop the training job early if low CPU utilization occurs.

A. 使用在Amazon CloudWatch 中捕獲的CPU 使用率指標。配置CloudWatch 警報以在CPU 使用率較低時提前停止訓練作業。

B. Use high-resolution custom metrics that are captured in Amazon CloudWatch. Configure an AWS Lambda function to analyze the metrics and to stop the training job early if issues are detected.

B. 使用在Amazon CloudWatch 中捕獲的高精度自訂指標。配置AWS Lambda 函數以分析指標並在偵測到問題時提前停止訓練作業。

C. Use the SageMaker Debugger `vanishing_gradient` and `LowGPUUtilization` built-in rules to detect issues and to launch the `StopTrainingJob` action if issues are detected.

C. 使用SageMaker Debugger `vanishing_gradient` 和 `LowGPUUtilization` 內建規則偵測問題並在偵測到問題時啟動`StopTrainingJob` 操作。

D. Use the SageMaker Debugger `confusion` and `feature_importance_overweight` built-in rules to detect issues and to launch the `StopTrainingJob` action if issues are detected.

D. 使用SageMaker Debugger confused 和feature_importance_overweight 內建規則偵測問題並在偵測到問題時啟動StopTrainingJob 操作。

問題# 226

A bank wants to launch a low-rate credit promotion campaign. The bank must identify which customers to target with the promotion and wants to make sure that each customer's full credit history is considered when an approval or denial decision is made. The bank's data science team used the XGBoost algorithm to train a classification model based on account transaction features. The data science team deployed the model by using the Amazon SageMaker model hosting service. The accuracy of the model is sufficient, but the data science team wants to be able to explain why the model denies the promotion to some customers. What should the data science team do to meet this requirement in the MOST operationally efficient manner?

一家銀行希望進行低利率信貸促銷活動。銀行必須確定促銷活動的目標客戶，並希望確保在做出批准或拒絕決定時考慮每個客戶的完整信用記錄。銀行的資料科學團隊使用XGBoost 演算法根據帳戶交易特徵訓練分類模型。資料科學團隊使用Amazon SageMaker 模型託管服務部署了此模型。該模型的準確性足夠高，但數據科學團隊希望能夠解釋為什麼該模型拒絕向某些客戶提供促銷。數據科學團隊該怎麼做才能以最有效率的方式滿足這項要求？

A. Create a SageMaker notebook instance. Upload the model artifact to the notebook. Use the plot_importance() method in the Python XGBoost interface to create a feature importance chart for the individual predictions.

A. 建立SageMaker 筆記本實例。將模型工件上傳到筆記本。使用Python XGBoost 介面中的plot_importance() 方法為各個預測建立特徵重要性圖表。

B. Retrain the model by using SageMaker Debugger. Configure Debugger to calculate and collect Shapley values. Create a chart that shows features and SHapley. Additive explanations (SHAP) values to explain how the features affect the model outcomes.

B. 使用SageMaker 偵錯器重新訓練模型。配置偵錯器以計算和收集Shapley 值。建立一個顯示特徵和SHapley 的圖表。附加解釋(SHAP) 值以解釋特徵如何影響模型結果。

C. Set up and run an explainability job powered by SageMaker Clarify to analyze the individual customer data, using the training data as a baseline. Create a chart that shows features and SHapley Additive explanations (SHAP) values to explain how the features affect the model outcomes.

C. 設定並執行由SageMaker Clarify 提供支援的可解釋性作業以分析單一客戶數據，並使用訓練數據作為基線。建立一個顯示特徵和SHapley 附加解釋(SHAP) 值的圖表以解釋特徵如何影響模型結果。

D. Use SageMaker Model Monitor to create Shapley values that help explain model behavior. Store the Shapley values in Amazon S3. Create a chart that shows features and SHapley Additive explanations (SHAP) values to explain how the features affect the model outcomes.

D. 使用SageMaker 模型監視器建立有助於解釋模型行為的Shapley 值。將Shapley 值儲存在Amazon S3 中。建立一個顯示特徵和SHapley 附加解釋(SHAP) 值的圖表以解釋特徵如何影響模型結果。

問題# 227

A company has hired a data scientist to create a loan risk model. The dataset contains loan amounts and variables such as loan type, region, and other demographic variables. The data scientist wants to use Amazon SageMaker to test bias regarding the loan amount distribution with respect to some of these categorical variables. Which pretraining bias metrics should the data scientist use to check the bias distribution? (Choose three.)

一家公司聘請了一位資料科學家來建立貸款風險模型。資料集包含貸款金額和變量，例如貸款類型、地區和其他人口統計變數。資料科學家希望使用Amazon SageMaker 來測試貸款金額分佈相對於其中一些分類變數的偏差。資料科學家應該使用哪些預訓練偏差指標來檢查偏差分佈？（選三個。）

A. Class imbalance.

A. 類別不平衡。

B. Conditional demographic disparity.

B. 條件人口差異。

C. Difference in proportions of labels.

C. 標籤比例差異。

D. Jensen-Shannon divergence.

D. Jensen-Shannon 散度。

E. Kullback-Leibler divergence.

E. Kullback-Leibler 散度。

F. Total variation distance.

F. 總變異距離。

問題# 228

A retail company wants to use Amazon Forecast to predict daily stock levels of inventory. The cost of running out of items in stock is much higher for the company than the cost of having excess inventory. The company has millions of data samples for multiple years for thousands of items. The company's purchasing department needs to predict demand for 30-day cycles for each item to ensure that restocking occurs. A machine learning (ML) specialist wants to use item-related features such as "category," "brand," and "safety stock count." The ML specialist also wants to use a binary time series feature that has "promotion applied?" as its name. Future promotion information is available only for the next 5 days. The ML specialist must choose an algorithm and an evaluation metric for a solution to produce prediction results that will maximize company profit. Which solution will meet these requirements?

一家零售公司希望使用Amazon Forecast 預測每日庫存水準。對公司而言，庫存商品缺貨的成本遠高於庫存過剩的成本。該公司擁有數千種商品的數百萬個數據樣本，這些樣本已經存在多年。該公司的採購部門需要預測每種商品30 天週期的需求，以確保補貨。機器學習(ML) 專家希望使用與商

品相關的功能，例如「類別」、「品牌」和「安全庫存數量」。ML 專家也希望使用名為「是否套用促銷？」的二進位時間序列功能。未來促銷訊息僅在接下來的5 天內可用。ML 專家必須選擇演算法和評估指標，以便解決方案產生能夠最大化公司利潤的預測結果。哪種解決方案可以滿足這些要求？

A. Train a model by using the Autoregressive Integrated Moving Average (ARIMA) algorithm. Evaluate the model by using the Weighted Quantile Loss (wQL) metric at 0.75 (P75).

A. 使用自迴歸綜合移動平均(ARIMA) 演算法訓練模型。使用0.75 (P75) 處的加權分位數損失(wQL) 指標評估模型。

B. Train a model by using the Autoregressive Integrated Moving Average (ARIMA) algorithm. Evaluate the model by using the Weighted Absolute Percentage Error (WAPE) metric.

B. 使用自迴歸綜合移動平均(ARIMA) 演算法訓練模型。使用加權絕對百分比誤差(WAPE) 指標評估模型。

C. Train a model by using the Convolutional Neural Network - Quantile Regression (CNN-QR) algorithm. Evaluate the model by using the Weighted Quantile Loss (wQL) metric at 0.75 (P75).

C. 使用卷積神經網路- 分位數迴歸(CNN-QR) 演算法訓練模型。使用0.75 (P75) 處的加權分位數損失(wQL) 指標評估模型。

D. Train a model by using the Convolutional Neural Network - Quantile Regression (CNN-QR) algorithm. Evaluate the model by using the Weighted Absolute Percentage Error (WAPE) metric.

D. 使用卷積神經網路- 分位數迴歸(CNN-QR) 演算法訓練模型。使用加權絕對百分比誤差(WAPE) 指標評估模型。

問題# 229

An online retail company wants to develop a natural language processing (NLP) model to improve customer service. A machine learning (ML) specialist is setting up distributed training of a Bidirectional Encoder Representations from Transformers (BERT) model on Amazon SageMaker. SageMaker will use eight compute instances for the distributed training. The ML specialist wants to ensure the security of the data during the distributed training. The data is stored in an Amazon S3 bucket. Which combination of steps should the ML specialist take to protect the data during the distributed training? (Choose three.)

一家線上零售公司希望開發一種自然語言處理(NLP) 模型來改善客戶服務。一位機器學習(ML) 專家正在Amazon SageMaker 上設定Transformers 雙向編碼器表示(BERT) 模型的分散式訓練。Sage Maker 將使用八個運算實例進行分散式訓練。ML 專家希望確保分散式訓練期間資料的安全性。資料儲存在Amazon S3 儲存桶中。ML 專家應採取哪些步驟組合來保護分散式訓練期間的資料？（選三個。）

A. Run distributed training jobs in a private VPC. Enable inter-container traffic encryption.

A. 在私有VPC 中執行分散式訓練作業。啟用容器間流量加密。

B. Run distributed training jobs across multiple VPCs. Enable VPC peering.

B. 跨多個VPC 執行分散式訓練作業。啟用VPC 對等連線。

C. Create an S3 VPC endpoint. Then configure network routes, endpoint policies, and S3 bucket policies.

C. 建立S3 VPC 終端節點。然後配置網路路由、終端節點策略和S3 儲存桶策略。

D. Grant read-only access to SageMaker resources by using an IAM role.

D. 使用IAM 角色授予對SageMaker 資源的唯讀存取權。

E. Create a NAT gateway. Assign an Elastic IP address for the NAT gateway.

E. 建立NAT 閘道。為NAT 網關指派彈性IP 位址。

F. Configure an inbound rule to allow traffic from a security group that is associated with the training instances.

F. 配置入站規則以允許來自與訓練執行個體關聯的安全群組的流量。

問題# 230

An analytics company has an Amazon SageMaker hosted endpoint for an image classification model. The model is a custom-built convolutional neural network (CNN) and uses the PyTorch deep learning framework. The company wants to increase throughput and decrease latency for customers that use the model. Which solution will meet these requirements MOST cost-effectively?

一家分析公司擁有一個用於影像分類模型的Amazon SageMaker 託管終端節點。該模型是一個客製化的捲積神經網路(CNN)，使用PyTorch 深度學習框架。該公司希望提高吞吐量並降低使用該模型的客戶的延遲。哪一種解決方案能夠以最具成本效益的方式滿足這些要求？

A. Use Amazon Elastic Inference on the SageMaker hosted endpoint.

A. 在SageMaker 託管終端節點上使用Amazon Elastic Inference。

B. Retrain the CNN with more layers and a larger dataset.

B. 使用更多層和更大的資料集重新訓練CNN。

C. Retrain the CNN with more layers and a smaller dataset.

C. 使用更多層和更小的資料集重新訓練CNN。

D. Choose a SageMaker instance type that has multiple GPUs.

D. 選擇具有多個GPU 的SageMaker 實例類型。

問題# 231

An ecommerce company is collecting structured data and unstructured data from its website, mobile apps, and IoT devices. The data is stored in several databases and Amazon S3 buckets. The company is implementing a scalable repository to store structured data and unstructured data. The company must implement a solution that provides a central data catalog, self-service access to the data, and granular data access policies and encryption to protect the data. Which combination of actions will meet these requirements with the LEAST amount of setup? (Choose three.)

一家電子商務公司正在從其網站、行動應用程式和物聯網設備收集結構化資料和非結構化資料。資料儲存在多個資料庫和Amazon S3 儲存桶中。該公司正在實施一個可擴展的儲存庫來儲存結構化資料和非結構化資料。該公司必須實施一個解決方案，提供中央資料目錄、自助存取資料以及細粒度的資料存取策略和加密以保護資料。哪種操作組合可以滿足這些要求且設定工作量最少？（選擇三項。）

A. Identify the existing data in the databases and S3 buckets. Link the data to AWS Lake Formation.

A. 識別資料庫和S3 儲存桶中的現有資料。將資料連結到AWS Lake Formation。

B. Identify the existing data in the databases and S3 buckets. Link the data to AWS Glue.

B. 識別資料庫和S3 儲存桶中的現有資料。將資料連結到AWS Glue。

C. Run AWS Glue crawlers on the linked data sources to create a central data catalog.

C. 在連結的資料來源上執行AWS Glue 爬網程式以建立中央資料目錄。

D. Apply granular access policies by using AWS Identity and Access Management (IAM). Configure server-side encryption on each data source.

D. 使用AWS Identity and Access Management (IAM) 應用細粒度存取策略。在每個資料來源上設定伺服器端加密。

E. Apply granular access policies and encryption by using AWS Lake Formation.

E. 使用AWS Lake Formation 應用細粒度存取策略和加密。

F. Apply granular access policies and encryption by using AWS Glue.

F. 使用AWS Glue 應用細粒度存取策略和加密。

問題# 232

A machine learning (ML) specialist is developing a deep learning sentiment analysis model that is based on data from movie reviews. After the ML specialist trains the model and reviews the model results on the validation set, the ML specialist discovers that the model is overfitting. Which solutions will MOST improve the model generalization and reduce overfitting? (Choose three.)

一位機器學習(ML) 專家正在開發一種基於電影評論資料的深度學習情緒分析模型。在ML 專家訓練模型並在驗證集上查看模型結果後，ML 專家發現該模型過度擬合。哪些解決方案最能提高模型泛化能力並減少過度擬合？（選擇三項。）

A. Shuffle the dataset with a different seed.

A. 使用不同的種子對資料集進行打亂。

B. Decrease the learning rate.

B. 降低學習率。

C. Increase the number of layers in the network.

C. 增加網路中的層數。

D. Add L1 regularization and L2 regularization.

D. 新增L1 正規化和L2 正規化。

E. Add dropout.

E. 新增dropout。

F. Decrease the number of layers in the network.

F. 減少網路中的層數。

問題# 233

An online advertising company is developing a linear model to predict the bid price of advertisements in real time with low-latency predictions. A data scientist has trained the linear model by using many features, but the model is overfitting the training dataset. The data scientist needs to prevent overfitting and must reduce the number of features. Which solution will meet these requirements?

一家線上廣告公司正在開發一種線性模型，以低延遲預測即時廣告的出價。一位資料科學家使用許多特徵訓練了線性模型，但該模型對訓練資料集過度擬合。資料科學家需要防止過度擬合，並且必須減少特徵數量。哪種解決方案可以滿足這些要求？

A. Retrain the model with L1 regularization applied.

A. 應用L1 正規化重新訓練模型。

B. Retrain the model with L2 regularization applied.

B. 應用L2 正規化重新訓練模型。

C. Retrain the model with dropout regularization applied.

C. 應用dropout 正規化重新訓練模型。

D. Retrain the model by using more data.

D. 使用更多資料重新訓練模型。

問題# 234

A credit card company wants to identify fraudulent transactions in real time. A data scientist builds a machine learning model for this purpose. The transactional data is captured and stored in Amazon S3. The historic data is already labeled with two classes: fraud (positive) and fair transactions (negative). The data scientist removes all the missing data and builds a classifier by using the XGBoost algorithm in Amazon SageMaker. The model produces the following results:

0.300

accuracy: 0.949 Which solution should the data scientist use to improve the performance of the model?

一家信用卡公司希望即時識別詐欺交易。一位資料科學家為此建構了一個機器學習模型。交易資料被捕獲並儲存在Amazon S3 中。歷史資料已標記為兩個類別：詐欺（正面）和公平交易（負面）。

數據科學家刪除所有缺失數據，並使用Amazon SageMaker 中的XGBoost 演算法建立分類器。

此模型產生以下結果：

- 真實例率(TPR)：0.700
- 假陰性率(FNR)：0.300
- 真陰性率(TNR)：0.977
- 假陽性率(FPR)：0.023
- 整體準確率：0.949

資料科學家應使用哪種解決方案來提高模型的效能？

A. Apply the Synthetic Minority Oversampling Technique (SMOTE) on the minority class in the training dataset. Retrain the model with the updated training data.

A. 對訓練資料集中的少數類應用合成少數類過採樣技術（SMOTE）。使用更新的訓練資料重新訓練模型。

B. Apply the Synthetic Minority Oversampling Technique (SMOTE) on the majority class in the training dataset. Retrain the model with the updated training data.

B. 對訓練資料集中的多數類應用合成少數類過採樣技術（SMOTE）。使用更新的訓練資料重新訓練模型。

C. Undersample the minority class.

C. 對少數類別進行欠採樣。

D. Oversample the majority class.

D. 對多數類別進行過採樣。

問題# 235

A company is training machine learning (ML) models on Amazon SageMaker by using 200 TB of data that is stored in Amazon S3 buckets. The training data consists of individual files that are each larger than 200 MB in size. The company needs a data access solution that offers the shortest processing time and the least amount of setup. Which solution will meet these requirements?

一家公司正在使用儲存在Amazon S3 儲存桶中的200 TB 資料在Amazon SageMaker 上訓練機器學習(ML) 模型。訓練資料由單一檔案組成，每個檔案的大小都大於200 MB。該公司需要一種提供最短處理時間和最少設定量的資料存取解決方案。哪種解決方案可以滿足這些要求？

A. Use File mode in SageMaker to copy the dataset from the S3 buckets to the ML instance storage.

A. 使用SageMaker 中的檔案模式將資料集從S3 儲存桶複製到ML 實例儲存。

B. Create an Amazon FSx for Lustre file system. Link the file system to the S3 buckets.

B. 建立Amazon FSx for Lustre 檔案系統。將檔案系統連結到S3 儲存桶。

C. Create an Amazon Elastic File System (Amazon EFS) file system. Mount the file system to the training instances.

C. 建立Amazon Elastic File System (Amazon EFS) 檔案系統。將檔案系統掛載到訓練實例。

D. Use FastFile mode in SageMaker to stream the files on demand from the S3 buckets.

D. 使用SageMaker 中的FastFile 模式按需從S3 儲存桶串流傳輸檔案。

問題# 236

An online store is predicting future book sales by using a linear regression model that is based on past sales data. The data includes duration, a numerical feature that represents the number of days that a book has been listed in the online store. A data scientist performs an exploratory data analysis and discovers that the relationship between book sales and duration is skewed and non-linear. Which data transformation step should the data scientist take to improve the predictions of the model?

一家網路商店正在使用基於過去銷售數據的線性迴歸模型來預測未來的圖書銷售量。資料包括持續時間，這是一個數值特徵，表示一本書在網路商店上架的天數。一位資料科學家進行了探索性資料分析，發現圖書銷售量和持續時間之間的關係是傾斜的和非線性的。資料科學家應該採取哪個資料轉換步驟來改善模型的預測？

A. One-hot encoding.

A. 獨熱編碼。

B. Cartesian product transformation.

B. 笛卡爾積變換。

C. Quantile binning.

C. 分位數分箱。

D. Normalization.

D. 標準化。

問題# 237

A company's data engineer wants to use Amazon S3 to share datasets with data scientists. The data scientists work in three departments: Finance, Marketing, and Human Resources. Each department has its own IAM user group. Some datasets contain sensitive information and should be accessed only by the data scientists from the Finance department. How can the data engineer set up access to meet these requirements?

一家公司的資料工程師希望使用Amazon S3 與資料科學家共享資料集。資料科學家在三個部門工

作：財務、行銷和人力資源。每個部門都有自己的IAM 使用者群組。有些資料集包含敏感訊息，只能由財務部門的資料科學家存取。資料工程師如何設定存取權限以滿足這些要求？

A. Create an S3 bucket for each dataset. Create an ACL for each S3 bucket. For each S3 bucket that contains a sensitive dataset, set the ACL to allow access only from the Finance department user group. Allow all three department user groups to access each S3 bucket that contains a non-sensitive dataset.

A. 為每個資料集建立一個S3 儲存桶。為每個S3 儲存桶建立一個ACL。對於包含敏感資料集的每個S3 儲存桶，將ACL 設定為僅允許財務部門使用者群組存取。允許所有三個部門使用者群組存取包含非敏感資料集的每個S3 儲存桶。

B. Create an S3 bucket for each dataset. For each S3 bucket that contains a sensitive dataset, set the bucket policy to allow access only from the Finance department user group. Allow all three department user groups to access each S3 bucket that contains a non-sensitive dataset.

B. 為每個資料集建立一個S3 儲存桶。對於包含敏感資料集的每個S3 儲存桶，將儲存桶策略設定為僅允許財務部門使用者群組存取。允許所有三個部門使用者群組存取包含非敏感資料集的每個S3 儲存桶。

C. Create a single S3 bucket that includes two folders to separate the sensitive datasets from the non-sensitive datasets. For the Finance department user group, attach an IAM policy that provides access to both folders. For the Marketing and Human Resources department user groups, attach an IAM policy that provides access to only the folder that contains the non-sensitive datasets.

C. 建立一個包含兩個資料夾的S3 儲存桶，以將敏感資料集與非敏感資料集分開。對於財務部門使用者群組，附加提供兩個資料夾存取權限的IAM 策略。對於行銷和人力資源部門使用者群組，附加僅提供對包含非敏感資料集的資料夾存取權限的IAM 策略。

D. Create a single S3 bucket that includes two folders to separate the sensitive datasets from the non-sensitive datasets. Set the policy for the S3 bucket to allow only the Finance department user group to access the folder that contains the sensitive datasets. Allow all three department user groups to access the folder that contains the non-sensitive datasets.

D. 建立一個包含兩個資料夾的S3 儲存桶，以將敏感資料集與非敏感資料集分開。設定S3 儲存桶的策略，僅允許財務部門使用者群組存取包含敏感資料集的資料夾。允許所有三個部門使用者群組存取包含非敏感資料集的資料夾。

問題# 238

A company operates an amusement park. The company wants to collect, monitor, and store real-time traffic data at several park entrances by using strategically placed cameras. The company's security team must be able to immediately access the data for viewing. Stored data must be indexed and must be accessible to the company's data science team. Which solution will meet these requirements MOST cost-effectively?

一家公司經營遊樂園。該公司希望透過使用策略性放置的攝影機來收集、監控和儲存多個遊樂園入口的即時交通數據。公司的安全團隊必須能夠立即存取這些數據以供查看。儲存的資料必須編入索引，並且必須可供公司的資料科學團隊存取。哪一種解決方案能夠以最具成本效益的方式滿足這些要求？

A. Use Amazon Kinesis Video Streams to ingest, index, and store the data. Use the built-in integration with Amazon Rekognition for viewing by the security team.

A. 使用Amazon Kinesis Video Streams 提取、索引和儲存資料。使用內建的Amazon Rekognition 集成，供安全團隊查看。

B. Use Amazon Kinesis Video Streams to ingest, index, and store the data. Use the built-in HTTP live streaming (HLS) capability for viewing by the security team.

B. 使用Amazon Kinesis Video Streams 提取、索引和儲存資料。使用內建的HTTP 即時串流(HLS) 功能，供安全團隊查看。

C. Use Amazon Rekognition Video and the GStreamer plugin to ingest the data for viewing by the security team. Use Amazon Kinesis Data Streams to index and store the data.

C. 使用Amazon Rekognition Video 和GStreamer 插件提取數據，供安全團隊查看。使用Amazon Kinesis Data Streams 索引和儲存資料。

D. Use Amazon Kinesis Data Firehose to ingest, index, and store the data. Use the built-in HTTP live streaming (HLS) capability for viewing by the security team.

D. 使用Amazon Kinesis Data Firehose 提取、索引和儲存資料。使用內建的HTTP 即時串流(HLS) 功能，供安全團隊查看。

問題# 239

An engraving company wants to automate its quality control process for plaques. The company performs the process before mailing each customized plaque to a customer. The company has created an Amazon S3 bucket that contains images of defects that should cause a plaque to be rejected. Low-confidence predictions must be sent to an internal team of reviewers who are using Amazon Augmented AI (Amazon A2I). Which solution will meet these requirements?

一家雕刻公司希望實現其牌匾品質控制流程的自動化。該公司在將每個客製化牌匾郵寄給客戶之前執行該流程。該公司創建了一個Amazon S3 儲存桶，其中包含可能導致牌匾被拒收的缺陷圖像。低置信度預測必須寄給使用Amazon Augmented AI (Amazon A2I) 的內部審核團隊。哪種解決方案可以滿足這些要求？

A. Use Amazon Textract for automatic processing. Use Amazon A2I with Amazon Mechanical Turk for manual review.

A. 使用Amazon Textract 進行自動處理。使用Amazon A2I 和Amazon Mechanical Turk 進行手動審核。

B. Use Amazon Rekognition for automatic processing. Use Amazon A2I with a private workforce option for manual review.

B. 使用Amazon Rekognition 進行自動處理。使用Amazon A2I 和私人員工選項進行人工審核。

C. Use Amazon Transcribe for automatic processing. Use Amazon A2I with a private workforce option for manual review.

C. 使用Amazon Transcribe 進行自動處理。使用Amazon A2I 和私人員工選項進行人工審核。

D. Use AWS Panorama for automatic processing. Use Amazon A2I with Amazon Mechanical Turk for manual review.

D. 使用AWS Panorama 進行自動處理。使用Amazon A2I 和Amazon Mechanical Turk 進行手動審核。

問題# 240

A machine learning (ML) engineer at a bank is building a data ingestion solution to provide transaction features to financial ML models. Raw transactional data is available in an Amazon Kinesis data stream. The solution must compute rolling averages of the ingested data from the data stream and must store the results in Amazon SageMaker Feature Store. The solution also must serve the results to the models in near real time. Which solution will meet these requirements?

一家銀行的機器學習(ML)工程師正在建立資料擷取解決方案，以便為金融ML模型提供交易功能。原始交易資料可在Amazon Kinesis 資料流中使用。此解決方案必須計算從資料流中提取的資料的捲動平均值，並且必須將結果儲存在Amazon SageMaker Feature Store 中。該解決方案還必須近乎即時地將結果提供給模型。哪種解決方案可以滿足這些要求？

A. Load the data into an Amazon S3 bucket by using Amazon Kinesis Data Firehose. Use a SageMaker Processing job to aggregate the data and to load the results into SageMaker Feature Store as an online feature group.

A. 使用Amazon Kinesis Data Firehose 將資料載入到Amazon S3 儲存桶中。使用SageMaker 處理作業聚合資料並將結果作為線上功能群組載入到SageMaker 功能儲存中。

B. Write the data directly from the data stream into SageMaker Feature Store as an online feature group. Calculate the rolling averages in place within SageMaker Feature Store by using the SageMaker GetRecord API operation.

B. 將資料從資料流直接寫入SageMaker 功能儲存作為線上功能組。使用SageMaker GetRecord API 操作在SageMaker 功能儲存中計算滾動平均值。

C. Consume the data stream by using an Amazon Kinesis Data Analytics SQL application that calculates the rolling averages. Generate a result stream. Consume the result stream by using a custom AWS Lambda function that publishes the results to SageMaker Feature Store as an online feature group.

C. 使用計算滾動平均值的Amazon Kinesis Data Analytics SQL 應用程式使用資料流。產生結果流。使用自訂AWS Lambda 函數使用結果流，該函數將結果作為線上功能組發佈到SageMaker 功能儲存。

D. Load the data into an Amazon S3 bucket by using Amazon Kinesis Data Firehose. Use a SageMaker Processing job to load the data into SageMaker Feature Store as an offline feature group. Compute the rolling averages at query time.

D. 使用Amazon Kinesis Data Firehose 將資料載入到Amazon S3 儲存桶中。使用SageMaker 處理作業將資料作為離線功能組載入到SageMaker 功能儲存中。在查詢時計算滾動平均值。

問題# 241

Each morning, a data scientist at a rental car company creates insights about the previous day's rental car reservation demands. The company needs to automate this process by streaming the data to Amazon S3 in near real time. The solution must detect high-demand rental cars at each of the company's locations. The solution also must create a visualization dashboard that automatically refreshes with the most recent data. Which solution will meet these requirements with the LEAST development time?

每天早上，租車公司的資料科學家都會對前一天的租車預訂需求進行分析。該公司需要透過近乎即時地將資料串流傳輸到Amazon S3 來實現此流程的自動化。此解決方案必須檢測公司每個地點的高需求租車。該解決方案還必須建立一個視覺化儀表板，該儀表板會自動刷新最新資料。哪種解決方案能夠以最少的開發時間滿足這些要求？

A. Use Amazon Kinesis Data Firehose to stream the reservation data directly to Amazon S3. Detect high-demand outliers by using Amazon QuickSight ML Insights. Visualize the data in QuickSight.

A. 使用Amazon Kinesis Data Firehose 將預留資料直接串流到Amazon S3。使用Amazon QuickSight ML Insights 偵測高需求異常值。在QuickSight 中可視化資料。

B. Use Amazon Kinesis Data Streams to stream the reservation data directly to Amazon S3. Detect high-demand outliers by using the Random Cut Forest (RCF) trained model in Amazon SageMaker. Visualize the data in Amazon QuickSight.

B. 使用Amazon Kinesis Data Streams 將預留資料直接串流到Amazon S3。使用Amazon SageMaker 中的隨機砍伐森林(RCF) 訓練模型偵測高需求異常值。在Amazon QuickSight 中視覺化資料。

C. Use Amazon Kinesis Data Firehose to stream the reservation data directly to Amazon S3. Detect high-demand outliers by using the Random Cut Forest (RCF) trained model in Amazon SageMaker. Visualize the data in Amazon QuickSight.

C. 使用Amazon Kinesis Data Firehose 將預留資料直接串流到Amazon S3。使用Amazon SageMaker 中的隨機砍伐森林(RCF) 訓練模型偵測高需求異常值。在Amazon QuickSight 中視覺化資料。

D. Use Amazon Kinesis Data Streams to stream the reservation data directly to Amazon S3. Detect high-demand outliers by using Amazon QuickSight ML Insights. Visualize the data in QuickSight.

D. 使用Amazon Kinesis Data Streams 將預留資料直接串流到Amazon S3。使用Amazon QuickSight ML Insights 偵測高需求異常值。在QuickSight 中可視化資料。

問題# 246

A machine learning (ML) engineer is integrating a production model with a customer metadata repository for real-time inference. The repository is hosted in Amazon SageMaker Feature Store. The engineer wants to retrieve only the latest version of the customer metadata record for a single customer at a time. Which solution will meet these requirements?

一位機器學習(ML) 工程師正在將生產模型與客戶元資料儲存庫整合以進行即時推理。此儲存庫託管在Amazon SageMaker Feature Store 中。這位工程師希望一次只檢索單一客戶的最新版本的客戶元資料記錄。哪種解決方案可以滿足這些要求？

A. Use the SageMaker Feature Store BatchGetRecord API with the record identifier. Filter to find the latest record.

A. 使用帶有記錄標識符的SageMaker Feature Store BatchGetRecord API。過濾以尋找最新記錄

。

B. Create an Amazon Athena query to retrieve the data from the feature table.

B. 建立Amazon Athena 查詢以從功能表中檢索資料。

C. Create an Amazon Athena query to retrieve the data from the feature table. Use the write_time value to find the latest record.

C. 建立Amazon Athena 查詢以從功能表中檢索資料。使用write_time 值尋找最新記錄。

D. Use the SageMaker Feature Store GetRecord API with the record identifier.

D. 使用帶有記錄標識符的SageMaker Feature Store GetRecord API。

問題# 247

A company's data scientist has trained a new machine learning model that performs better on test data than the company's existing model performs in the production environment. The data scientist wants to replace the existing model that runs on an Amazon SageMaker endpoint in the production environment. However, the company is concerned that the new model might not work well on the production environment data. The data scientist needs to perform A/B testing in the production environment to evaluate whether the new model performs well on production environment data. Which combination of steps must the data scientist take to perform the A/B testing? (Choose two.)

一家公司的資料科學家訓練了一種新的機器學習模型，該模型在測試資料上的表現優於公司現有模型在生產環境中的表現。資料科學家想要替換在生產環境中在Amazon SageMaker 終端節點上運行的現有模型。但是，該公司擔心新模型可能無法很好地處理生產環境數據。資料科學家需要在生產環境中執行A/B 測試，以評估新模型在生產環境資料上的表現是否良好。資料科學家必須採取哪些步驟組合來執行A/B 測試？（選兩個。）

A. Create a new endpoint configuration that includes a production variant for each of the two models.

A. 建立一個包含兩個模型各自生產變體的新端點配置。

B. Create a new endpoint configuration that includes two target variants that point to different endpoints.

B. 建立一個包含指向不同端點的兩個目標變體的新端點配置。

C. Deploy the new model to the existing endpoint.

C. 將新模型部署到現有端點。

D. Update the existing endpoint to activate the new model.

D. 更新現有端點以啟動新模型。

E. Update the existing endpoint to use the new endpoint configuration.

E. 更新現有端點以使用新端點配置。

問題# 248

A data scientist is working on a forecast problem by using a dataset that consists of.csv files that are stored in Amazon S3. The files contain a timestamp variable in the following format:March 1st, 2020, 08:14pm -There is a hypothesis about seasonal differences in the dependent variable. This number could be higher or lower for weekdays because some days and hours present varying values, so the day of the week, month, or hour could be an important factor. As a result, the data scientist needs to transform the timestamp into weekdays, month, and day as three separate variables to conduct an analysis.Which solution requires the LEAST operational overhead to create a new dataset with the added features?

一位資料科學家正在使用由儲存在Amazon S3 中的.csv 檔案組成的資料集來解決預測問題。這些文件包含以下格式的時間戳變數：2020 年3 月1 日，晚上8:14 - 假設因變數存在季節性差異。工作日的這個數字可能會更高或更低，因為有些日子和幾小時的值會有所不同，因此星期幾、月份或小時可能是一個重要因素。因此，資料科學家需要將時間戳轉換為工作日、月份和日期作為三個單獨的變數來進行分析。哪種解決方案需要最少的營運開銷來建立具有附加功能的新資料集？

A. Create an Amazon EMR cluster. Develop PySpark code that can read the timestamp variable as a string, transform and create the new variables, and save the dataset as a new file in Amazon S3.

A. 建立Amazon EMR 群集。開發PySpark 程式碼，該程式碼可以將時間戳變數讀取為字串，轉換並建立新變量，並將資料集保存為Amazon S3 中的新檔案。

B. Create a processing job in Amazon SageMaker. Develop Python code that can read the timestamp variable as a string, transform and create the new variables, and save the dataset as a new file in Amazon S3.

B. 在Amazon SageMaker 中建立處理作業。開發Python 程式碼，該程式碼可以將時間戳變數讀取為字串，轉換並建立新變量，並將資料集保存為Amazon S3 中的新檔案。

C. Create a new flow in Amazon SageMaker Data Wrangler. Import the S3 file, use the Featurize date/time transform to generate the new variables, and save the dataset as a new file in Amazon S3.

C. 在Amazon SageMaker Data Wrangler 中建立新流程。匯入S3 文件，使用Featurize 日期/時間轉換產生新變量，並將資料集儲存為Amazon S3 中的新文件。

D. Create an AWS Glue job. Develop code that can read the timestamp variable as a string, transform and create the new variables, and save the dataset as a new file in Amazon S3.

D. 建立AWS Glue 作業。開發程式碼，該程式碼可以將時間戳變數讀取為字串，轉換並建立新變量，並將資料集儲存為Amazon S3 中的新檔案。

問題# 249

A manufacturing company has a production line with sensors that collect hundreds of quality metrics. The company has stored sensor data and manual inspection results in a data lake for several months. To automate quality control, the machine learning team must build an automated mechanism that determines whether the produced goods are good quality, replacement market quality, or scrap quality based on the manual inspection results.Which modeling approach will deliver the MOST accurate prediction of product quality?

一家製造公司擁有一條生產線，生產線上裝有感測器，可收集數百個品質指標。該公司已將感測器資料和人工檢查結果儲存在資料湖中數月。為了實現品質控制自動化，機器學習團隊必須建立一個

自動化機制，根據人工檢查結果確定生產的商品是優質、替代市場品質還是廢品品質。哪一種建模方法可以最準確地預測產品品質？

A. Amazon SageMaker DeepAR forecasting algorithm.

A. Amazon SageMaker DeepAR 預測演算法。

B. Amazon SageMaker XGBoost algorithm.

B. Amazon SageMaker XGBoost 演算法。

C. Amazon SageMaker Latent Dirichlet Allocation (LDA) algorithm.

C. Amazon SageMaker 潛在狄利克雷分配(LDA) 演算法。

D. A convolutional neural network (CNN) and ResNet.

D. 卷積神經網路(CNN) 和ResNet。

問題# 250

A healthcare company wants to create a machine learning (ML) model to predict patient outcomes. A data science team developed an ML model by using a custom ML library. The company wants to use Amazon SageMaker to train this model. The data science team creates a custom SageMaker image to train the model. When the team tries to launch the custom image in SageMaker Studio, the data scientists encounter an error within the application. Which service can the data scientists use to access the logs for this error?

一家醫療保健公司希望創建一個機器學習(ML) 模型來預測患者的結果。資料科學團隊使用自訂ML 庫開發了一個ML 模型。該公司希望使用Amazon SageMaker 來訓練此模型。資料科學團隊創建了一個自訂SageMaker 映像來訓練模型。當團隊嘗試在SageMaker Studio 中啟動自訂映像時，資料科學家在應用程式中遇到錯誤。資料科學家可以使用哪項服務來存取此錯誤的日誌？

A. Amazon S3.

A. Amazon S3。

B. Amazon Elastic Block Store (Amazon EBS).

B. Amazon Elastic Block Store (Amazon EBS)。

C. AWS CloudTrail.

C. AWS CloudTrail。

D. Amazon CloudWatch.

D. Amazon CloudWatch。

問題# 251

A data scientist wants to build a financial trading bot to automate investment decisions. The financial bot should recommend the quantity and price of an asset to buy or sell to maximize long-term profit. The data scientist will continuously stream financial transactions to the bot for training purposes. The data scientist must select the appropriate machine learning (ML) algorithm to develop the financial trading bot. Which type of ML algorithm will meet these requirements?

一位資料科學家想要建立一個金融交易機器人來自動化投資決策。金融機器人應該建議購買或出售資產的數量和價格，以最大化長期利潤。資料科學家將持續將金融交易傳輸到機器人進行訓練。資料科學家必須選擇合適的機器學習(ML)演算法來開發金融交易機器人。哪種類型的ML演算法可以滿足這些要求？

A. Supervised learning.

A. 監督學習。

B. Unsupervised learning.

B. 無監督學習。

C. Semi-supervised learning.

C. 半監督學習。

D. Reinforcement learning.

D. 強化學習。

問題# 252

A manufacturing company wants to create a machine learning (ML) model to predict when equipment is likely to fail. A data science team already constructed a deep learning model by using TensorFlow and a custom Python script in a local environment. The company wants to use Amazon SageMaker to train the model. Which TensorFlow estimator configuration will train the model MOST cost-effectively?

一家製造公司希望創建一個機器學習(ML)模型來預測設備何時可能發生故障。資料科學團隊已經在本地環境中使用TensorFlow和自訂Python腳本建立了一個深度學習模型。該公司希望使用Amazon SageMaker來訓練該模型。哪種TensorFlow估算器配置可以最經濟高效地訓練模型？

A. Turn on SageMaker Training Compiler by adding `compiler_config=TrainingCompilerConfig()` as a parameter. Pass the script to the estimator in the call to the TensorFlow `fit()` method.

A. 透過新增`compiler_config=TrainingCompilerConfig()`作為參數來開啟SageMaker Training Compiler。在對TensorFlow `fit()`方法的呼叫中將腳本傳遞給估算器。

B. Turn on SageMaker Training Compiler by adding `compiler_config=TrainingCompilerConfig()` as a parameter. Turn on managed spot training by setting the `use_spot_instances` parameter to `True`. Pass the script to the estimator in the call to the TensorFlow `fit()` method.

B. 透過新增 `compiler_config=TrainingCompilerConfig()` 作為參數來開啟 SageMaker Training Compiler。透過將 `use_spot_instances` 參數設定為 `True` 來開啟託管點訓練。在對 TensorFlow `fit()` 方法的呼叫中將腳本傳遞給估算器。

C. Adjust the training script to use distributed data parallelism. Specify appropriate values for the distribution parameter. Pass the script to the estimator in the call to the TensorFlow `fit()` method.

C. 調整訓練腳本以使用分散式資料並行性。為分佈參數指定適當的值。在對 TensorFlow `fit()` 方法的呼叫中將腳本傳遞給估算器。

D. Turn on SageMaker Training Compiler by adding `compiler_config=TrainingCompilerConfig()` as a parameter. Set the `MaxWaitTimeInSeconds` parameter to be equal to the `MaxRuntimeInSeconds` parameter. Pass the script to the estimator in the call to the TensorFlow `fit()` method.

D. 透過新增 `compiler_config=TrainingCompilerConfig()` 作為參數來開啟 SageMaker Training Compiler。將 `MaxWaitTimeInSeconds` 參數設定為等於 `MaxRuntimeInSeconds` 參數。在對 TensorFlow `fit()` 方法的呼叫中將腳本傳遞給估算器。

問題# 253

An automotive company uses computer vision in its autonomous cars. The company trained its object detection models successfully by using transfer learning from a convolutional neural network (CNN). The company trained the models by using PyTorch through the Amazon SageMaker SDK. The vehicles have limited hardware and compute power. The company wants to optimize the model to reduce memory, battery, and hardware consumption without a significant sacrifice in accuracy. Which solution will improve the computational efficiency of the models?

一家汽車公司在其自動駕駛汽車中使用電腦視覺。該公司透過使用卷積神經網路(CNN) 的遷移學習成功訓練了其物體偵測模型。該公司透過 Amazon SageMaker SDK 使用 PyTorch 訓練了模型。這些車輛的硬體和運算能力有限。該公司希望優化模型以減少記憶體、電池和硬體消耗，而不會顯著犧牲準確性。哪一種解決方案可以提高模型的計算效率？

A. Use Amazon CloudWatch metrics to gain visibility into the SageMaker training weights, gradients, biases, and activation outputs. Compute the filter ranks based on the training information. Apply pruning to remove the low-ranking filters. Set new weights based on the pruned set of filters. Run a new training job with the pruned model.

A. 使用 Amazon CloudWatch 指標來了解 SageMaker 訓練權重、梯度、偏差和啟動輸出。根據訓練資訊計算過濾器排名。應用修剪以刪除排名較低的過濾器。根據修剪後的過濾器集設定新權重。使用修剪後的模型執行新的訓練作業。

B. Use Amazon SageMaker Ground Truth to build and run data labeling workflows. Collect a larger labeled dataset with the labelling workflows. Run a new training job that uses the new labeled data with previous training data.

B. 使用 Amazon SageMaker Ground Truth 建置和執行資料標記工作流程。使用標記工作流程收集更大的標記資料集。執行使用新標記資料和先前訓練資料的新訓練作業。

C. Use Amazon SageMaker Debugger to gain visibility into the training weights, gradients, biases, and activation outputs. Compute the filter ranks based on the training information. Apply pruning to remove the low-ranking filters. Set the new weights based on the pruned set of filters. Run a new training job with the pruned model.

C. 使用Amazon SageMaker Debugger 來了解訓練權重、梯度、偏差和啟動輸出。根據訓練資訊計算過濾器排名。應用修剪以刪除排名較低的過濾器。根據修剪後的過濾器集設定新權重。使用修剪後的模型執行新的訓練作業。

D. Use Amazon SageMaker Model Monitor to gain visibility into the ModelLatency metric and OverheadLatency metric of the model after the company deploys the model. Increase the model learning rate. Run a new training job.

D. 使用Amazon SageMaker Model Monitor 在公司部署模型後查看模型的ModelLatency 指標和OverheadLatency 指標。提高模型學習率。運行新的訓練作業。

問題# 254

A data scientist wants to improve the fit of a machine learning (ML) model that predicts house prices. The data scientist makes a first attempt to fit the model, but the fitted model has poor accuracy on both the training dataset and the test dataset. Which steps must the data scientist take to improve model accuracy? (Choose three.)

一位資料科學家想要提高預測房價的機器學習(ML) 模型的適配度。資料科學家首次嘗試擬合模型，但擬合模型在訓練資料集和測試資料集上的準確率都很低。資料科學家必須採取哪些步驟來提高模型準確度？（選擇三項。）

A. Increase the amount of regularization that the model uses.

A. 增加模型使用的正規化量。

B. Decrease the amount of regularization that the model uses.

B. 減少模型使用的正規化量。

C. Increase the number of training examples that that model uses.

C. 增加此模型所使用的訓練範例的數量。

D. Increase the number of test examples that the model uses.

D. 增加模型使用的測試範例的數量。

E. Increase the number of model features that the model uses.

E. 增加模型所使用的模型特徵的數量。

F. Decrease the number of model features that the model uses.

F. 減少模型所使用的模型特徵的數量。

問題# 255

A car company is developing a machine learning solution to detect whether a car is present in an image. The image dataset consists of one million images. Each image in the dataset is 200 pixels in height by 200 pixels in width. Each image is labeled as either having a car or not having a car. Which architecture is MOST likely to produce a model that detects whether a car is present in an image with the highest accuracy?

一家汽車公司正在開發一種機器學習解決方案，用於檢測影像中是否存在汽車。圖像資料集包含一百萬張圖像。資料集中的每張影像高200 像素，寬200 像素。每張圖像都標記為有車或無車。哪種架構最有可能產生一個以最高準確率檢測影像中是否存在汽車的模型？

A. Use a deep convolutional neural network (CNN) classifier with the images as input. Include a linear output layer that outputs the probability that an image contains a car.

A. 使用深度卷積神經網路(CNN) 分類器，將影像作為輸入。包括一個線性輸出層，輸出影像中包含汽車的機率。

B. Use a deep convolutional neural network (CNN) classifier with the images as input. Include a softmax output layer that outputs the probability that an image contains a car.

B. 使用深度卷積神經網路(CNN) 分類器，將影像作為輸入。包括一個softmax 輸出層，輸出影像中包含汽車的機率。

C. Use a deep multilayer perceptron (MLP) classifier with the images as input. Include a linear output layer that outputs the probability that an image contains a car.

C. 使用深度多層感知器(MLP) 分類器，將影像作為輸入。包括一個線性輸出層，輸出影像中包含汽車的機率。

D. Use a deep multilayer perceptron (MLP) classifier with the images as input. Include a softmax output layer that outputs the probability that an image contains a car.

D. 使用深度多層感知器(MLP) 分類器，將影像作為輸入。包括一個softmax 輸出層，輸出影像中包含汽車的機率。

問題# 256

A company is creating an application to identify, count, and classify animal images that are uploaded to the company's website. The company is using the Amazon SageMaker image classification algorithm with an ImageNetV2 convolutional neural network (CNN). The solution works well for most animal images but does not recognize many animal species that are less common. The company obtains 10,000 labeled images of less common animal species and stores the images in Amazon S3. A machine learning (ML) engineer needs to incorporate the images into the model by using Pipe mode in SageMaker. Which combination of steps should the ML engineer take to train the model? (Choose two.)

一家公司正在開發一款應用程序，用於識別、計數和分類上傳到公司網站的動物圖像。該公司正在使用帶有ImageNetV2 卷積神經網路(CNN) 的Amazon SageMaker 影像分類演算法。此解決方案適用於大多數動物影像，但無法識別許多不太常見的動物物種。該公司獲取了10,000 張不太常見的動物物種的標籤圖像，並將這些圖像儲存在Amazon S3 中。機器學習(ML) 工程師需要使用SageMaker 中的管道模式將影像合併到模型中。ML 工程師應採取哪些步驟組合來訓練模型？（選兩個。）

A. Use a ResNet model. Initiate full training mode by initializing the network with random weights.

A. 使用ResNet 模型。透過使用隨機權重初始化網路來啟動完整訓練模式。

B. Use an Inception model that is available with the SageMaker image classification algorithm.

B. 使用SageMaker 影像分類演算法提供的Inception 模型。

C. Create a .lst file that contains a list of image files and corresponding class labels. Upload the .lst file to Amazon S3.

C. 建立包含映像檔清單和對應類別標籤的 .lst 檔案。將 .lst 檔案上傳到Amazon S3。

D. Initiate transfer learning. Train the model by using the images of less common species.

D. 啟動遷移學習。使用較不常見物種的影像訓練模型。

E. Use an augmented manifest file in JSON Lines format.

E. 使用JSON 行格式的增強清單檔。

問題# 257

A music streaming company is building a pipeline to extract features. The company wants to store the features for offline model training and online inference. The company wants to track feature history and to give the company's data science teams access to the features. Which solution will meet these requirements with the MOST operational efficiency?

一家音樂串流媒體公司正在建立一個提取特徵的管道。該公司希望儲存這些特徵，用於離線模型訓練和線上推理。該公司希望追蹤特徵歷史記錄，並讓公司的數據科學團隊存取這些特徵。哪種解決方案能夠以最高的營運效率滿足這些要求？

A. Use Amazon SageMaker Feature Store to store features for model training and inference. Create an online store for online inference. Create an offline store for model training. Create an IAM role for data scientists to access and search through feature groups.

A. 使用Amazon SageMaker 功能儲存來儲存用於模型訓練和推理的功能。建立用於線上推理的線上商店。建立用於模型訓練的離線商店。為資料科學家建立IAM 角色以存取和搜尋功能組。

B. Use Amazon SageMaker Feature Store to store features for model training and inference. Create an online store for both online inference and model training. Create an IAM role for data scientists to access and search through feature groups.

B. 使用Amazon SageMaker 功能儲存來儲存用於模型訓練和推理的功能。建立用於線上推理和模型訓練的線上商店。為資料科學家建立IAM 角色以存取和搜尋功能組。

C. Create one Amazon S3 bucket to store online inference features. Create a second S3 bucket to store offline model training features. Turn on versioning for the S3 buckets and use tags to specify which tags are for online inference features and which are for offline model training features. Use Amazon Athena to query the S3 bucket for online inference. Connect the S3 bucket for offline model training to a SageMaker training job. Create an IAM policy that allows data scientists to access both buckets.

C. 建立一個Amazon S3 儲存桶來儲存線上推理功能。建立第二個S3 儲存桶來儲存離線模型訓練功

能。為S3 儲存桶啟用版本控制並使用標籤指定哪些標籤用於線上推理功能，哪些標籤用於離線模型訓練功能。使用Amazon Athena 查詢用於線上推理的S3 儲存桶。將用於離線模型訓練的S3 儲存桶連接到SageMaker 訓練作業。創建允許資料科學家存取這兩個儲存桶的IAM 策略。

D. Create two separate Amazon DynamoDB tables to store online inference features and offline model training features. Use time-based versioning on both tables. Query the DynamoDB table for online inference. Move the data from DynamoDB to Amazon S3 when a new SageMaker training job is launched. Create an IAM policy that allows data scientists to access both tables.

D. 建立兩個單獨的Amazon DynamoDB 表來儲存線上推理功能和離線模型訓練功能。在兩個表上使用基於時間的版本控制。查詢DynamoDB 表以進行線上推理。啟動新的SageMaker 訓練作業時，將資料從DynamoDB 移至Amazon S3。建立允許資料科學家存取這兩個表的IAM 策略。

問題# 258

A beauty supply store wants to understand some characteristics of visitors to the store. The store has security video recordings from the past several years. The store wants to generate a report of hourly visitors from the recordings. The report should group visitors by hair style and hair color. Which solution will meet these requirements with the LEAST amount of effort?

一家美容用品店希望了解店內訪客的一些特徵。這家店有過去幾年的安防錄影。這家店希望根據錄影產生每小時訪客報告。該報告應按髮型和髮色對訪客進行分組。哪種解決方案能夠以最少的努力滿足這些要求？

A. Use an object detection algorithm to identify a visitor's hair in video frames. Pass the identified hair to an ResNet-50 algorithm to determine hair style and hair color.

A. 使用物體偵測演算法辨識影片影格中的訪客頭髮。將識別出的頭髮傳遞給ResNet-50 演算法來確定髮型和頭髮顏色。

B. Use an object detection algorithm to identify a visitor's hair in video frames. Pass the identified hair to an XGBoost algorithm to determine hair style and hair color.

B. 使用物體偵測演算法辨識影片畫面中的訪客頭髮。將辨識出的頭髮傳遞給XGBoost 演算法來決定髮型和髮色。

C. Use a semantic segmentation algorithm to identify a visitor's hair in video frames. Pass the identified hair to an ResNet-50 algorithm to determine hair style and hair color.

C. 使用語意分割演算法辨識視訊影格中的訪客頭髮。將識別出的頭髮傳遞給ResNet-50 演算法來確定髮型和頭髮顏色。

D. Use a semantic segmentation algorithm to identify a visitor's hair in video frames. Pass the identified hair to an XGBoost algorithm to determine hair style and hair.

D. 使用語意分割演算法辨識視訊影格中的訪客頭髮。將識別出的頭髮傳遞給XGBoost 演算法來確定髮型和頭髮。

問題# 259

A financial services company wants to automate its loan approval process by building a machine learning (ML) model. Each loan data point contains credit history from a third-party data source and demographic information about the customer. Each loan approval prediction must come with a report that contains an explanation for why the customer was approved for a loan or was denied for a loan. The company will use Amazon SageMaker to build the model. Which solution will meet these requirements with the LEAST development effort?

一家金融服務公司希望透過建立機器學習(ML)模型來實現貸款審批流程的自動化。每個貸款資料點都包含來自第三方資料來源的信用記錄和有關客戶的人口統計資料。每項貸款核准預測都必須附有一份報告，其中包含客戶貸款核准或被拒絕的原因說明。該公司將使用Amazon SageMaker來建立模型。哪一種解決方案能夠以最少的開發工作量來滿足這些要求？

A. Use SageMaker Model Debugger to automatically debug the predictions, generate the explanation, and attach the explanation report.

A. 使用SageMaker Model Debugger 自動偵錯預測、產生解釋並附加解釋報告。

B. Use AWS Lambda to provide feature importance and partial dependence plots. Use the plots to generate and attach the explanation report.

B. 使用AWS Lambda 提供功能重要性和部分依賴圖。使用這些圖產生並附加解釋報告。

C. Use SageMaker Clarify to generate the explanation report. Attach the report to the predicted results.

C. 使用SageMaker Clarify 產生解釋報告。將報告附加到預測結果。

D. Use custom Amazon CloudWatch metrics to generate the explanation report. Attach the report to the predicted results.

D. 使用自訂Amazon CloudWatch 指標產生解釋報告。將報告附加到預測結果。

問題# 260

A financial company sends special offers to customers through weekly email campaigns. A bulk email marketing system takes the list of email addresses as an input and sends the marketing campaign messages in batches. Few customers use the offers from the campaign messages. The company does not want to send irrelevant offers to customers. A machine learning (ML) team at the company is using Amazon SageMaker to build a model to recommend specific offers to each customer based on the customer's profile and the offers that the customer has accepted in the past. Which solution will meet these requirements with the MOST operational efficiency?

一家金融公司透過每週的電子郵件活動向客戶發送特別優惠。大量電子郵件行銷系統將電子郵件地址清單作為輸入，並分批發送行銷活動訊息。很少有客戶使用活動訊息中的優惠。該公司不想向客戶發送不相關的優惠。該公司的機器學習(ML)團隊正在使用Amazon SageMaker 建立一個模型，根據客戶的個人資料和客戶過去接受的優惠向每位客戶推薦特定優惠。哪種解決方案能夠以最高的營運效率滿足這些要求？

A. Use the Factorization Machines algorithm to build a model that can generate personalized offer recommendations for customers. Deploy a SageMaker endpoint to generate offer recommendations. Feed the offer recommendations into the bulk email marketing system.

A. 使用分解機演算法建立一個可以為客戶產生個人化優惠建議的模型。部署SageMaker 終端節點以產生優惠建議。將優惠建議輸入大量電子郵件行銷系統。

B. Use the Neural Collaborative Filtering algorithm to build a model that can generate personalized offer recommendations for customers. Deploy a SageMaker endpoint to generate offer recommendations. Feed the offer recommendations into the bulk email marketing system.

B. 使用神經協同過濾演算法建立一個可以為客戶產生個人化優惠建議的模型。部署SageMaker 終端節點以產生優惠建議。將優惠建議輸入大量電子郵件行銷系統。

C. Use the Neural Collaborative Filtering algorithm to build a model that can generate personalized offer recommendations for customers. Deploy a SageMaker batch inference job to generate offer recommendations. Feed the offer recommendations into the bulk email marketing system.

C. 使用神經協同過濾演算法建立一個可以為客戶產生個人化優惠建議的模型。部署SageMaker 批次推理作業以產生優惠建議。將優惠建議輸入大量電子郵件行銷系統。

D. Use the Factorization Machines algorithm to build a model that can generate personalized offer recommendations for customers. Deploy a SageMaker batch inference job to generate offer recommendations. Feed the offer recommendations into the bulk email marketing system.

D. 使用分解機演算法建立一個可以為客戶產生個人化優惠建議的模型。部署SageMaker 批次推理作業以產生優惠建議。將優惠建議輸入大量電子郵件行銷系統。

問題# 261

A social media company wants to develop a machine learning (ML) model to detect inappropriate or offensive content in images. The company has collected a large dataset of labeled images and plans to use the built-in Amazon SageMaker image classification algorithm to train the model. The company also intends to use SageMaker pipe mode to speed up the training. The company splits the dataset into training, validation, and testing datasets. The company stores the training and validation images in folders that are named Training and Validation, respectively. The folders contain subfolders that correspond to the names of the dataset classes. The company resizes the images to the same size and generates two input manifest files named training.lst and validation.lst, for the training dataset and the validation dataset, respectively. Finally, the company creates two separate Amazon S3 buckets for uploads of the training dataset and the validation dataset. Which additional data preparation steps should the company take before uploading the files to Amazon S3?

一家社群媒體公司希望開發一種機器學習(ML)模型來偵測影像中不當或冒犯性的內容。該公司收集了大量標籤的圖像資料集，並計劃使用內建的Amazon SageMaker 圖像分類演算法來訓練該模型。該公司還打算使用SageMaker 管道模式來加快訓練速度。該公司將資料集分為訓練、驗證和測試資料集。該公司將訓練和驗證圖像分別儲存在名為Training 和Validation 的資料夾中。資料夾包含與資料集類別名稱相對應的子資料夾。該公司將圖像調整為相同大小，並分別為訓練資料集和驗證資料集產生兩個名為training.lst 和validation.lst 的輸入清單檔案。最後，該公司創建兩個單獨的Amazon S3 儲存桶，用於上傳訓練資料集和驗證資料集。在將文件上傳到Amazon S3 之前，公司應該採取哪些額外的資料準備步驟？

A. Generate two Apache Parquet files, training.parquet and validation.parquet, by reading the images into a Pandas data frame and storing the data frame as a Parquet file. Upload the Parquet files to the training S3 bucket.

A. 將圖像讀入Pandas 資料框並將資料框儲存為Parquet 文件，產生兩個Apache Parquet 文件，training.parquet 和validation.parquet。將Parquet 檔案上傳到訓練S3 儲存桶。

B. Compress the training and validation directories by using the Snappy compression library. Upload the manifest and compressed files to the training S3 bucket.

B. 使用Snappy 壓縮函式庫壓縮訓練和驗證目錄。將清單和壓縮檔案上傳到訓練S3 儲存桶。

C. Compress the training and validation directories by using the gzip compression library. Upload the manifest and compressed files to the training S3 bucket.

C. 使用gzip 壓縮庫壓縮訓練和驗證目錄。將清單和壓縮檔案上傳到訓練S3 儲存桶。

D. Generate two RecordIO files, training.rec and validation.rec, from the manifest files by using the im2rec Apache MXNet utility tool. Upload the RecordIO files to the training S3 bucket.

D. 使用im2rec Apache MXNet 實用工具從清單檔案產生兩個RecordIO 文件，training.rec 和validation.rec。將RecordIO 檔案上傳到訓練S3 儲存桶。

問題# 262

A media company wants to create a solution that identifies celebrities in pictures that users upload. The company also wants to identify the IP address and the timestamp details from the users so the company can prevent users from uploading pictures from unauthorized locations. Which solution will meet these requirements with LEAST development effort?

一家媒體公司希望創建一個解決方案，以識別用戶上傳圖片中的名人。該公司還希望識別用戶的IP 位址和時間戳詳細信息，以便公司可以阻止用戶從未經授權的位置上傳圖片。哪一種解決方案可以以最少的開發工作量來滿足這些要求？

A. Use AWS Panorama to identify celebrities in the pictures. Use AWS CloudTrail to capture IP address and timestamp details.

A. 使用AWS Panorama 辨識圖片中的名人。使用AWS CloudTrail 擷取IP 位址和時間戳詳細資料。

B. Use AWS Panorama to identify celebrities in the pictures. Make calls to the AWS Panorama Device SDK to capture IP address and timestamp details.

B. 使用AWS Panorama 辨識圖片中的名人。呼叫AWS Panorama Device SDK 擷取IP 位址和時間戳詳細資訊。

C. Use Amazon Rekognition to identify celebrities in the pictures. Use AWS CloudTrail to capture IP address and timestamp details.

C. 使用Amazon Rekognition 辨識圖片中的名人。使用AWS CloudTrail 擷取IP 位址和時間戳詳細資料。

D. Use Amazon Rekognition to identify celebrities in the pictures. Use the text detection feature to capture IP address and timestamp details.

D. 使用Amazon Rekognition 辨識圖片中的名人。使用文字偵測功能擷取IP 位址和時間戳詳細資訊。

問題# 263

A pharmaceutical company performs periodic audits of clinical trial sites to quickly resolve critical findings. The company stores audit documents in text format. Auditors have requested help from a data science team to quickly analyze the documents. The auditors need to discover the 10 main topics within the documents to prioritize and distribute the review work among the auditing team members. Documents that describe adverse events must receive the highest priority. A data scientist will use statistical modeling to discover abstract topics and to provide a list of the top words for each category to help the auditors assess the relevance of the topic. Which algorithms are best suited to this scenario? (Choose two.)

一家製藥公司定期對臨床試驗地點進行審計，以快速解決關鍵發現。該公司以文字格式儲存審計文件。審計人員已請求數據科學團隊的協助來快速分析文件。審計人員需要發現文件中的10個主要主題，以便確定優先順序並在審計團隊成員之間分配審查工作。描述不良事件的文件必須獲得最高優先順序。資料科學家將使用統計模型來發現抽象主題並提供每個類別的熱門單字列表，以幫助審計人員評估主題的相關性。哪種演算法最適合這種情況？（選兩個。）

A. Latent Dirichlet allocation (LDA).

A. 潛在狄利克雷分佈（LDA）。

B. Random forest classifier.

B. 隨機森林分類器。

C. Neural topic modeling (NTM).

C. 神經主題模型（NTM）。

D. Linear support vector machine.

D. 線性支援向量機。

E. Linear regression.

E. 線性迴歸。

問題# 264

A company needs to deploy a chatbot to answer common questions from customers. The chatbot must base its answers on company documentation. Which solution will meet these requirements with the LEAST development effort?

一家公司需要部署一個聊天機器人來回答客戶的常見問題。聊天機器人必須根據公司文件來回答問

題。哪一種解決方案能夠以最少的開發工作量來滿足這些要求？

A. Index company documents by using Amazon Kendra. Integrate the chatbot with Amazon Kendra by using the Amazon Kendra Query API operation to answer customer questions.

A. 使用Amazon Kendra 索引公司文件。使用Amazon Kendra 查詢API 操作將聊天機器人與Amazon Kendra 整合以回答客戶問題。

B. Train a Bidirectional Attention Flow (BiDAF) network based on past customer questions and company documents. Deploy the model as a real-time Amazon SageMaker endpoint. Integrate the model with the chatbot by using the SageMaker Runtime InvokeEndpoint API operation to answer customer questions.

B. 根據過去的客戶問題和公司文件訓練雙向注意力流(BiDAF) 網路。將模型部署為即時Amazon SageMaker 終端節點。使用SageMaker Runtime InvokeEndpoint API 操作將模型與聊天機器人整合以回答客戶問題。

C. Train an Amazon SageMaker Blazing Text model based on past customer questions and company documents. Deploy the model as a real-time SageMaker endpoint. Integrate the model with the chatbot by using the SageMaker Runtime InvokeEndpoint API operation to answer customer questions.

C. 根據過去的客戶問題和公司文件訓練Amazon SageMaker Blazing Text 模型。將模型部署為即時SageMaker 終端節點。使用SageMaker Runtime InvokeEndpoint API 操作將模型與聊天機器人整合以回答客戶問題。

D. Index company documents by using Amazon OpenSearch Service. Integrate the chatbot with OpenSearch Service by using the OpenSearch Service k-nearest neighbors (k-NN) Query API operation to answer customer questions.

D. 使用Amazon OpenSearch Service 索引公司文件。使用OpenSearch Service k-最近鄰(k-NN) 查詢API 操作將聊天機器人與OpenSearch Service 整合以回答客戶問題。

問題# 265

A company wants to conduct targeted marketing to sell solar panels to homeowners. The company wants to use machine learning (ML) technologies to identify which houses already have solar panels. The company has collected 8,000 satellite images as training data and will use Amazon SageMaker Ground Truth to label the data. The company has a small internal team that is working on the project. The internal team has no ML expertise and no ML experience. Which solution will meet these requirements with the LEAST amount of effort from the internal team?

一家公司希望進行有針對性的行銷，向房主銷售太陽能板。該公司希望使用機器學習(ML) 技術來識別哪些房屋已經安裝了太陽能板。該公司已收集了8,000 張衛星圖像作為訓練數據，並將使用Amazon SageMaker Ground Truth 來標記數據。該公司有一個小型內部團隊正在從事該專案。內部團隊沒有ML 專業知識和ML 經驗。哪一種解決方案可以滿足這些要求，而內部團隊需要付出的努力最少？

A. Set up a private workforce that consists of the internal team. Use the private workforce and the SageMaker Ground Truth active learning feature to label the data. Use Amazon Rekognition Custom Labels for model training and hosting.

A. 設置由內部團隊組成的私人員工團隊。使用私人員工團隊和SageMaker Ground Truth 主動學習功能來標記資料。使用Amazon Rekognition 自訂標籤進行模型訓練和託管。

B. Set up a private workforce that consists of the internal team. Use the private workforce to label the data. Use Amazon Rekognition Custom Labels for model training and hosting.

B. 設置由內部團隊組成的私人員工團隊。使用私人員工來標記資料。使用Amazon Rekognition 自訂標籤進行模型訓練和託管。

C. Set up a private workforce that consists of the internal team. Use the private workforce and the SageMaker Ground Truth active learning feature to label the data. Use the SageMaker Object Detection algorithm to train a model. Use SageMaker batch transform for inference.

C. 設定由內部團隊組成的私人員工隊伍。使用私人員工團隊和SageMaker Ground Truth 主動學習功能來標記資料。使用SageMaker 物件偵測演算法訓練模型。使用SageMaker 批次轉換進行推理。

D. Set up a public workforce. Use the public workforce to label the data. Use the SageMaker Object Detection algorithm to train a model. Use SageMaker batch transform for inference.

D. 設定公共員工隊伍。使用公共員工來標記資料。使用SageMaker 物件偵測演算法訓練模型。使用SageMaker 批次轉換進行推理。

問題# 266

A company hosts a machine learning (ML) dataset repository on Amazon S3. A data scientist is preparing the repository to train a model. The data scientist needs to redact personally identifiable information (PII) from the dataset. Which solution will meet these requirements with the LEAST development effort?

一家公司在Amazon S3 上託管了一個機器學習(ML) 資料集儲存庫。一位資料科學家正在準備該儲存庫來訓練模型。資料科學家需要從資料集中刪除個人識別資訊(PII)。哪一種解決方案能夠以最少的開發工作量來滿足這些要求？

A. Use Amazon SageMaker Data Wrangler with a custom transformation to identify and redact the PII.

A. 使用Amazon SageMaker Data Wrangler 進行自訂轉換來識別和編輯PII

B. Create a custom AWS Lambda function to read the files, identify the PII, and redact the PII.

B. 建立自訂AWS Lambda 函數來讀取檔案、識別PII, 並編輯PII

C. Use AWS Glue DataBrew to identify and redact the PII.

C. 使用AWS Glue DataBrew 來識別和編輯PII

D. Use an AWS Glue development endpoint to implement the PII redaction from within a notebook.

D. 使用AWS Glue 開發終端節點在筆記本中實作PII 編輯。

問題# 267

A company is deploying a new machine learning (ML) model in a production environment. The company is concerned that the ML model will drift over time, so the company creates a script to aggregate all inputs and predictions into a single file at the end of each day. The company stores the file as an object in an Amazon S3 bucket. The total size of the daily file is 100 GB. The daily file size will increase over time. Four times a year, the company samples the data from the previous 90 days to check the ML model for drift. After the 90-day period, the company must keep the files for compliance reasons. The company needs to use S3 storage classes to minimize costs. The company wants to maintain the same storage durability of the data. Which solution will meet these requirements?

一家公司正在生產環境中部署新的機器學習(ML)模型。該公司擔心ML模型會隨著時間的推移而發生漂移，因此該公司創建了一個腳本，在每天結束時將所有輸入和預測匯總到一個文件中。該公司將文件作為物件儲存在Amazon S3儲存桶中。每日檔案的總大小為100 GB。每日文件大小將隨著時間的推移而增加。該公司每年四次從前90天的數據中抽樣，以檢查ML模型是否有漂移。90天后，出於合規原因，該公司必須保留這些文件。該公司需要使用S3儲存類別來最大限度地降低成本。該公司希望保持資料的相同儲存持久性。哪種解決方案可以滿足這些要求？

A. Store the daily objects in the S3 Standard-InfrequentAccess (S3 Standard-IA) storage class. Configure an S3 Lifecycle rule to move the objects to S3 Glacier Flexible Retrieval after 90 days.

A. 將每日物件儲存在S3 Standard-InfrequentAccess (S3 Standard-IA) 儲存類別中。配置S3生命週期規則，在90天後將物件移至S3 Glacier Flexible Retrieval。

B. Store the daily objects in the S3 One Zone-Infrequent Access (S3 One Zone-IA) storage class. Configure an S3 Lifecycle rule to move the objects to S3 Glacier Flexible Retrieval after 90 days.

B. 將每日物件儲存在S3 One Zone-Infrequent Access (S3 One Zone-IA) 儲存類別中。配置S3生命週期規則，在90天後將物件移至S3 Glacier Flexible Retrieval。

C. Store the daily objects in the S3 Standard-InfrequentAccess (S3 Standard-IA) storage class. Configure an S3 Lifecycle rule to move the objects to S3 Glacier Deep Archive after 90 days.

C. 將每日物件儲存在S3 Standard-InfrequentAccess (S3 Standard-IA) 儲存類別中。配置S3生命週期規則，在90天後將物件移到S3 Glacier Deep Archive。

D. Store the daily objects in the S3 One Zone-Infrequent Access (S3 One Zone-IA) storage class. Configure an S3 Lifecycle rule to move the objects to S3 Glacier Deep Archive after 90 days.

D. 將每日物件儲存在S3 One Zone-Infrequent Access (S3 One Zone-IA) 儲存類別中。配置S3生命週期規則，在90天後將物件移到S3 Glacier Deep Archive。

問題# 268

A company wants to enhance audits for its machine learning (ML) systems. The auditing system must be able to perform metadata analysis on the features that the ML models use. The audit solution must generate a report that analyzes the metadata. The solution also must be able to set the data sensitivity and authorship of features. Which solution will meet these requirements with the LEAST development effort?

一家公司希望增強其機器學習(ML)系統的審計。審計系統必須能夠對ML模型使用的功能執行元資料分析。審計解決方案必須產生分析元資料的報告。該解決方案還必須能夠設定功能的資料敏感度

和作者身份。哪一種解決方案能夠以最少的開發工作量來滿足這些要求？

A. Use Amazon SageMaker Feature Store to select the features. Create a data flow to perform feature-level metadata analysis. Create an Amazon DynamoDB table to store feature-level metadata. Use Amazon QuickSight to analyze the metadata.

A. 使用Amazon SageMaker Feature Store 選擇功能。建立資料流以執行功能級元資料分析。建立Amazon DynamoDB 表來儲存功能級元資料。使用Amazon QuickSight 分析元資料。

B. Use Amazon SageMaker Feature Store to set feature groups for the current features that the ML models use. Assign the required metadata for each feature. Use SageMaker Studio to analyze the metadata.

B. 使用Amazon SageMaker Feature Store 為ML 模型使用的目前功能設定功能組。為每個功能分配所需的元資料。使用SageMaker Studio 分析元資料。

C. Use Amazon SageMaker Features Store to apply custom algorithms to analyze the feature-level metadata that the company requires. Create an Amazon DynamoDB table to store feature-level metadata. Use Amazon QuickSight to analyze the metadata.

C. 使用Amazon SageMaker Features Store 應用自訂演算法來分析公司所需的元資料。建立Amazon DynamoDB 表來儲存功能級元資料。使用Amazon QuickSight 分析元資料。

D. Use Amazon SageMaker Feature Store to set feature groups for the current features that the ML models use. Assign the required metadata for each feature. Use Amazon QuickSight to analyze the metadata.

D. 使用Amazon SageMaker Feature Store 為ML 模型使用的目前功能設定功能組。為每個功能分配所需的元資料。使用Amazon QuickSight 分析元資料。

問題# 269

A machine learning (ML) specialist uploads a dataset to an Amazon S3 bucket that is protected by server-side encryption with AWS KMS keys (SSE-KMS). The ML specialist needs to ensure that an Amazon SageMaker notebook instance can read the dataset that is in Amazon S3. Which solution will meet these requirements?

機器學習(ML) 專家將資料集上傳到Amazon S3 儲存桶，該儲存桶受AWS KMS 金鑰(SSE-KMS) 的伺服器端加密保護。ML 專家需要確保Amazon SageMaker 筆記本實例可以讀取Amazon S3 中的資料集。哪種解決方案可以滿足這些要求？

A. Define security groups to allow all HTTP inbound and outbound traffic. Assign the security groups to the SageMaker notebook instance.

A. 定義安全群組以允許所有HTTP 入站和出站流量。將安全性群組指派給SageMaker 筆記本實例。

B. Configure the SageMaker notebook instance to have access to the VPC. Grant permission in the AWS Key Management Service (AWS KMS) key policy to the notebook's VPC.

B. 配置SageMaker 筆記本實例以存取VPC。在AWS Key Management Service (AWS KMS) 金鑰策略中向筆記本的VPC 授予權限。

C. Assign an IAM role that provides S3 read access for the dataset to the SageMaker notebook. Grant permission in the KMS key policy to the IAM role.

C. 指派一個IAM 角色，該角色為SageMaker 筆記本提供對資料集的S3 讀取存取權。在KMS 金鑰策略中向IAM 角色授予權限。

D. Assign the same KMS key that encrypts the data in Amazon S3 to the SageMaker notebook instance.

D. 將加密Amazon S3 中資料的相同KMS 金鑰指派給SageMaker 筆記本實例。

問題# 270

A company has a podcast platform that has thousands of users. The company implemented an algorithm to detect low podcast engagement based on a 10-minute running window of user events such as listening to, pausing, and closing the podcast. A machine learning (ML) specialist is designing the ingestion process for these events. The ML specialist needs to transform the data to prepare the data for inference. How should the ML specialist design the transformation step to meet these requirements with the LEAST operational effort?

一家公司擁有一個擁有數千名用戶的播客平台。該公司實施了一種演算法，根據10 分鐘的用戶事件運行視窗（例如收聽、暫停和關閉播客）來偵測播客參與度低的情況。機器學習(ML) 專家正在設計這些事件的提取過程。ML 專家需要轉換資料以準備資料進行推理。ML 專家應如何設計轉換步驟以最少的營運工作量來滿足這些要求？

A. Use an Amazon Managed Streaming for Apache Kafka (Amazon MSK) cluster to ingest event data. Use Amazon Kinesis Data Analytics to transform the most recent 10 minutes of data before inference.

A. 使用Amazon Managed Streaming for Apache Kafka (Amazon MSK) 叢集擷取事件資料。使用Amazon Kinesis Data Analytics 在推理之前轉換最近10 分鐘的數據。

B. Use Amazon Kinesis Data Streams to ingest event data. Store the data in Amazon S3 by using Amazon Kinesis Data Firehose. Use AWS Lambda to transform the most recent 10 minutes of data before inference.

B. 使用Amazon Kinesis Data Streams 擷取事件資料。使用Amazon Kinesis Data Firehose 將資料儲存在Amazon S3 中。使用AWS Lambda 在推理之前轉換最近10 分鐘的資料。

C. Use Amazon Kinesis Data Streams to ingest event data. Use Amazon Kinesis Data Analytics to transform the most recent 10 minutes of data before inference.

C. 使用Amazon Kinesis Data Streams 擷取事件資料。使用Amazon Kinesis Data Analytics 在推理之前轉換最近10 分鐘的數據。

D. Use an Amazon Managed Streaming for Apache Kafka (Amazon MSK) cluster to ingest event data. Use AWS Lambda to transform the most recent 10 minutes of data before inference.

D. 使用Amazon Managed Streaming for Apache Kafka (Amazon MSK) 叢集擷取事件資料。使用AWS Lambda 在推理之前轉換最近10 分鐘的資料。

問題# 271

A machine learning (ML) specialist is training a multilayer perceptron (MLP) on a dataset with multiple classes. The target class of interest is unique compared to the other classes in the dataset, but it does not achieve an acceptable recall metric. The ML specialist varies the number and size of the MLP's hidden layers, but the results do not improve significantly. Which solution will improve recall in the LEAST amount of time?

一位機器學習(ML)專家正在一個包含多個類別的資料集上訓練多層感知器(MLP)。目標類別與資料集中的其他類別相比是獨一無二的，但它沒有達到可接受的召回率指標。ML專家改變了MLP隱藏層的數量和大小，但結果並沒有顯著改善。哪種解決方案可以在最短的時間內提高召回率？

A. Add class weights to the MLP's loss function, and then retrain.

A. 將類別權重加入MLP的損失函數中，然後重新訓練。

B. Gather more data by using Amazon Mechanical Turk, and then retrain.

B. 使用Amazon Mechanical Turk收集更多數據，然後重新訓練。

C. Train a k-means algorithm instead of an MLP.

C. 訓練k-means演算法而不是MLP。

D. Train an anomaly detection model instead of an MLP.

D. 訓練異常檢測模型而不是MLP。

問題# 272

A machine learning (ML) specialist uploads 5 TB of data to an Amazon SageMaker Studio environment. The ML specialist performs initial data cleansing. Before the ML specialist begins to train a model, the ML specialist needs to create and view an analysis report that details potential bias in the uploaded data. Which combination of actions will meet these requirements with the LEAST operational overhead? (Choose two.)

機器學習(ML)專家將5 TB的資料上傳到Amazon SageMaker Studio環境。ML專家執行初始資料清理。在ML專家開始訓練模型之前，ML專家需要建立並查看分析報告，詳細說明上傳資料中的潛在偏差。哪種操作組合可以以最少的營運開銷滿足這些要求？（選兩個。）

A. Use SageMaker Clarify to automatically detect data bias.

A. 使用SageMaker Clarify自動偵測資料偏差。

B. Turn on the bias detection option in SageMaker Ground Truth to automatically analyze data features.

B. 開啟SageMaker Ground Truth中的偏差偵測選項，自動分析資料特徵。

C. Use SageMaker Model Monitor to generate a bias drift report.

C. 使用SageMaker Model Monitor產生偏差漂移報告。

D. Configure SageMaker Data Wrangler to generate a bias report.

D. 設定SageMaker Data Wrangler 產生偏差報告。

E. Use SageMaker Experiments to perform a data check.

E. 使用SageMaker Experiments 執行資料檢查。

問題# 273

A network security vendor needs to ingest telemetry data from thousands of endpoints that run all over the world. The data is transmitted every 30 seconds in the form of records that contain 50 fields. Each record is up to 1 KB in size. The security vendor uses Amazon Kinesis Data Streams to ingest the data. The vendor requires hourly summaries of the records that Kinesis Data Streams ingests. The vendor will use Amazon Athena to query the records and to generate the summaries. The Athena queries will target 7 to 12 of the available data fields. Which solution will meet these requirements with the LEAST amount of customization to transform and store the ingested data?

網路安全供應商需要從世界各地運行的數千個端點提取遙測資料。資料每30秒以包含50個欄位的記錄形式傳輸一次。每個記錄的大小最多為1 KB。安全供應商使用Amazon Kinesis Data Streams 提取資料。供應商需要Kinesis Data Streams 提取的記錄的每小時摘要。供應商將使用Amazon Athena 查詢記錄並產生摘要。Athena 查詢將針對7到12個可用資料欄位。哪種解決方案可以滿足這些要求，並且轉換和儲存提取的資料所需的自訂量最少？

A. Use AWS Lambda to read and aggregate the data hourly. Transform the data and store it in Amazon S3 by using Amazon Kinesis Data Firehose.

A. 使用AWS Lambda 每小時讀取和聚合資料。使用Amazon Kinesis Data Firehose 轉換資料並將其儲存在Amazon S3 中。

B. Use Amazon Kinesis Data Firehose to read and aggregate the data hourly. Transform the data and store it in Amazon S3 by using a short-lived Amazon EMR cluster.

B. 使用Amazon Kinesis Data Firehose 每小時讀取和聚合資料。使用短期Amazon EMR 叢集轉換資料並將其儲存在Amazon S3 中。

C. Use Amazon Kinesis Data Analytics to read and aggregate the data hourly. Transform the data and store it in Amazon S3 by using Amazon Kinesis Data Firehose.

C. 使用Amazon Kinesis Data Analytics 每小時讀取和聚合資料。使用Amazon Kinesis Data Firehose 轉換資料並將其儲存在Amazon S3 中。

D. Use Amazon Kinesis Data Firehose to read and aggregate the data hourly. Transform the data and store it in Amazon S3 by using AWS Lambda.

D. 使用Amazon Kinesis Data Firehose 每小時讀取和聚合資料。使用AWS Lambda 轉換資料並將其儲存在Amazon S3 中。

問題# 274

A medical device company is building a machine learning (ML) model to predict the likelihood of device recall based on customer data that the company collects from a plain text survey. One of the survey questions asks which medications the customer is taking. The data for this field contains the names of medications that customers enter manually. Customers misspell some of the medication names. The column that contains the medication name data gives a categorical feature with high cardinality but redundancy. What is the MOST effective way to encode this categorical feature into a numeric feature?

一家醫療設備公司正在建立一個機器學習(ML)模型，以根據公司從純文字調查中收集的客戶資料來預測設備召回的可能性。調查問題之一是詢問客戶正在服用哪些藥物。此欄位的資料包含客戶手動輸入的藥物名稱。客戶拼錯了一些藥物名稱。包含藥物名稱資料的欄位給出了具有高基數但冗餘的分類特徵。將此分類特徵編碼為數字特徵的最有效方法是什麼？

A. Spell check the column. Use Amazon SageMaker one-hot encoding on the column to transform a categorical feature to a numerical feature.

A. 對列進行拼字檢查。使用Amazon SageMaker 單熱編碼對列進行拼字檢查，將分類特徵轉換為數值特徵。

B. Fix the spelling in the column by using char-RNN. Use Amazon SageMaker Data Wrangler one-hot encoding to transform a categorical feature to a numerical feature.

B. 使用char-RNN 修復列中的拼字。使用Amazon SageMaker Data Wrangler 單熱編碼對列進行拼字檢查，將分類特徵轉換為數值特徵。

C. Use Amazon SageMaker Data Wrangler similarity encoding on the column to create embeddings of vectors of real numbers.

C. 使用Amazon SageMaker Data Wrangler 相似性編碼對列進行拼字檢查，以建立實數向量的嵌入。

D. Use Amazon SageMaker Data Wrangler ordinal encoding on the column to encode categories into an integer between 0 and the total number of categories in the column.

D. 使用Amazon SageMaker Data Wrangler 序數編碼對列進行拼字檢查，將類別編碼為介於0 和列中類別總數之間的整數。

問題# 275

A machine learning (ML) engineer has created a feature repository in Amazon SageMaker Feature Store for the company. The company has AWS accounts for development, integration, and production. The company hosts a feature store in the development account. The company uses Amazon S3 buckets to store feature values offline. The company wants to share features and to allow the integration account and the production account to reuse the features that are in the feature repository. Which combination of steps will meet these requirements? (Choose two.)

一位機器學習(ML)工程師在Amazon SageMaker Feature Store 中為公司建立了一個功能儲存庫。該公司擁有用於開發、整合和生產的AWS 帳戶。該公司在開發帳戶中託管一個功能儲存。該公司使用Amazon S3 儲存桶離線儲存功能值。該公司希望共享功能，並允許整合帳戶和生產帳戶重複使

用功能性儲存庫中的功能。哪一種步驟組合可以滿足這些要求？（選兩個。）

A. Create an IAM role in the development account that the integration account and production account can assume. Attach IAM policies to the role that allow access to the feature repository and the S3 buckets.

A. 在開發帳戶中建立整合帳戶和生產帳戶可以承擔的IAM 角色。將IAM 原則附加到允許存取功能儲存庫和S3 儲存桶的角色。

B. Share the feature repository that is associated the S3 buckets from the development account to the integration account and the production account by using AWS Resource Access Manager (AWS RAM).

B. 使用AWS 資源存取管理器(AWS RAM) 將與開發帳戶中的S3 儲存桶關聯的功能儲存庫共用到整合帳戶和生產帳戶。

C. Use AWS Security Token Service (AWS STS) from the integration account and the production account to retrieve credentials for the development account.

C. 使用整合帳戶和生產帳戶中的AWS 安全令牌服務(AWS STS) 檢索開發帳戶的憑證。

D. Set up S3 replication between the development S3 buckets and the integration and production S3 buckets.

D. 在開發S3 儲存桶與整合和生產S3 儲存桶之間設定S3 複製。

E. Create an AWS PrivateLink endpoint in the development account for SageMaker.

E. 在SageMaker 的開發帳戶中建立AWS PrivateLink 終端節點。

問題# 276

A company is building a new supervised classification model in an AWS environment. The company's data science team notices that the dataset has a large quantity of variables. All the variables are numeric. The model accuracy for training and validation is low. The model's processing time is affected by high latency. The data science team needs to increase the accuracy of the model and decrease the processing time. What should the data science team do to meet these requirements?

一家公司正在AWS 環境中建立一個新的監督分類模型。該公司的數據科學團隊注意到數據集包含大量變數。所有變數都是數字。訓練和驗證的模型準確率較低。高延遲影響了模型的處理時間。數據科學團隊需要提高模型的準確率並減少處理時間。數據科學團隊該怎麼做才能滿足這些要求？

A. Create new features and interaction variables.

A. 建立新特徵和互動變數。

B. Use a principal component analysis (PCA) model.

B. 使用主成分分析(PCA) 模型。

C. Apply normalization on the feature set.

C. 對特徵集應用規範化。

D. Use a multiple correspondence analysis (MCA) model.

D. 使用多重對應分析(MCA) 模型。

問題# 277

An exercise analytics company wants to predict running speeds for its customers by using a dataset that contains multiple health-related features for each customer. Some of the features originate from sensors that provide extremely noisy values. The company is training a regression model by using the built-in Amazon SageMaker linear learner algorithm to predict the running speeds. While the company is training the model, a data scientist observes that the training loss decreases to almost zero, but validation loss increases. Which technique should the data scientist use to optimally fit the model?

一家運動分析公司希望使用包含每個客戶的多個健康相關特徵的資料集來預測客戶的跑步速度。其中一些特徵來自提供極度雜訊值的感測器。該公司正在使用內建的Amazon SageMaker 線性學習器演算法訓練迴歸模型以預測跑步速度。在公司訓練模型時，資料科學家觀察到訓練損失減少到幾乎為零，但驗證損失增加。資料科學家應該使用哪種技術來最佳地擬合模型？

A. Add L1 regularization to the linear learner regression model.

A. 在線性學習者迴歸模型加入L1 正規化。

B. Perform a principal component analysis (PCA) on the dataset. Use the linear learner regression model.

B. 對資料集執行主成分分析(PCA)。使用線性學習者迴歸模型。

C. Perform feature engineering by including quadratic and cubic terms. Train the linear learner regression model.

C. 透過包含二次和三次項來執行特徵工程。訓練線性學習者迴歸模型。

D. Add L2 regularization to the linear learner regression model.

D. 在線性學習者迴歸模型中加入L2 正規化。

問題# 278

A company's machine learning (ML) specialist is building a computer vision model to classify 10 different traffic signs. The company has stored 100 images of each class in Amazon S3, and the company has another 10,000 unlabeled images. All the images come from dash cameras and are a size of 224 pixels $\times$ 224 pixels. After several training runs, the model is overfitting on the training data. Which actions should the ML specialist take to address this problem? (Choose two.)

一家公司的機器學習(ML) 專家正在建立一個電腦視覺模型，以對10 種不同的交通標誌進行分類。該公司已在Amazon S3 中儲存了每個類別的100 張圖像，並且該公司還有另外10,000 張未標記的圖像。所有影像均來自自行車記錄儀，尺寸為224 像素 $\times$ 224 像素。經過幾次訓練運行後，模型對訓練資料過度擬合。ML 專家應採取哪些措施來解決這個問題？（選兩個。）

A. Use Amazon SageMaker Ground Truth to label the unlabeled images.

A. 使用Amazon SageMaker Ground Truth 標記未標記的影像。

B. Use image preprocessing to transform the images into grayscale images.

B. 使用影像預處理將影像轉換為灰階影像。

C. Use data augmentation to rotate and translate the labeled images.

C. 使用資料增強來旋轉和平移標記的影像。

D. Replace the activation of the last layer with a sigmoid.

D. 用S 形函數取代最後一層的激活。

E. Use the Amazon SageMaker k-nearest neighbors (k-NN) algorithm to label the unlabeled images.

E. 使用Amazon SageMaker k 最近鄰(k-NN) 演算法標記未標記的影像。

問題# 279

A data science team is working with a tabular dataset that the team stores in Amazon S3. The team wants to experiment with different feature transformations such as categorical feature encoding. Then the team wants to visualize the resulting distribution of the dataset. After the team finds an appropriate set of feature transformations, the team wants to automate the workflow for feature transformations. Which solution will meet these requirements with the MOST operational efficiency?

一個資料科學團隊正在處理儲存在Amazon S3 中的表格資料集。團隊希望嘗試不同的特徵轉換，例如分類特徵編碼。然後，該團隊希望可視化資料集的結果分佈。在找到一組合適的特徵轉換後，團隊希望自動化特徵轉換的工作流程。哪種解決方案能夠以最高的營運效率滿足這些要求？

A. Use Amazon SageMaker Data Wrangler preconfigured transformations to explore feature transformations. Use SageMaker Data Wrangler templates for visualization. Export the feature processing workflow to a SageMaker pipeline for automation.

A. 使用Amazon SageMaker Data Wrangler 預先配置轉換來探索特徵轉換。使用SageMaker Data Wrangler 範本進行視覺化。將特徵處理工作流程匯出到SageMaker 管道以實現自動化。

B. Use an Amazon SageMaker notebook instance to experiment with different feature transformations. Save the transformations to Amazon S3. Use Amazon QuickSight for visualization. Package the feature processing steps into an AWS Lambda function for automation.

B. 使用Amazon SageMaker 筆記本實例來試驗不同的特徵轉換。將轉換儲存到Amazon S3。使用Amazon QuickSight 進行視覺化。將特徵處理步驟打包到AWS Lambda 函數中以實現自動化。

C. Use AWS Glue Studio with custom code to experiment with different feature transformations. Save the transformations to Amazon S3. Use Amazon QuickSight for visualization. Package the feature processing steps into an AWS Lambda function for automation.

C. 使用自訂程式碼的AWS Glue Studio 來試驗不同的特徵轉換。將轉換儲存到Amazon S3。使用Amazon QuickSight 進行視覺化。將特徵處理步驟打包到AWS Lambda 函數中以實現自動化。

D. Use Amazon SageMaker Data Wrangler preconfigured transformations to experiment with different feature transformations. Save the transformations to Amazon S3. Use Amazon QuickSight for visualization. Package each feature transformation step into a separate AWS Lambda function. Use AWS Step Functions for workflow automation.

D. 使用Amazon SageMaker Data Wrangler 預先配置轉換來試驗不同的特徵轉換。將轉換儲存到Amazon S3。使用Amazon QuickSight 進行視覺化。將每個特徵轉換步驟打包到單獨的AWS Lambda 函數中。使用AWS Step Functions 實現工作流程自動化。

問題# 280

A company plans to build a custom natural language processing (NLP) model to classify and prioritize user feedback. The company hosts the data and all machine learning (ML) infrastructure in the AWS Cloud. The ML team works from the company's office, which has an IPsec VPN connection to one VPC in the AWS Cloud. The company has set both the `enableDnsHostnames` attribute and the `enableDnsSupport` attribute of the VPC to true. The company's DNS resolvers point to the VPC DNS. The company does not allow the ML team to access Amazon SageMaker notebooks through connections that use the public internet. The connection must stay within a private network and within the AWS internal network. Which solution will meet these requirements with the LEAST development effort?

一家公司計劃建立一個自訂自然語言處理(NLP) 模型來對使用者回饋進行分類和優先排序。該公司在AWS 雲端中託管資料和所有機器學習(ML) 基礎設施。ML 團隊在該公司的辦公室工作，該辦公室與AWS 雲端中的一個VPC 建立了IPsec VPN 連線。該公司已將VPC 的`enableDnsHostnames` 屬性和`enableDnsSupport` 屬性都設為true。公司的DNS 解析器指向VPC DNS。該公司不允許ML 團隊透過使用公共互聯網的連接來存取Amazon SageMaker 筆記型電腦。連線必須保持在私有網路和AWS 內部網路內。哪一種解決方案可以以最少的開發工作量來滿足這些要求？

A. Create a VPC interface endpoint for the SageMaker notebook in the VPC. Access the notebook through a VPN connection and the VPC endpoint.

A. 在VPC 中為SageMaker 筆記本建立VPC 介面終端節點。透過VPN 連接和VPC 終端節點存取筆記本。

B. Create a bastion host by using Amazon EC2 in a public subnet within the VPC. Log in to the bastion host through a VPN connection. Access the SageMaker notebook from the bastion host.

B. 使用VPC 中公有子網路中的Amazon EC2 建立堡壘主機。透過VPN 連線登入堡壘主機。從堡壘主機存取SageMaker 筆記本。

C. Create a bastion host by using Amazon EC2 in a private subnet within the VPC with a NAT gateway. Log in to the bastion host through a VPN connection. Access the SageMaker notebook from the bastion host.

C. 使用VPC 中帶有NAT 閘道的私有子網路中的Amazon EC2 建立堡壘主機。透過VPN 連線登入堡壘主機。從堡壘主機存取SageMaker 筆記本。

D. Create a NAT gateway in the VPC. Access the SageMaker notebook HTTPS endpoint through a VPN connection and the NAT gateway.

D. 在VPC 中建立NAT 閘道。透過VPN 連線和NAT 閘道存取SageMaker 筆電HTTPS 終端節點。

問題# 281

A data scientist is using Amazon Comprehend to perform sentiment analysis on a dataset of one million social media posts. Which approach will process the dataset in the LEAST time?

一位資料科學家正在使用Amazon Comprehend 對一百萬個社群媒體貼文的資料集進行情緒分析。哪種方法可以在最短的時間內處理資料集？

A. Use a combination of AWS Step Functions and an AWS Lambda function to call the DetectSentiment API operation for each post synchronously.

A. 使用AWS Step Functions 和AWS Lambda 函數的組合來同步呼叫每個貼文的DetectSentiment API 操作。

B. Use a combination of AWS Step Functions and an AWS Lambda function to call the BatchDetectSentiment API operation with batches of up to 25 posts at a time.

B. 使用AWS Step Functions 和AWS Lambda 函數的組合來呼叫BatchDetectSentiment API 操作，每次最多處理25 個貼文。

C. Upload the posts to Amazon S3. Pass the S3 storage path to an AWS Lambda function that calls the StartSentimentDetectionJob API operation.

C. 將貼文上傳到Amazon S3。將S3 儲存路徑傳遞給呼叫StartSentimentDetectionJob API 作業的AWS Lambda 函數。

D. Use an AWS Lambda function to call the BatchDetectSentiment API operation with the whole dataset.

D. 使用AWS Lambda 函數對整個資料集呼叫BatchDetectSentiment API 操作。

問題# 282

A machine learning (ML) specialist at a retail company must build a system to forecast the daily sales for one of the company's stores. The company provided the ML specialist with sales data for this store from the past 10 years. The historical dataset includes the total amount of sales on each day for the store. Approximately 10% of the days in the historical dataset are missing sales data. The ML specialist builds a forecasting model based on the historical dataset. The specialist discovers that the model does not meet the performance standards that the company requires. Which action will MOST likely improve the performance for the forecasting model?

一家零售公司的機器學習(ML) 專家必須建立一個系統來預測該公司一家商店的每日銷售額。該公司向ML 專家提供了這家商店過去10 年的銷售數據。歷史資料集包括該商店每天的總銷售額。歷史資料集中大約有10% 的天數缺少銷售資料。ML 專家根據歷史資料集建立了預測模型。專家發現模型不符合公司要求的性能標準。哪種措施最有可能提高預測模型的效能？

A. Aggregate sales from stores in the same geographic area.

A. 總結同一地理區域內各商店的銷售額。

B. Apply smoothing to correct for seasonal variation.

B. 應用平滑法來校正季節性變化。

C. Change the forecast frequency from daily to weekly.

C. 將預測頻率從每日更改為每週。

D. Replace missing values in the dataset by using linear interpolation.

D. 使用線性內插法取代資料集中的缺失值。

問題# 283

A mining company wants to use machine learning (ML) models to identify mineral images in real time. A data science team built an image recognition model that is based on convolutional neural network (CNN). The team trained the model on Amazon SageMaker by using GPU instances. The team will deploy the model to a SageMaker endpoint. The data science team already knows the workload traffic patterns. The team must determine instance type and configuration for the workloads. Which solution will meet these requirements with the LEAST development effort?

一家礦業公司希望使用機器學習(ML)模型來即時識別礦物影像。資料科學團隊建構了一個基於卷積神經網路(CNN)的影像辨識模型。該團隊使用GPU實例在Amazon SageMaker上訓練該模型。該團隊將模型部署到SageMaker終端節點。數據科學團隊已經了解工作負載流量模式。該團隊必須確定工作負載的執行個體類型和配置。哪一種解決方案能夠以最少的開發工作量來滿足這些要求?

A. Register the model artifact and container to the SageMaker Model Registry. Use the SageMaker Inference Recommender Default job type. Provide the known traffic pattern for load testing to select the best instance type and configuration based on the workloads.

A. 將模型工件和容器註冊到SageMaker模型登錄。使用SageMaker推理推薦器預設作業類型。提供已知的負載測試流量模式，以根據工作負載選擇最佳執行個體類型和配置。

B. Register the model artifact and container to the SageMaker Model Registry. Use the SageMaker Inference Recommender Advanced job type. Provide the known traffic pattern for load testing to select the best instance type and configuration based on the workloads.

B. 將模型工件和容器註冊到SageMaker模型登錄。使用SageMaker推理推薦器進階作業類型。提供已知的負載測試流量模式，以根據工作負載選擇最佳執行個體類型和配置。

C. Deploy the model to an endpoint by using GPU instances. Use AWS Lambda and Amazon API Gateway to handle invocations from the web. Use open-source tools to perform load testing against the endpoint and to select the best instance type and configuration.

C. 使用GPU實例將模型部署到終端節點。使用AWS Lambda和Amazon API Gateway處理來自Web的呼叫。使用開源工具對終端節點執行負載測試並選擇最佳執行個體類型和配置。

D. Deploy the model to an endpoint by using CPU instances. Use AWS Lambda and Amazon API Gateway to handle invocations from the web. Use open-source tools to perform load testing against the endpoint and to select the best instance type and configuration.

D. 使用CPU實例將模型部署到終端節點。使用AWS Lambda和Amazon API Gateway處理來自Web的呼叫。使用開源工具對終端節點執行負載測試並選擇最佳執行個體類型和配置。

b 的呼叫。使用開源工具對終端節點執行負載測試並選擇最佳執行個體類型和配置。

問題# 284

A company is building custom deep learning models in Amazon SageMaker by using training and inference containers that run on Amazon EC2 instances. The company wants to reduce training costs but does not want to change the current architecture. The SageMaker training job can finish after interruptions. The company can wait days for the results. Which combination of resources should the company use to meet these requirements MOST cost-effectively? (Choose two.)

一家公司正在使用在Amazon EC2 執行個體上運行的訓練和推理容器在Amazon SageMaker 中建立自訂深度學習模型。該公司希望降低培訓成本，但不想改變目前的架構。SageMaker 訓練工作可以在中斷後完成。該公司可以等待幾天才能獲得結果。公司應該使用哪種資源組合來最經濟有效率地滿足這些要求？（選兩個。）

A. On-Demand Instances.

A. 按需實例。

B. Checkpoints.

B. 檢查點。

C. Reserved Instances.

C. 預留實例。

D. Incremental training.

D. 增量訓練。

E. Spot instances.

E. 競價實例。

問題# 285

A company hosts a public web application on AWS. The application provides a user feedback feature that consists of free-text fields where users can submit text to provide feedback. The company receives a large amount of free-text user feedback from the online web application. The product managers at the company classify the feedback into a set of fixed categories including user interface issues, performance issues, new feature request, and chat issues for further actions by the company's engineering teams. A machine learning (ML) engineer at the company must automate the classification of new user feedback into these fixed categories by using Amazon SageMaker. A large set of accurate data is available from the historical user feedback that the product managers previously classified. Which solution should the ML engineer apply to perform multi-class text classification of the user feedback?

一家公司在AWS 上託管一個公共Web 應用程式。該應用程式提供用戶回饋功能，該功能由自由文字欄位組成，用戶可以在其中提交文字以提供回饋。該公司從線上Web 應用程式收到大量自由文字

使用者回饋。公司的產品經理將回饋分為一組固定類別，包括使用者介面問題、效能問題、新功能請求和聊天問題，以便公司工程團隊採取進一步行動。公司的機器學習(ML)工程師必須使用Amazon SageMaker 自動將新使用者回饋分類到這些固定類別。產品經理先前分類的歷史使用者回饋中提供了大量準確的數據。ML工程師應應用哪種解決方案來對使用者回饋執行多類文字分類？

A. Use the SageMaker Latent Dirichlet Allocation (LDA) algorithm.

A. 使用SageMaker 潛在狄利克雷分配(LDA) 演算法。

B. Use the SageMaker BlazingText algorithm.

B. 使用SageMaker BlazingText 演算法。

C. Use the SageMaker Neural Topic Model (NTM) algorithm.

C. 使用SageMaker 神經主題模型(NTM) 演算法。

D. Use the SageMaker CatBoost algorithm.

D. 使用SageMaker CatBoost 演算法。

問題# 286

A digital media company wants to build a customer churn prediction model by using tabular data. The model should clearly indicate whether a customer will stop using the company's services. The company wants to clean the data because the data contains some empty fields, duplicate values, and rare values. Which solution will meet these requirements with the LEAST development effort?

一家數位媒體公司希望使用表格資料建立客戶流失預測模型。此模型應明確指出客戶是否會停止使用本公司的服務。該公司希望清理數據，因為數據包含一些空白字段、重複值和稀有值。哪一種解決方案能夠以最少的開發工作量來滿足這些要求？

A. Use SageMaker Canvas to automatically clean the data and to prepare a categorical model.

A. 使用SageMaker Canvas 自動清理資料並準備分類模型。

B. Use SageMaker Data Wrangler to clean the data. Use the built-in SageMaker XGBoost algorithm to train a classification model.

B. 使用SageMaker Data Wrangler 清理資料。使用內建的SageMaker XGBoost 演算法訓練分類模型。

C. Use SageMaker Canvas automatic data cleaning and preparation tools. Use the built-in SageMaker XGBoost algorithm to train a regression model.

C. 使用SageMaker Canvas 自動資料清理和準備工具。使用內建的SageMaker XGBoost 演算法訓練迴歸模型。

D. Use SageMaker Data Wrangler to clean the data. Use the SageMaker Autopilot to train a regression model.

D. 使用SageMaker Data Wrangler 清理資料。使用SageMaker Autopilot 訓練迴歸模型。

問題# 287

A data engineer is evaluating customer data in Amazon SageMaker Data Wrangler. The data engineer will use the customer data to create a new model to predict customer behavior. The engineer needs to increase the model performance by checking for multicollinearity in the dataset. Which steps can the data engineer take to accomplish this with the LEAST operational effort? (Choose two.)

一名資料工程師正在Amazon SageMaker Data Wrangler 中評估客戶資料。資料工程師將使用客戶資料建立新模型來預測客戶行為。工程師需要透過檢查資料集中的多重共線性來提高模型效能。資料工程師可以採取哪些步驟以最少的營運工作量來實現此目標？（選兩個。）

A. Use SageMaker Data Wrangler to refit and transform the dataset by applying one-hot encoding to category-based variables.

A. 使用SageMaker Data Wrangler 透過對基於類別的變數應用獨熱編碼來重新擬合和轉換資料集。

B. Use SageMaker Data Wrangler diagnostic visualization. Use principal components analysis (PCA) and singular value decomposition (SVD) to calculate singular values.

B. 使用SageMaker Data Wrangler 診斷視覺化。使用主成分分析(PCA) 和奇異值分解(SVD) 計算奇異值。

C. Use the SageMaker Data Wrangler Quick Model visualization to quickly evaluate the dataset and to produce importance scores for each feature.

C. 使用SageMaker Data Wrangler 快速模型視覺化快速評估資料集並為每個特徵產生重要性分數。

D. Use the SageMaker Data Wrangler Min Max Scaler transform to normalize the data.

D. 使用SageMaker Data Wrangler Min Max Scaler 轉換來標準化資料。

E. Use SageMaker Data Wrangler diagnostic visualization. Use least absolute shrinkage and selection operator (LASSO) to plot coefficient values from a LASSO model that is trained on the dataset.

E. 使用SageMaker Data Wrangler 診斷視覺化。使用最小絕對收縮和選擇運算子(LASSO) 繪製在資料集上訓練的LASSO 模型的係數值。

問題# 288

A company processes millions of orders every day. The company uses Amazon DynamoDB tables to store order information. When customers submit new orders, the new orders are immediately added to the DynamoDB tables. New orders arrive in the DynamoDB tables continuously. A data scientist must build a peak-time prediction solution. The data scientist must also create an Amazon QuickSight dashboard to display near real-time order insights. The data scientist needs to build a solution that will give QuickSight access to the data as soon as new order information arrives. Which solution will meet these requirements with the LEAST delay between when a new order is processed and when QuickSight can access the new order information?

一家公司每天處理數百萬個訂單。該公司使用Amazon DynamoDB 表來儲存訂單資訊。當客戶提交新訂單時，新訂單會立即加入DynamoDB 表中。新訂單不斷到達DynamoDB 表中。資料科學家

必須建立一個高峰時間預測解決方案。資料科學家還必須建立一個Amazon QuickSight 儀表板來顯示近乎即時的訂單洞察。資料科學家需要建立一個解決方案，讓QuickSight 在新訂單資訊到達後立即存取資料。哪種解決方案可以滿足這些要求，並且處理新訂單和QuickSight 可以存取新訂單資訊之間的延遲最少？

A. Use AWS Glue to export the data from Amazon DynamoDB to Amazon S3. Configure QuickSight to access the data in Amazon S3.

A. 使用AWS Glue 將資料從Amazon DynamoDB 匯出到Amazon S3。配置QuickSight 以存取Amazon S3 中的資料。

B. Use Amazon Kinesis Data Streams to export the data from Amazon DynamoDB to Amazon S3. Configure QuickSight to access the data in Amazon S3.

B. 使用Amazon Kinesis Data Streams 將資料從Amazon DynamoDB 匯出到Amazon S3。配置QuickSight 以存取Amazon S3 中的資料。

C. Use an API call from QuickSight to access the data that is in Amazon DynamoDB directly.

C. 使用QuickSight 的API 呼叫直接存取Amazon DynamoDB 中的資料。

D. Use Amazon Kinesis Data Firehose to export the data from Amazon DynamoDB to Amazon S3. Configure QuickSight to access the data in Amazon S3.

D. 使用Amazon Kinesis Data Firehose 將資料從Amazon DynamoDB 匯出到Amazon S3。配置QuickSight 以存取Amazon S3 中的資料。

問題# 289

A data engineer is preparing a dataset that a retail company will use to predict the number of visitors to stores. The data engineer created an Amazon S3 bucket. The engineer subscribed the S3 bucket to an AWS Data Exchange data product for general economic indicators. The data engineer wants to join the economic indicator data to an existing table in Amazon Athena to merge with the business data. All these transformations must finish running in 30-60 minutes. Which solution will meet these requirements MOST cost-effectively?

一名資料工程師正在準備一個資料集，零售公司將使用該資料集來預測商店的訪客數量。資料工程師創建了一個Amazon S3 儲存桶。工程師將S3 儲存桶訂閱到AWS Data Exchange 資料產品以取得一般經濟指標。資料工程師希望將經濟指標資料加入Amazon Athena 中的現有表中，以與業務資料合併。所有這些轉換必須在30-60 分鐘內完成。哪一種解決方案能夠以最具成本效益的方式滿足這些要求？

A. Configure the AWS Data Exchange product as a producer for an Amazon Kinesis data stream. Use an Amazon Kinesis Data Firehose delivery stream to transfer the data to Amazon S3. Run an AWS Glue job that will merge the existing business data with the Athena table. Write the result set back to Amazon S3.

A. 將AWS Data Exchange 產品配置為Amazon Kinesis 資料流的生產者。使用Amazon Kinesis Data Firehose 傳輸流將資料傳輸到Amazon S3。執行AWS Glue 作業，將現有業務資料與Athena 表合併。將結果集寫回Amazon S3。

B. Use an S3 event on the AWS Data Exchange S3 bucket to invoke an AWS Lambda function. Program the Lambda function to use Amazon SageMaker Data Wrangler to merge the existing business data with the Athena table. Write the result set back to Amazon S3.

B. 使用AWS Data Exchange S3 儲存桶上的S3 事件呼叫AWS Lambda 函數。對Lambda 函數進行程式設計以使用Amazon SageMaker Data Wrangler 將現有業務資料與Athena 表合併。將結果集寫回Amazon S3。

C. Use an S3 event on the AWS Data Exchange S3 bucket to invoke an AWS Lambda function. Program the Lambda function to run an AWS Glue job that will merge the existing business data with the Athena table. Write the results back to Amazon S3.

C. 使用AWS Data Exchange S3 儲存桶上的S3 事件呼叫AWS Lambda 函數。對Lambda 函數進行程式設計以執行AWS Glue 作業，將現有業務資料與Athena 表合併。將結果寫回Amazon S3。

D. Provision an Amazon Redshift cluster. Subscribe to the AWS Data Exchange product and use the product to create an Amazon Redshift table. Merge the data in Amazon Redshift. Write the results back to Amazon S3.

D. 配置Amazon Redshift 叢集。訂閱AWS Data Exchange 產品並使用該產品建立Amazon Redshift 表。合併Amazon Redshift 中的資料。將結果寫回Amazon S3。

問題# 290

A company operates large cranes at a busy port. The company plans to use machine learning (ML) for predictive maintenance of the cranes to avoid unexpected breakdowns and to improve productivity. The company already uses sensor data from each crane to monitor the health of the cranes in real time. The sensor data includes rotation speed, tension, energy consumption, vibration, pressure, and temperature for each crane. The company contracts AWS ML experts to implement an ML solution. Which potential findings would indicate that an ML-based solution is suitable for this scenario? (Choose two.)

一家公司在繁忙的港口運營大型起重機。該公司計劃使用機器學習(ML) 對起重機進行預測性維護，以避免意外故障並提高生產力。該公司已經使用來自每台起重機的傳感器數據來即時監控起重機的運作狀況。感測器數據包括每台起重機的轉速、張力、能耗、振動、壓力和溫度。該公司聘請AWS ML 專家來實施ML 解決方案。哪些潛在發現表明基於ML 的解決方案適合此場景？（選兩個。）

A. The historical sensor data does not include a significant number of data points and attributes for certain time periods.

A. 歷史感測器資料不包含特定時間段的大量資料點和屬性。

B. The historical sensor data shows that simple rule-based thresholds can predict crane failures.

B. 歷史感測器數據顯示，簡單的基於規則的閾值可以預測起重機故障。

C. The historical sensor data contains failure data for only one type of crane model that is in operation and lacks failure data of most other types of crane that are in operation.

C. 歷史感測器數據僅包含一種正在運行的起重機模型的故障數據，缺少大多數其他正在運行的起重機類型的故障數據。

D. The historical sensor data from the cranes are available with high granularity for the last 3 years.

D. 過去3年，起重機的歷史感測器資料具有高粒度。

E. The historical sensor data contains most common types of crane failures that the company wants to predict.

E. 歷史感測器資料包含公司想要預測的最常見起重機故障類型。

問題# 291

A company wants to create an artificial intelligence (AI) yoga instructor that can lead large classes of students. The company needs to create a feature that can accurately count the number of students who are in a class. The company also needs a feature that can differentiate students who are performing a yoga stretch correctly from students who are performing a stretch incorrectly. Determine whether students are performing a stretch correctly, the solution needs to measure the location and angle of each student's arms and legs. A data scientist must use Amazon SageMaker to access video footage of a yoga class by extracting image frames and applying computer vision models. Which combination of models will meet these requirements with the LEAST effort? (Choose two.)

一家公司想要創建一個人工智慧(AI) 瑜珈教練，可以帶領大量學生上課。該公司需要創建一項功能，可以準確計算課堂上學生的數量。該公司還需要一項功能，可以區分正確進行瑜珈伸展的學生和不正確進行伸展的學生。確定學生是否正確進行伸展，解決方案需要測量每個學生手臂和腿的位置和角度。資料科學家必須使用Amazon SageMaker 透過提取圖像幀並應用電腦視覺模型來存取瑜珈課程的影片片段。哪一種模型組合可以以最少的努力滿足這些要求？（選兩個。）

A. Image Classification.

A. 影像分類。

B. Optical Character Recognition (OCR).

B. 光學字元辨識(OCR)。

C. Object Detection.

C. 物體檢測。

D. Pose estimation.

D. 姿勢估計。

E. Image Generative Adversarial Networks (GANs).

E. 影像生成對抗網路(GAN)。

問題# 292

An ecommerce company has used Amazon SageMaker to deploy a factorization machines (FM) model to suggest products for customers. The company's data science team has developed two new models by using the TensorFlow and PyTorch deep learning frameworks. The company needs to use A/B testing to evaluate the new models against the deployed model. The required A/B testing setup is as follows:

- 15% of traffic to the TensorFlow model, and 15% of traffic to the PyTorch model.
- customers who are from Europe, send all traffic to the TensorFlow model.

Which architecture can the company use to implement the required A/B testing setup?

一家電子商務公司已使用Amazon SageMaker 部署分解機(FM) 模型來向客戶推薦產品。該公司的數據科學團隊使用TensorFlow 和PyTorch 深度學習框架開發了兩個新模型。該公司需要使用A/B 測試來根據已部署的模型評估新模型。所需的A/B 測試設定如下：

- 將70% 的流量傳送至FM 模型
- 將15% 的流量傳送至TensorFlow 模型
- 將15% 的流量傳送至PyTorch 模型。
- 對於來自歐洲的客戶，將所有流量傳送至TensorFlow 模型。

公司可以使用哪種架構來實現所需的A/B 測試設定？

A. Create two new SageMaker endpoints for the TensorFlow and PyTorch models in addition to the existing SageMaker endpoint. Create an Application Load Balancer. Create a target group for each endpoint. Configure listener rules and add weight to the target groups. To send traffic to the TensorFlow model for customers who are from Europe, create an additional listener rule to forward traffic to the TensorFlow target group.

A. 除了現有的SageMaker 終端節點之外，還為TensorFlow 和PyTorch 模型建立兩個新的SageMaker 終端節點。建立應用程式負載平衡器。為每個終端節點建立一個目標群組。配置偵聽器規則並為目標群組新增權重。若要將流量傳送至來自歐洲的客戶的TensorFlow 模型，請建立額外的偵聽器規則以將流量轉送至TensorFlow 目標群組。

B. Create two production variants for the TensorFlow and PyTorch models. Create an auto scaling policy and configure the desired A/B weights to direct traffic to each production variant. Update the existing SageMaker endpoint with the auto scaling policy. To send traffic to the TensorFlow model for customers who are from Europe, set the TargetVariant header in the request to point to the variant name of the TensorFlow model.

B. 為TensorFlow 和PyTorch 模型建立兩個生產變體。建立自動擴展策略並配置所需的A/B 權重以將流量引導到每個生產變體。使用自動擴充策略更新現有的SageMaker 終端節點。若要將流量傳送至來自歐洲的客戶的TensorFlow 模型，請將請求中的TargetVariant 標頭設定為指向TensorFlow 模型的變體名稱。

C. Create two new SageMaker endpoints for the TensorFlow and PyTorch models in addition to the existing SageMaker endpoint. Create a Network Load Balancer. Create a target group for each endpoint. Configure listener rules and add weight to the target groups. To send traffic to the TensorFlow model for customers who are from Europe, create an additional listener rule to forward traffic to the TensorFlow target group.

C. 除了現有的SageMaker 終端節點之外，還為TensorFlow 和PyTorch 模型建立兩個新的SageMaker 終端節點。建立網路負載平衡器。為每個終端節點建立一個目標群組。配置偵聽器規則並為目標群組新增權重。若要將流量傳送至來自歐洲的客戶的TensorFlow 模型，請建立額外的偵聽器規則以將流量轉送至TensorFlow 目標群組。

D. Create two production variants for the TensorFlow and PyTorch models. Specify the weight for each production variant in the SageMaker endpoint configuration. Update the existing SageMaker endpoint with the new configuration. To send traffic to the TensorFlow model for customers who are from Europe, set the TargetVariant header in the request to point to the variant name of the TensorFlow model.

D. 為TensorFlow 和PyTorch 模型建立兩個生產變體。在SageMaker 終端節點配置中為每個生產變體指定權重。使用新配置更新現有的SageMaker 終端節點。若要將流量傳送至來自歐洲的客戶的T

問題# 293

A data scientist stores financial datasets in Amazon S3. The data scientist uses Amazon Athena to query the datasets by using SQL. The data scientist uses Amazon SageMaker to deploy a machine learning (ML) model. The data scientist wants to obtain inferences from the model at the SageMaker endpoint. However, when the data scientist attempts to invoke the SageMaker endpoint, the data scientist receives SQL statement failures. The data scientist's IAM user is currently unable to invoke the SageMaker endpoint. Which combination of actions will give the data scientist's IAM user the ability to invoke the SageMaker endpoint? (Choose three.)

資料科學家將財務資料集儲存在Amazon S3 中。資料科學家使用Amazon Athena 透過SQL 查詢資料集。資料科學家使用Amazon SageMaker 部署機器學習(ML) 模型。資料科學家希望從SageMaker 終端節點的模型中獲取推理。但是，當資料科學家嘗試呼叫SageMaker 終端節點時，資料科學家會收到SQL 語句失敗。資料科學家的IAM 使用者目前無法呼叫SageMaker 終端節點。哪種操作組合將使資料科學家的IAM 使用者能夠呼叫SageMaker 終端節點？（選三個。）

A. Attach the AmazonAthenaFullAccess AWS managed policy to the user identity.

A. 將AmazonAthenaFullAccess AWS 託管策略附加到使用者身分。

B. Include a policy statement for the data scientist's IAM user that allows the IAM user to perform the sagemaker:InvokeEndpoint action.

B. 為資料科學家的IAM 使用者新增一條策略聲明，允許該IAM 使用者執行sagemaker:InvokeEndpoint 操作。

C. Include an inline policy for the data scientist's IAM user that allows SageMaker to read S3 objects.

C. 為資料科學家的IAM 使用者新增一條內聯策略，允許SageMaker 讀取S3 物件。

D. Include a policy statement for the data scientist's IAM user that allows the IAM user to perform the sagemaker:GetRecord action.

D. 為資料科學家的IAM 使用者新增一條策略聲明，允許該IAM 使用者執行sagemaker:GetRecord 操作。

E. Include the SQL statement "USING EXTERNAL FUNCTION ml_function_name" in the Athena SQL query.

E. 在Athena SQL 查詢中加入SQL 語句「USING EXTERNAL FUNCTION ml_function_name」。

F. Perform a user remapping in SageMaker to map the IAM user to another IAM user that is on the hosted endpoint.

F. 在SageMaker 中執行使用者重新映射，以將IAM 使用者對應到託管終端節點上的另一個IAM 使用者。

問題# 294

A data scientist is building a linear regression model. The scientist inspects the dataset and notices that the mode of the distribution is lower than the median, and the median is lower than the mean. Which data transformation will give the data scientist the ability to apply a linear regression model?

一位資料科學家正在建立一個線性迴歸模型。科學家檢查資料集並注意到分佈的眾數低於中位數，而中位數低於平均值。哪一種資料轉換將使資料科學家能夠應用線性迴歸模型？

A. Exponential transformation.

A. 指數變換。

B. Logarithmic transformation.

B. 對數轉換。

C. Polynomial transformation.

C. 多項式變換。

D. Sinusoidal transformation.

D. 正弦變換。

問題# 295

A data scientist receives a collection of insurance claim records. Each record includes a claim ID, the final outcome of the insurance claim, and the date of the final outcome. The final outcome of each claim is a selection from among 200 outcome categories. Some claim records include only partial information. However, incomplete claim records include only 3 or 4 outcome categories from among the 200 available outcome categories. The collection includes hundreds of records for each outcome category. The records are from the previous 3 years. The data scientist must create a solution to predict the number of claims that will be in each outcome category every month, several months in advance. Which solution will meet these requirements?

資料科學家收到一組保險索賠記錄。每筆記錄都包含索賠ID、保險索賠的最終結果以及最終結果的日期。每項索賠的最終結果是從200個結果類別中選擇的。有些索賠記錄僅包含部分資訊。但是，不完整的索賠記錄僅包含200個可用結果類別中的3或4個結果類別。該集合包含每個結果類別的數百筆記錄。這些記錄來自前3年。資料科學家必須提前幾個月創建一個解決方案來預測每個結果類別每月的索賠數量。哪種解決方案可以滿足這些要求？

A. Perform classification every month by using supervised learning of the 200 outcome categories based on claim contents.

A. 每月使用基於索賠內容的200個結果類別的監督學習進行分類。

B. Perform reinforcement learning by using claim IDs and dates. Instruct the insurance agents who submit the claim records to estimate the expected number of claims in each outcome category every month.

B. 使用索賠ID和日期進行強化學習。指示提交索賠記錄的保險代理人估計每月每個結果類別的預期索賠數量。

C. Perform forecasting by using claim IDs and dates to identify the expected number of claims in each outcome category every month.

C. 使用索賠ID 和日期進行預測，以確定每月每個結果類別的預期索賠數量。

D. Perform classification by using supervised learning of the outcome categories for which partial information on claim contents is provided. Perform forecasting by using claim IDs and dates for all other outcome categories.

D. 使用監督學習對索賠內容提供部分資訊的結果類別進行分類。使用所有其他結果類別的索賠ID 和日期進行預測。

問題# 296

A retail company stores 100 GB of daily transactional data in Amazon S3 at periodic intervals. The company wants to identify the schema of the transactional data. The company also wants to perform transformations on the transactional data that is in Amazon S3. The company wants to use a machine learning (ML) approach to detect fraud in the transformed data. Which combination of solutions will meet these requirements with the LEAST operational overhead? (Choose three.)

一家零售公司定期在Amazon S3 中儲存100 GB 的每日交易資料。該公司希望識別交易資料的架構。該公司還希望對Amazon S3 中的交易資料執行轉換。該公司希望使用機器學習(ML) 方法來偵測轉換後的資料中的詐欺行為。哪種解決方案組合可以滿足這些要求且營運開銷最少？（選擇三項。）

A. Use Amazon Athena to scan the data and identify the schema.

A. 使用Amazon Athena 掃描資料並識別模式。

B. Use AWS Glue crawlers to scan the data and identify the schema.

B. 使用AWS Glue 爬蟲掃描資料並識別模式。

C. Use Amazon Redshift to store procedures to perform data transformations.

C. 使用Amazon Redshift 預存程序以執行資料轉換。

D. Use AWS Glue workflows and AWS Glue jobs to perform data transformations.

D. 使用AWS Glue 工作流程和AWS Glue 作業執行資料轉換。

E. Use Amazon Redshift ML to train a model to detect fraud.

E. 使用Amazon Redshift ML 訓練模型以偵測詐欺。

F. Use Amazon Fraud Detector to train a model to detect fraud.

F. 使用Amazon Fraud Detector 訓練模型以偵測詐欺。

問題# 297

A data scientist uses Amazon SageMaker Data Wrangler to define and perform transformations and feature engineering on historical data. The data scientist saves the transformations to SageMaker Feature Store. The historical data is periodically uploaded to an Amazon S3 bucket. The data scientist needs to transform the new historic data and add it to the online feature store. The data scientist needs to prepare the new historic data for training and inference by using native integrations. Which solution will meet these requirements with the LEAST development effort?

資料科學家使用Amazon SageMaker Data Wrangler 定義和執行歷史資料的轉換和特徵工程。資料科學家將轉換儲存到SageMaker Feature Store。歷史資料定期上傳到Amazon S3 儲存桶。資料科學家需要轉換新的歷史資料並將其添加到線上特徵儲存。資料科學家需要使用本機整合來準備新的歷史資料以進行訓練和推理。哪一種解決方案能夠以最少的開發工作量來滿足這些要求？

A. Use AWS Lambda to run a predefined SageMaker pipeline to perform the transformations on each new dataset that arrives in the S3 bucket.

A. 使用AWS Lambda 執行預先定義的SageMaker 管道，對到達S3 儲存桶的每個新資料集執行轉換。

B. Run an AWS Step Functions step and a predefined SageMaker pipeline to perform the transformations on each new dataset that arrives in the S3 bucket.

B. 執行AWS Step Functions 步驟和預先定義的SageMaker 管道，對到達S3 儲存桶的每個新資料集執行轉換。

C. Use Apache Airflow to orchestrate a set of predefined transformations on each new dataset that arrives in the S3 bucket.

C. 使用Apache Airflow 對到達S3 儲存桶的每個新資料集編排一組預先定義的轉換。

D. Configure Amazon EventBridge to run a predefined SageMaker pipeline to perform the transformations when a new data is detected in the S3 bucket.

D. 設定Amazon EventBridge 以執行預先定義的SageMaker 管道，以便在S3 儲存桶中偵測到新資料時執行轉換。

問題# 298

An insurance company developed a new experimental machine learning (ML) model to replace an existing model that is in production. The company must validate the quality of predictions from the new experimental model in a production environment before the company uses the new experimental model to serve general user requests. New one model can serve user requests at a time. The company must measure the performance of the new experimental model without affecting the current live traffic. Which solution will meet these requirements?

一家保險公司開發了一種新的實驗性機器學習(ML) 模型來取代現有的生產模型。該公司必須在生產環境中驗證新實驗模型的預測質量，然後才能使用該模型來滿足一般用戶請求。新模型一次只能滿足一個使用者請求。該公司必須在不影響當前即時流量的情況下測量新實驗模型的性能。哪種解決方案可以滿足這些要求？

A. A/B testing.

A. A/B 測試。

B. Canary release.

B. 金絲雀發布。

C. Shadow deployment.

C. 影子部署。

D. Blue/green deployment.

D. 藍/綠部署。

問題# 299

A company deployed a machine learning (ML) model on the company website to predict real estate prices. Several months after deployment, an ML engineer notices that the accuracy of the model has gradually decreased. The ML engineer needs to improve the accuracy of the model. The engineer also needs to receive notifications for any future performance issues. Which solution will meet these requirements?

一家公司在公司網站上部署了一個機器學習(ML)模型來預測房地產價格。部署幾個月後，ML工程師注意到模型的準確性逐漸下降。ML工程師需要提高模型的準確性。工程師還需要收到有關任何未來性能問題的通知。哪種解決方案可以滿足這些要求？

A. Perform incremental training to update the model. Activate Amazon SageMaker Model Monitor to detect model performance issues and to send notifications.

A. 執行增量訓練以更新模型。啟動Amazon SageMaker 模型監控器以偵測模型效能問題並發送通知。

B. Use Amazon SageMaker Model Governance. Configure Model Governance to automatically adjust model hyperparameters. Create a performance threshold alarm in Amazon CloudWatch to send notifications.

B. 使用Amazon SageMaker 模型管理。配置模型管理以自動調整模型超參數。在Amazon CloudWatch 中建立效能閾值警報以發送通知。

C. Use Amazon SageMaker Debugger with appropriate thresholds. Configure Debugger to send Amazon CloudWatch alarms to alert the team. Retrain the model by using only data from the previous several months.

C. 使用具有適當閾值的Amazon SageMaker 偵錯器。配置調試器以發送Amazon CloudWatch 警報以提醒團隊。僅使用前幾個月的資料重新訓練模型。

D. Use only data from the previous several months to perform incremental training to update the model. Use Amazon SageMaker Model Monitor to detect model performance issues and to send notifications.

D. 僅使用前幾個月的資料執行增量訓練以更新模型。使用Amazon SageMaker 模型監視器偵測模型效能問題並發送通知。

問題# 300

A university wants to develop a targeted recruitment strategy to increase new student enrollment. A data scientist gathers information about the academic performance history of students. The data scientist wants to use the data to build student profiles. The university will use the profiles to direct resources to recruit students who are likely to enroll in the university. Which combination of steps should the data scientist take to predict whether a particular student applicant is likely to enroll in the university? (Choose two.)

一所大學希望制定有針對性的招生策略，以增加新生入學率。一位資料科學家收集有關學生學業成績歷史的資訊。資料科學家希望使用這些資料來建立學生檔案。大學將利用這些檔案來指導資源招募可能入讀該大學的學生。資料科學家應該採取哪些步驟組合來預測特定學生申請人是否有可能入讀該大學？（選兩個。）

A. Use Amazon SageMaker Ground Truth to sort the data into two groups named "enrolled" or "not enrolled".

A. 使用Amazon SageMaker Ground Truth 將資料分為兩個名為「已註冊」或「未註冊」的群組。

B. Use a forecasting algorithm to run predictions.

B. 使用預測演算法運行預測。

C. Use a regression algorithm to run predictions.

C. 使用迴歸演算法運行預測。

D. Use a classification algorithm to run predictions.

D. 使用分類演算法運行預測。

E. Use the built-in Amazon SageMaker k-means algorithm to cluster the data into two groups named "enrolled" or "not enrolled".

E. 使用內建的Amazon SageMaker k-means 演算法將資料聚類為名為「已註冊」或「未註冊」的兩個群組。

問題# 301

A machine learning (ML) specialist is using the Amazon SageMaker DeepAR forecasting algorithm to train a model on CPU-based Amazon EC2 On-Demand instances. The model currently takes multiple hours to train. The ML specialist wants to decrease the training time of the model. Which approaches will meet this requirement? (Choose two.)

一位機器學習(ML)專家正在使用Amazon SageMaker DeepAR 預測演算法在基於CPU 的Amazon EC2 按需實例上訓練模型。該模型目前需要幾個小時才能訓練完成。ML 專家希望減少模型的訓練時間。哪些方法可以滿足這項要求？（選兩個。）

A. Replace On-Demand Instances with Spot Instances.

A. 以競價型實例取代隨選實例。

B. Configure model auto scaling dynamically to adjust the number of instances automatically.

B. 動態配置模型自動擴展以自動調整實例數量。

C. Replace CPU-based EC2 instances with GPU-based EC2 instances.

C. 以基於GPU 的EC2 執行個體取代基於CPU 的EC2 執行個體。

D. Use multiple training instances.

D. 使用多個訓練實例。

E. Use a pre-trained version of the model. Run incremental training.

E. 使用預先訓練的模型版本。運行增量訓練。

問題# 302

A chemical company has developed several machine learning (ML) solutions to identify chemical process abnormalities. The time series values of independent variables and the labels are available for the past 2 years and are sufficient to accurately model the problem. The regular operation label is marked as 0. The abnormal operation label is marked as 1. Process abnormalities have a significant negative effect on the company's profits. The company must avoid these abnormalities. Which metrics will indicate an ML solution that will provide the GREATEST probability of detecting an abnormality?

一家化學公司開發了幾種機器學習(ML) 解決方案來識別化學過程異常。獨立變數和標籤的時間序列值在過去2 年內可用，足以準確地模擬問題。常規操作標籤標示為0，異常操作標籤標示為1。過程異常對公司的利潤有顯著的負面影響。公司必須避免這些異常。哪些指標將顯示ML 解決方案將提供檢測異常的最大機率？

A. Precision = 0.91 - Recall = 0.6.

A. 精確度= 0.91 - 召回率= 0.6

B. Precision = 0.61 - Recall = 0.98.

B. 精確度= 0.61 - 召回率= 0.98

C. Precision = 0.7 - Recall = 0.9.

C. 精確度= 0.7 - 召回率= 0.9

D. Precision = 0.98 - Recall = 0.8.

D. 精確度= 0.98 - 召回率= 0.8 .

問題# 303

An online delivery company wants to choose the fastest courier for each delivery at the moment an order is placed. The company wants to implement this feature for existing users and new users of its application. Data scientists have trained separate models with XGBoost for this purpose, and the models are stored in Amazon S3. There is one model for each city where the company operates. Operation engineers are hosting these models in Amazon EC2 for responding to the web client requests, with one instance for each model, but the instances have only a 5% utilization in CPU and memory. The operation engineers want to avoid managing unnecessary resources. Which solution will enable the company to achieve its goal with the LEAST operational overhead?

一家線上配送公司希望在下訂單時為每次配送選擇最快的快遞員。該公司希望為其應用程式的現有用戶和新用戶實現此功能。資料科學家為此使用XGBoost 訓練了單獨的模型，這些模型儲存在Amazon S3 中。該公司營運的每個城市都有一個模型。營運工程師將這些模型託管在Amazon EC2 中以回應Web 用戶端請求，每個模型一個實例，但這些執行個體的CPU 和記憶體利用率僅為5%。營運工程師希望避免管理不必要的資源。哪種解決方案能讓公司以最少的營運開銷實現其目標？

A. Create an Amazon SageMaker notebook instance for pulling all the models from Amazon S3 using the boto3 library. Remove the existing instances and use the notebook to perform a SageMaker batch transform for performing inferences offline for all the possible users in all the cities. Store the results in different files in Amazon S3. Point the web client to the files.

A. 使用boto3 函式庫建立一個Amazon SageMaker 筆記本實例，以便從Amazon S3 中擷取所有模型。刪除現有實例並使用筆記本執行SageMaker 批次轉換，以便為所有城市的所有可能使用者離線執行推理。將結果儲存在Amazon S3 中的不同檔案中。將Web 用戶端指向這些檔案。

B. Prepare an Amazon SageMaker Docker container based on the open-source multi-model server. Remove the existing instances and create a multi-model endpoint in SageMaker instead, pointing to the S3 bucket containing all the models. Invoke the endpoint from the web client at runtime, specifying the TargetModel parameter according to the city of each request.

B. 基於開源多模型伺服器準備一個Amazon SageMaker Docker 容器。刪除現有實例並在SageMaker 中建立一個多模型終端節點，指向包含所有模型的S3 儲存桶。在運行時從Web 用戶端呼叫終端節點，根據每個請求的城市指定TargetModel 參數。

C. Keep only a single EC2 instance for hosting all the models. Install a model server in the instance and load each model by pulling it from Amazon S3. Integrate the instance with the web client using Amazon API Gateway for responding to the requests in real time, specifying the target resource according to the city of each request.

C. 僅保留一個EC2 執行個體來託管所有模型。在實例中安裝模型伺服器並透過從Amazon S3 中提取每個模型來載入它。使用Amazon API Gateway 將實例與Web 用戶端集成，以即時回應請求，根據每個請求的城市指定目標資源。

D. Prepare a Docker container based on the prebuilt images in Amazon SageMaker. Replace the existing instances with separate SageMaker endpoints, one for each city where the company operates. Invoke the endpoints from the web client, specifying the URL and EndpointName parameter according to the city of each request.

D. 根據Amazon SageMaker 中預先建置的映像檔準備Docker 容器。將現有實例替換為單獨的SageMaker 終端節點，每個終端節點對應公司營運所在的每個城市。從Web 用戶端呼叫終端節點，根據每個要求的城市指定URL 和EndpointName 參數。

問題# 304

A company builds computer-vision models that use deep learning for the autonomous vehicle industry. A machine learning (ML) specialist uses an Amazon EC2 instance that has a CPU:GPU ratio of 12:1 to train the models. The ML specialist examines the instance metric logs and notices that the GPU is idle half of the time. The ML specialist must reduce training costs without increasing the duration of the training jobs. Which solution will meet these requirements?

一家公司為自動駕駛汽車行業構建了使用深度學習的電腦視覺模型。機器學習(ML)專家使用CPU:GPU 比率為12:1 的Amazon EC2 執行個體來訓練模型。ML 專家檢查實例指標日誌並注意到GPU 有一半時間處於空閒狀態。ML 專家必須在不增加訓練作業持續時間的情況下降低訓練成本。哪種解決方案可以滿足這些要求？

A. Switch to an instance type that has only CPUs.

A. 切換到僅具有CPU 的執行個體類型。

B. Use a heterogeneous cluster that has two different instances groups.

B. 使用具有兩個不同實例組的異質叢集。

C. Use memory-optimized EC2 Spot Instances for the training jobs.

C. 對訓練作業使用記憶體最佳化的EC2 Spot 實例。

D. Switch to an instance type that has a CPU:GPU ratio of 6:1.

D. 切換到CPU:GPU 比率為6:1 的實例類型。

問題# 305

A company wants to forecast the daily price of newly launched products based on 3 years of data for older product prices, sales, and rebates. The time-series data has irregular timestamps and is missing some values. Data scientist must build a dataset to replace the missing values. The data scientist needs a solution that resamples the data daily and exports the data for further modeling. Which solution will meet these requirements with the LEAST implementation effort?

一家公司希望根據3 年的舊產品價格、銷售額和折扣數據來預測新推出產品的每日價格。時間序列資料的時間戳記不規則，並且缺少一些值。資料科學家必須建立一個資料集來替換缺失的值。資料科學家需要一個解決方案，每天重新採樣資料並匯出資料以進行進一步建模。哪一種解決方案能夠以最少的實施工作量來滿足這些要求？

A. Use Amazon EMR Serverless with PySpark.

A. 將Amazon EMR Serverless 與PySpark 結合使用。

B. Use AWS Glue DataBrew.

B. 使用AWS Glue DataBrew。

C. Use Amazon SageMaker Studio Data Wrangler.

C. 使用Amazon SageMaker Studio Data Wrangler。

D. Use Amazon SageMaker Studio Notebook with Pandas.

D. 將Amazon SageMaker Studio Notebook 與Pandas 結合使用。

問題# 306

A data scientist is building a forecasting model for a retail company by using the most recent 5 years of sales records that are stored in a data warehouse. The dataset contains sales records for each of the company's stores across five commercial regions. The data scientist creates a working dataset with StoreID, Region, Date, and Sales Amount as columns. The data scientist wants to analyze yearly average sales for each region. The scientist also wants to compare how each region performed compared to average sales across all commercial regions. Which visualization will help the data scientist better understand the data trend?

一位資料科學家正在使用儲存在資料倉儲中的最近5年的銷售記錄為一家零售公司建立預測模型。此資料集包含該公司五個商業區域內每家商店的銷售記錄。資料科學家建立了一個工作資料集，其中StoreID、Region、Date 和Sales Amount 作為欄位。資料科學家想要分析每個地區的年平均銷售額。科學家也想比較每個地區的表現與所有商業區域的平均銷售額。哪種視覺化可以幫助資料科學家更了解資料趨勢？

A. Create an aggregated dataset by using the Pandas GroupBy function to get average sales for each year for each store. Create a bar plot, faceted by year, of average sales for each store. Add an extra bar in each facet to represent average sales.

A. 使用Pandas GroupBy 函數建立一個聚合資料集，以取得每個商店每年的平均銷售額。建立一個按年份分面的長條圖，顯示每個商店的平均銷售額。在每個方面添加一條額外的長條圖以表示平均銷售額。

B. Create an aggregated dataset by using the Pandas GroupBy function to get average sales for each year for each store. Create a bar plot, colored by region and faceted by year, of average sales for each store. Add a horizontal line in each facet to represent average sales.

B. 使用Pandas GroupBy 函數建立一個聚合資料集，以取得每個商店每年的平均銷售額。建立一個按地區著色、按年份分面的長條圖，顯示每個商店的平均銷售額。在每個方面添加一條水平線以表示平均銷售額。

C. Create an aggregated dataset by using the Pandas GroupBy function to get average sales for each year for each region. Create a bar plot of average sales for each region. Add an extra bar in each facet to represent average sales.

C. 使用Pandas GroupBy 函數建立一個聚合資料集，以取得每個地區每年的平均銷售額。建立一個顯示每個地區平均銷售額的長條圖。在每個方面添加一條額外的長條圖以表示平均銷售額。

D. Create an aggregated dataset by using the Pandas GroupBy function to get average sales for each year for each region. Create a bar plot, faceted by year, of average sales for each region. Add a horizontal line in each facet to represent average sales.

D. 使用Pandas GroupBy 函數建立一個聚合資料集，以取得每個地區每年的平均銷售額。建立一個

按年份分面的長條圖，顯示每個地區的平均銷售額。在每個方面添加一條水平線以表示平均銷售額。

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問題# 307

A company uses sensors on devices such as motor engines and factory machines to measure parameters, temperature and pressure. The company wants to use the sensor data to predict equipment malfunctions and reduce services outages. Machine learning (ML) specialist needs to gather the sensors data to train a model to predict device malfunctions. The ML specialist must ensure that the data does not contain outliers before training the model. How can the ML specialist meet these requirements with the LEAST operational overhead?

一家公司在汽車引擎和工廠機器等設備上使用感測器來測量參數、溫度和壓力。該公司希望使用感測器數據來預測設備故障並減少服務中斷。機器學習(ML)專家需要收集感測器資料來訓練模型以預測設備故障。ML專家在訓練模型之前必須確保資料不包含異常值。ML專家如何以最少的營運開銷滿足這些要求？

A. Load the data into an Amazon SageMaker Studio notebook. Calculate the first and third quartile. Use a SageMaker Data Wrangler data flow to remove only values that are outside of those quartiles.

A. 將資料載入到Amazon SageMaker Studio 筆記本中。計算第一和第三四分位數。使用SageMaker Data Wrangler 資料流僅刪除這些四分位數以外的值。

B. Use an Amazon SageMaker Data Wrangler bias report to find outliers in the dataset. Use a Data Wrangler data flow to remove outliers based on the bias report.

B. 使用Amazon SageMaker Data Wrangler 偏差報告找出資料集中的異常值。使用Data Wrangler 資料流根據偏差報告刪除異常值。

C. Use an Amazon SageMaker Data Wrangler anomaly detection visualization to find outliers in the dataset. Add a transformation to a Data Wrangler data flow to remove outliers.

C. 使用Amazon SageMaker Data Wrangler 異常偵測來視覺化尋找資料集中的異常值。在Data Wrangler 資料流中新增轉換以刪除異常值。

D. Use Amazon Lookout for Equipment to find and remove outliers from the dataset.

D. 使用Amazon Lookout for Equipment 尋找並刪除資料集中的異常值。

問題# 308

A data scientist obtains a tabular dataset that contains 150 correlated features with different ranges to build a regression model. The data scientist needs to achieve more efficient model training by implementing a solution that minimizes impact on the model's performance. The data scientist decides to perform a principal component analysis (PCA) preprocessing step to reduce the number of features to a smaller set of independent features before the data scientist uses the new features in the regression model. Which preprocessing step will meet these requirements?

資料科學家獲得一個表格資料集，其中包含150個具有不同範圍的相關特徵，用於建立迴歸模型。

資料科學家需要透過實施解決方案來實現更有效率的模型訓練，以最大限度地減少對模型效能的影響。資料科學家決定執行主成分分析(PCA) 預處理步驟，以在資料科學家在迴歸模型中使用新特徵之前將特徵數量減少為一組較小的獨立特徵。哪個預處理步驟可以滿足這些要求？

A. Use the Amazon SageMaker built-in algorithm for PCA on the dataset to transform the data.

A. 使用Amazon SageMaker 內建的PCA 演算法對資料集進行資料轉換。

B. Load the data into Amazon SageMaker Data Wrangler. Scale the data with a Min Max Scaler transformation step. Use the SageMaker built-in algorithm for PCA on the scaled dataset to transform the data.

B. 將資料載入到Amazon SageMaker Data Wrangler 中。使用Min Max Scaler 轉換步驟縮放資料。使用SageMaker 內建的PCA 演算法對縮放後的資料集進行資料轉換。

C. Reduce the dimensionality of the dataset by removing the features that have the highest correlation. Load the data into Amazon SageMaker Data Wrangler. Perform a Standard Scaler transformation step to scale the data. Use the SageMaker built-in algorithm for PCA on the scaled dataset to transform the data.

C. 透過刪除相關性最高的特徵來降低資料集的維數。將資料載入到Amazon SageMaker Data Wrangler 中。執行標準縮放器轉換步驟來縮放資料。使用SageMaker 內建的PCA 演算法對縮放後的資料集進行資料轉換。

D. Reduce the dimensionality of the dataset by removing the features that have the lowest correlation. Load the data into Amazon SageMaker Data Wrangler. Perform a Min Max Scaler transformation step to scale the data. Use the SageMaker built-in algorithm for PCA on the scaled dataset to transform the data.

D. 透過刪除相關性最低的特徵來降低資料集的維數。將資料載入到Amazon SageMaker Data Wrangler 中。執行Min Max Scaler 轉換步驟來縮放資料。使用SageMaker 內建的PCA 演算法對縮放後的資料集進行資料轉換。

問題# 309

An online retailer collects the following data on customer orders: demographics, behaviors, location, shipment progress, and delivery time. A data scientist joins all the collected datasets. The result is a single dataset that includes 980 variables. The data scientist must develop a machine learning (ML) model to identify groups of customers who are likely to respond to a marketing campaign. Which combination of algorithms should the data scientist use to meet this requirement? (Choose two.)

一家線上零售商收集了以下有關客戶訂單的數據：人口統計、行為、地點、出貨進度和交貨時間。資料科學家將所有收集到的資料集合併起來。結果是一個包含980 個變數的資料集。資料科學家必須開發一個機器學習(ML) 模型來識別可能對行銷活動做出反應的客戶群。資料科學家應該使用哪種演算法組合來滿足這項要求？（選兩個。）

A. Latent Dirichlet Allocation (LDA).

A. 潛在狄利克雷分配(LDA)。

B. K-means.

B. K-均值。

C. Semantic segmentation.

C. 語意分割。

D. Principal component analysis (PCA).

D. 主成分分析(PCA)。

E. Factorization machines (FM).

E. 因式分解機(FM)。

問題# 310

A machine learning engineer is building a bird classification model. The engineer randomly separates a dataset into a training dataset and a validation dataset. During the training phase, the model achieves very high accuracy. However, the model did not generalize well during validation of the validation dataset. The engineer realizes that the original dataset was imbalanced. What should the engineer do to improve the validation accuracy of the model?

一位機器學習工程師正在建立鳥類分類模型。工程師將資料集隨機分為訓練資料集和驗證資料集。在訓練階段，該模型實現了非常高的準確率。然而，在驗證資料集的驗證過程中，該模型的泛化效果不佳。工程師意識到原始資料集是不平衡的。工程師該怎麼做才能提高模型的驗證準確率？

A. Perform stratified sampling on the original dataset.

A. 對原始資料集進行分層抽樣。

B. Acquire additional data about the majority classes in the original dataset.

B. 取得更多有關原始資料集中多數類別的資料。

C. Use a smaller, randomly sampled version of the training dataset.

C. 使用較小的、隨機抽樣的訓練資料集版本。

D. Perform systematic sampling on the original dataset.

D. 對原始資料集進行系統抽樣。

問題# 311

A data engineer wants to perform exploratory data analysis (EDA) on a petabyte of data. The data engineer does not want to manage compute resources and wants to pay only for queries that are run. The data engineer must write the analysis by using Python from a Jupyter notebook. Which solution will meet these requirements?

一位資料工程師想要對PB級資料執行探索性資料分析(EDA)。這位資料工程師不想管理運算資源，只想為執行的查詢付費。這位資料工程師必須使用Jupyter筆記本中的Python編寫分析。哪種解決方案可以滿足這些要求？

A. Use Apache Spark from within Amazon Athena.

A. 在Amazon Athena 中使用Apache Spark。

B. Use Apache Spark from within Amazon SageMaker.

B. 在Amazon SageMaker 中使用Apache Spark。

C. Use Apache Spark from within an Amazon EMR cluster.

C. 在Amazon EMR 叢集中使用Apache Spark。

D. Use Apache Spark through an integration with Amazon Redshift.

D. 透過與Amazon Redshift 整合使用Apache Spark。

問題# 312

A data scientist receives a new dataset in.csv format and stores the dataset in Amazon S3. The data scientist will use the dataset to train a machine learning (ML) model. The data scientist first needs to identify any potential data quality issues in the dataset. The data scientist must identify values that are missing or values that are not valid. The data scientist must also identify the number of outliers in the dataset. Which solution will meet these requirements with the LEAST operational effort?

資料科學家收到一個.csv 格式的新資料集，並將該資料集儲存在Amazon S3 中。資料科學家將使用此資料集訓練機器學習(ML) 模型。資料科學家首先需要確定資料集中是否存在任何潛在的資料品質問題。資料科學家必須確定缺失的值或無效的值。資料科學家還必須確定資料集中的異常值數量。哪種解決方案能夠以最少的營運工作量滿足這些要求？

A. Create an AWS Glue job to transform the data from.csv format to Apache Parquet format. Use an AWS Glue crawler and Amazon Athena with appropriate SQL queries to retrieve the required information.

A. 建立一個AWS Glue 作業，將資料從.csv 格式轉換為Apache Parquet 格式。使用AWS Glue 爬蟲程式和Amazon Athena 以及適當的SQL 查詢來檢索所需的資訊。

B. Leave the dataset in.csv format. Use an AWS Glue crawler and Amazon Athena with appropriate SQL queries to retrieve the required information.

B. 將資料集保留為.csv 格式。使用AWS Glue 爬蟲程式和Amazon Athena 以及適當的SQL 查詢來檢索所需的資訊。

C. Create an AWS Glue job to transform the data from.csv format to Apache Parquet format. Import the data into Amazon SageMaker Data Wrangler. Use the Data Quality and Insights Report to retrieve the required information.

C. 建立一個AWS Glue 作業，將資料從.csv 格式轉換為Apache Parquet 格式。將資料導入Amazon SageMaker Data Wrangler。使用數據品質和洞察報告來檢索所需的資訊。

D. Leave the dataset in.csv format. Import the data into Amazon SageMaker Data Wrangler. Use the Data Quality and Insights Report to retrieve the required information.

D. 將資料集保留為.csv 格式。將資料導入Amazon SageMaker Data Wrangler。使用數據品質和洞察報告來檢索所需的資訊。

問題# 313

An ecommerce company has developed a XGBoost model in Amazon SageMaker to predict whether a customer will return a purchased item. The dataset is imbalanced. Only 5% of customers return items. A data scientist must find the hyperparameters to capture as many instances of returned items as possible. The company has a small budget for compute. How should the data scientist meet these requirements MOST cost-effectively?

一家電子商務公司在Amazon SageMaker 中開發了一個XGBoost 模型，用於預測客戶是否會退回已購買的商品。資料集不平衡。只有5% 的客戶會退貨。資料科學家必須找到超參數來捕獲盡可能的退貨實例。該公司的計算預算很少。資料科學家應如何以最具成本效益的方式滿足這些要求？

A. Tune all possible hyperparameters by using automatic model tuning (AMT). Optimize on {"HyperParameterTuningJobObjective": {"MetricName": "validation:accuracy", "Type": "Maximize"}}.

A. 使用自動模型調整(AMT) 調整所有可能的超參數。在{"HyperParameterTuningJobObjective": {"MetricName": "validation:accuracy", "Type": "Maximize"}} 上進行最佳化。

B. Tune the csv_weight hyperparameter and the scale_pos_weight hyperparameter by using automatic model tuning (AMT). Optimize on {"HyperParameterTuningJobObjective": {"MetricName": "validation:ll", "Type": "Maximize"}}.

B. 使用自動模型調整(AMT) 調整csv_weight 超參數和scale_pos_weight 超參數。在{"HyperParameterTuningJobObjective": {"MetricName": "validation:ll", "Type": "Maximize"}} 上進行最佳化。

C. Tune all possible hyperparameters by using automatic model tuning (AMT). Optimize on {"HyperParameterTuningJobObjective": {"MetricName": "validation:f1", "Type": "Maximize"}}.

C. 使用自動模型調整(AMT) 調整所有可能的超參數。在{"HyperParameterTuningJobObjective": {"MetricName": "validation:f1", "Type": "Maximize"}} 上進行最佳化。

D. Tune the csv_weight hyperparameter and the scale_pos_weight hyperparameter by using automatic model tuning (AMT). Optimize on {"HyperParameterTuningJobObjective": {"MetricName": "validation:f1", "Type": "Minimize"}}.

D. 使用自動模型調整(AMT) 調整csv_weight 超參數和scale_pos_weight 超參數。對{"HyperParameterTuningJobObjective": {"MetricName": "validation:f1", "Type": "Minimize"}} 進行最佳化。

問題# 314

A data scientist is trying to improve the accuracy of a neural network classification model. The data scientist wants to run a large hyperparameter tuning job in Amazon SageMaker. However, previous smaller tuning jobs on the same model often ran for several weeks. The ML specialist wants to reduce the computation time required to run the tuning job. Which actions will MOST reduce the computation time for the hyperparameter tuning job? (Choose two.)

一位資料科學家正在嘗試提高神經網路分類模型的準確性。資料科學家希望在Amazon SageMaker中執行大型超參數調整作業。但是，先前在同一模型上進行的小型調整作業通常要執行數週。ML專家希望減少執行調整作業所需的計算時間。哪些操作將最大程度地減少超參數調整作業的計算時間？（選兩個。）

A. Use the Hyperband tuning strategy.

A. 使用Hyperband 調優策略。

B. Increase the number of hyperparameters.

B. 增加超參數數。

C. Set a lower value for the MaxNumberOfTrainingJobs parameter.

C. 為MaxNumberOfTrainingJobs 參數設定較低的值。

D. Use the grid search tuning strategy.

D. 使用網格搜尋調優策略。

E. Set a lower value for the MaxParallelTrainingJobs parameter.

E. 為MaxParallelTrainingJobs 參數設定較低的值。

問題# 315

A machine learning (ML) specialist needs to solve a binary classification problem for a marketing dataset. The ML specialist must maximize the Area Under the ROC Curve (AUC) of the algorithm by training an XGBoost algorithm. The ML specialist must find values for the eta, alpha, min_child_weight, and max_depth hyperparameters that will generate the most accurate model. Which approach will meet these requirements with the LEAST operational overhead?

機器學習(ML)專家需要解決行銷資料集的二元分類問題。ML專家必須透過訓練XGBoost演算法來最大化演算法的ROC曲線下面積(AUC)。ML專家必須找到eta、alpha、min_child_weight和max_depth超參數的值，以產生最準確的模型。哪種方法可以以最少的營運開銷滿足這些要求？

A. Use a bootstrap script to install scikit-learn on an Amazon EMR cluster. Deploy the EMR cluster. Apply k-fold cross-validation methods to the algorithm.

A. 使用引導腳本在Amazon EMR叢集上安裝scikit-learn。部署EMR叢集。將k倍交叉驗證方法應用於演算法。

B. Deploy Amazon SageMaker prebuilt Docker images that have scikit-learn installed. Apply k-fold cross-validation methods to the algorithm.

B. 部署安裝了scikit-learn 的Amazon SageMaker 預先建置Docker 映像。將k 倍交叉驗證方法應用於演算法。

C. Use Amazon SageMaker automatic model tuning (AMT). Specify a range of values for each hyperparameter.

C. 使用Amazon SageMaker 自動模型調整(AMT)。為每個超參數指定一個值範圍。

D. Subscribe to an AUC algorithm that is on AWS Marketplace. Specify a range of values for each hyperparameter.

D. 訂閱AWS Marketplace 上的AUC 演算法。為每個超參數指定一個值範圍。

問題# 316

A machine learning (ML) developer for an online retailer recently uploaded a sales dataset into Amazon SageMaker Studio. The ML developer wants to obtain importance scores for each feature of the dataset. The ML developer will use the importance scores to feature engineer the dataset. Which solution will meet this requirement with the LEAST development effort?

一家線上零售商的機器學習(ML) 開發人員最近將銷售資料集上傳到Amazon SageMaker Studio。ML 開發人員希望獲得資料集每個特徵的重要性分數。ML 開發人員將使用重要性分數對資料集進行特徵工程。哪種解決方案可以以最少的開發工作量滿足此要求？

A. Use SageMaker Data Wrangler to perform a Gini importance score analysis.

A. 使用SageMaker Data Wrangler 執行基尼重要性分數分析。

B. Use a SageMaker notebook instance to perform principal component analysis (PCA).

B. 使用SageMaker 筆記本實例執行主成分分析(PCA)。

C. Use a SageMaker notebook instance to perform a singular value decomposition analysis.

C. 使用SageMaker 筆記本實例執行奇異值分解分析。

D. Use the multicollinearity feature to perform a lasso feature selection to perform an importance scores analysis.

D. 使用多重共線性功能執行套索特徵選擇以執行重要性分數分析。

問題# 317

A company is setting up a mechanism for data scientists and engineers from different departments to access an Amazon SageMaker Studio domain. Each department has a unique SageMaker Studio domain. The company wants to build a central proxy application that data scientists and engineers can log in to by using their corporate credentials. The proxy application will authenticate users by using the company's existing Identity provider (IdP). The application will then route users to the appropriate SageMaker Studio domain. The company plans to maintain a table in Amazon DynamoDB that contains SageMaker domains for each department. How should the company meet these requirements?

一家公司正在為不同部門的資料科學家和工程師建立一種機制，以存取Amazon SageMaker Studio 網域。每個部門都有一個唯一的SageMaker Studio 網域。該公司希望建立一個中央代理應用程式，讓資料科學家和工程師使用他們的公司憑證登入。代理應用程式將使用公司現有的身分提供者(IdP) 對使用者進行身份驗證。然後，該應用程式將使用者路由到適當的SageMaker Studio 網域。該公司計劃在Amazon DynamoDB 中維護一個包含每個部門的SageMaker 網域的表格。公司應如何滿足這些要求？

A. Use the SageMaker CreatePresignedDomainUrl API to generate a presigned URL for each domain according to the DynamoDB table. Pass the presigned URL to the proxy application.

A. 使用SageMaker CreatePresignedDomainUrl API 根據DynamoDB 表為每個網域產生預簽URL。將預簽URL 傳遞給代理應用程式。

B. Use the SageMaker CreateHumanTaskUi API to generate a UI URL. Pass the URL to the proxy application.

B. 使用SageMaker CreateHumanTaskUi API 產生UI URL。將URL 傳遞給代理應用程式。

C. Use the Amazon SageMaker ListHumanTaskUis API to list all UI URLs. Pass the appropriate URL to the DynamoDB table so that the proxy application can use the URL.

C. 使用Amazon SageMaker ListHumanTaskUis API 列出所有UI URL。將適當的URL 傳遞給DynamoDB 表，以便代理應用程式可以使用該URL。

D. Use the SageMaker CreatePresignedNotebookInstanceUrl API to generate a presigned URL. Pass the presigned URL to the proxy application.

D. 使用SageMaker CreatePresignedNotebookInstanceUrl API 產生預簽名URL。將預簽URL 傳遞給代理應用程式。

問題# 318

An insurance company is creating an application to automate car insurance claims. A machine learning (ML) specialist used an Amazon SageMaker Object Detection - TensorFlow built-in algorithm to train a model to detect scratches and dents in images of cars. After the model was trained, the ML specialist noticed that the model performed better on the training dataset than on the testing dataset. Which approach should the ML specialist use to improve the performance of the model on the testing data?

一家保險公司正在開發一款應用程式，用於自動處理汽車保險索賠。一位機器學習(ML) 專家使用Amazon SageMaker 物件偵測- TensorFlow 內建演算法來訓練模型，以偵測汽車影像中的刮痕和凹痕。訓練模型後，ML 專家注意到模型在訓練資料集上的表現優於測試資料集。ML 專家應該使用哪種方法來提高模型在測試資料上的效能？

A. Increase the value of the momentum hyperparameter.

A. 增加動量超參數的值。

B. Reduce the value of the dropout_rate hyperparameter.

B. 降低dropout_rate超參數的值。

C. Reduce the value of the learning_rate hyperparameter.

C. 降低learning_rate超參數的值。

D. Increase the value of the L2 hyperparameter.

D. 增加L2超參數的值。

問題# 319

A developer at a retail company is creating a daily demand forecasting model. The company stores the historical hourly demand data in an Amazon S3 bucket. However, the historical data does not include demand data for some hours. The developer wants to verify that an autoregressive integrated moving average (ARIMA) approach will be a suitable model for the use case. How should the developer verify the suitability of an ARIMA approach?

一家零售公司的開發人員正在創建每日需求預測模型。該公司將歷史每小時需求資料儲存在Amazon S3 儲存桶中。但是，歷史資料不包括某些時段的需求資料。開發人員想要驗證自回歸綜合移動平均(ARIMA) 方法是否適合該用例。開發人員應如何驗證ARIMA 方法的適用性？

A. Use Amazon SageMaker Data Wrangler. Import the data from Amazon S3. Impute hourly missing data. Perform a Seasonal Trend decomposition.

A. 使用Amazon SageMaker Data Wrangler。從Amazon S3 匯入資料。估算每小時缺失的資料。執行季節性趨勢分解。

B. Use Amazon SageMaker Autopilot. Create a new experiment that specifies the S3 data location. Choose ARIMA as the machine learning (ML) problem. Check the model performance.

B. 使用Amazon SageMaker Autopilot。建立一個指定S3 資料位置的新實驗。選擇ARIMA 作為機器學習(ML) 問題。檢查模型性能。

C. Use Amazon SageMaker Data Wrangler. Import the data from Amazon S3. Resample data by using the aggregate daily total. Perform a Seasonal Trend decomposition.

C. 使用Amazon SageMaker Data Wrangler。從Amazon S3 匯入資料。使用每日總計重新採樣數據。執行季節性趨勢分解。

D. Use Amazon SageMaker Autopilot. Create a new experiment that specifies the S3 data location. Impute missing hourly values. Choose ARIMA as the machine learning (ML) problem. Check the model performance.

D. 使用Amazon SageMaker Autopilot。建立一個指定S3 資料位置的新實驗。估算每小時缺失的值。選擇ARIMA 作為機器學習(ML) 問題。檢查模型性能。

問題# 320

A company decides to use Amazon SageMaker to develop machine learning (ML) models. The company will host SageMaker notebook instances in a VPC. The company stores training data in an Amazon S3 bucket. Company security policy states that SageMaker notebook instances must not have internet connectivity. Which solution will meet the company's security requirements?

一家公司決定使用Amazon SageMaker 開發機器學習(ML) 模型。該公司將在VPC 中託管SageMaker 筆記本實例。該公司將訓練資料儲存在Amazon S3 儲存桶中。公司安全策略規定SageMaker 筆記本實例不得有網路連線。哪一種解決方案可以滿足公司的安全要求？

A. Connect the SageMaker notebook instances that are in the VPC by using AWS Site-to-Site VPN to encrypt all internet-bound traffic. Configure VPC flow logs. Monitor all network traffic to detect and prevent any malicious activity.

A. 使用AWS Site-to-Site VPN 連線VPC 中的SageMaker 筆記型電腦實例，以加密所有Internet 綁定流量。配置VPC 流日誌。監控所有網路流量以偵測和防止任何惡意活動。

B. Configure the VPC that contains the SageMaker notebook instances to use VPC interface endpoints to establish connections for training and hosting. Modify any existing security groups that are associated with the VPC interface endpoint to allow only outbound connections for training and hosting.

B. 配置包含SageMaker 筆記型電腦實例的VPC 以使用VPC 介面終端節點建立用於訓練和託管的連線。修改與VPC 介面終端節點關聯的任何現有安全群組，以僅允許用於訓練和託管的出站連線。

C. Create an IAM policy that prevents access the internet. Apply the IAM policy to an IAM role. Assign the IAM role to the SageMaker notebook instances in addition to any IAM roles that are already assigned to the instances.

C. 建立阻止存取Internet 的IAM 策略。將IAM 策略應用於IAM 角色。除了已指派給實例的任何IAM 角色之外，還將IAM 角色指派給SageMaker 筆記本實例。

D. Create VPC security groups to prevent all incoming and outgoing traffic. Assign the security groups to the SageMaker notebook instances.

D. 建立VPC 安全群組以阻止所有傳入和傳出流量。將安全性群組指派給SageMaker 筆記本實例。

問題# 321

A machine learning (ML) engineer uses Bayesian optimization for a hyperparameter tuning job in Amazon SageMaker. The ML engineer uses precision as the objective metric. The ML engineer wants to use recall as the objective metric. The ML engineer also wants to expand the hyperparameter range for a new hyperparameter tuning job. The new hyperparameter range will include the range of the previously performed tuning job. Which approach will run the new hyperparameter tuning job in the LEAST amount of time?

機器學習(ML) 工程師使用貝葉斯優化在Amazon SageMaker 中執行超參數調整作業。ML 工程師使用精確度作為客觀指標。ML 工程師希望使用召回率作為客觀指標。ML 工程師也希望擴充新超參數調整作業的超參數範圍。新的超參數範圍將包括先前執行的調整作業的範圍。哪一種方法可以在最短的時間內執行新的超參數調整作業？

A. Use a warm start hyperparameter tuning job.

A. 使用熱啟動超參數調整作業。

B. Use a checkpointing hyperparameter tuning job.

B. 使用檢查點超參數調整作業。

C. Use the same random seed for the hyperparameter tuning job.

C. 對超參數調整作業使用相同的隨機種子。

D. Use multiple jobs in parallel for the hyperparameter tuning job.

D. 對超參數調整作業並行使用多個作業。

問題# 322

A news company is developing an article search tool for its editors. The search tool should look for the articles that are most relevant and representative for particular words that are queried among a corpus of historical news documents. The editors test the first version of the tool and report that the tool seems to look for word matches in general. The editors have to spend additional time to filter the results to look for the articles where the queried words are most important. A group of data scientists must redesign the tool so that it isolates the most frequently used words in a document. The tool also must capture the relevance and importance of words for each document in the corpus. Which solution meets these requirements?

一家新聞公司正在為其編輯開發一篇文章搜尋工具。該搜尋工具應在歷史新聞文檔語料庫中找到與查詢的特定單字最相關且最具代表性的文章。編輯測試了該工具的第一個版本，並報告說該工具似乎在尋找一般的單字匹配。編輯必須花費額外的時間來過濾結果，以查找查詢單字最重要的文章。一組資料科學家必須重新設計該工具，以便它能隔離文件中最常用的單字。該工具還必須捕獲語料庫中每個文件的單字相關性和重要性。哪種解決方案符合這些要求？

A. Extract the topics from each article by using Latent Dirichlet Allocation (LDA) topic modeling. Create a topic table by assigning the sum of the topic counts as a score for each word in the articles. Configure the tool to retrieve the articles where this topic count score is higher for the queried words.

A. 使用潛在狄利克雷分配(LDA)主題模型從每篇文章中提取主題。透過將主題計數總和分配為文章中每個單字的分數來建立主題表。配置該工具以檢索查詢詞的主題計數分數較高的文章。

B. Build a term frequency for each word in the articles that is weighted with the article's length. Build an inverse document frequency for each word that is weighted with all articles in the corpus. Define a final highlight score as the product of both of these frequencies. Configure the tool to retrieve the articles where this highlight score is higher for the queried words.

B. 為文章中的每個單字建立一個以文章長度加權的詞頻。為每個單字建立一個以語料庫中的所有文章加權的逆文檔頻率。將最終的突出顯示分數定義為這兩個頻率的乘積。配置該工具以檢索查詢詞的突出顯示分數較高的文章。

C. Download a pretrained word-embedding lookup table. Create a titles-embedding table by averaging the title's word embedding for each article in the corpus. Define a highlight score for each word as inversely proportional to the distance between its embedding and the title embedding. Configure the tool to retrieve the articles where this highlight score is higher for the queried words.

C. 下載預訓練的字詞嵌入查找表。透過語料庫中每篇文章的標題詞嵌入取平均值來建立標題嵌入表。將每個單字的突出顯示分數定義為與其嵌入和標題嵌入之間的距離成反比。配置工具以檢索查詢詞的突出顯示分數較高的文章。

D. Build a term frequency score table for each word in each article of the corpus. Assign a score of zero to all stop words. For any other words, assign a score as the word's frequency in the article. Configure the tool to retrieve the articles where this frequency score is higher for the queried words.

D. 為語料庫中每篇文章中的每個單字建立一個詞頻分數表。為所有停用詞分配零分。對於任何其他詞，分配一個分數作為該詞在文章中的頻率。配置工具以檢索查詢詞的此頻率分數較高的文章。

問題# 323

A growing company has a business-critical key performance indicator (KPI) for the uptime of a machine learning (ML) recommendation system. The company is using Amazon SageMaker hosting services to develop a recommendation model in a single Availability Zone within an AWS Region. A machine learning (ML) specialist must develop a solution to achieve high availability. The solution must have a recovery time objective (RTO) of 5 minutes. Which solution will meet these requirements with the LEAST effort?

一家成長中的公司有一個業務關鍵績效指標(KPI)，用於衡量機器學習(ML)推薦系統的正常運作時間。該公司正在使用Amazon SageMaker 託管服務在AWS 區域內的單一可用區中開發推薦模型。機器學習(ML) 專家必須開發一種解決方案來實現高可用性。此解決方案的恢復時間目標(RTO) 必須為5 分鐘。哪種解決方案可以以最少的努力滿足這些要求？

A. Deploy multiple instances for each endpoint in a VPC that spans at least two Regions.

A. 在跨至少兩個區域的VPC 中為每個終端節點部署多個執行個體。

B. Use the SageMaker auto scaling feature for the hosted recommendation models.

B. 對託管的推薦模型使用SageMaker 自動擴充功能。

C. Deploy multiple instances for each production endpoint in a VPC that spans least two subnets that are in a second Availability Zone.

C. 在跨第二個可用區中的至少兩個子網路的VPC 中為每個生產終端節點部署多個實例。

D. Frequently generate backups of the production recommendation model. Deploy the backups in a second Region.

D. 頻繁產生生產推薦模型的備份。在第二個區域中部署備份。

問題# 324

A global company receives and processes hundreds of documents daily. The documents are in printed.pdf format or.jpg format.A machine learning (ML) specialist wants to build an automated document processing workflow to extract text from specific fields from the documents and to classify the documents. The ML specialist wants a solution that requires low maintenance.Which solution will meet these requirements with the LEAST operational effort?

一家跨國公司每天接收和處理數百份文件。這些文件採用印刷版.pdf 格式或.jpg 格式。一位機器學習(ML) 專家希望建立一個自動化文件處理工作流程，以從文件中的特定欄位中提取文字並對文件進行分類。 ML 專家想要一種維護成本低的解決方案。哪種解決方案能夠以最少的營運工作量滿足這些要求？

A. Use a PaddleOCR model in Amazon SageMaker to detect and extract the required text and fields. Use a SageMaker text classification model to classify the document.

A. 使用Amazon SageMaker 中的PaddleOCR 模型偵測並擷取所需的文字和欄位。使用SageMaker 文字分類模型對文件進行分類。

B. Use a PaddleOCR model in Amazon SageMaker to detect and extract the required text and fields. Use Amazon Comprehend to classify the document.

B. 使用Amazon SageMaker 中的PaddleOCR 模型偵測並擷取所需的文字和欄位。使用Amazon Comprehend 對文件進行分類。

C. Use Amazon Textract to detect and extract the required text and fields. Use Amazon Rekognition to classify the document.

C. 使用Amazon Textract 檢測並提取所需的文字和欄位。使用Amazon Rekognition 對文件進行分類。

D. Use Amazon Textract to detect and extract the required text and fields. Use Amazon Comprehend to classify the document.

D. 使用Amazon Textract 檢測並提取所需的文字和欄位。使用Amazon Comprehend 對文件進行分類。

問題# 325

A company wants to detect credit card fraud. The company has observed that an average of 2% of credit card transactions are fraudulent. A data scientist trains a classifier on a year's worth of credit card transaction data. The classifier needs to identify the fraudulent transactions. The company wants to accurately capture as many fraudulent transactions as possible.Which metrics should the data scientist use to optimize the classifier? (Choose two.)

一家公司想要偵測信用卡詐欺行為。該公司發現，平均有2% 的信用卡交易是詐騙的。一位資料科學家根據一年的信用卡交易資料訓練分類器。分類器需要識別詐欺交易。該公司希望盡可能準確地捕獲詐欺交易。資料科學家應該使用哪些指標來優化分類器？（選兩個。）

A. Specificity.

A. 特異性。

B. False positive rate.

B. 假陽性率。

C. Accuracy.

C. 準確度。

D. F1 score.

D. F1 分數。

E. True positive rate.

E. 真陽性率。

問題# 326

A data scientist is designing a repository that will contain many images of vehicles. The repository must scale automatically in size to store new images every day. The repository must support versioning of the images. The data scientist must implement a solution that maintains multiple immediately accessible copies of the data in different AWS Regions. Which solution will meet these requirements?

一位資料科學家正在設計一個包含許多車輛影像的儲存庫。儲存庫必須自動擴展大小以每天存儲新圖像。儲存庫必須支援影像的版本控制。資料科學家必須實施一種解決方案，在不同的AWS區域中維護多個可立即存取的資料副本。哪種解決方案可以滿足這些要求？

A. Amazon S3 with S3 Cross-Region Replication (CRR).

A. 具有S3跨區域複製(CRR)的Amazon S3。

B. Amazon Elastic Block Store (Amazon EBS) with snapshots that are shared in a secondary Region.

B. 具有在輔助區域中共享的快照的Amazon Elastic Block Store (Amazon EBS)。

C. Amazon Elastic File System (Amazon EFS) Standard storage that is configured with Regional availability.

C. 具有區域可用性的Amazon Elastic File System (Amazon EFS) 標準儲存。

D. AWS Storage Gateway Volume Gateway.

D. AWS Storage Gateway 磁碟區閘道。

問題# 327

An ecommerce company wants to update a production real-time machine learning (ML) recommendation engine API that uses Amazon SageMaker. The company wants to release a new model but does not want to make changes to applications that rely on the API. The company also wants to evaluate the performance of the new model in production traffic before the company fully rolls out the new model to all users. Which solution will meet these requirements with the LEAST operational overhead?

一家電子商務公司希望更新使用Amazon SageMaker 的生產即時機器學習(ML) 推薦引擎API。該公司希望發布一個新模型，但不想對依賴該API 的應用程式進行更改。該公司還希望在向所有用戶全面推出新模型之前，評估新模型在生產流量中的表現。哪種解決方案可以以最少的營運開銷滿足這些要求？

A. Create a new SageMaker endpoint for the new model. Configure an Application Load Balancer (ALB) to distribute traffic between the old model and the new model.

A. 為新模型建立新的SageMaker 終端節點。配置應用程式負載平衡器(ALB) 以在舊模型和新模型之間分配流量。

B. Modify the existing endpoint to use SageMaker production variants to distribute traffic between the old model and the new model.

B. 修改現有終端節點以使用SageMaker 生產變體在舊模型和新模型之間分配流量。

C. Modify the existing endpoint to use SageMaker batch transform to distribute traffic between the old model and the new model.

C. 修改現有終端節點以使用SageMaker 批次轉換在舊模型和新模型之間分配流量。

D. Create a new SageMaker endpoint for the new model. Configure a Network Load Balancer (NLB) to distribute traffic between the old model and the new model.

D. 為新模型建立新的SageMaker 終端節點。配置網路負載平衡器(NLB) 以在舊模型和新模型之間分配流量。

問題# 328

A machine learning (ML) specialist at a manufacturing company uses Amazon SageMaker DeepAR to forecast input materials and energy requirements for the company. Most of the data in the training dataset is missing values for the target variable. The company stores the training dataset as JSON files. The ML specialist develop a solution by using Amazon SageMaker DeepAR to account for the missing values in the training dataset. Which approach will meet these requirements with the LEAST development effort?

一家製造公司的機器學習(ML) 專家使用Amazon SageMaker DeepAR 預測公司的輸入材料和能源需求。訓練資料集中的大多數資料都缺少目標變數的值。該公司將訓練資料集儲存為JSON 檔案。ML 專家使用Amazon SageMaker DeepAR 開發解決方案來解釋訓練資料集中的缺失值。哪一種方法可以以最少的開發工作量來滿足這些要求？

A. Impute the missing values by using the linear regression method. Use the entire dataset and the imputed values to train the DeepAR model.

A. 使用線性迴歸方法估算缺失值。使用整個資料集和估算值來訓練DeepAR 模型。

B. Replace the missing values with not a number (NaN). Use the entire dataset and the encoded missing values to train the DeepAR model.

B. 將缺失值替換為非數字(NaN)。使用整個資料集和編碼的缺失值來訓練DeepAR 模型。

C. Impute the missing values by using a forward fill. Use the entire dataset and the imputed values to train the DeepAR model.

C. 使用前向填充估算缺失值。使用整個資料集和估算值來訓練DeepAR 模型。

D. Impute the missing values by using the mean value. Use the entire dataset and the imputed values to train the DeepAR model.

D. 使用平均值估算缺失值。使用整個資料集和估算值來訓練DeepAR 模型。

問題# 329

A law firm handles thousands of contracts every day. Every contract must be signed. Currently, a lawyer manually checks all contracts for signatures. The law firm is developing a machine learning (ML) solution to automate signature detection for each contract. The ML solution must also provide a confidence score for each contract page. Which Amazon Textract API action can the law firm use to generate a confidence score for each page of each contract?

一家律師事務所每天處理數千份合約。每份合約都必須簽署。目前，律師手動檢查所有合約的簽名。該律師事務所正在開發一種機器學習(ML) 解決方案，以自動偵測每份合約的簽名。ML 解決方案還必須為每個合約頁面提供置信度分數。律師事務所可以使用哪種Amazon Textract API 操作來為每個合約的每一頁產生置信度分數？

A. Use the AnalyzeDocument API action. Set the FeatureTypes parameter to SIGNATURES. Return the confidence scores for each page.

A. 使用AnalyzeDocument API 操作。將FeatureTypes 參數設定為SIGNATURES。傳回每頁的置信度分數。

B. Use the Prediction API call on the documents. Return the signatures and confidence scores for each page.

B. 對文件使用Prediction API 呼叫。傳回每頁的簽名和置信度分數。

C. Use the StartDocumentAnalysis API action to detect the signatures. Return the confidence scores for each page.

C. 使用StartDocumentAnalysis API 操作偵測簽章。傳回每頁的置信度分數。

D. Use the GetDocumentAnalysis API action to detect the signatures. Return the confidence scores for each page.

D. 使用GetDocumentAnalysis API 操作檢測簽章。傳回每頁的置信度分數。

問題# 330

A company that operates oil platforms uses drones to photograph locations on oil platforms that are difficult for humans to access to search for corrosion. Experienced engineers review the photos to determine the severity of corrosion. There can be several corroded areas in a single photo. The engineers determine whether the identified corrosion needs to be fixed immediately, scheduled for future maintenance, or requires no action. The corrosion appears in an average of 0.1% of all photos. A data science team needs to create a solution that automates the process of reviewing the photos and classifying the need for maintenance. Which combination of steps will meet these requirements? (Choose three.)

一家經營石油平台的公司使用無人機拍攝石油平台上人類難以到達的位置以尋找腐蝕。經驗豐富的工程師會查看照片以確定腐蝕的嚴重程度。一張照片中可能會有多个腐蝕區域。工程師會確定已發現的腐蝕是否需要立即修復、安排將來維護或無需採取任何措施。腐蝕出現在所有照片中的比例平均為0.1%。資料科學團隊需要創建一個解決方案，以自動執行查看照片和分類維護需求的過程。哪一種步驟組合可以滿足這些要求？（選三個。）

A. Use an object detection algorithm to train a model to identify corrosion areas of a photo.

A. 使用物件偵測演算法訓練模型來辨識照片中的腐蝕區域。

B. Use Amazon Rekognition with label detection on the photos.

B. 使用Amazon Rekognition 對照片進行標籤偵測。

C. Use a k-means clustering algorithm to train a model to classify the severity of corrosion in a photo.

C. 使用k 均值聚類演算法訓練模型來對照片中的腐蝕嚴重程度進行分類。

D. Use an XGBoost algorithm to train a model to classify the severity of corrosion in a photo.

D. 使用XGBoost 演算法訓練模型來對照片中的腐蝕嚴重程度進行分類。

E. Perform image augmentation on photos that contain corrosion.

E. 對包含腐蝕的照片執行影像增強。

F. Perform image augmentation on photos that do not contain corrosion.

F. 對不包含腐蝕的照片執行影像增強。

問題# 331

A company maintains a 2 TB dataset that contains information about customer behaviors. The company stores the dataset in Amazon S3. The company stores a trained model container in Amazon Elastic Container Registry (Amazon ECR). A machine learning (ML) specialist needs to score a batch model for the dataset to predict customer behavior. The ML specialist must select a scalable approach to score the model. Which solution will meet these requirements MOST cost-effectively?

一家公司維護一個2 TB 的資料集，其中包含有關客戶行為的資訊。該公司將資料集儲存在Amazon S3 中。該公司將經過訓練的模型容器存放在Amazon Elastic Container Registry (Amazon ECR) 中。機器學習(ML) 專家需要對資料集的批次模型進行評分，以預測客戶行為。ML 專家必須選擇一種可擴展的方法來對模型進行評分。哪一種解決方案能夠以最具成本效益的方式滿足這些要求？

A. Score the model by using AWS Batch managed Amazon EC2 Reserved Instances. Create an Amazon EC2 instance store volume and mount it to the Reserved Instances.

A. 使用AWS Batch 管理的Amazon EC2 預留執行個體對模型進行評分。建立Amazon EC2 執行個體儲存磁碟區並將其掛載到預留執行個體。

B. Score the model by using AWS Batch managed Amazon EC2 Spot Instances. Create an Amazon FSx for Lustre volume and mount it to the Spot Instances.

B. 使用AWS Batch 管理的Amazon EC2 Spot 執行個體對模型進行評分。建立Amazon FSx for Lustre 磁碟區並將其掛載到Spot 實例。

C. Score the model by using an Amazon SageMaker notebook on Amazon EC2 Reserved Instances. Create an Amazon EBS volume and mount it to the Reserved Instances.

C. 使用Amazon EC2 預留執行個體上的Amazon SageMaker 筆記本對模型進行評分。建立Amazon EBS 磁碟區並將其掛載到預留實例。

D. Score the model by using Amazon SageMaker notebook on Amazon EC2 Spot Instances. Create an Amazon Elastic File System (Amazon EFS) file system and mount it to the Spot Instances.

D. 使用Amazon EC2 Spot 執行個體上的Amazon SageMaker 筆記本對模型進行評分。建立Amazon Elastic File System (Amazon EFS) 檔案系統並將其掛載到Spot 實例。

問題# 332

A data scientist is implementing a deep learning neural network model for an object detection task on images. The data scientist wants to experiment with a large number of parallel hyperparameter tuning jobs to find hyperparameters that optimize compute time. The data scientist must ensure that jobs that underperform are stopped. The data scientist must allocate computational resources to well-performing hyperparameter configurations. The data scientist is using the hyperparameter tuning job to tune the stochastic gradient descent (SGD) learning rate, momentum, epoch, and mini-batch size. Which technique will meet these requirements with LEAST computational time?

一位資料科學家正在為影像上的物件偵測任務實施深度學習神經網路模型。資料科學家想要試驗大量平行超參數調整作業，以找到最佳化計算時間的超參數。資料科學家必須確保停止表現不佳的作業。資料科學家必須將計算資源分配給表現良好的超參數配置。資料科學家正在使用超參數調整作業來調整隨機梯度下降(SGD)學習率、動量、時期和小批量大小。哪種技術可以以最少的計算時間來滿足這些要求？

A. Grid search.

A. 網格搜尋。

B. Random search.

B. 隨機搜尋。

C. Bayesian optimization.

C. 貝葉斯優化。

D. Hyperband.

D. 超級頻帶。

問題# 333

An agriculture company wants to improve crop yield forecasting for the upcoming season by using crop yields from the last three seasons. The company wants to compare the performance of its new scikit-learn model to the benchmark. A data scientist needs to package the code into a container that computes both the new model forecast and the benchmark. The data scientist wants AWS to be responsible for the operational maintenance of the container. Which solution will meet these requirements?

一家農業公司希望利用過去三個季節的農作物產量來改善下一季節農作物產量的預測。該公司希望將其新的scikit-learn 車型的性能與基準進行比較。資料科學家需要將程式碼打包到一個容器中，該容器既可以計算新模型預測，又可以計算基準。資料科學家希望AWS 負責容器的營運維護。哪種解決方案可以滿足這些要求？

A. Package the code as the training script for an Amazon SageMaker scikit-learn container.

A. 將程式碼打包為Amazon SageMaker scikit-learn 容器的訓練腳本。

B. Package the code into a custom-built container. Push the container to Amazon Elastic Container Registry (Amazon ECR).

B. 將程式碼打包到客製化容器中。將容器推送到Amazon Elastic Container Registry (Amazon ECR)。

C. Package the code into a custom-built container. Push the container to AWS Fargate.

C. 將程式碼打包到客製化容器中。將容器推送到AWS Fargate。

D. Package the code by extending an Amazon SageMaker scikit-learn container.

D. 透過擴充Amazon SageMaker scikit-learn 容器來打包程式碼。

問題# 334

A cybersecurity company is collecting on-premises server logs, mobile app logs, and IoT sensor data. The company backs up the ingested data in an Amazon S3 bucket and sends the ingested data to Amazon OpenSearch Service for further analysis. Currently, the company has a custom ingestion pipeline that is running on Amazon EC2 instances. The company needs to implement a new serverless ingestion pipeline that can automatically scale to handle sudden changes in the data flow. Which solution will meet these requirements MOST cost-effectively?

一家網路安全公司正在收集本地伺服器日誌、行動應用程式日誌和物聯網感測器資料。該公司將提取的資料備份到Amazon S3 儲存桶中，並將提取的資料傳送到Amazon OpenSearch Service 進行進一步分析。目前，該公司有一個在Amazon EC2 執行個體上運行的自訂提取管道。該公司需要實施一條新的無伺服器提取管道，該管道可以自動擴展以處理資料流的突然變化。哪一種解決方案能夠以最具成本效益的方式滿足這些要求？

A. Create two Amazon Data Firehose delivery streams to send data to the S3 bucket and OpenSearch Service. Configure the data sources to send data to the delivery streams.

A. 建立兩個Amazon Data Firehose 傳輸流，將資料傳送到S3 儲存體桶和OpenSearch 服務。配置資料來源以將資料傳送到傳輸流。

B. Create one Amazon Kinesis data stream. Create two Amazon Data Firehose delivery streams to send data to the S3 bucket and OpenSearch Service. Connect the delivery streams to the data stream. Configure the data sources to send data to the data stream.

B. 建立一個Amazon Kinesis 資料流。建立兩個Amazon Data Firehose 傳輸流，將資料傳送到S3 儲存桶和OpenSearch 服務。將傳輸流連接到資料流。配置資料來源以將資料傳送到資料流。

C. Create one Amazon Data Firehose delivery stream to send data to OpenSearch Service. Configure the delivery stream to back up the raw data to the S3 bucket. Configure the data sources to send data to the delivery stream.

C. 建立一個Amazon Data Firehose 傳輸流，將資料傳送到OpenSearch 服務。配置傳輸流以將原始資料備份到S3 儲存桶。配置資料來源以將資料傳送到傳輸流。

D. Create one Amazon Kinesis data stream. Create one Amazon Data Firehose delivery stream to send data to OpenSearch Service. Configure the delivery stream to back up the data to the S3 bucket. Connect the delivery stream to the data stream. Configure the data sources to send data to the data stream.

D. 建立一個Amazon Kinesis 資料流。建立一個Amazon Data Firehose 傳輸流，將資料傳送到OpenSearch 服務。配置傳輸流以將資料備份到S3 儲存桶。將傳輸流連接到資料流。配置資料來源以將資料傳送到資料流。

問題# 335

A bank has collected customer data for 10 years in CSV format. The bank stores the data in an on-premises server. A data science team wants to use Amazon SageMaker to build and train a machine learning (ML) model to predict churn probability. The team will use the historical data. The data scientists want to perform data transformations quickly and to generate data insights before the team builds a model for production. Which solution will meet these requirements with the LEAST development effort?

一家銀行以CSV 格式收集了10 年的客戶資料。銀行將資料儲存在本地伺服器中。資料科學團隊希望使用Amazon SageMaker 建立和訓練機器學習(ML) 模型來預測客戶流失機率。團隊將使用歷史數據。資料科學家希望在團隊建立生產模型之前快速執行資料轉換並產生資料洞察。哪一種解決方案能夠以最少的開發工作量來滿足這些要求？

A. Upload the data into the SageMaker Data Wrangler console directly. Perform data transformations and generate insights within Data Wrangler.

A. 將資料直接上傳到SageMaker Data Wrangler 控制台。在Data Wrangler 中執行資料轉換並產生見解。

B. Upload the data into an Amazon S3 bucket. Allow SageMaker to access the data that is in the bucket. Import the data from the S3 bucket into SageMaker Data Wrangler. Perform data transformations and generate insights within Data Wrangler.

B. 將資料上傳到Amazon S3 儲存桶。允許SageMaker 存取儲存桶中的資料。將資料從S3 儲存桶導入SageMaker Data Wrangler。在Data Wrangler 中執行資料轉換並產生見解。

C. Upload the data into the SageMaker Data Wrangler console directly. Allow SageMaker and Amazon QuickSight to access the data that is in an Amazon S3 bucket. Perform data transformations in Data Wrangler and save the transformed data into a second S3 bucket. Use QuickSight to generate data insights.

C. 將資料直接上傳到SageMaker Data Wrangler 控制台。允許SageMaker 和Amazon QuickSight 存取Amazon S3 儲存桶中的資料。在Data Wrangler 中執行資料轉換並將轉換後的資料儲存到第二個S3 儲存桶中。使用QuickSight 產生資料見解。

D. Upload the data into an Amazon S3 bucket. Allow SageMaker to access the data that is in the bucket. Import the data from the bucket into SageMaker Data Wrangler. Perform data transformations in Data Wrangler. Save the data into a second S3 bucket. Use a SageMaker Studio notebook to generate data insights.

D. 將資料上傳到Amazon S3 儲存桶。允許SageMaker 存取儲存桶中的資料。將資料從儲存桶導入SageMaker Data Wrangler。在Data Wrangler 中執行資料轉換。將資料儲存到第二個S3 儲存桶中。使用SageMaker Studio 筆記本產生資料見解。

問題# 336

A media company wants to deploy a machine learning (ML) model that uses Amazon SageMaker to recommend new articles to the company's readers. The company's readers are primarily located in a single city. The company notices that the heaviest reader traffic predictably occurs early in the morning, after lunch, and again after work hours. There is very little traffic at other times of day. The media company needs to minimize the time required to deliver recommendations to its readers. The expected amount of data that the API call will return for inference is less than 4 MB. Which solution will meet these requirements in the MOST cost-effective way?

一家媒體公司希望部署一個機器學習(ML) 模型，該模型使用Amazon SageMaker 向公司的讀者推薦新文章。該公司的讀者主要位於一個城市。該公司注意到，讀者流量最大的時間可預測為清晨、午餐後以及下班後。一天中其他時間的流量很少。媒體公司需要盡量縮短提供讀者推薦所需的時間。API 呼叫將傳回用於推理的預期資料量小於4 MB。哪一種解決方案能夠以最具成本效益的方式滿足這些要求？

A. Real-time inference with auto scaling.

A. 具有自動擴展的即時推理。

B. Serverless inference with provisioned concurrency.

B. 具有預先配置並發性的無伺服器推理。

C. Asynchronous inference.

C. 非同步推理。

D. A batch transform task.

D. 批次轉換任務。

問題# 337

A machine learning (ML) engineer is using Amazon SageMaker automatic model tuning (AMT) to optimize a model's hyperparameters. The ML engineer notices that the tuning jobs take a long time to run. The tuning jobs continue even when the jobs are not significantly improving against the objective metric. The ML engineer needs the training jobs to optimize the hyperparameters more quickly. How should the ML engineer configure the SageMaker AMT data types to meet these requirements?

機器學習(ML)工程師正在使用Amazon SageMaker 自動模型調整(AMT) 來最佳化模型的超參數。ML工程師注意到調整作業需要很長時間才能運作。即使作業沒有顯著改善目標指標，調整作業仍會持續。ML工程師需要訓練作業來更快地最佳化超參數。ML工程師應如何配置SageMaker AMT 資料類型以滿足這些要求？

A. Set Strategy to the Bayesian value.

A. 將Strategy 設定為貝葉斯值。

B. Set RetryStrategy to a value of 1.

B. 將RetryStrategy 設定為值1。

C. Set ParameterRanges to the narrow range Inferred from previous hyperparameter jobs.

C. 將ParameterRanges 設定為從先前的超參數作業推斷出的窄範圍。

D. Set TrainingJobEarlyStoppingType to the AUTO value.

D. 將TrainingJobEarlyStoppingType 設定為AUTO 值。

問題# 338

A global bank requires a solution to predict whether customers will leave the bank and choose another bank. The bank is using a dataset to train a model to predict customer loss. The training dataset has 1,000 rows. The training dataset includes 100 instances of customers who left the bank. A machine learning (ML) specialist is using Amazon SageMaker Data Wrangler to train a churn prediction model by using a SageMaker training job. After training, the ML specialist notices that the model returns only false results. The ML specialist must correct the model so that it returns more accurate predictions. Which solution will meet these requirements?

一家全球性銀行需要一種解決方案來預測客戶是否會離開該銀行並選擇另一家銀行。該銀行正在使用資料集來訓練模型以預測客戶流失。訓練資料集有1,000 行。訓練資料集包括100 個離開銀行的客戶實例。機器學習(ML) 專家正在使用Amazon SageMaker Data Wrangler 透過SageMaker 訓練作業來訓練客戶流失預測模型。訓練後，ML 專家注意到模型只傳回錯誤結果。ML 專家必須修正模型，使其傳回更準確的預測。哪種解決方案可以滿足這些要求？

A. Apply anomaly detection to remove outliers from the training dataset before training.

A. 在訓練之前應用異常檢測從訓練資料集中移除異常值。

B. Apply Synthetic Minority Oversampling Technique (SMOTE) to the training dataset before training.

B. 在訓練之前對訓練資料集應用合成少數過採樣技術（SMOTE）。

C. Apply normalization to the features of the training dataset before training.

C. 在訓練之前對訓練資料集的特徵應用標準化。

D. Apply undersampling to the training dataset before training.

D. 在訓練之前對訓練資料集應用欠採樣。

問題# 339

A banking company provides financial products to customers around the world. A machine learning (ML) specialist collected transaction data from internal customers. The ML specialist split the dataset into training, testing, and validation datasets. The ML specialist analyzed the training dataset by using Amazon SageMaker Clarify. The analysis found that the training dataset contained fewer examples of customers in the 40 to 55 year-old age group compared to the other age groups. Which type of pretraining bias did the ML specialist observe in the training dataset?

一家銀行為全球客戶提供金融產品。一位機器學習(ML)專家收集了內部客戶的交易資料。ML專家將資料集分為訓練、測試和驗證資料集。ML專家使用Amazon SageMaker Clarify分析了訓練資料集。分析發現，與其他年齡層相比，訓練資料集中包含的40至55歲年齡層客戶的範例較少。ML專家在訓練資料集中觀察到了哪種類型的預訓練偏差？

A. Difference in proportions of labels (DPL).

A. 標籤比例差異(DPL)。

B. Class imbalance (CI).

B. 類別不平衡(CI)。

C. Conditional demographic disparity (CDD).

C. 條件人口差異(CDD)。

D. Kolmogorov-Smirnov (KS).

D. 柯爾莫哥洛夫-斯米爾諾夫(KS)。

問題# 340

A tourism company uses a machine learning (ML) model to make recommendations to customers. The company uses an Amazon SageMaker environment and set hyperparameter tuning completion criteria to MaxNumberOfTrainingJobs. An ML specialist wants to change the hyperparameter tuning completion criteria. The ML specialist wants to stop tuning immediately after an internal algorithm determines that tuning job is unlikely to improve more than 1% over the objective metric from the best training job. Which completion criteria will meet this requirement?

一家旅遊公司使用機器學習(ML)模型向客戶提供建議。該公司使用Amazon SageMaker環境並將超參數調整完成標準設定為MaxNumberOfTrainingJobs。ML專家想要更改超參數調整完成標準。在內部演算法確定調整作業不太可能比最佳訓練作業的客觀指標提高1%以上後，ML專家希望立即停止調整。哪個完成標準可以滿足此要求？

A. MaxRuntimeInSeconds.

A. 最大運轉時間（秒）。

B. TargetObjectiveMetricValue.

B. 目標目標指標值。

C. CompleteOnConvergence.

C. 收斂完成。

D. MaxNumberOfTrainingJobsNotImproving.

D. 未改進的最大訓練作業數。

問題# 341

A car company has dealership locations in multiple cities. The company uses a machine learning (ML) recommendation system to market cars to its customers. An ML engineer trained the ML recommendation model on a dataset that includes multiple attributes about each car. The dataset includes attributes such as car brand, car type, fuel efficiency, and price. The ML engineer uses Amazon SageMaker Data Wrangler to analyze and visualize data. The ML engineer needs to identify the distribution of car prices for a specific type of car. Which type of visualization should the ML engineer use to meet these requirements?

一家汽車公司在多個城市設有經銷店。該公司使用機器學習(ML)推薦系統向客戶推銷汽車。ML工程師在包含每輛汽車的多個屬性的資料集上訓練了ML推薦模型。資料集包括汽車品牌、汽車類型、燃油效率和價格等屬性。ML工程師使用Amazon SageMaker Data Wrangler來分析和視覺化資料。ML工程師需要確定特定類型汽車的價格分佈。ML工程師應該使用哪種類型的視覺化來滿足這些要求？

A. Use the SageMaker Data Wrangler scatter plot visualization to inspect the relationship between the car price and type of car.

A. 使用SageMaker Data Wrangler散點圖視覺化檢查汽車價格和汽車類型之間的關係。

B. Use the SageMaker Data Wrangler quick model visualization to quickly evaluate the data and produce importance scores for the car price and type of car.

B. 使用SageMaker Data Wrangler 快速模型視覺化快速評估資料並產生汽車價格和汽車類型的重要性分數。

C. Use the SageMaker Data Wrangler anomaly detection visualization to Identify outliers for the specific features.

C. 使用SageMaker Data Wrangler 異常檢測視覺化識別特定特徵的異常值。

D. Use the SageMaker Data Wrangler histogram visualization to inspect the range of values for the specific feature.

D. 使用SageMaker Data Wrangler 直方圖視覺化檢查特定特徵的值範圍。

問題# 342

A media company is building a computer vision model to analyze images that are on social media. The model consists of CNNs that the company trained by using images that the company stores in Amazon S3. The company used an Amazon SageMaker training job in File mode with a single Amazon EC2 On-Demand Instance. Every day, the company updates the model by using about 10,000 images that the company has collected in the last 24 hours. The company configures training with only one epoch. The company wants to speed up training and lower costs without the need to make any code changes. Which solution will meet these requirements?

一家媒體公司正在建立一個電腦視覺模型來分析社群媒體上的圖像。該模型由該公司使用儲存在Amazon S3 中的圖像訓練的CNN 組成。該公司在檔案模式下使用單一Amazon EC2 按需實例執行Amazon SageMaker 訓練作業。每天，該公司都會使用過去24 小時內收集的約10,000 張圖像來更新模型。該公司僅使用一個時期來配置訓練。該公司希望加快訓練速度並降低成本，而無需進行任何程式碼變更。哪種解決方案可以滿足這些要求？

A. Instead of File mode, configure the SageMaker training job to use Pipe mode. Ingest the data from a pipe.

A. 設定SageMaker 訓練作業以使用管道模式，而不是檔案模式。從管道中提取資料。

B. Instead of File mode, configure the SageMaker training job to use FastFile mode with no other changes.

B. 設定SageMaker 訓練作業以使用FastFile 模式，而不是檔案模式，無需進行其他變更。

C. Instead of On-Demand Instances, configure the SageMaker training job to use Spot Instances. Make no other changes,.

C. 設定SageMaker 訓練作業以使用Spot 實例，而不是隨選實例。不做其他更改。

D. Instead of On-Demand Instances, configure the SageMaker training job to use Spot Instances, implement model checkpoints.

D. 設定SageMaker 訓練作業以使用Spot 實例，而不是按需實例，並實施模型檢查點。

問題# 343

A telecommunications company has deployed a machine learning model using Amazon SageMaker. The model identifies customers who are likely to cancel their contract when calling customer service. These customers are then directed to a specialist service team. The model has been trained on historical data from multiple years relating to customer contracts and customer service interactions in a single geographic region. The company is planning to launch a new global product that will use this model. Management is concerned that the model might incorrectly direct a large number of calls from customers in regions without historical data to the specialist service team. Which approach would MOST effectively address this issue?

一家電信公司使用Amazon SageMaker 部署了一個機器學習模型。此模型識別出在致電客戶服務時可能取消合約的客戶。然後這些客戶被引導至專業服務團隊。該模型已根據多年與單一地理區域的客戶合約和客戶服務互動相關的歷史資料進行了訓練。該公司計劃推出一款將使用此模型的全球新產品。管理層擔心該模型可能會錯誤地將大量來自沒有歷史數據的地區客戶的電話引導至專業服務團隊。哪一種方法可以最有效地解決這個問題？

A. Enable Amazon SageMaker Model Monitor data capture on the model endpoint. Create a monitoring baseline on the training dataset. Schedule monitoring jobs. Use Amazon CloudWatch to alert the data scientists when the numerical distance of regional customer data fails the baseline drift check. Reevaluate the training set with the larger data source and retrain the model.

A. 在模型終端節點上啟用Amazon SageMaker Model Monitor 資料擷取。在訓練資料集上建立監控基線。安排監控作業。當區域客戶資料的數值距離未通過基線漂移檢查時，使用Amazon Cloud Watch 提醒資料科學家。使用更大的資料來源重新評估訓練集並重新訓練模型。

B. Enable Amazon SageMaker Debugger on the model endpoint. Create a custom rule to measure the variance from the baseline training dataset. Use Amazon CloudWatch to alert the data scientists when the rule is invoked. Reevaluate the training set with the larger data source and retrain the model.

B. 在模型終端節點上啟用Amazon SageMaker Debugger。建立自訂規則以測量與基線訓練資料集的差異。使用Amazon CloudWatch 在呼叫規則時提醒資料科學家。使用更大的資料來源重新評估訓練集並重新訓練模型。

C. Capture all customer calls routed to the specialist service team in Amazon S3. Schedule a monitoring job to capture all the true positives and true negatives, correlate them to the training dataset, and calculate the accuracy. Use Amazon CloudWatch to alert the data scientists when the accuracy decreases. Reevaluate the training set with the additional data from the specialist service team and retrain the model.

C. 在Amazon S3 中捕獲路由到專家服務團隊的所有客戶呼叫。安排監控作業以捕捉所有真陽性和真陰性，將它們與訓練資料集關聯，並計算準確度。當準確度下降時，使用Amazon CloudWatch 提醒資料科學家。使用專家服務團隊的其他數據重新評估訓練集並重新訓練模型。

D. Enable Amazon CloudWatch on the model endpoint. Capture metrics using Amazon CloudWatch Logs and send them to Amazon S3. Analyze the monitored results against the training data baseline. When the variance from the baseline exceeds the regional customer variance, reevaluate the training set and retrain the model.

D. 在模型終端節點上啟用Amazon CloudWatch。使用Amazon CloudWatch Logs 捕獲指標並將其發送到Amazon S3。根據訓練資料基線分析監控結果。當與基線的差異超過區域客戶差異時，重新評估訓練集並重新訓練模型。

問題# 344

A machine learning (ML) engineer is creating a binary classification model. The ML engineer will use the model in a highly sensitive environment. There is no cost associated with missing a positive label. However, the cost of making a false positive inference is extremely high. What is the most important metric to optimize the model for in this scenario?

一位機器學習(ML)工程師正在創建一個二元分類模型。ML工程師將在高度敏感的環境中使用該模型。漏掉一個正標籤不會產生任何成本。但是，做出假正推論的成本極高。在這種情況下，優化模型最重要的指標是什麼？

A. Accuracy.

A. 準確度

B. Precision.

B. 精確度

C. Recall.

C. 召回率

D. F1.

D. F1.

問題# 345

An ecommerce company discovers that the search tool for the company's website is not presenting the top search results to customers. The company needs to resolve the issue so the search tool will present results that customers are most likely to want to purchase. Which solution will meet this requirement with the LEAST operational effort?

一家電子商務公司發現，公司網站的搜尋工具沒有提供客戶最佳搜尋結果。該公司需要解決此問題，以便搜尋工具能夠顯示客戶最有可能購買的結果。哪種解決方案能夠以最少的營運工作量滿足此要求？

A. Use the Amazon SageMaker BlazingText algorithm to add context to search results through query expansion.

A. 使用Amazon SageMaker BlazingText 演算法透過查詢擴展為搜尋結果添加上下文。

B. Use the Amazon SageMaker XGBoost algorithm to improve candidate ranking.

B. 使用Amazon SageMaker XGBoost 演算法提高候選排名。

C. Use Amazon CloudSearch and sort results by the search relevance score.

C. 使用Amazon CloudSearch 並依搜尋相關性分數對結果進行排序。

D. Use Amazon CloudSearch and sort results by the geographic location.

D. 使用Amazon CloudSearch 並依地理位置對結果進行排序。

問題# 346

A machine learning (ML) specialist collected daily product usage data for a group of customers. The ML specialist appended customer metadata such as age and gender from an external data source. The ML specialist wants to understand product usage patterns for each day of the week for customers in specific age groups. The ML specialist creates two categorical features named `dayofweek` and `binned_age`, respectively. Which approach should the ML specialist use to discover the relationship between the two new categorical features?

一位機器學習(ML)專家收集了一組客戶的每日產品使用資料。ML專家從外部資料來源附加了客戶元數據，例如年齡和性別。ML專家希望了解特定年齡層客戶一週中每一天的產品使用模式。ML專家分別建立了兩個名為`dayofweek`和`binned_age`的分類特徵。ML專家應該使用哪一種方法來發現這兩個新分類特徵之間的關係？

A. Create a scatterplot for `day_of_week` and `binned_age`.

A. 為`day_of_week`和`binned_age`建立散佈圖。

B. Create crosstabs for `day_of_week` and `binned_age`.

B. 為`day_of_week`和`binned_age`建立交叉表。

C. Create word clouds for `day_of_week` and `binned_age`.

C. 為`day_of_week`和`binned_age`建立詞雲。

D. Create a boxplot for `day_of_week` and `binned_age`.

D. 為`day_of_week`和`binned_age`建立箱線圖。

問題# 347

A company needs to develop a model that uses a machine learning (ML) model for risk analysis. An ML engineer needs to evaluate the contribution each feature of a training dataset makes to the prediction of the target variable before the ML engineer selects features. How should the ML engineer predict the contribution of each feature?

一家公司需要開發一個使用機器學習(ML)模型進行風險分析的模型。在選擇特徵之前，ML工程師需要評估訓練資料集的每個特徵對目標變數預測的貢獻。ML工程師應該如何預測每個特徵的貢獻？

A. Use the Amazon SageMaker Data Wrangler multicollinearity measurement features and the principal component analysis (PCA) algorithm to calculate the variance of the dataset along multiple directions in the feature space.

A. 使用Amazon SageMaker Data Wrangler 多重共線性測量功能和主成分分析(PCA)演算法計算特徵空間中多個方向上資料集的變異數。

B. Use an Amazon SageMaker Data Wrangler quick model visualization to find feature importance scores that are between 0.5 and 1.

B. 使用Amazon SageMaker Data Wrangler 快速模型視覺化尋找介於0.5和1之間的特徵重要性分數。

C. Use the Amazon SageMaker Data Wrangler bias report to identify potential biases in the data related to feature engineering.

C. 使用Amazon SageMaker Data Wrangler 偏差報告識別與特徵工程相關的數據中的潛在偏差。

D. Use an Amazon SageMaker Data Wrangler data flow to create and modify a data preparation pipeline. Manually add the feature scores.

D. 使用Amazon SageMaker Data Wrangler 資料流建立和修改資料準備管道。手動新增特徵分數。

問題# 348

A company is building a predictive maintenance system using real-time data from devices on remote sites. There is no AWS Direct Connect connection or VPN connection between the sites and the company's VPC. The data needs to be ingested in real time from the devices into Amazon S3. Transformation is needed to convert the raw data into clean.csv data to be fed into the machine learning (ML) model. The transformation needs to happen during the ingestion process. When transformation fails, the records need to be stored in a specific location in Amazon S3 for human review. The raw data before transformation also needs to be stored in Amazon S3. How should an ML specialist architect the solution to meet these requirements with the LEAST effort?

一家公司正在使用來自遠端站點設備的即時數據來建立預測性維護系統。網站與公司的VPC 之間沒有AWS Direct Connect 連線或VPN 連線。需要將資料從設備即時提取到Amazon S3 中。需要進行轉換以將原始數據轉換為乾淨的.csv 數據，然後輸入到機器學習(ML) 模型中。轉換需要在提取過程中進行。當轉換失敗時，記錄需要儲存在Amazon S3 中的特定位置以供手動審核。轉換前的原始資料也需要儲存在Amazon S3 中。ML 專家應如何建立解決方案以最少的努力滿足這些要求？

A. Use Amazon Data Firehose with Amazon S3 as the destination. Configure Firehose to invoke an AWS Lambda function for data transformation. Enable source record backup on Firehose.

A. 使用Amazon Data Firehose 並將Amazon S3 用作目標。配置Firehose 以呼叫AWS Lambda 函數進行資料轉換。在Firehose 上啟用來源記錄備份。

B. Use Amazon Managed Streaming for Apache Kafka. Set up workers in Amazon Elastic Container Service (Amazon ECS) to move data from Kafka brokers to Amazon S3 while transforming it. Configure workers to store raw and unsuccessfully transformed data in different S3 buckets.

B. 使用Amazon Managed Streaming for Apache Kafka。在Amazon Elastic Container Service (Amazon ECS) 中設定工作程序，以在轉換資料的同時將資料從Kafka 代理移至Amazon S3。配置工作程序以將原始資料和未成功轉換的資料儲存在不同的S3 儲存桶中。

C. Use Amazon Data Firehose with Amazon S3 as the destination. Configure Firehose to invoke an Apache Spark job in AWS Glue for data transformation. Enable source record backup and configure the error prefix.

C. 使用Amazon Data Firehose 並將Amazon S3 用作目標。配置Firehose 以呼叫AWS Glue 中的Apache Spark 作業進行資料轉換。啟用來源記錄備份並配置錯誤前綴。

D. Use Amazon Kinesis Data Streams in front of Amazon Data Firehose. Use Kinesis Data Streams with AWS Lambda to store raw data in Amazon S3. Configure Firehose to invoke a Lambda function for data transformation with Amazon S3 as the destination.

D. 在Amazon Data Firehose 前面使用Amazon Kinesis Data Streams。使用Kinesis Data Streams 和AWS Lambda 將原始資料儲存在Amazon S3 中。配置Firehose 以呼叫Lambda 函數進行資料轉換，並將Amazon S3 用作目標。

問題# 349

A company wants to use machine learning (ML) to improve its customer churn prediction model. The company stores data in an Amazon Redshift data warehouse. A data science team wants to use Amazon Redshift machine learning (Amazon Redshift ML) to build a model and run predictions for new data directly within the data warehouse. Which combination of steps should the company take to use Amazon Redshift ML to meet these requirements? (Choose three.)

一家公司希望使用機器學習(ML) 來改進其客戶流失預測模型。該公司將資料儲存在Amazon Redshift 資料倉儲中。資料科學團隊希望使用Amazon Redshift 機器學習(Amazon Redshift ML) 建立模型並直接在資料倉儲中執行新資料的預測。該公司應採取哪些步驟組合來使用Amazon Redshift ML 來滿足這些要求？（選三個。）

A. Define the feature variables and target variable for the churn prediction model.

A. 為客戶流失預測模型定義特徵變數和目標變數。

B. Use the SQL EXPLAIN_MODEL function to run predictions.

B. 使用SQL EXPLAIN_MODEL 函數執行預測。

C. Write a CREATE MODEL SQL statement to create a model.

C. 編寫CREATE MODEL SQL 語句來建立模型。

D. Use Amazon Redshift Spectrum to train the model.

D. 使用Amazon Redshift Spectrum 訓練模型。

E. Manually export the training data to Amazon S3.

E. 手動將訓練資料匯出到Amazon S3。

F. Use the SQL prediction function to run predictions.

F. 使用SQL 預測函數來執行預測。

問題# 350

A company's machine learning (ML) team needs to build a system that can detect whether people in a collection of images are wearing the company's logo. The company has a set of labeled training data. Which algorithm should the ML team use to meet this requirement?

一家公司的機器學習(ML)團隊需要建立一個系統，該系統可以偵測影像集合中的人物是否佩戴了公司的徽標。該公司擁有一組標籤的訓練資料。ML 團隊應該使用哪種演算法來滿足此要求？

A. Principal component analysis (PCA).

A. 主成分分析 (PCA)。

B. Recurrent neural network (RNN).

B. 循環神經網路 (RNN)。

C. K-nearest neighbors (k-NN).

C. K-最近鄰 (k-NN)。

D. Convolutional neural network (CNN).

D. 卷積神經網路 (CNN)。

問題# 351

A data scientist uses Amazon SageMaker Data Wrangler to obtain a feature summary from a dataset that the data scientist imported from Amazon S3. The data scientist notices that the prediction power for a dataset feature has a score of 1. What is the cause of the score?

資料科學家使用Amazon SageMaker Data Wrangler 從其從Amazon S3 導入的資料集中取得特徵摘要。資料科學家注意到資料集特徵的預測能力得分為1。該得分的原因是什麼？

A. Target leakage occurred in the imported dataset.

A. 匯入的資料集中發生了目標洩漏。

B. The data scientist did not fine-tune the training and validation split.

B. 資料科學家沒有對訓練和驗證分割進行微調。

C. The SageMaker Data Wrangler algorithm that the data scientist used did not find an optimal model fit for each feature to calculate the prediction power.

C. 資料科學家使用的SageMaker Data Wrangler 演算法沒有找到適合每個特徵的最佳模型來計算預測能力。

D. The data scientist did not process the features enough to accurately calculate prediction power.

D. 資料科學家沒有對特徵進行足夠的處理以準確計算預測能力。

問題# 352

A data scientist is conducting exploratory data analysis (EDA) on a dataset that contains information about product suppliers. The dataset records the country where each product supplier is located as a two-letter text code. For example, the code for New Zealand is "NZ." The data scientist needs to transform the country codes for model training. The data scientist must choose the solution that will result in the smallest increase in dimensionality. The solution must not result in any information loss. Which solution will meet these requirements?

一位資料科學家正在對包含產品供應商資訊的資料集進行探索性資料分析(EDA)。此資料集以兩個字母的文字代碼記錄每個產品供應商所在的國家。例如，紐西蘭的代碼為“NZ”。資料科學家需要轉換國家/地區代碼以進行模型訓練。資料科學家必須選擇導致維數增加最小的解決方案。該解決方案不得導致任何資訊遺失。哪種解決方案可以滿足這些要求？

A. Add a new column of data that includes the full country name.

A. 新增包含完整國家名稱的新資料列。

B. Encode the country codes into numeric variables by using similarity encoding.

B. 使用相似性編碼將國家代碼編碼為數字變數。

C. Map the country codes to continent names.

C. 將國家代碼對應到大洲名稱。

D. Encode the country codes into numeric variables by using one-hot encoding.

D. 使用獨熱編碼將國家代碼編碼為數字變數。

問題# 353

A data scientist is building a new model for an ecommerce company. The model will predict how many minutes it will take to deliver a package. During model training, the data scientist needs to evaluate model performance. Which metrics should the data scientist use to meet this requirement? (Choose two.)

一位資料科學家正在為一家電子商務公司建立一個新模型。該模型將預測遞送包裹需要多少分鐘。在模型訓練期間，資料科學家需要評估模型效能。資料科學家應該使用哪些指標來滿足此要求？（選兩個。）

A. Inference Latency.

A. 推理延遲

B. Mean squared error (MSE).

B. 均方誤差(MSE)

C. Root mean squared error (RMSE).

C. 均方根誤差(RMSE)

D. Precision.

D. 精度

E. Accuracy.

E. 準確度。

問題# 354

A machine learning (ML) specialist is developing a model for a company. The model will classify and predict sequences of objects that are displayed in a video. The ML specialist decides to use a hybrid architecture that consists of a convolutional neural network (CNN) followed by a classifier three-layer recurrent neural network (RNN). The company developed a similar model previously but trained the model to classify a different set of objects. The ML specialist wants to save time by using the previously trained model and adapting the model for the current use case and set of objects. Which combination of steps will accomplish this goal with the LEAST amount of effort? (Choose two.)

一位機器學習(ML)專家正在為一家公司開發一個模型。該模型將對影片中顯示的物件序列進行分類和預測。ML專家決定使用由卷積神經網路(CNN)和分類器三層循環神經網路(RNN)組成的混合架構。該公司之前開發了一個類似的模型，但訓練該模型來對不同的物件集進行分類。ML專家希望透過使用先前訓練的模型並針對當前用例和物件集調整模型來節省時間。哪種步驟組合可以以最少的努力實現此目標？（選兩個。）

A. Reinitialize the weights of the entire CNN. Retrain the CNN on the classification task by using the new set of objects.

A. 重新初始化整個CNN的權重。使用新的物件集在分類任務上重新訓練CNN。

B. Reinitialize the weights of the entire network. Retrain the entire network on the prediction task by using the new set of objects.

B. 重新初始化整個網路的權重。使用新的物件集在預測任務上重新訓練整個網路。

C. Reinitialize the weights of the entire RNN. Retrain the entire model on the prediction task by using the new set of objects.

C. 重新初始化整個RNN的權重。使用新的物件集在預測任務上重新訓練整個模型。

D. Reinitialize the weights of the last fully connected layer of the CNN. Retrain the CNN on the classification task by using the new set of objects.

D. 重新初始化CNN的最後一個全連接層的權重。使用新的物件集在分類任務上重新訓練CNN。

E. Reinitialize the weights of the last layer of the RNN. Retrain the entire model on the prediction task by using the new set of objects.

E. 重新初始化RNN最後一層的權重。使用新的物件集在預測任務上重新訓練整個模型。

問題# 355

A company distributes an online multiple-choice survey to several thousand people. Respondents to the survey can select multiple options for each question. A machine learning (ML) engineer needs to comprehensively represent every response from all respondents in a dataset. The ML engineer will use the dataset to train a logistic regression model. Which solution will meet these requirements?

一家公司向數千人分發了一份線上多項選擇調查問卷。問卷調查的受訪者可以為每個問題選擇多個選項。機器學習(ML)工程師需要將所有受訪者的每個答案都綜合到一個資料集中。ML工程師將使用該資料集來訓練邏輯迴歸模型。哪種解決方案可以滿足這些要求？

A. Perform one-hot encoding on every possible option for each question of the survey.

A. 對調查中每個問題的每個可能選項執行獨熱編碼。

B. Perform binning on all the answers each respondent selected for each question.

B. 將每個問題的每個受訪者選擇的所有答案進行分箱。

C. Use Amazon Mechanical Turk to create categorical labels for each set of possible responses.

C. 使用Amazon Mechanical Turk 為每組可能的答案建立分類標籤。

D. Use Amazon Textract to create numeric features for each set of possible responses.

D. 使用Amazon Textract 為每組可能的答案創建數字特徵。

問題# 356

A manufacturing company stores production volume data in a PostgreSQL database. The company needs an end-to-end solution that will give business analysts the ability to prepare data for processing and to predict future production volume based the previous year's production volume. The solution must not require the company to have coding knowledge. Which solution will meet these requirements with the LEAST effort?

一家製造公司將生產量資料儲存在PostgreSQL 資料庫中。該公司需要一個端到端解決方案，使業務分析師能夠準備資料進行處理，並根據前一年的生產量預測未來的生產量。該解決方案必須不要公司具備編碼知識。哪種解決方案能夠以最少的努力滿足這些要求？

A. Use AWS Database Migration Service (AWS DMS) to transfer the data from the PostgreSQL database to an Amazon S3 bucket. Create an Amazon EMR cluster to read the S3 bucket and perform the data preparation. Use Amazon SageMaker Studio for the prediction modeling.

A. 使用AWS Database Migration Service (AWS DMS) 將資料從PostgreSQL 資料庫傳輸到Amazon S3 儲存桶。建立Amazon EMR 叢集以讀取S3 儲存桶並執行資料準備。使用Amazon SageMaker Studio 進行預測建模。

B. Use AWS Glue DataBrew to read the data that is in the PostgreSQL database and to perform the data preparation. Use Amazon SageMaker Canvas for the prediction modeling.

B. 使用AWS Glue DataBrew 讀取PostgreSQL 資料庫中的資料並執行資料準備。使用Amazon Sag

eMaker Canvas 進行預測建模。

C. Use AWS Database Migration Service (AWS DMS) to transfer the data from the PostgreSQL database to an Amazon S3 bucket. Use AWS Glue to read the data in the S3 bucket and to perform the data preparation. Use Amazon SageMaker Canvas for the prediction modeling.

C. 使用AWS Database Migration Service (AWS DMS) 將資料從PostgreSQL 資料庫傳輸到Amazon S3 儲存桶。使用AWS Glue 讀取S3 儲存桶中的資料並執行資料準備。使用Amazon SageMaker Canvas 進行預測建模。

D. Use AWS Glue DataBrew to read the data that is in the PostgreSQL database and to perform the data preparation. Use Amazon SageMaker Studio for the prediction modeling.

D. 使用AWS Glue DataBrew 讀取PostgreSQL 資料庫中的資料並執行資料準備。使用Amazon SageMaker Studio 進行預測建模。

問題# 357

A data scientist needs to create a model for predictive maintenance. The model will be based on historical data to identify rare anomalies in the data. The historical data is stored in an Amazon S3 bucket. The data scientist needs to use Amazon SageMaker Data Wrangler to ingest the data. The data scientist also needs to perform exploratory data analysis (EDA) to understand the statistical properties of the data. Which solution will meet these requirements with the LEAST amount of compute resources?

資料科學家需要創建一個預測性維護模型。該模型將基於歷史數據來識別數據中的罕見異常。歷史資料儲存在Amazon S3 儲存桶中。資料科學家需要使用Amazon SageMaker Data Wrangler 來擷取資料。資料科學家還需要執行探索性資料分析(EDA) 來了解資料的統計特性。哪一種解決方案能夠以最少的計算資源滿足這些要求？

A. Import the data by using the None option.

A. 使用None 選項匯入資料。

B. Import the data by using the Stratified option.

B. 使用Stratified 選項匯入資料。

C. Import the data by using the First K option. Infer the value of K from domain knowledge.

C. 使用First K 選項匯入資料。根據領域知識推論K 的值。

D. Import the data by using the Randomized option. Infer the random size from domain knowledge.

D. 使用Randomized 選項匯入資料。根據領域知識推斷隨機大小。

問題# 358

An ecommerce company has observed that customers who use the company's website rarely view items that the website recommends to customers. The company wants to recommend items to customers that customers are more likely to want to purchase. Which solution will meet this requirement in the SHORTEST amount of time?

一家電子商務公司發現，使用公司網站的客戶很少查看網站向客戶推薦的商品。該公司希望向客戶推薦客戶更有可能購買的商品。哪種解決方案可以在最短的時間內滿足這項要求？

- A. Host the company's website on Amazon EC2 Accelerated Computing instances to increase the website response speed.
A. 將公司的網站託管在Amazon EC2 加速運算執行個體上，以提高網站回應速度。
 - B. Host the company's website on Amazon EC2 GPU-based instances to increase the speed of the website's search tool.
B. 將公司的網站託管在基於Amazon EC2 GPU 的執行個體上，以提高網站搜尋工具的速度。
 - C. Integrate Amazon Personalize into the company's website to provide customers with personalized recommendations.
C. 將Amazon Personalize 整合到公司的網站中，為客戶提供個人化推薦。
 - D. Use Amazon SageMaker to train a Neural Collaborative Filtering (NCF) model to make product recommendations.
D. 使用Amazon SageMaker 訓練神經協同過濾(NCF) 模型以進行產品推薦。
-

問題# 359

A machine learning (ML) engineer is preparing a dataset for a classification model. The ML engineer notices that some continuous numeric features have a significantly greater value than most other features. A business expert explains that the features are independently informative and that the dataset is representative of the target distribution. After training, the model's inferences accuracy is lower than expected. Which preprocessing technique will result in the GREATEST increase of the model's inference accuracy?

一位機器學習(ML) 工程師正在為分類模型準備資料集。 ML 工程師注意到一些連續數值特徵的值明顯高於大多數其他特徵。一位業務專家解釋說，這些特徵具有獨立的資訊量，並且資料集代表了目標分佈。經過訓練後，模型的推理準確度低於預期。哪種預處理技術將最大程度地提高模型的推理準確度？

- A. Normalize the problematic features.
A. 規範化有問題的特徵。
- B. Bootstrap the problematic features.
B. 引導有問題的特徵。
- C. Remove the problematic features.
C. 刪除有問題的特徵。
- D. Extrapolate synthetic features.

問題# 360

A manufacturing company produces 100 types of steel rods. The rod types have varying material grades and dimensions. The company has sales data for the steel rods for the past 50 years. A data scientist needs to build a machine learning (ML) model to predict future sales of the steel rods. Which solution will meet this requirement in the MOST operationally efficient way?

一家製造公司生產100種鋼棒。這些鋼棒的材料等級和尺寸各不相同。該公司擁有過去50年的鋼棒銷售數據。資料科學家需要建立一個機器學習(ML)模型來預測鋼棒的未來銷售情況。哪一種解決方案能夠以最高效的方式滿足這項要求？

A. Use the Amazon SageMaker DeepAR forecasting algorithm to build a single model for all the products.

A. 使用Amazon SageMaker DeepAR 預測演算法為所有產品建立單一模型。

B. Use the Amazon SageMaker DeepAR forecasting algorithm to build separate models for each product.

B. 使用Amazon SageMaker DeepAR 預測演算法為每種產品建立單獨的模型。

C. Use Amazon SageMaker Autopilot to build a single model for all the products.

C. 使用Amazon SageMaker Autopilot 為所有產品建立單一模型。

D. Use Amazon SageMaker Autopilot to build separate models for each product.

D. 使用Amazon SageMaker Autopilot 為每種產品建立單獨的模型。

問題# 361

A machine learning (ML) specialist is building a credit score model for a financial institution. The ML specialist has collected data for the previous 3 years of transactions and third-party metadata that is related to the transactions. After the ML specialist builds the initial model, the ML specialist discovers that the model has low accuracy for both the training data and the test data. The ML specialist needs to improve the accuracy of the model. Which solutions will meet this requirement? (Choose two.)

一位機器學習(ML)專家正在為一家金融機構建立信用評分模型。ML專家收集了過去3年的交易資料以及與交易相關的第三方元資料。在建立初始模型後，ML專家發現該模型對於訓練資料和測試資料的準確率都很低。ML專家需要提高模型的準確度。哪些解決方案可以滿足此要求？（選兩個。）

A. Increase the number of passes on the existing training data. Perform more hyperparameter tuning.

A. 增加現有訓練資料的傳遞次數。執行更多超參數調整。

B. Increase the amount of regularization. Use fewer feature combinations.

B. 增加正規化量。使用較少的特徵組合。

C. Add new domain-specific features. Use more complex models.

C. 新增的領域特定功能。使用更複雜的模型。

D. Use fewer feature combinations. Decrease the number of numeric attribute bins.

D. 使用較少的特徵組合。減少數值屬性箱的數量。

E. Decrease the amount of training data examples. Reduce the number of passes on the existing training data.

E. 減少訓練資料範例的數量。減少對現有訓練資料的傳遞次數。

問題# 362

A data scientist uses Amazon SageMaker to perform hyperparameter tuning for a prototype machine learning (ML) model. The data scientist's domain knowledge suggests that the hyperparameter is highly sensitive to changes. The optimal value, x , is in the $0.5 < x < 1.0$ range. The data scientist's domain knowledge suggests that the optimal value is close to 1.0. The data scientist needs to find the optimal hyperparameter value with a minimum number of runs and with a high degree of consistent tuning conditions. Which hyperparameter scaling type should the data scientist use to meet these requirements?

資料科學家使用Amazon SageMaker 為原型機器學習(ML) 模型執行超參數調整。資料科學家的領域知識表明，超參數對變化高度敏感。最佳值 x 在 $0.5 < x < 1.0$ 範圍內。資料科學家的領域知識表明，最佳值接近1.0。資料科學家需要找到具有最少運行次數和高度一致的調整條件的最佳超參數值。資料科學家應該使用哪種超參數擴展類型來滿足這些要求？

A. Auto scaling.

A. 自動縮放。

B. Linear scaling.

B. 線性縮放。

C. Logarithmic scaling.

C. 對數縮放。

D. Reverse logarithmic scaling.

D. 反對數縮放。

問題# 363

A data scientist uses Amazon SageMaker Data Wrangler to analyze and visualize data. The data scientist wants to refine a training dataset by selecting predictor variables that are strongly predictive of the target variable. The target variable correlates with other predictor variables. The data scientist wants to understand the variance in the data along various directions in the feature space. Which solution will meet these requirements?

資料科學家使用Amazon SageMaker Data Wrangler 分析和視覺化資料。資料科學家希望透過選擇對目標變數具有強預測性的預測變數來優化訓練資料集。目標變數與其他預測變數相關。資料科學家希望了解特徵空間中各個方向上的資料差異。哪種解決方案可以滿足這些要求？

A. Use the SageMaker Data Wrangler multicollinearity measurement features with a variance inflation factor (VIF) score. Use the VIF score as a measurement of how closely the variables are related to each other.

A. 將SageMaker Data Wrangler 多重共線性測量功能與變異數膨脹因子(VIF) 分數結合使用。使用VIF 分數作為變數之間關聯程度的測量。

B. Use the SageMaker Data Wrangler Data Quality and Insights Report quick model visualization to estimate the expected quality of a model that is trained on the data.

B. 使用SageMaker Data Wrangler 資料品質和洞察報告快速模型視覺化來估計在資料上訓練的模型的預期品質。

C. Use the SageMaker Data Wrangler multicollinearity measurement features with the principal component analysis (PCA) algorithm to provide a feature space that includes all of the predictor variables.

C. 將SageMaker Data Wrangler 多重共線性測量功能與主成分分析(PCA) 演算法結合使用，以提供包含所有預測變數的特徵空間。

D. Use the SageMaker Data Wrangler Data Quality and Insights Report feature to review features by their predictive power.

D. 使用SageMaker Data Wrangler 資料品質和洞察報告功能按預測能力審查特徵。

問題# 364

A business to business (B2B) ecommerce company wants to develop a fair and equitable risk mitigation strategy to reject potentially fraudulent transactions. The company wants to reject fraudulent transactions despite the possibility of losing some profitable transactions or customers. Which solution will meet these requirements with the LEAST operational effort?

一家企業對企業(B2B) 電子商務公司希望制定公平合理的風險緩解策略，以拒絕潛在的詐欺交易。儘管可能會失去一些有利可圖的交易或客戶，但該公司仍希望拒絕詐欺交易。哪種解決方案能夠以最少的營運工作量滿足這些要求？

A. Use Amazon SageMaker to approve transactions only for products the company has sold in the past.

A. 使用Amazon SageMaker 僅核准公司過去銷售過的產品的交易。

B. Use Amazon SageMaker to train a custom fraud detection model based on customer data.

B. 使用Amazon SageMaker 根據客戶資料訓練自訂詐欺偵測模型。

C. Use the Amazon Fraud Detector prediction API to approve or deny any activities that Fraud Detector identifies as fraudulent.

C. 使用Amazon Fraud Detector 預測API 批准或拒絕Fraud Detector 識別為欺詐的任何活動。

D. Use the Amazon Fraud Detector prediction API to identify potentially fraudulent activities so the company can review the activities and reject fraudulent transactions.

D. 使用Amazon Fraud Detector 預測API 識別潛在的詐欺活動，以便公司可以審查這些活動並拒絕詐欺交易。

問題# 365

A data scientist needs to develop a model to detect fraud. The data scientist has less data for fraudulent transactions than for legitimate transactions. The data scientist needs to check for bias in the model before finalizing the model. The data scientist needs to develop the model quickly. Which solution will meet these requirements with the LEAST operational overhead?

資料科學家需要開發一個模型來偵測詐欺行為。資料科學家掌握的詐欺交易資料比合法交易資料少。資料科學家需要在最終確定模型之前檢查模型中的偏差。資料科學家需要快速開發模型。哪種解決方案能夠以最少的營運開銷滿足這些要求？

A. Process and reduce bias by using the synthetic minority oversampling technique (SMOTE) in Amazon EMR. Use Amazon SageMaker Studio Classic to develop the model. Use Amazon Augmented AI (Amazon A2I) to check the model for bias before finalizing the model.

A. 使用Amazon EMR 中的合成少數過採樣技術(SMOTE) 處理並減少偏差。使用Amazon SageMaker Studio Classic 開發模型。使用Amazon Augmented AI (Amazon A2I) 在最終確定模型之前檢查模型是否有偏差。

B. Process and reduce bias by using the synthetic minority oversampling technique (SMOTE) in Amazon EMR. Use Amazon SageMaker Clarify to develop the model. Use Amazon Augmented AI (Amazon A2I) to check the model for bias before finalizing the model.

B. 使用Amazon EMR 中的合成少數過採樣技術(SMOTE) 處理並減少偏差。使用Amazon SageMaker Clarify 開發模型。使用Amazon Augmented AI (Amazon A2I) 在最終確定模型之前檢查模型是否有偏差。

C. Process and reduce bias by using the synthetic minority oversampling technique (SMOTE) in Amazon SageMaker Studio. Use Amazon SageMaker JumpStart to develop the model. Use Amazon SageMaker Clarify to check the model for bias before finalizing the model.

C. 使用Amazon SageMaker Studio 中的合成少數過採樣技術(SMOTE) 處理並減少偏差。使用Amazon SageMaker JumpStart 發展模型。使用Amazon SageMaker Clarify 在最終確定模型之前檢查模型是否有偏差。

D. Process and reduce bias by using an Amazon SageMaker Studio notebook. Use Amazon SageMaker JumpStart to develop the model. Use Amazon SageMaker Model Monitor to check the model for bias before finalizing the model.

D. 使用Amazon SageMaker Studio 筆記型電腦處理並減少偏差。使用Amazon SageMaker Jump Start 發展模型。使用Amazon SageMaker Model Monitor 在最終確定模型之前檢查模型是否有偏差。

問題# 366

A company has 2,000 retail stores. The company needs to develop a new model to predict demand based on holidays and weather conditions. The model must predict demand in each geographic area where the retail stores are located. Before deploying the newly developed model, the company wants to test the model for 2 to 3 days. The model needs to be robust enough to adapt to supply chain and retail store requirements. Which combination of steps should the company take to meet these requirements with the LEAST operational overhead? (Choose two.)

一家公司有2,000 家零售店。該公司需要開發一種新模型，根據假日和天氣狀況預測需求。此模型必須預測零售店所在每個地理區域的需求。在部署新開發的模型之前，該公司希望測試模型2 到3 天。該模型需要足夠強大，以適應供應鏈和零售店的要求。公司應採取哪些步驟組合來以最少的營運開銷滿足這些要求？（選兩個。）

A. Develop the model by using the Amazon Forecast Prophet model.

A. 使用Amazon Forecast Prophet 模型開發模型。

B. Develop the model by using the Amazon Forecast holidays featurization and weather index.

B. 使用Amazon Forecast 假期特徵和天氣指數開發模型。

C. Deploy the model by using a canary strategy that uses Amazon SageMaker and AWS Step Functions.

C. 使用利用Amazon SageMaker 和AWS Step Functions 的Canary 策略部署模型。

D. Deploy the model by using an A/B testing strategy that uses Amazon SageMaker Pipelines.

D. 使用利用Amazon SageMaker Pipelines 的A/B 測試策略部署模型。

E. Deploy the model by using an A/B testing strategy that uses Amazon SageMaker and AWS Step Functions.

E. 使用利用Amazon SageMaker 和AWS Step Functions 的A/B 測試策略部署模型。

問題# 367

A finance company has collected stock return data for 5,000 publicly traded companies. A financial analyst has a dataset that contains 2,000 attributes for each company. The financial analyst wants to use Amazon SageMaker to identify the top 15 attributes that are most valuable to predict future stock returns. Which solution will meet these requirements with the LEAST operational overhead?

一家金融公司收集了5,000 家上市公司的股票收益資料。一位財務分析師擁有一個資料集，其中包

含每家公司的2,000 個屬性。這位財務分析師希望使用Amazon SageMaker 確定對預測未來股票收益最有價值的15 個屬性。哪種解決方案能夠以最少的營運開銷滿足這些要求？

A. Use the linear learner algorithm in SageMaker to train a linear regression model to predict the stock returns. Identify the most predictive features by ranking absolute coefficient values.

A. 使用SageMaker 中的線性精簡演算法訓練線性迴歸模型來預測股票報酬。透過對絕對係數值進行排序來確定最具預測性的特徵。

B. Use random forest regression in SageMaker to train a model to predict the stock returns. Identify the most predictive features based on Gini importance scores.

B. 使用SageMaker 中的隨機森林迴歸訓練模型來預測股票報酬。根據基尼重要性分數確定最具預測性的特徵。

C. Use an Amazon SageMaker Data Wrangler quick model visualization to predict the stock returns. Identify the most predictive features based on the quick mode's feature importance scores.

C. 使用Amazon SageMaker Data Wrangler 快速模型視覺化來預測股票報酬。根據快速模式的功能重要性得分來確定最具預測性的特徵。

D. Use Amazon SageMaker Autopilot to build a regression model to predict the stock returns. Identify the most predictive features based on an Amazon SageMaker Clarify report.

D. 使用Amazon SageMaker Autopilot 建立迴歸模型來預測股票報酬。根據Amazon SageMaker Clarify 報告確定最具預測性的特徵。
